1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

SCHEM,X646,MLB

X646 MLB

REV	ECN	DESCRIPTION OF REVISION	CK APPD / DATE
			2016-10-06

LAST_MODIFICATION=Thu Oct 6 17:00:29 2016

PAGE	CSA	CONTENTS	SYNC	DATE
1	1	SCHEM,X646,MLB	J95_mlb	10/21/2015
2	2	BOM Configuration	j135_mlb	06/28/2016
3	3	DEBUG: LEDs	ANDREW	02/18/2016
4	4	MECHANICAL: Holes/PD parts	FIYIN	07/15/2016
5	5	CPU & CHIPSET: CPU DMI/PEG/FDI/RSVD	DDRESSLER	02/03/2016
6	6	CPU & CHIPSET: CPU Clock/Misc/JTAG/CFG	DDRESSLER	02/17/2016
7	7	CPU & CHIPSET: DDR4 Interfaces	DDRESSLER	02/03/2016
8	8	CPU & CHIPSET: CPU Power	DDRESSLER	02/03/2016
9	9	CPU & CHIPSET: CPU Ground	DDRESSLER	02/03/2016
10	10	CPU & CHIPSET: CPU Decoupling	DDRESSLER	06/20/2016
11	11	CPU & CHIPSET: Clocks/HDA/JTAG	ANDREW	05/31/2016
12	12	CPU & CHIPSET: PCH RTC/DMI/PM/CPU_Misc	ANDREW	08/29/2016
13	13	CPU & CHIPSET: PCH PCI-E/USB	ANDREW	08/29/2016
14	14	CPU & CHIPSET: PCH GPIO/Misc	ANDREW	10/04/2016
15	15	CPU & CHIPSET: PCH Power/Decoupling	ANDREW	02/19/2016
16	16	CPU & CHIPSET: PCH Grounds	kchooMLB	10/30/2015
17	18	CPU & CHIPSET: XDP	ANDREW	08/29/2016
18	19	CPU & CHIPSET: Chipset Support	FIYIN	05/23/2016
19	20	CPU & CHIPSET: Project Chipset Support	ARMIN	06/22/2016
20	22	DRAM:VREF/VTT EN	DDRESSLER	02/03/2016
21	23	DRAM: SO-DIMM Connector A Slot0	DDRESSLER	02/03/2016
22	24	DRAM: SO-DIMM Connector A Slot1	DDRESSLER	02/03/2016
23	25	DRAM: SO-DIMM CONNECTOR B SLOT0	DDRESSLER	02/03/2016
24	26	DRAM: SO-DIMM CONNECTOR B SLOT1	DDRESSLER	02/03/2016
25	27	DRAM: ALIASES AND BITSWAPS	DDRESSLER	02/03/2016
26	28	USB-C HIGH SPEED 1	armin	07/11/2016
27	29	USB-C HIGH SPEED 2	ARMIN	08/29/2016
28	30	USB-C Support	ARMIN	10/03/2016
29	31	USB-C PORT CONTROLLER A	armin	07/11/2016
30	32	USB-C PORT CONTROLLER B	armin	07/11/2016
31	33	USB-C CONNECTOR	ARMIN	08/29/2016
32	34	USB-C Support 2	armin	07/11/2016
33	35	WIRELESS: Airport/Bluetooth	FIYIN	09/15/2016
34	37	SSD:CONNECTOR	JERRY	02/13/2016
35	38	HDD: CONN & Temp Sense	JERRY	02/01/2016
36	39	ETHERNET: PHY (CAESAR IV)	FIYIN	01/19/2016
37	40	ETHERNET: Support & Connector	FIYIN	09/09/2016
38	41	SD CARD: Connector	FIYIN	01/19/2016
39	42	CAMERA: Controller	ANDREW	08/29/2016
40	43	CAMERA: Controller Support	DDRESSLER	08/27/2016
41	44	DISPLAY: Support	ANDREW	10/04/2016
42	45	DISPLAY: MUXing	ANDREW	02/17/2016
43	46	USB-A: EXTERNAL USB PORTS A & B	FIYIN	06/20/2016
44	47	USB-A: EXTERNAL USB PORTS C & D	FIYIN	06/20/2016
45	50	SMC: Controller	JERRY	06/08/2016
46	51	SMC: Controller Support	JERRY	03/22/2016
47	52	CPU & CHIPSET: SPI and Debug Connector	ANDREW	02/23/2016
48	53	SMC: SMBus Connections	KEVIN	02/04/2016
49	54	SENSORS: I and V Sense	JERRY	06/22/2016
50	55	SENSORS: I and V Sense(Continued)	JERRY	04/15/2016
51	56	SENSORS: Temperature Sensors	JERRY	04/14/2016
52	57	SENSORS: PSU/PCC	JERRY	04/14/2016
53	60	FAN: System Fan	ANDREW	04/06/2016
54	62	AUDIO: HDA BRIDGE	DAVID (AUDIO)	09/07/2016
55	63	AUDIO: AUDIO JACK CODEC	JAMES(AUDIO)	09/05/2016
56	64	AUDIO: LEFT SPKR AMP	DAVID (AUDIO)	09/07/2016
57	65	AUDIO: RIGHT SPKR AMP	DAVID (AUDIO)	09/07/2016
58	66	AUDIO: AUDIO JACK CONN	JAMES(AUDIO)	09/05/2016
59	67	AUDIO: EMPTY	JAMES(AUDIO)	03/30/2016
60	68	AUDIO: EMPTY	JAMES(AUDIO)	02/23/2016
61	69	PLATFORM POWER: Connectors / VReg G3Hot	SUNIL	08/23/2016
62	70	CPU & CHIPSET: CPU CORE VR	SUNIL	09/21/2016
63	71	CPU & CHIPSET: CPU CORE VR (VCC)	SUNIL	04/06/2016
64	72	CPU & CHIPSET: CPU CORE VR (VCCGT)	SUNIL	04/06/2016
65	73	CPU & CHIPSET: CPU VDDQ VR	SUNIL	08/29/2016
66	75	CPU & CHIPSET: CPU VCCIO VR	sunil	07/11/2016
67	76	PLATFORM POWER: 3.3V S5/5V S4 VR	SUNIL	08/23/2016
68	77	CPU & CHIPSET: PCH 1V0 & 1V8 VR	SUNIL	05/31/2016
69	78	CPU & CHIPSET: CPU CORE VR (VCCSA)	SUNIL	08/22/2016
70	81	DISPLAY: LCD Backlight Driver (LP8565)	kchooMLB	10/30/2015
71	82	DISPLAY: Backlight Driver 2	kchooMLB	10/30/2015
72	84	PLATFORM POWER: FET-Controlled S0 and S4	kchooMLB	10/30/2015
73	85	PLATFORM POWER: Regulator Enables	kchooMLB	10/30/2015
74	86	PLATFORM POWER: PM Power Good	kchooMLB	10/30/2015
75	87	GFX: GPU PCIE INTF	PETE	03/07/2016
76	88	GFX: GPU VDDC DECOUPLING	PETE	08/30/2016
77	89	GFX: GPU VMEMIO/VDDCI DECOUPLING	PETE	07/17/2016
78	90	GFX: GPU MEM INTF A/B	PETE	02/04/2016
79	91	GFX: GPU MEM INTF C/D	PETE	02/04/2016
80	92	GFX: GDDR5 FRAME BUFFER A	PETE	08/30/2016
81	93	GFX: GDDR5 FRAME BUFFER B	PETE	08/30/2016
82	94	GFX: GDDR5 FRAME BUFFER C	PETE	08/30/2016
83	95	GFX: GDDR5 FRAME BUFFER D	PETE	08/30/2016
84	96	GFX: DisplayPort INTF	PETE	07/17/2016
85	97	GFX: GPU GPIOS/PLLS/VBIOS ROM	PETE	07/17/2016
86	98	GFX: GPU STRAPS/TEMP SENSORS	PETE	02/19/2016
87	99	GFX: GPU GNDS	PETE	02/16/2016
88	101	GRAPHICS: GPU CORE VR	SUNIL	10/06/2016
89	102	GRAPHICS: GPU CORE VR (PHASES 1-3)	sunil	07/11/2016
90	103	GRAPHICS: GPU CORE VR (PHASES 4-6)	SUNIL	07/15/2016
91	104	GRAPHICS: GPU VDDQ VR	SUNIL	07/18/2016
92	105	GRAPHICS: GPU VDDCI VR	SUNIL	05/31/2016
93	106	GRAPHICS: GPU 0V8 VR	SUNIL	06/06/2016
94	107	GRAPHICS: GPU 1V8 VR	SUNIL	07/14/2016
95	108	GRAPHICS: GPU CORE PCC	SUNIL	08/23/2016
96	109	GRAPHICS: GPU VR MISC	sunil	01/13/2016
97	110	Power Connectors/Aliases	sunil	11/06/2015
98	111	POWER BLOCK DIAGRAM	JERRY	04/15/2016
99	112	Signal Aliases	ANDREW	07/14/2016
100	114	Unused Signal Aliases	ANDREW	04/14/2016
101	115	ALTERNATE PARTS	ANDREW	07/12/2016
102	121	X646 RULE DEFINITIONS	J78_MLB	06/30/2014

DRAWING

TITLE=X646_MLB
ABBREV=DRAWING
LAST_MODIFIED=Thu Oct 6 17:00:29 2016

SCHEM,X646,MLB

DRAWING TITLE: SCHEM, MLB, X646

Apple Inc.

DRAWING NUMBER: 051-01506 SIZE: D

REVISION: 5.19.0

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PAGE: 1 OF 121

SHEET: 1 OF 102

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DRAWING

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-00160	PCBA,DEV,MLB,X646	DEVELOPMENT,X646_DEVEL
685-00106	PCBA,MLB,COMMON PARTS,X646	X646_COMMON
639-03362	PCBA,MLB,CTO,EL XTA,VRAM_SAMSUNG,8GB,X646	X646,MLB_CMNPTS,CPU:CTO,GPU_SKU:EL_XTA,GPU:EL,FB_8G_SAM_EL,EEEE:HNL6,TRANS:DLT,ENET:TRANS
639-03363	PCBA,MLB,CTO,EL XTA,VRAM_HYNIX,8GB,X646	X646,MLB_CMNPTS,CPU:CTO,GPU_SKU:EL_XTA,GPU:EL,FB_8G_HYN_EL,EEEE:HNL7,TRANS:DLT,ENET:TRANS
639-03364	PCBA,MLB,CTO,EL XTA,VRAM_MICRON,8GB,X646	X646,MLB_CMNPTS,CPU:CTO,GPU_SKU:EL_XTA,GPU:EL,FB_8G_ELP_EL,EEEE:HNL8,TRANS:DLT,ENET:TRANS
639-03365	PCBA,MLB,ULT,EL PROA,VRAM_SAMSUNG,4GB,X646	X646,MLB_CMNPTS,CPU:ULT,GPU_SKU:EL_PROA,GPU:EL,FB_4G7_SAM_EL,EEEE:HNL9,TRANS:DLT,ENET:TRANS
639-03366	PCBA,MLB,ULT,EL PROA,VRAM_HYNIX,4GB,X646	X646,MLB_CMNPTS,CPU:ULT,GPU_SKU:EL_PROA,GPU:EL,FB_4G7_HYN_EL,EEEE:HNLC,TRANS:DLT,ENET:TRANS
639-03367	PCBA,MLB,ULT,EL PROA,VRAM_MICRON,4GB,X646	X646,MLB_CMNPTS,CPU:ULT,GPU_SKU:EL_PROA,GPU:EL,FB_4G7_ELP_EL,EEEE:HNLD,TRANS:DLT,ENET:TRANS
639-03368	PCBA,MLB,BEST,EL LEA,VRAM_SAMSUNG,4GB,X646	X646,MLB_CMNPTS,CPU:BEST,GPU_SKU:EL_LEA,GPU:EL,FB_4G7_SAM_EL,EEEE:HNLF,TRANS:TDK,ENET:TRANS
639-03369	PCBA,MLB,BEST,EL LEA,VRAM_HYNIX,4GB,X646	X646,MLB_CMNPTS,CPU:BEST,GPU_SKU:EL_LEA,GPU:EL,FB_4G7_HYN_EL,EEEE:HNLG,TRANS:TDK,ENET:TRANS
639-03370	PCBA,MLB,BEST,EL LEA,VRAM_MICRON,4GB,X646	X646,MLB_CMNPTS,CPU:BEST,GPU_SKU:EL_LEA,GPU:EL,FB_4G7_ELP_EL,EEEE:HNLH,TRANS:TDK,ENET:TRANS
639-03371	PCBA,MLB,BETTER,BF XPA,VRAM_SAMSUNG,2GB,X646	X646,MLB_CMNPTS,CPU:BTR,GPU_SKU:BF_XPA,GPU:BF,FB_4G7_SAM_BF,EEEE:HNLJ,TRANS:TDK,ENET:TRANS
639-03372	PCBA,MLB,BETTER,BF XPA,VRAM_HYNIX,2GB,X646	X646,MLB_CMNPTS,CPU:BTR,GPU_SKU:BF_XPA,GPU:BF,FB_4G7_HYN_BF,EEEE:HNLK,TRANS:TDK,ENET:TRANS
639-03373	PCBA,MLB,BETTER,BF XPA,VRAM_MICRON,2GB,X646	X646,MLB_CMNPTS,CPU:BTR,GPU_SKU:BF_XPA,GPU:BF,FB_4G7_ELP_BF,EEEE:HNLL,TRANS:TDK,ENET:TRANS
639-02813	PCBA,MLB EL,TEST BOARD,X646	X646,MLB_CMNPTS,GPU:EL,EEEE:HD97,TRANS:DLT,ENET:TRANS
639-02970	PCBA,MLB BF,TEST BOARD,X646	X646,MLB_CMNPTS,GPU:BF,EEEE:HHCG,TRANS:TDK

BOM Groups

BOM GROUP	BOM OPTIONS
X646_COMMON	COMMON,ALTERNATE,X646_PROGPARTS,XDP,CPU_DRMOS:CURRENT,GPUCLK:OSC,SSD:Y,VR_BULKCAP:CURRENT,1V8_AUDIO_DLY:N
X646_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG
X646_DEVEL	XDP_CONN,T112
FB_8G_SAM_EL	FB_BD_8G:SAM,FB_AC_8G:SAM,VR_VRAM_SAM_1V55
FB_8G_ELP_EL	FB_BD_8G:ELP,FB_AC_8G:ELP,VR_VRAM_ELP_1V50
FB_8G_HYN_EL	FB_BD_8G:HYN,FB_AC_8G:HYN,VR_VRAM_HYN_1V50
FB_4G7_SAM_EL	FB_BD_4G7:SAM,FB_AC_4G7:SAM,VR_VRAM_SAM_1V50
FB_4G7_ELP_EL	FB_BD_4G7:ELP,FB_AC_4G7:ELP,VR_VRAM_ELP_1V55
FB_4G7_HYN_EL	FB_BD_4G7:HYN,FB_AC_4G7:HYN,VR_VRAM_HYN_1V50
FB_4G7_SAM_BF	FB_BD_4G7:SAM,VR_VRAM_SAM_1V50
FB_4G7_ELP_BF	FB_BD_4G7:ELP,VR_VRAM_ELP_1V55
FB_4G7_HYN_BF	FB_BD_4G7:HYN,VR_VRAM_HYN_1V50

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00344	1	CPU,KBL,QKHB,B0,QS,4.2,91W,1.15,L1151	CPU	CRITICAL	CPU:CTO
337S00345	1	CPU,KBL,QKH9,B0,QS,3.8,91W,1.15,L1151	CPU	CRITICAL	CPU:ULT
337S00343	1	CPU,KBL,QKHB,B0,QS,3.5,65W,1.15,L1151	CPU	CRITICAL	CPU:BEST
337S00340	1	CPU,KBL,QKHC,B0,QS,3.4,65W,1.1,L1151	CPU	CRITICAL	CPU:BTR

SOCKET& CONNECTOR

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S00002	1	CONN,CPU,SOCKET,LGA1151,SKYLAKE,HF	U0500	CRITICAL	
806-07339	1	COVER,SHIELD,USB-C,X635	EMI_SHIELD	CRITICAL	

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00187	1	IC,PCH-H,SKL,SSKGI170,SR2C9,D1,PRQ,BGB37	U1100	CRITICAL	
338S00254	1	IC,TBT,ALPINE RIDGE,SLLSM,PRQ,C1,CSP337	U2800	CRITICAL	
343S0616	1	IC,BCM57766A1,ENET&SD,8X8	U3900	CRITICAL	
353S00961	2	IC,CD3215,ACE,C00,USB PWR SW,BLNK,BGA96	U3100,U3200	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S00734	1	IC,EFI ROM,VTBD,EVT,X646	U5210	CRITICAL	BOOTROM:PROG
335S00006	1	IC,64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S00568	1	IC,SMC-B1,EXTERNAL,VTBD,X646	U5000	CRITICAL	SMC:PROG
338S1214	1	IC,SMC12-B1,40MHZ/50DMIPS,MCU,157BGA	U5000	CRITICAL	SMC:BLANK
341S00740	1	IC,AR,V9.3,EVT,X646	U2890	CRITICAL	TBTROM:PROG
335S00133	1	IC,SPI FLASH ROM,8MBIT,50MHZ,USON8	U2890	CRITICAL	TBTROM:BLANK
341S3912	1	IC,ENET ROM,NUMONYX,V1.15,J16/J16G/J17	U3990	CRITICAL	CIVROM:PROG
335S0854	1	1-MBIT,SPI FLASH ROM SOIC, HF	U3990	CRITICAL	CIVROM:BLANK
341S00667	1	IC,CAMERA,FLASH,V72.52,X646	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S00555	1	IC,GPU ROM,ELM,VTBD,X646	U9700	CRITICAL	GPUROM:PROG
335S0724	1	1MBIT SERIAL FLASH 2X3X0.6MM	U9700	CRITICAL	GPUROM:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL,MLB,2D	EEEE_HNL6	CRITICAL	EEEE:HNL6
825-7896	1	LABEL,MLB,2D	EEEE_HNL7	CRITICAL	EEEE:HNL7
825-7896	1	LABEL,MLB,2D	EEEE_HNL8	CRITICAL	EEEE:HNL8
825-7896	1	LABEL,MLB,2D	EEEE_HNL9	CRITICAL	EEEE:HNL9
825-7896	1	LABEL,MLB,2D	EEEE_HNLC	CRITICAL	EEEE:HNLC
825-7896	1	LABEL,MLB,2D	EEEE_HNLD	CRITICAL	EEEE:HNLD
825-7896	1	LABEL,MLB,2D	EEEE_HNLF	CRITICAL	EEEE:HNLF
825-7896	1	LABEL,MLB,2D	EEEE_HNLG	CRITICAL	EEEE:HNLG
825-7896	1	LABEL,MLB,2D	EEEE_HNLH	CRITICAL	EEEE:HNLH
825-7896	1	LABEL,MLB,2D	EEEE_HNLJ	CRITICAL	EEEE:HNLJ
825-7896	1	LABEL,MLB,2D	EEEE_HNLK	CRITICAL	EEEE:HNLK
825-7896	1	LABEL,MLB,2D	EEEE_HNLL	CRITICAL	EEEE:HNLL
825-7896	1	LABEL,MLB,2D	EEEE_HD97	CRITICAL	EEEE:HD97
825-7896	1	LABEL,MLB,2D	EEEE_HHCG	CRITICAL	EEEE:HHCG

ALTERNATE PARTS TABLE LOCATED ON CSA 115

J135 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-01506	1	SCH,MLB,X646	SCH1	CRITICAL	X646
820-00609	1	PCBF,MLB,X646	PCB1	CRITICAL	X646
685-00106	1	PCBA,MLB COMMON PARTS,X646	CMNPTS	CRITICAL	MLB_CMNPTS

GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00277	1	IC,GPU,AMD,ELLESMERE,XTA,Q5,A1,LF,BGA1401	U8700	CRITICAL	GPU_SKU:EL_XTA
337S00279	1	IC,GPU,AMD,ELLESMERE,PROA,Q5,A1,LF,BGA1401	U8700	CRITICAL	GPU_SKU:EL_PROA
337S00347	1	IC,GPU,AMD,ELLESMERE,LEA,Q5,A1,LF,BGA1401	U8700	CRITICAL	GPU_SKU:EL_LEA
337S00346	1	IC,GPU,AMD,BAFFIN,XPA,A1,LF,Q5,BGA1401	U8700	CRITICAL	GPU_SKU:BF_XPA
333S00100	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_8G:HYN
333S00100	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_8G:HYN
333S00075	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_8G:ELP
333S00075	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_8G:ELP
333S00074	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_8G:SAM
333S00074	4	IC,GDDR5,8GBIT,7GBPS,20NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_8G:SAM
333S00043	4	IC,GDDR5,4GBIT,7GBPS,25NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_4G7:HYN
333S00043	4	IC,GDDR5,4GBIT,7GBPS,25NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_4G7:HYN
333S00044	4	IC,GDDR5,4GBIT,7GBPS,25NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_4G7:ELP
333S00044	4	IC,GDDR5,4GBIT,7GBPS,25NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_4G7:ELP
333S00078	4	IC,GDDR5,4GBIT,7GBPS,20NM,BGA170	U9300,U9350,U9500,U9550	CRITICAL	FB_BD_4G7:SAM
333S00078	4	IC,GDDR5,4GBIT,7GBPS,20NM,BGA170	U9200,U9250,U9400,U9450	CRITICAL	FB_AC_4G7:SAM

SYNC_MASTER=j135_m1b SYNC_DATE=06/28/2016

PAGE TITLE

BOM Configuration

Apple Inc.

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PAGE 2 OF 121

SHEET 2 OF 102

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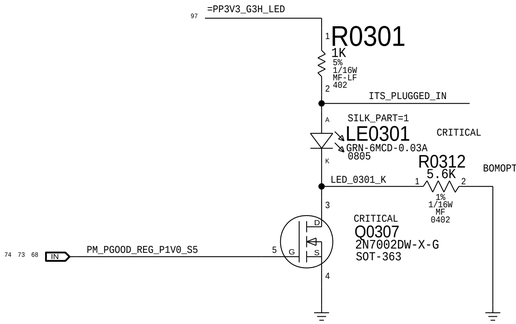
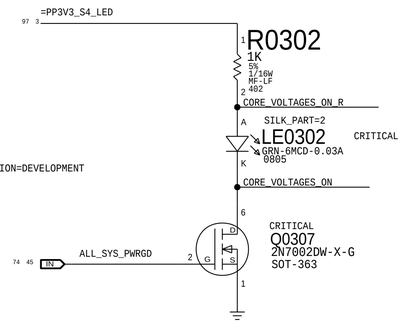
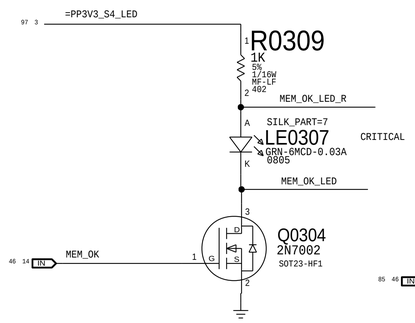
S5 Pwr LED



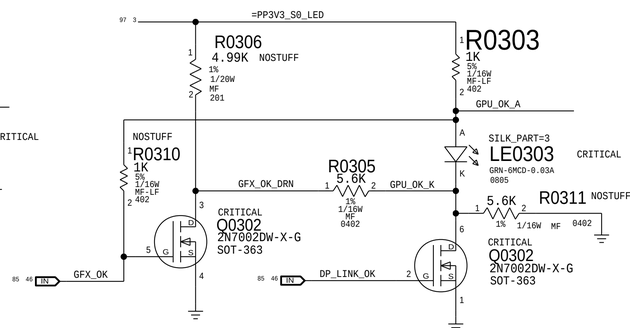
ALL_SYS_PWRGD LED



MEM_OK LED

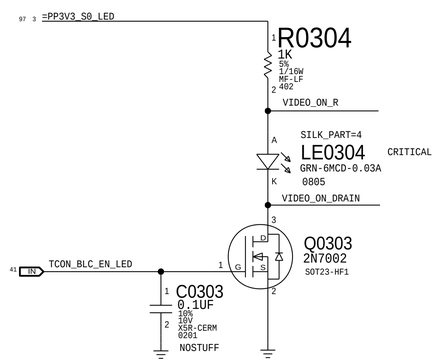


GPU OK LED

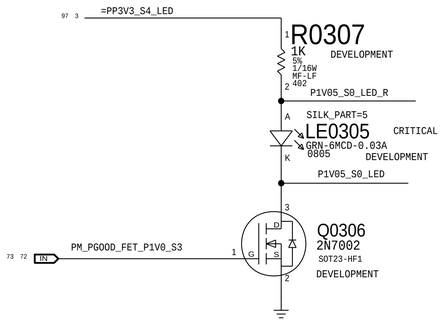


THIS LED IS DRIVEN BY GPU:
GFX_OK HIGH LIGHTS THE LED DIMLY
DP_LINK_OK HIGH LIGHTS THE LED FULLY

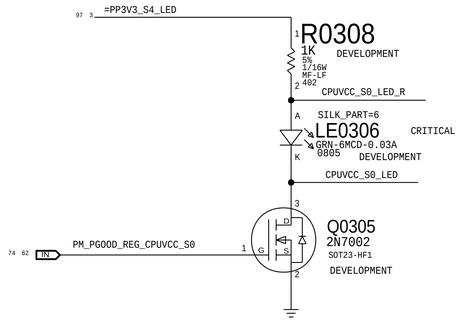
TCON OK LED



P1V0_S3_PWRGD LED



CPUVCC_S0_PWRGD LED



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=ANDREW
SYNC_DATE=02/18/2016

PAGE TITLE
DEBUG: LEDs

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SIZE D
REVISION 5.19.0
BRANCH
PAGE 3 OF 121
SHEET 3 OF 102

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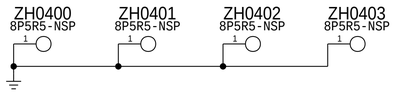
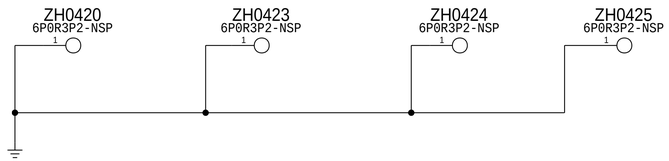
CPU Heatsink

4MM PLATED HOLES (998-4158)



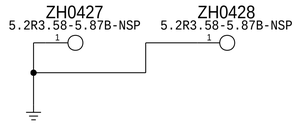
GPU HEATSINK MOUNTING FEATURES

(998-5013. PLATED HOLE, 3.2MM DIA, 6MM PAD TOP/BOT)



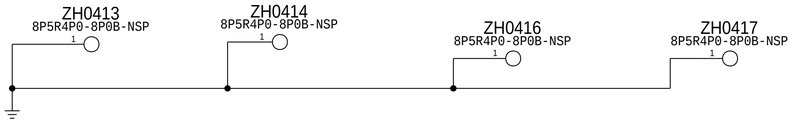
GPU HEATPIPE MOUNTING FEATURES

998-06191 (PLATED HOLES, 3.58MM INNER DIAMETER)

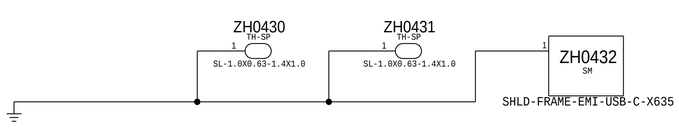


Rear Cover

998-5014 (PLATED HOLES, 4MM DRILL, 8.5MM TOP, 8MM BOT)

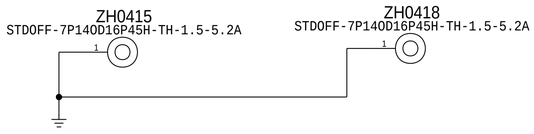


USB-C Alignment Holes and EMI Can



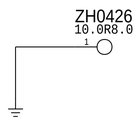
Rear Cover

860-5674 (PCB STANDOFF)



HEATPIPE MTG HOLES

998-5527 (PLATED HOLES, 8MM DIA, 10MM PAD)

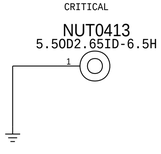


WIRELESS CARD MTG HOLES

998-06186 (PLATED HOLES, 3.3mm INNER DIAMETER, 5mm Top pad)

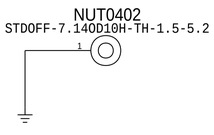
SSD STANDOFF

APN: 860-00314



LOL Boss

860-5675 (PLATED HOLE, 8.41mm pad Top, 8mm pad Bot)



BOM_COST_GROUP=MECHANICALS

SYNC_MASTER=SYDYA SYNC_DATE=07/15/2019

PAGE TITLE
MECHANICAL: Holes/PD parts

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REVISION 5.19.0

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PAGE 4 OF 121

SHEET 4 OF 102

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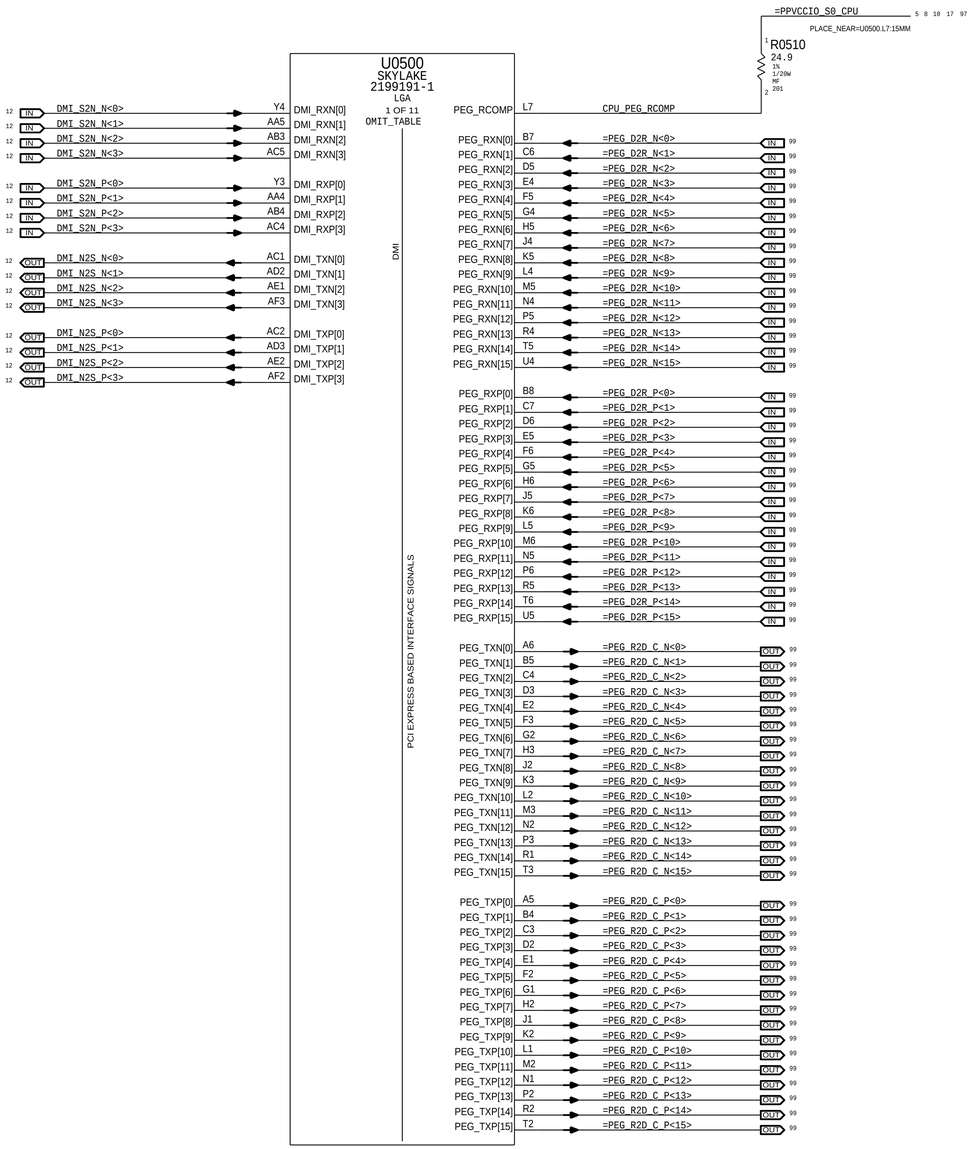

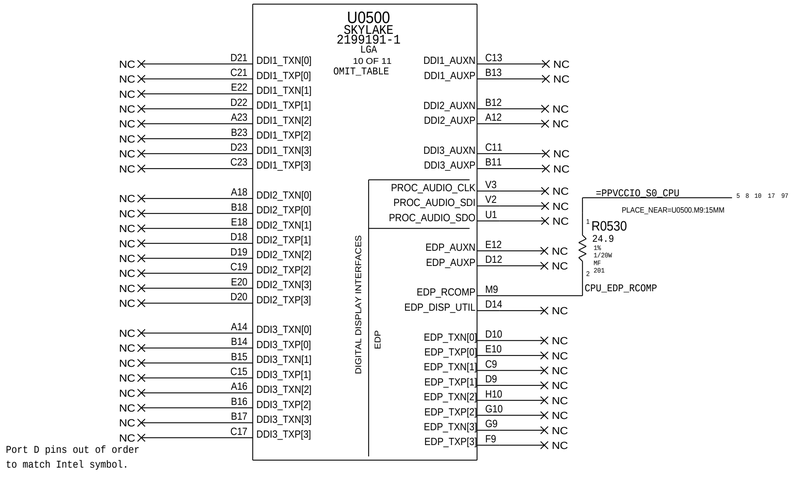

Port D pins out of order
to match Intel symbol.

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DRESSLER SYNC_DATE=02/03/2016

PAGE TITLE
CPU & CHIPSET: CPU DMI/PEG/FDI/RSVD

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SIZE D
REVISION 5.19.0
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PAGE 5 OF 121
SHEET 5 OF 102

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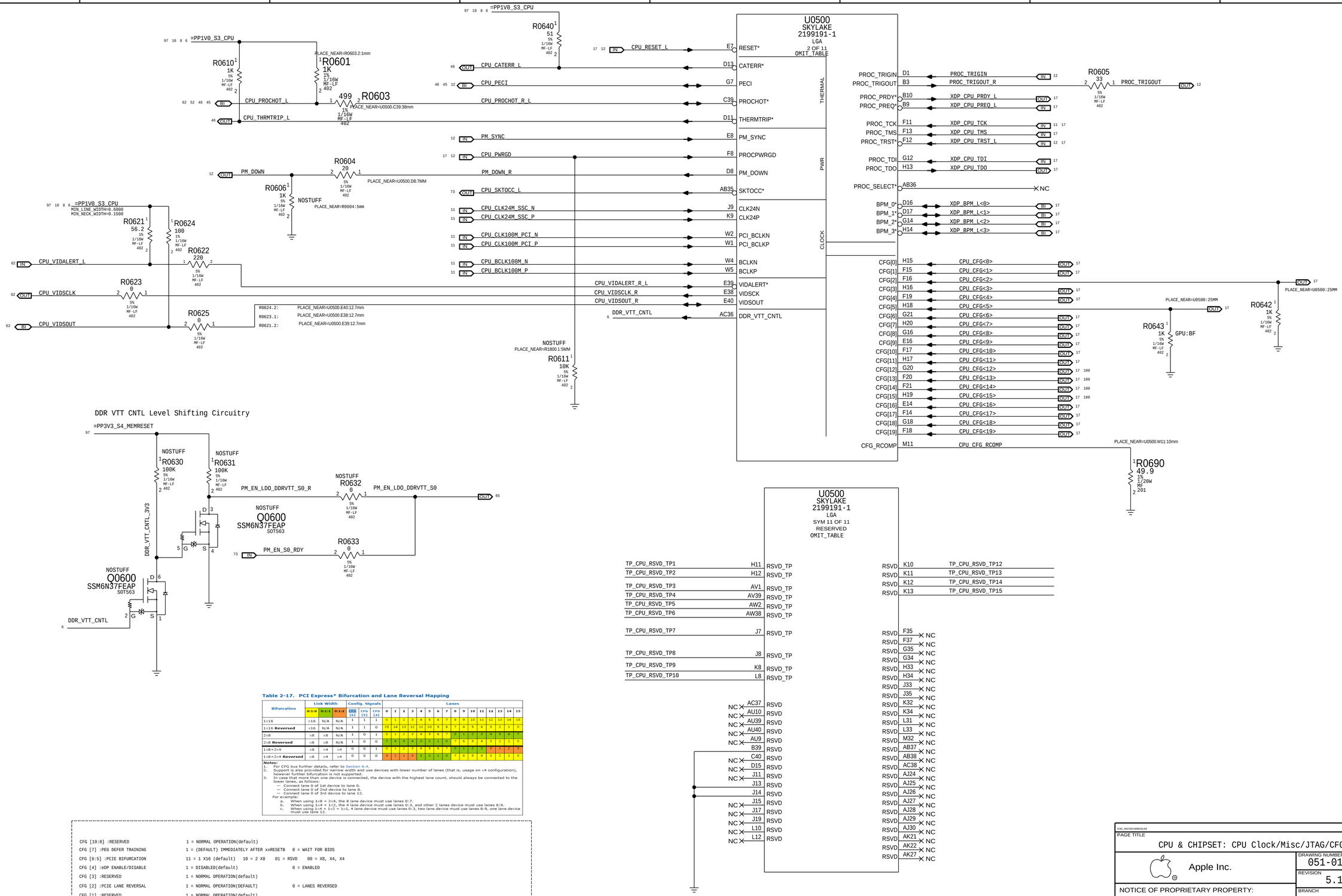
U0500
SKYLAKE
2199191-1
LGA
2 OF 11
OMIT_TABLE
THERMAL
PWR
CLOCK
RESET*
CATERR*
PECI
PROCHOT*
THERMTRIP*
PM_SYNC
PROCPWRGD
PM_DOWN
SKTOCC*
CLK24N
CLK24P
PCI_BCLKN
PCI_BCLKP
BCLKN
BCLKP
VIDALERT*
VIDSCK
VIDSOUT
DDR_VTT_CNTL
CPU_RESET_L
CPU_CATERR_L
CPU_PECI
CPU_PROCHOT_R_L
CPU_THRMTRIP_L
PM_SYNC
CPU_PWRGD
PM_DOWN_R
CPU_SKTOCC_L
CPU_CLK24M_SSC_N
CPU_CLK24M_SSC_P
CPU_CLK100M_PCI_N
CPU_CLK100M_PCI_P
CPU_BCLK100M_N
CPU_BCLK100M_P
CPU_VIDALERT_R_L
CPU_VIDSCLK_R
CPU_VIDSOUT_R
DDR_VTT_CNTL
=PP1V0_S3_CPU
R0640 51
R0610 1K
R0601 1K
R0603 499
R0604 20
R0606 1K
NOSTUFF
PLACE_NEAR=R0603.2:1mm
PLACE_NEAR=U0500.C39:38mm
PLACE_NEAR=U0500.D8:7MM
PLACE_NEAR=R0604:5mm
R0621 56.2
R0624 100
R0622 220
R0623 0
R0625 0
CPU_VIDALERT_L
CPU_VIDSCLK
CPU_VIDSOUT
R0624.2: PLACE_NEAR=U0500.E40:12.7mm
R0623.1: PLACE_NEAR=U0500.E38:12.7mm
R0621.2: PLACE_NEAR=U0500.E39:12.7mm
R0611 10K
PLACE_NEAR=R1800.1:5MM
PROC_TRIGIN
PROC_TRIGOUT
PROC_PRDY*
PROC_PREQ*
PROC_TCK
PROC_TMS
PROC_TRST*
PROC_TDI
PROC_TDO
PROC_SELECT*
BPM_0*
BPM_1*
BPM_2*
BPM_3*
PROC_TRIGOUT_R
XDP_CPU_PRDY_L
XDP_CPU_PREQ_L
XDP_CPU_TCK
XDP_CPU_TMS
XDP_CPU_TRST_L
XDP_CPU_TDI
XDP_CPU_TDO
XDP_BPM_L<0>
XDP_BPM_L<1>
XDP_BPM_L<2>
XDP_BPM_L<3>
R0605 33
CFG[0]
CFG[1]
CFG[2]
CFG[3]
CFG[4]
CFG[5]
CFG[6]
CFG[7]
CFG[8]
CFG[9]
CFG[10]
CFG[11]
CFG[12]
CFG[13]
CFG[14]
CFG[15]
CFG[16]
CFG[17]
CFG[18]
CFG[19]
CFG_RCOMP
R0642 1K
R0643 1K
GPU:BF
PLACE_NEAR=U0500:25MM
R0690 49.9
PLACE_NEAR=U0500.M11:10mm
DDR VTT CNTL Level Shifting Circuitry
=PP3V3_S4_MEMRESET
R0630 100K
R0631 100K
R0632 0
R0633 0
Q0600 SSM6N37FEAP SOT563
DDR_VTT_CNTL_3V3
PM_EN_LDO_DDRVTT_S0_R
PM_EN_LDO_DDRVTT_S0
PM_EN_S0_RDY
U0500
SKYLAKE
2199191-1
LGA
SYM 11 OF 11
RESERVED
OMIT_TABLE
TP_CPU_RSVD_TP1
TP_CPU_RSVD_TP2
TP_CPU_RSVD_TP3
TP_CPU_RSVD_TP4
TP_CPU_RSVD_TP5
TP_CPU_RSVD_TP6
TP_CPU_RSVD_TP7
TP_CPU_RSVD_TP8
TP_CPU_RSVD_TP9
TP_CPU_RSVD_TP10
TP_CPU_RSVD_TP12
TP_CPU_RSVD_TP13
TP_CPU_RSVD_TP14
TP_CPU_RSVD_TP15
Table 2-17. PCI Express* Bifurcation and Lane Reversal Mapping
Bifurcation
Link Width
Config. Signals
Lanes
1x16
1x16 Reversed
2x8
2x8 Reversed
1x8+2x4
1x8+2x4 Reversed
Notes:
1. For CFG bus further details, refer to Section 6.4.
2. Support is also provided for narrow width and use devices with lower number of lanes (that is, usage on x4 configuration), however further bifurcation is not supported.
3. In case that more than one device is connected, the device with the highest lane count, should always be connected to the lower lanes, as follows:
— Connect lane 0 of 1st device to lane 0.
— Connect lane 0 of 2nd device to lane 8.
— Connect lane 0 of 3rd device to lane 12.
For example:
a. When using 1x8 + 2x4, the 8 lane device must use lanes 0:7.
b. When using 1x4 + 1x2, the 4 lane device must use lanes 0:3, and other 2 lanes device must use lanes 8:9.
c. When using 1x4 + 1x2 + 1x1, 4 lane device must use lanes 0:3, two lane device must use lanes 8:9, one lane device must use lane 12.
CFG [19:8] :RESERVED
CFG [7] :PEG DEFER TRAINING
CFG [6:5] :PCIE BIFURCATION
CFG [4] :eDP ENABLE/DISABLE
CFG [3] :RESERVED
CFG [2] :PCIE LANE REVERSAL
CFG [1] :RESERVED
CFG [0] :STALL RESET SEQUENCE
1 = NORMAL OPERATION(default)
1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
11 = 1 X16 (default) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
1 = DISABLED(default) 0 = ENABLED
1 = NORMAL OPERATION(default)
1 = NORMAL OPERATION(DEFAULT) 0 = LANES REVERSED
1 = NORMAL OPERATION(default)
1 = NORMAL OPERATION(default) 0 = STALL
**ALL CFG PINS DEFAULT TO HIGH WITH 3K INTERNAL PULLUPS UNLESS PULLED LOW ON BOARD.
BOM_COST_GROUP=CPU & CHIPSET
PAGE TITLE
CPU & CHIPSET: CPU Clock/Misc/JTAG/CFG
Apple Inc.
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SIZE D
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PAGE 6 OF 121
SHEET 6 OF 102
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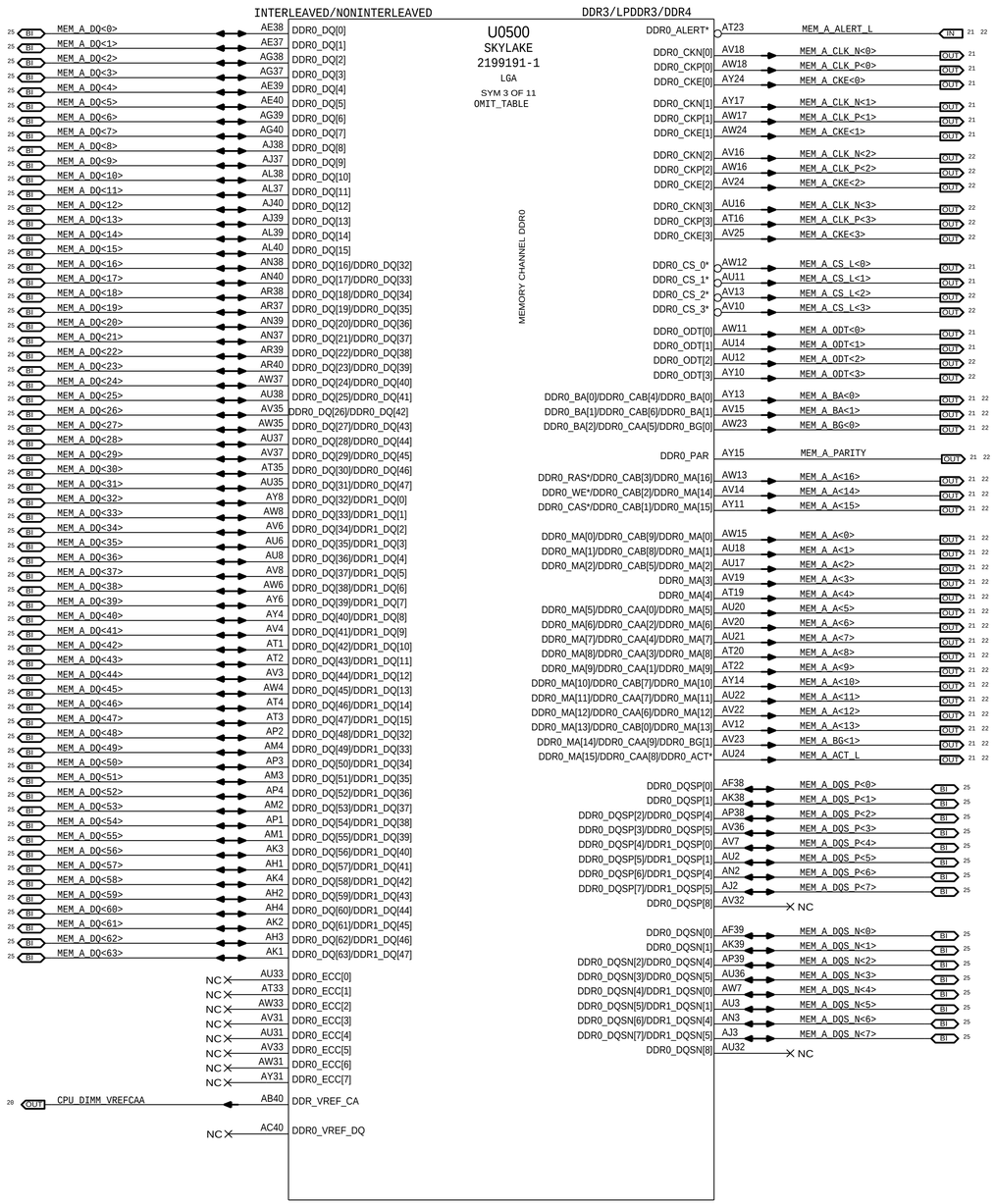

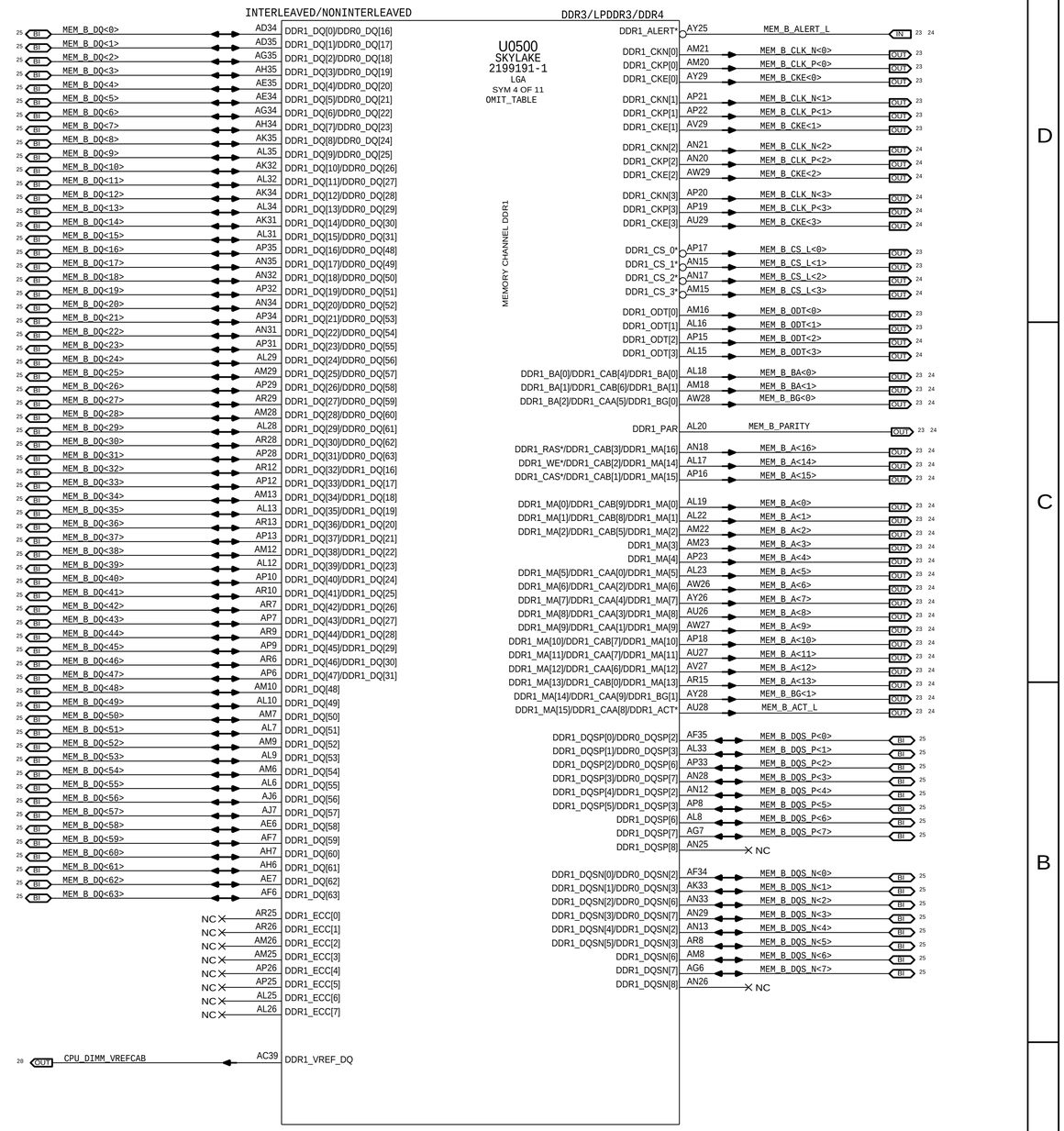

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	CPU & CHIPSET: DDR4 Interfaces
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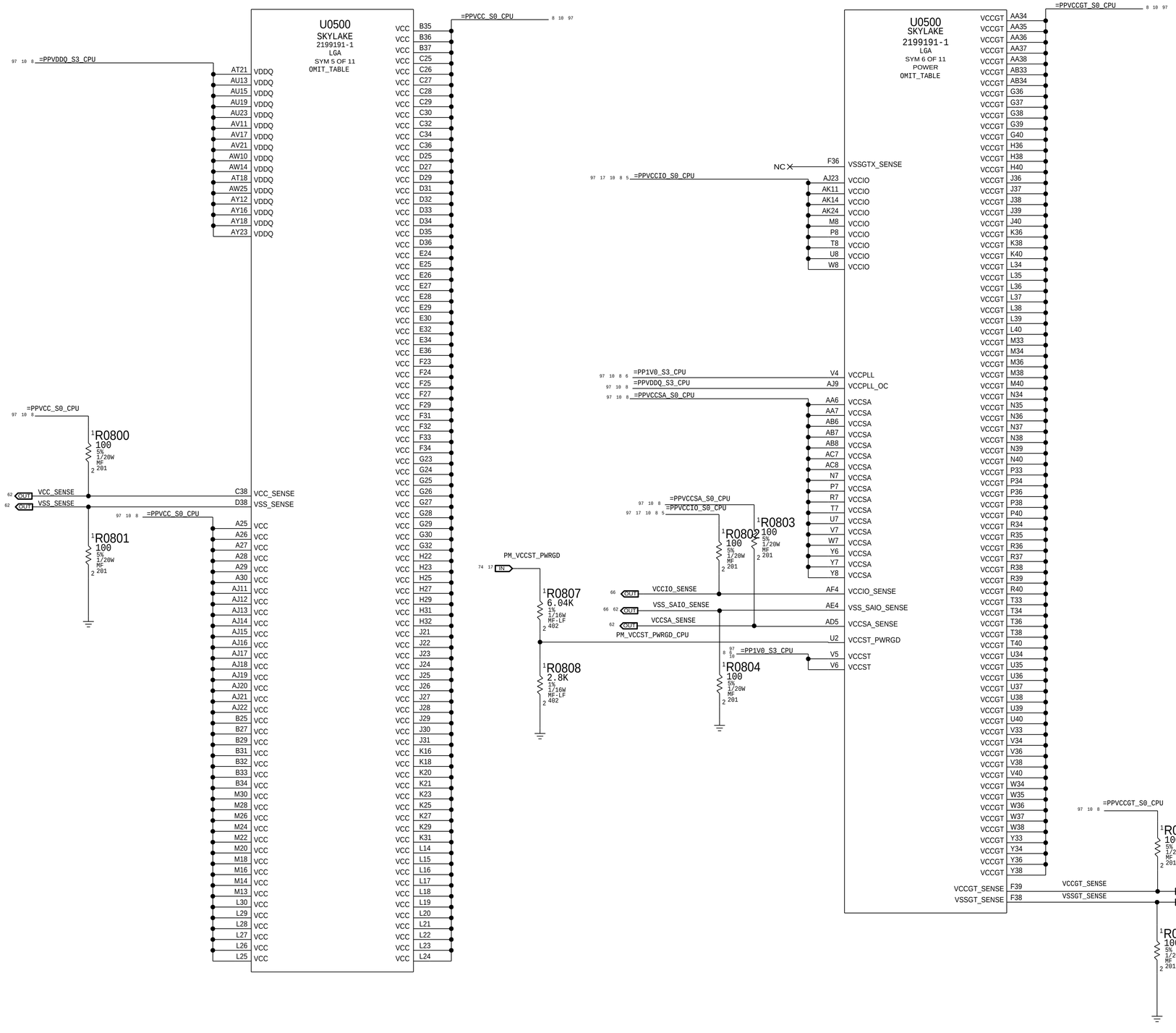
U0500
SKYLAKE
2199191-1
LGA
SYM 5 OF 11
OMIT_TABLE
=PPVDDQ_S3_CPU
AT21 VDDQ
AU13 VDDQ
AU15 VDDQ
AU19 VDDQ
AU23 VDDQ
AV11 VDDQ
AV17 VDDQ
AV21 VDDQ
AW10 VDDQ
AW14 VDDQ
AT18 VDDQ
AW25 VDDQ
AY12 VDDQ
AY16 VDDQ
AY18 VDDQ
AY23 VDDQ
=PPVCC_S0_CPU
B35 VCC
B36 VCC
B37 VCC
C25 VCC
C26 VCC
C27 VCC
C28 VCC
C29 VCC
C30 VCC
C32 VCC
C34 VCC
C36 VCC
D25 VCC
D27 VCC
D29 VCC
D31 VCC
D32 VCC
D33 VCC
D34 VCC
D35 VCC
D36 VCC
E24 VCC
E25 VCC
E26 VCC
E27 VCC
E28 VCC
E29 VCC
E30 VCC
E32 VCC
E34 VCC
E36 VCC
F23 VCC
F24 VCC
F25 VCC
F27 VCC
F29 VCC
F31 VCC
F32 VCC
F33 VCC
F34 VCC
G23 VCC
G24 VCC
G25 VCC
G26 VCC
G27 VCC
G28 VCC
G29 VCC
G30 VCC
G32 VCC
H22 VCC
H23 VCC
H25 VCC
H27 VCC
H29 VCC
H31 VCC
H32 VCC
J21 VCC
J22 VCC
J23 VCC
J24 VCC
J25 VCC
J26 VCC
J27 VCC
J28 VCC
J29 VCC
J30 VCC
J31 VCC
K16 VCC
K18 VCC
K20 VCC
K21 VCC
K23 VCC
K25 VCC
K27 VCC
K29 VCC
K31 VCC
L14 VCC
L15 VCC
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L17 VCC
L18 VCC
L19 VCC
L20 VCC
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L22 VCC
L23 VCC
L24 VCC
A25 VCC
A26 VCC
A27 VCC
A28 VCC
A29 VCC
A30 VCC
AJ11 VCC
AJ12 VCC
AJ13 VCC
AJ14 VCC
AJ15 VCC
AJ16 VCC
AJ17 VCC
AJ18 VCC
AJ19 VCC
AJ20 VCC
AJ21 VCC
AJ22 VCC
B25 VCC
B27 VCC
B29 VCC
B31 VCC
B32 VCC
B33 VCC
B34 VCC
M30 VCC
M28 VCC
M26 VCC
M24 VCC
M22 VCC
M20 VCC
M18 VCC
M16 VCC
M14 VCC
M13 VCC
L30 VCC
L29 VCC
L28 VCC
L27 VCC
L26 VCC
L25 VCC
=PPVCC_S0_CPU
R0800 100 5% 1/20W MF 201
VCC_SENSE
VSS_SENSE
C38 VCC_SENSE
D38 VSS_SENSE
R0801 100 5% 1/20W MF 201
U0500
SKYLAKE
2199191-1
LGA
SYM 6 OF 11
POWER
OMIT_TABLE
=PPVCCGT_S0_CPU
AA34 VCCGT
AA35 VCCGT
AA36 VCCGT
AA37 VCCGT
AA38 VCCGT
AB33 VCCGT
AB34 VCCGT
G36 VCCGT
G37 VCCGT
G38 VCCGT
G39 VCCGT
G40 VCCGT
H36 VCCGT
H38 VCCGT
H40 VCCGT
J36 VCCGT
J37 VCCGT
J38 VCCGT
J39 VCCGT
J40 VCCGT
K36 VCCGT
K38 VCCGT
K40 VCCGT
L34 VCCGT
L35 VCCGT
L36 VCCGT
L37 VCCGT
L38 VCCGT
L39 VCCGT
L40 VCCGT
M33 VCCGT
M34 VCCGT
M36 VCCGT
M38 VCCGT
M40 VCCGT
N34 VCCGT
N35 VCCGT
N36 VCCGT
N37 VCCGT
N38 VCCGT
N39 VCCGT
N40 VCCGT
P33 VCCGT
P34 VCCGT
P36 VCCGT
P38 VCCGT
P40 VCCGT
R34 VCCGT
R35 VCCGT
R36 VCCGT
R37 VCCGT
R38 VCCGT
R39 VCCGT
R40 VCCGT
T33 VCCGT
T34 VCCGT
T36 VCCGT
T38 VCCGT
T40 VCCGT
U34 VCCGT
U35 VCCGT
U36 VCCGT
U37 VCCGT
U38 VCCGT
U39 VCCGT
U40 VCCGT
V33 VCCGT
V34 VCCGT
V36 VCCGT
V38 VCCGT
V40 VCCGT
W34 VCCGT
W35 VCCGT
W36 VCCGT
W37 VCCGT
W38 VCCGT
Y33 VCCGT
Y34 VCCGT
Y36 VCCGT
Y38 VCCGT
VCCGT_SENSE F39
VSSGT_SENSE F38
F36 VSSGTX_SENSE
NC
=PPVCCIO_S0_CPU
AJ23 VCCIO
AK11 VCCIO
AK14 VCCIO
AK24 VCCIO
M8 VCCIO
P8 VCCIO
T8 VCCIO
U8 VCCIO
W8 VCCIO
=PP1V0_S3_CPU
=PPVDDQ_S3_CPU
=PPVCCSA_S0_CPU
V4 VCCPLL
AJ9 VCCPLL_OC
AA6 VCCSA
AA7 VCCSA
AB6 VCCSA
AB7 VCCSA
AB8 VCCSA
AC7 VCCSA
AC8 VCCSA
N7 VCCSA
P7 VCCSA
R7 VCCSA
T7 VCCSA
U7 VCCSA
V7 VCCSA
W7 VCCSA
Y6 VCCSA
Y7 VCCSA
Y8 VCCSA
=PPVCCSA_S0_CPU
=PPVCCIO_S0_CPU
R0802 100 5% 1/20W MF 201
R0803 100 5% 1/20W MF 201
PM_VCCST_PWRGD
R0807 6.04K 1% 1/10W MF-LF 402
R0808 2.8K 1% 1/10W MF-LF 402
VCCIO_SENSE
VSS_SAIO_SENSE
VCCSA_SENSE
PM_VCCST_PWRGD_CPU
AF4 VCCIO_SENSE
AE4 VSS_SAIO_SENSE
AD5 VCCSA_SENSE
U2 VCCST_PWRGD
=PP1V0_S3_CPU
V5 VCCST
V6 VCCST
R0804 100 5% 1/20W MF 201
=PPVCCGT_S0_CPU
R0805 100 5% 1/20W MF 201
VCCGT_SENSE
VSSGT_SENSE
R0806 100 5% 1/20W MF 201
BOM_COST_GROUP=CPU & CHIPSET
SYNC_MASTER=DORESSLER
SYNC_DATE=02/03/2016
PAGE TITLE
CPU & CHIPSET: CPU Power
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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
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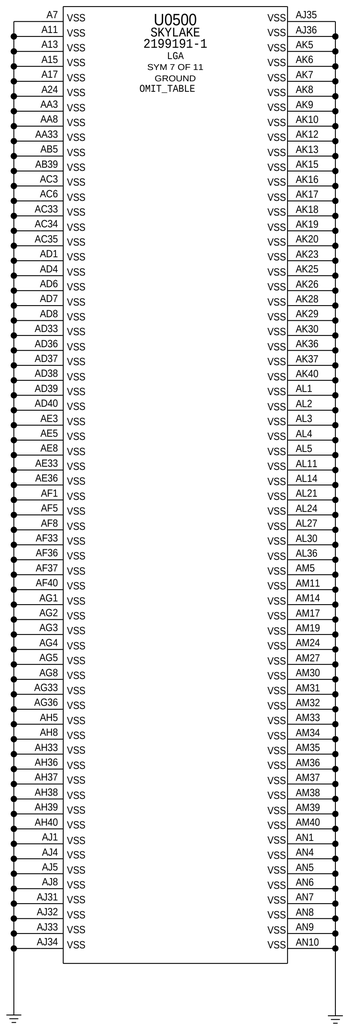

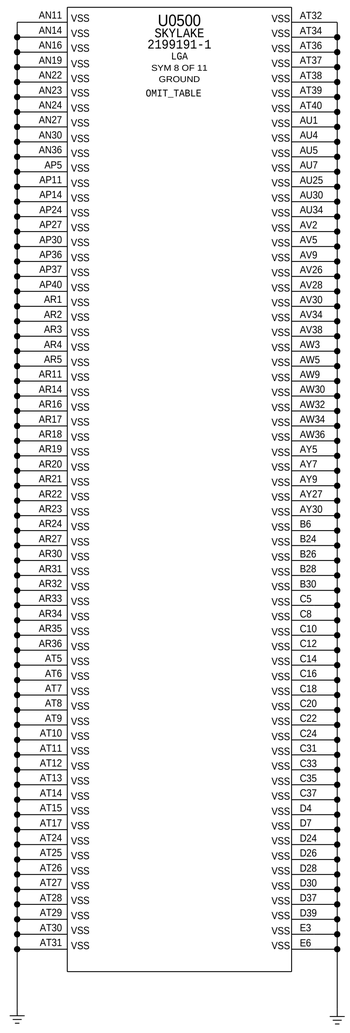

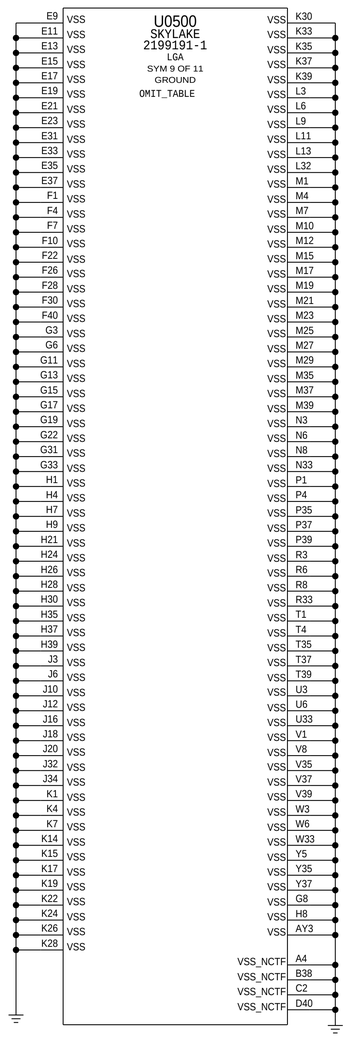

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DORESSLER SYNC_DATE=02/03/2016

PAGE TITLE

CPU & CHIPSET: CPU Ground

Apple Inc.

DRAWING NUMBER 051-01506

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PAGE 9 OF 121

SHEET 9 OF 102

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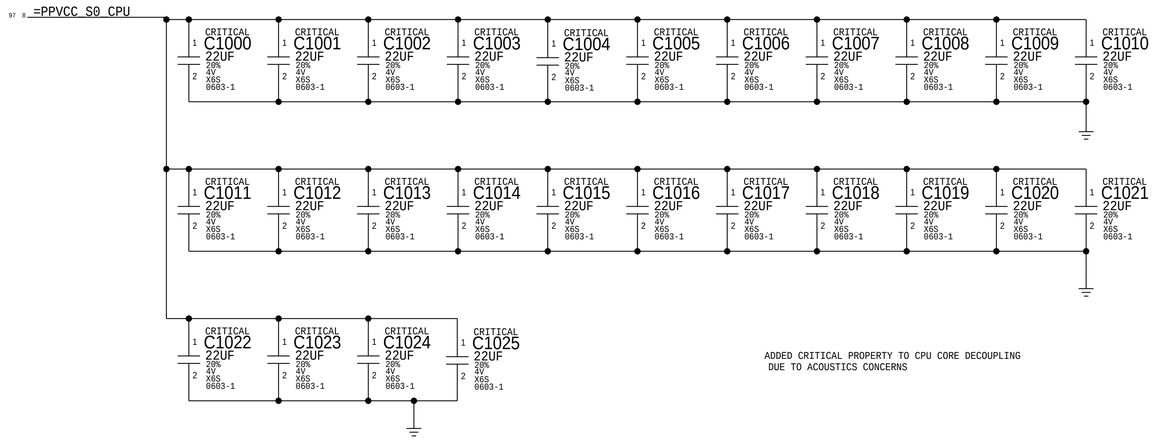
CPU VCORE DECOUPLING

Intel Recommendation:12x 22UF 0805 (top side cavity)
6x 22UF 0603 (top side cavity)
5x 22UF 0805 (top side outside cavity)

Apple Implementation:26x 22UF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



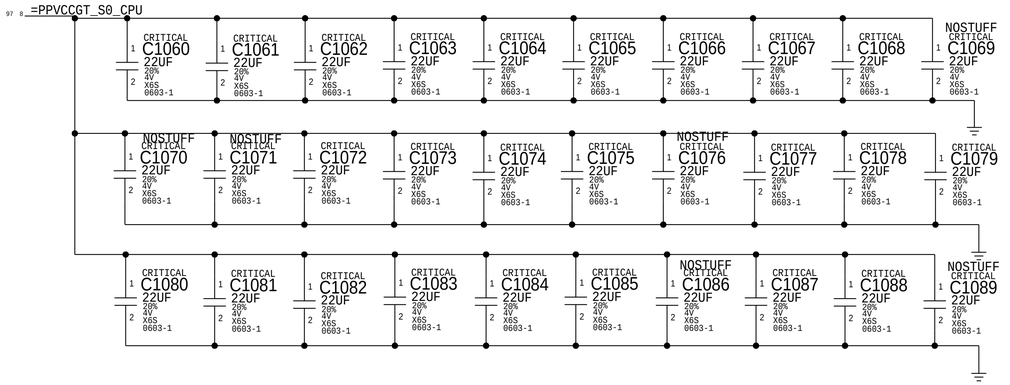
ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

CPU GT DECOUPLING

INTEL RECOMMENDATION:8X 47UF 0805 (TOP SIDE CAVITY)
4x 47UF 0805 (top side outside cavity)

Apple Implementation: 30x 22uF 0603

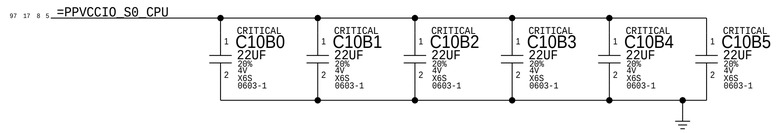
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



CPU VCCIO DECOUPLING

Intel Recommendation:5x 22UF 0603 (top side cavity)
1x 22UF 0805 (top side cavity)

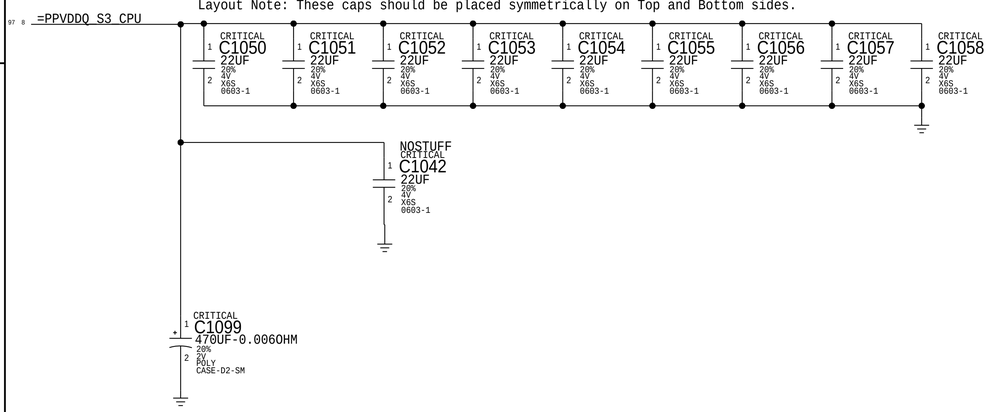
Apple Implementation:(following Intel recommendation w/ 0603)
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



Memory (CPU VCCDDR) DECOUPLING

INTEL RECOMMENDATION:4X 22UF 0603 (TOP SIDE OUTSIDE SOCKET CAP)

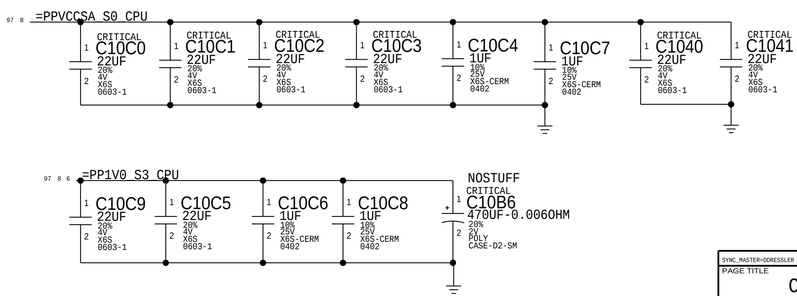
Apple Implementation:9x 22UF 0603 (J78 carry over)
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



CPU VCCSA / VCCST+VCCPLL DECOUPLING

Intel Recommendation:2x 22UF 0603 near top side cavity

Apple Implementation: VCCST/VCCPLL: 1X 22UF 0603/2X 1UF 0402
Layout Note: These caps should be placed on top side cavity.



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=DRESSLER SYNC_DATE=06/20/2016

PAGE TITLE
CPU & CHIPSET: CPU Decoupling

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SIZE D
REVISION 5.19.0
BRANCH
PAGE 10 OF 121
SHEET 10 OF 102

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SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

U1100
SKL-PCH-H
TBD
FCBGA
SYM 1 OF 12

CRITICAL
OMIT_TABLE

RSVD & TP PINS

NC C1 RSVD
NC D1 RSVD
NC N29 RSVD
NC N31 RSVD
NC P24 RSVD
NC P27 RSVD
NC P29 RSVD
NC P31 RSVD
NC R24 RSVD
NC R27 RSVD
NC U13 RSVD
NC W13 RSVD
NC AB13 RSVD
NC AE17 RSVD
NC AF17 RSVD
NC AG14 RSVD
NC AG15 RSVD
NC AN29 RSVD
NC AR22 RSVD
NC BD1 RSVD
NC BE2 RSVD

TP_PCH_1 AN17 TP1
100 TP_PCH_2 AR19 TP2

JTAG

17 11 OUT PCH_ITP_PMODE AT2 ITP_PMODE
17 OUT XDP_PCH_TCK AN3 JTAG_TCK
17 IN XDP_PCH_TDI AP2 JTAG_TDI
17 OUT XDP_PCH_TDO AP1 JTAG_TDO
17 IN XDP_PCH_TMS AR2 JTAG_TMS
11 PCH_JTAGX AR3 JTAGX

AUDIO

HDA_BCLK BA9 HDA_BIT_CLK_R — R1109 33 5% 1/20W MF 201 — HDA_BIT_CLK OUT 54
HDA_SYNC BD9 HDA_SYNC_R — R1110 33 5% 1/20W MF 201 — HDA_SYNC OUT 54
PLACE_NEAR=U1100.BA9:5MM
PLACE_NEAR=U1100.BD9:5.3MM
HDA_SDI0 BE7 HDA_SDIN0 IN 54
HDA_SDI1 BC8 NC_HDA_SDIN1
HDA_SDO BB7 HDA_SDOUT_R — R1111 33 5% 1/20W MF 201 — HDA_SDOUT OUT 18 54
HDA_RST* BD8 HDA_RST_R_L — R1112 33 5% 1/20W MF 201 — HDA_RST_L OUT 54
PLACE_NEAR=U1100.BB7:6MM
PLACE_NEAR=U1100.BD8:7MM

DISPA_BCLK AM2 NC
DISPA_SDI AN2 NC
DISPA_SDO AM1 NC

GPP_D5/I2S0_SFRM AJ33 NC
GPP_D6/I2S0_TXD AM43 NC
GPP_D7/I2S0_RXD AN42 NC
GPP_D8/I2S0_SCLK AL42 NC

GPP_D17/DMIC_CLK1 AJ42 GPP_D17 — TP-PG TP — TP1100
GPP_D18/DMIC_DATA1 AJ38 GPP_D18 — TP-PG TP — TP1101
GPP_D19/DMIC_CLK0 AJ35 GPP_D19 — TP-PG TP — TP1102
GPP_D20/DMIC_DATA0 AH44 GPP_D20 — TP-PG TP — TP1103

R1117 0 5% 1/20W MF 0201
17 6 IN XDP_CPU_TCK — XDP_CPU_TCK

R1118 0 5% 1/20W MF 0201
PCH_XTAL24_IN_R

R1102 1M 5% 1/20W MF 201
PCH_XTAL24_OUT_R

Y1101 1.6MX1.2MM 24.000MHZ-20PPM-9.5PF-60OHM

C1101 10PF 5% 25V CER 0201
C1102 10PF 5% 25V CER 0201

R1119 0 5% 1/20W MF 0201

U1100
SKL-PCH-H
TBD
FCBGA
SYM 2 OF 12

CRITICAL
OMIT_TABLE

CLOCK SIGNALS

33 OUT PCIE_CLK100M_AP_N N7 CLKOUT_SRC_N0
33 OUT PCIE_CLK100M_AP_P N8 CLKOUT_SRC_P0
36 OUT PCIE_CLK100M_ENET_N L7 CLKOUT_SRC_N1
36 OUT PCIE_CLK100M_ENET_P L5 CLKOUT_SRC_P1
NC D3 CLKOUT_SRC_N2
NC F2 CLKOUT_SRC_P2
26 OUT PCIE_CLK100M_TBT_X_N E5 CLKOUT_SRC_N3
26 OUT PCIE_CLK100M_TBT_X_P G4 CLKOUT_SRC_P3
NC D5 CLKOUT_SRC_N4
NC E6 CLKOUT_SRC_P4
34 OUT PCIE_CLK100M_SSD_N D8 CLKOUT_SRC_N5
34 OUT PCIE_CLK100M_SSD_P D7 CLKOUT_SRC_P5
NC R8 CLKOUT_SRC_N6
NC R7 CLKOUT_SRC_P6
75 OUT PEG_CLK100M_N U5 CLKOUT_SRC_N7
75 OUT PEG_CLK100M_P U7 CLKOUT_SRC_P7
NC W10 CLKOUT_SRC_N8
NC W11 CLKOUT_SRC_P8
NC N3 CLKOUT_SRC_N9
NC N2 CLKOUT_SRC_P9
NC P3 CLKOUT_SRC_N10
NC P2 CLKOUT_SRC_P10
NC R3 CLKOUT_SRC_N11
NC R4 CLKOUT_SRC_P11
NC U2 CLKOUT_SRC_N12
NC U3 CLKOUT_SRC_P12
NC W7 CLKOUT_SRC_N13
NC Y5 CLKOUT_SRC_P13
NC P1 CLKOUT_SRC_N14
NC R2 CLKOUT_SRC_P14
NC R13 CLKOUT_SRC_N15
NC R11 CLKOUT_SRC_P15

XTAL24_IN A6 PCH_XTAL24_IN
XTAL24_OUT A5 PCH_XTAL24_OUT

PLACE_NEAR=U1100.E1:5MM
XCLK_BIASREF E1 PCH_DIFFCLK_BIASREF — R1115 2.7K 0.1% 1/16W MF 0402 — =PP1V0_S5_PCH_VCC 11 15 97

PLACE_NEAR=U1100.BC17:10MM
GPP_A9/CLKOUT_LPC0 /ESPI_CLK BC17 LPC_CLK24M_SMC_R — R1120 33 5% 1/16W MF-LF 402 — LPC_CLK24M_SMC OUT 45
GPP_A10/CLKOUT_LPC1 AV19 NC 24MHz SE output
GPP_A16/CLKOUT_48 AR17 NC CLKOUT IS FOR SERVER ONLY
C1103 10PF 5% 25V CER 0201 NOSTUFF
PLACE_NEAR=U1100.BC17:12MM

CLKOUT_ITPXDP_N L1 ITPXDP_CLK100M_N OUT 17 100
CLKOUT_ITPXDP_P L2 ITPXDP_CLK100M_P OUT 17 100

CLKOUT_CPUBCLK_N H2 CPU_BCLK100M_N OUT 6
CLKOUT_CPUBCLK_P G2 CPU_BCLK100M_P OUT 6
direct 100MHz clock dif core ref clock from PCH to processor

CLKOUT_CPUNSSC_N F1 CPU_CLK24M_SSC_N OUT 6
CLKOUT_CPUNSSC_P G1 CPU_CLK24M_SSC_P OUT 6
direct 24MHz diff crystal ref clock from PCH to processor

CLKOUT_CPUPCIBCLK_N J1 CPU_CLK100M_PCI_N OUT 6
CLKOUT_CPUPCIBCLK_P J2 CPU_CLK100M_PCI_P OUT 6

PCI EXPRESS
CLOCKS & CONTROL

GPP_B5/SRCCLKREQ0* BC24 AP_CLKREQ_PCH_L — R1116 0 1/16W MF-LF 5% 402 — AP_CLKREQ_L IN 11 33
GPP_B6/SRCCLKREQ1* AW24 ENET_CLKREQ_L IN 11 36
GPP_B7/SRCCLKREQ2* AT24 NC
GPP_B8/SRCCLKREQ3* BD25 TBT_X_CLKREQ_PCH_L — R1155 1K 1/20W MF 5% 201 — TBT_X_CLKREQ_L IN 11 26
GPP_B9/SRCCLKREQ4* BB24 NC
GPP_B10/SRCCLKREQ5* BE25 SSD_CLKREQ_PCH_L — R1121 0 1/16W MF-LF 5% 402 — SSD_CLKREQ_L IN 11 34
GPP_H0/SRCCLKREQ6* AT33 NC
GPP_H1/SRCCLKREQ7* AR31 PEG_CLKREQ_L IN 11 75
GPP_H2/SRCCLKREQ8* BD32 NC
GPP_H3/SRCCLKREQ9* BC32 NC
GPP_H4/SRCCLKREQ10* BB31 TBT_PWR_EN_PCH OUT 11 19
GPP_H5/SRCCLKREQ11* BC33 NC
GPP_H6/SRCCLKREQ12* BA33 NC
GPP_H7/SRCCLKREQ13* AW33 TBT_X_USB_PWR_EN OUT 11 28
GPP_H8/SRCCLKREQ14* BB33 NC
GPP_H9/SRCCLKREQ15* BD33 NC

CLKREQs

=PP1V0_S5_PCH_VCC 11 15 97
R1105 1K 5% 1/20W MF 201 — PCH_ITP_PMODE 11 17
R1106 NOSTUFF 51 5% 1/20W MF 201 — PCH_JTAGX 11

=PP3V3_S0_PCH_VCC 12 14 15 18 97
R1150 10K — TBT_PWR_EN_PCH 11 19
R1151 10K — TBT_X_USB_PWR_EN 11 28
R1100 10K — SSD_CLKREQ_L 11 34
R1104 47K — TBT_X_CLKREQ_L 11 26
R1101 10K — ENET_CLKREQ_L 11 36
R1113 10K — AP_CLKREQ_L 11 33
R1108 10K — PEG_CLKREQ_L 11 75
R1125 NOSTUFF 10K
5% 1/20W MF 201

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU & CHIPSET: Clocks/HDA/JTAG

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REVISION 5.19.0
BRANCH
PAGE 11 OF 121
SHEET 11 OF 102

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SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

CRITICAL

U1100
SKL-PCH-H
TBD
FCBGA
SYM 3 OF 12

OMIT_TABLE

SPI

SPI_CLK_R	BC31	SPI0_CLK	
SPI_CS0_R_L	BD31	SPI0_CS0*	
NC	AW31	SPI0_CS1*	
NC	AT31	SPI0_CS2*	
SPI_MOSI_R	BB29	SPI0_MOSI	
SPI_MISO_R	BE30	SPI0_MISO	
SPI_IO_R<2>	BC29	SPI0_IO2	
SPI_IO_R<3>	BD30	SPI0_IO3	

GSPI

GPP_B15/GSPI0_CS*	AR24	NC
GPP_B16/GSPI0_CLK	AW27	NC
GPP_B17/GSPI0_MISO	BD27	NC
GPP_B18/GSPI0_MOSI	BD28	PCH_STRP_NOREBOOT
GPP_B19/GSPI1_CS*	BC27	NC
GPP_B20/GSPI1_CLK	AV29	NC
GPP_B21/GSPI1_MISO	AR29	NC
GPP_B22/GSPI1_MOSI	AT29	PCH_STRP_BOOT_LPC

C-LINK

CL_CLK	AV2	TP_CLINK_CLK
CL_DATA	AV3	TP_CLINK_DATA
CL_RST*	AW2	TP_CLINK_RESET_L

U1100
SKL-PCH-H
TBD
FCBGA
SYM 4 OF 12

CRITICAL

OMIT_TABLE

SYSTEM POWER MANAGEMENT

PCH_SUSACK_L	BB19	GPP_A15/SUSACK*	
PM_SYSRST_L	AW1	SYS_RESET*	
PM_PCH_SYS_PWROK	AY1	SYS_PWROK	
PM_PCH_PWROK	AW11	PCH_PWROK	
PLT_RESET_L	BB27	GPP_B13/PLTRST*	
PM_RSMRST_PCH_L	BA11	RSMRST*	
PCH_SUSWARN_L	BD19	GPP_A13/SUSWARN*/SUSPWRDNACK	
PM_PWRBTN_L	AT13	GPD3/PWRBTN*	
SSD_SR_EN_L	BB15	GPD1/ACPRESENT	
PCH_BATLOW_L	BD13	GPD0/BATLOW*	
PM_SLP_S0_L	BC26	GPP_B12/SLP_S0*	
TP_PCH_SLP_WLAN_L	AV13	GPD9/SLP_WLAN*	
PCH_PME_L	BD17	GPP_A11/PME*	
TP_PCH_GPP_A12	BB17	GPP_A12/BMBUSY*/ISH_GP6/SX_EXIT_HOLDOFF*	
PCH_CLK32K_RTCX1	BC9	RTCX1	
PCH_CLK32K_RTCX2	BD10	RTCX2	
PCH_INTRUDER_L	BE11	INTRUDER*	RTC
PCH_SRTCRST_L	BB10	SRTCRST*	
RTC_RESET_L	BC10	RTCRST*	

DRAM_RESET*	BC14	PCH_MEM_RESET_L	
DSW_PWROK	AV11	PM_RSMRST_PCH_L	
WAKE*	BC13	PCIE_WAKE_L	
GPP_A8/CLKRUN*	AW22	LPC_CLKRUN_L	
GPP_A14/SUS_STAT*/ESPI_RESET*	BC18	LPC_PWRDWN_L	
GPD8/SUSCLK	AN15	PM_CLK32K_SUSCLK_R	
GPD10/SLP_S5*	BA13	PM_SLP_S5_L	
GPD5/SLP_S4*	BD15	PM_SLP_S4_L	
GPD4/SLP_S3*	AW15	PM_SLP_S3_L	
GPD6/SLP_A*	BC15	TP_PM_SLP_A_L	
SLP_SUS*	BB13	PM_SLP_SUS_L	
SLP_LAN*	AV15	TP_PCH_SLP_LAN_L	
GPD2/LAN_WAKE*	BD11	SMC_WAKE_SCI_L	
GPD11/LANPHYPC	AR15	NC	
GPP_B2/VRALERT*	BD23	BT_DEV_WAKE	

GPP_A0/RCIN*/ESPI_ALERT1*	AT17	PCH_RCIN_TERM	
GPP_A1/LAD0/ESPI_IO0	AT22	LPC_AD_R<0>	
GPP_A2/LAD1/ESPI_IO1	AV22	LPC_AD_R<1>	
GPP_A3/LAD2/ESPI_IO2	AT19	LPC_AD_R<2>	
GPP_A4/LAD3/ESPI_IO3	BD16	LPC_AD_R<3>	
GPP_A5/LFRAME*/ESPI_CS0*	BE16	LPC_FRAME_R_L	
GPP_A6/SERIRQ/ESPI_CS1*	BA17	LPC_SERIRQ	
GPP_A7/PIRQA*/ESPI_ALERT0*	AW17	SMC_RUNTIME_SCI_L	
GPD7/RSVD	BD14	NC	

ESPI/LPC

BATLOW only in mobile package

=PPVDDQ_S3_MEMRESET

=PP3V3_G3_PCH_RTC

=PP3V3_S0_PCH_VCC

R1230	10K	LPC_SERIRQ	
R1234	10K	PCH_RCIN_TERM	
R1292	10K	SMC_RUNTIME_SCI_L	
R1235 NOSTUFF	10K	BT_DEV_WAKE	

=PP3V3_S4_PCH_VCC

R1231	100K	SSD_SR_EN_L	
R1227 NOSTUFF	100K		

=PP3V3_S5_PCH_VCC

R1203	1K	PCH_STRP_NOREBOOT	
R1236	8.2K	LPC_CLKRUN_L	
R1233	10K	PCH_PME_L	
R1210	1K	PCIE_WAKE_L	
R1211	3.0K	PM_PWRBTN_L	
R1232	10K	PCH_BATLOW_L	
R1290 NOSTUFF	100K	TP_PCH_GPP_A12	
R1204 NOSTUFF	1K	PCH_STRP_BOOT_LPC	
R1291	10K	SMC_WAKE_SCI_L	

R1237 NOSTUFF	100K	PCH_STRP_GPP_B23	
R1206	1K	PCH_PECI	
R1212	100K	PM_SLP_S3_L	
R1213	100K	PM_SLP_S4_L	
R1214	100K	PM_SLP_S5_L	
R1215 NOSTUFF	100K	PM_SLP_SUS_L	
R1239	100K	BT_TIMESTAMP	
R1238	100K	PM_SLP_S0_L	

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U1100
SKL-PCH-H
TBD
FCBGA
SYM 5 OF 12

CRITICAL

OMIT_TABLE

DMI

CPU/MISC

DMI_S2N_N<0>	C27	DMI_TXN0
DMI_S2N_P<0>	B27	DMI_TXP0
DMI_S2N_N<1>	B28	DMI_TXN1
DMI_S2N_P<1>	A28	DMI_TXP1
DMI_S2N_N<2>	B29	DMI_TXN2
DMI_S2N_P<2>	C29	DMI_TXP2
DMI_S2N_N<3>	B30	DMI_TXN3
DMI_S2N_P<3>	A30	DMI_TXP3
DMI_N2S_N<0>	L27	DMI_RXN0
DMI_N2S_P<0>	N27	DMI_RXP0
DMI_N2S_N<1>	E24	DMI_RXN1
DMI_N2S_P<1>	G24	DMI_RXP1
DMI_N2S_N<2>	G27	DMI_RXN2
DMI_N2S_P<2>	E26	DMI_RXP2
DMI_N2S_N<3>	L29	DMI_RXN3
DMI_N2S_P<3>	K29	DMI_RXP3

PROCPWRGD	AM3	PCH_PROCPWRGD	
THERMTRIP*	AJ3	PM_THRMTRIP_PCH_R_L	
PECI	AL3	PCH_PECI	
GPP_B3/CPU_GP2	BC23	NC	
GPP_B4/CPU_GP3	BD24	BT_TIMESTAMP	
GPP_E3/CPU_GP0	AF41	XDP_PCH_OBSDATA_A2	
GPP_E7/CPU_GP1	AE44	XDP_PCH_OBSDATA_B2	
GPP_B23/SML1ALERT*/PCHHOT*	AT27	PCH_STRP_GPP_B23	
CPU_TRST*	AY5	PCH_TRST_L	
PLTRST_PROC*	AK2	CPU_RESET_L	
PRDY*	AT4	PCH_PRDY_L	
PREQ*	AT3	PCH_PREQ_L	
PCH_TRIGIN	AK1	PROC_TRIGOUT	
PCH_TRIGOUT	AL2	PROC_TRIGIN_R	PROC_TRIGIN
PM_DOWN	AH2		PM_DOWN
PM_SYNC	AJ4	PM_SYNC_R	PM_SYNC

=PP1V0_S3_VCCST

PLACE R1217 NEAR TRACE T-POINT

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

CPU & CHIPSET: PCH RTC/DMI/PM/CPU_Misc

Apple Inc.

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SIZE D

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PAGE 12 OF 121

SHEET 12 OF 102

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OMIT_TABLE | CRITICAL

U1100 SKL-PCH-H TBD FCBGA SYM 7 OF 12

NC	B7	USB3_1_RXN	USB2N_1	AF5	NC		
NC	A7	USB3_1_RXP	USB2P_1	AG7	NC		
NC	C11	USB3_1_TXN	USB2N_2	AD5	USB_EXTC_N	BI	44
NC	B11	USB3_1_TXP	USB2P_2	AD7	USB_EXTC_P	BI	44
44 IN USB3_EXTD_RX_F_N	C8	USB3_2_RXN	USB2N_3	AG8	USB_EXTB_N	BI	43
44 IN USB3_EXTD_RX_F_P	B8	USB3_2_RXP	USB2P_3	AG10	USB_EXTB_P	BI	43
44 OUT USB3_EXTD_TX_N	B12	USB3_2_TXN	USB2N_4	AE1	USB_EXTD_N	BI	44
44 OUT USB3_EXTD_TX_P	A12	USB3_2_TXP	USB2P_4	AE2	USB_EXTD_P	BI	44
NC	B10	USB3_3_RXN	USB2N_5	AC2	USB_CAMERA_N	BI	39
NC	A9	USB3_3_RXP	USB2P_5	AC3	USB_CAMERA_P	BI	39
NC	C13	USB3_3_TXN	USB2N_6	AF2	USB_BT_N	BI	33
NC	D13	USB3_3_TXP	USB2P_6	AF3	USB_BT_P	BI	33
43 IN USB3_EXTA_RX_F_N	E11	USB3_4_RXN	USB2N_7	AB3	NC		
43 IN USB3_EXTA_RX_F_P	G11	USB3_4_RXP	USB2P_7	AB2	NC		
43 OUT USB3_EXTA_TX_N	A14	USB3_4_TXN	USB2N_8	AL8	NC		
43 OUT USB3_EXTA_TX_P	B13	USB3_4_TXP	USB2P_8	AL7	NC		
43 IN USB3_EXTB_RX_F_N	G13	USB3_5_RXN	USB2N_9	AA1	NC		
43 IN USB3_EXTB_RX_F_P	H13	USB3_5_RXP	USB2P_9	AA2	NC		
43 OUT USB3_EXTB_TX_N	B14	USB3_5_TXN	USB2N_10	AJ8	USB_EXTA_N	BI	43
43 OUT USB3_EXTB_TX_P	C14	USB3_5_TXP	USB2P_10	AJ7	USB_EXTA_P	BI	43
44 IN USB3_EXTC_RX_F_N	K15	USB3_6_RXN	USB2N_11	W2	NC		
44 IN USB3_EXTC_RX_F_P	K13	USB3_6_RXP	USB2P_11	W3	NC		
44 OUT USB3_EXTC_TX_N	B15	USB3_6_TXN	USB2N_12	AD3	NC		
44 OUT USB3_EXTC_TX_P	C15	USB3_6_TXP	USB2P_12	AD2	NC		
			USB2N_13	V2	USB_UPC_PCH_XA_N	BI	32
			USB2P_13	V1	USB_UPC_PCH_XA_P	BI	32
			USB2N_14	AJ11	USB_UPC_PCH_XB_N	BI	32
			USB2P_14	AJ13	USB_UPC_PCH_XB_P	BI	32

USB3 USB2

GPP_E9/USB2_OC0*	AD43	USB_EXTA_OC_L	IN 13 17 43
GPP_E10/USB2_OC1*	AD42	USB_EXTB_OC_L	IN 13 17 43
GPP_E11/USB2_OC2*	AD39	USB_EXTC_OC_L	IN 13 17 44
GPP_E12/USB2_OC3*	AC44	USB_EXTD_OC_L	IN 13 17 44
GPP_F15/USB2_OC4*	Y43	USB_EXTE_OC_L	IN 13 32
GPP_F16/USB2_OC5*	Y41	USB_EXTF_OC_L	IN 13 32
GPP_F17/USB2_OC6*	W44	USB_EXTC_HILIM_EN	OUT 44
GPP_F18/USB2_OC7*	W43	USB_EXTD_HILIM_EN	OUT 44
USB2_COMP	AG3	PCH_USB_RBIAS	PLACE_NEAR=U1100.AG3:11.4MM
USB2_ID	AG2	PCH_USB2_ID	R1314 113 1% 1/20W MF 201
USB2_VBUSSENSE	AD10	PCH_USB_VBUSSENSE	R1318 1K 5% 1/20W MF 201

USB2_ID HAS WEAK PULLUP

R1301 1K 5% 1/16W MF-LF 402

=PP3V3_S4_PCH_VCC 12 14 97

R1303	10K	1	2	5% 1/20W MF 201	USB_EXTA_OC_L	13 17 43
R1304	10K	1	2	5% 1/20W MF 201	USB_EXTC_OC_L	13 17 44
R1307	10K	2	1	5% 1/20W MF 201	USB_EXTB_OC_L	13 17 43
R1308	10K	1	2	5% 1/20W MF 201	USB_EXTD_OC_L	13 17 44
R1309	10K	1	2	5% 1/20W MF 201	USB_EXTE_OC_L	13 32
R1310	10K	1	2	5% 1/20W MF 201	USB_EXTF_OC_L	13 32
R1333 NOSTUFF	10K	1	2	5% 1/20W MF 201	PCH_SSD_RESET_L	13 34
R1334	10K	1	2	5% 1/20W MF 201	SSD_PWR_EN	13 34

=PP3V3_S5_PCH_VCC 12 14 15 19 97

R1331	10K	1	2	5% 1/20W MF 201	HDD_PWR_EN	13 35
R1316	10K	1	2	5% 1/20W MF 201	PCH_SATALED_L	13 17 35
R1322	10K	1	2	5% 1/20W MF 201	PCH_CAM_EXT_BOOT_L	13
R1323	10K	1	2	5% 1/20W MF 201	PCH_CAM_RESET	13
R1330	10K	1	2	5% 1/20W MF 201	TBT_POC_RESET	13 28
R1332	10K	1	2	5% 1/20W MF 201	WOL_EN	13 37

SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

CRITICAL | OMIT_TABLE

U1100 SKL-PCH-H TBD FCBGA SYM 8 OF 12 PCIE/SATA/USB3

28 IN PCIE_TBT_X_D2R_N<0>	K19	PCIE5_RXN	G31	NC			
28 IN PCIE_TBT_X_D2R_P<0>	L19	PCIE5_RXP	H31	NC			
28 OUT PCIE_TBT_X_R2D_C_N<0>	D22	PCIE5_TXN	C31	NC			
28 OUT PCIE_TBT_X_R2D_C_P<0>	C22	PCIE5_TXP	B31	NC			
28 IN PCIE_TBT_X_D2R_N<1>	G22	PCIE6_RXN	G29	NC			
28 IN PCIE_TBT_X_D2R_P<1>	E22	PCIE6_RXP	E29	NC			
28 OUT PCIE_TBT_X_R2D_C_N<1>	B22	PCIE6_TXN	C32	NC			
28 OUT PCIE_TBT_X_R2D_C_P<1>	A23	PCIE6_TXP	B32	NC			
28 IN PCIE_TBT_X_D2R_N<2>	L22	PCIE7_RXN	G35	SATA_HDD_D2R_N	IN	35
28 IN PCIE_TBT_X_D2R_P<2>	K22	PCIE7_RXP	E35	SATA_HDD_D2R_P	IN	35
28 OUT PCIE_TBT_X_R2D_C_N<2>	C23	PCIE7_TXN	C36	SATA_HDD_R2D_C_N	OUT	35
28 OUT PCIE_TBT_X_R2D_C_P<2>	B23	PCIE7_TXP	B36	SATA_HDD_R2D_C_P	OUT	35
28 IN PCIE_TBT_X_D2R_N<3>	K24	PCIE8_RXN	D39	NC		
28 IN PCIE_TBT_X_D2R_P<3>	L24	PCIE8_RXP	E37	NC		
28 OUT PCIE_TBT_X_R2D_C_N<3>	C24	PCIE8_TXN	B38	NC		
28 OUT PCIE_TBT_X_R2D_C_P<3>	B24	PCIE8_TXP	C38	NC		
NC	L31	PCIE11_RXN	F41	NC		
NC	K31	PCIE11_RXP	E41	NC		
NC	C33	PCIE11_TXN	B39	NC		
NC	B33	PCIE11_TXP	A39	NC		
NC	G33	PCIE12_RXN	D43	NC		
NC	H33	PCIE12_RXP	E42	NC		
NC	B35	PCIE12_TXN	A41	NC		
NC	A35	PCIE12_TXP	A40	NC		
33 IN PCIE_AP_D2R_N	H15	PCIE1_RXN/USB3_7_RXN	H42	PCIE_SSD_D2R_N<0>	IN	34
33 IN PCIE_AP_D2R_P	G15	PCIE1_RXP/USB3_7_RXP	H40	PCIE_SSD_D2R_P<0>	IN	34
33 OUT PCIE_AP_R2D_C_N	A16	PCIE1_TXN/USB3_7_TXN	E45	PCIE_SSD_R2D_N<0>	OUT	34
33 OUT PCIE_AP_R2D_C_P	B16	PCIE1_TXP/USB3_7_TXP	F45	PCIE_SSD_R2D_P<0>	OUT	34
36 IN PCIE_ENET_D2R_N	E17	PCIE2_RXN/USB3_8_RXN	K37	PCIE_SSD_D2R_N<1>	IN	34
36 IN PCIE_ENET_D2R_P	G17	PCIE2_RXP/USB3_8_RXP	G37	PCIE_SSD_D2R_P<1>	IN	34
36 OUT PCIE_ENET_R2D_C_N	B19	PCIE2_TXN/USB3_8_TXN	G45	PCIE_SSD_R2D_N<1>	OUT	34
36 OUT PCIE_ENET_R2D_C_P	C19	PCIE2_TXP/USB3_8_TXP	G44	PCIE_SSD_R2D_P<1>	OUT	34
NC	L17	PCIE3_RXN/USB3_9_RXN	PCIE19_RXN	L37	PCIE_SSD_D2R_N<2>	IN	34
NC	K17	PCIE3_RXP/USB3_9_RXP	PCIE19_RXP	L39	PCIE_SSD_D2R_P<2>	IN	34
NC	B20	PCIE3_TXN/USB3_9_TXN	PCIE19_TXN	H43	PCIE_SSD_R2D_N<2>	OUT	34
NC	C20	PCIE3_TXP/USB3_9_TXP	PCIE19_TXP	H44	PCIE_SSD_R2D_P<2>	OUT	34
NC	E20	PCIE4_RXN/USB3_10_RXN	PCIE20_RXN	N39	PCIE_SSD_D2R_N<3>	IN	34
NC	G19	PCIE4_RXP/USB3_10_RXP	PCIE20_RXP	N38	PCIE_SSD_D2R_P<3>	IN	34
NC	B21	PCIE4_TXN/USB3_10_TXN	PCIE20_TXN	K44	PCIE_SSD_R2D_N<3>	OUT	34
NC	A21	PCIE4_TXP/USB3_10_TXP	PCIE20_TXP	J45	PCIE_SSD_R2D_P<3>	OUT	34

17 OUT PCH_SATAXPCIE0_SATAGP0	AG36	GPP_E0/SATAXPCIE0/SATAGP0	GPP_E4/DEVSLP0	AG42	PCH_DEVSLP0	BI 17
17 OUT PCH_SATAXPCIE1_SATAGP1	AG35	GPP_E1/SATAXPCIE1/SATAGP1	GPP_E5/DEVSLP1	AG43	PCH_DEVSLP1	BI 17
17 OUT PCH_SATAXPCIE2_SATAGP2	AG39	GPP_E2/SATAXPCIE2/SATAGP2	GPP_E6/DEVSLP2	AE45	PCH_DEVSLP2	BI 17
17 OUT PCH_SATAXPCIE3_SATAGP3	AD35	GPP_F0/SATAXPCIE3/SATAGP3	GPP_F5/DEVSLP3	AB41	PCH_PLT_RST_TBT_X_L	
17 OUT PCH_SATAXPCIE4_SATAGP4	AD31	GPP_F1/SATAXPCIE4/SATAGP4	GPP_F6/DEVSLP4	AB42	NC	
17 OUT PCH_SATAXPCIE5_SATAGP5	AD38	GPP_F2/SATAXPCIE5/SATAGP5	GPP_F7/DEVSLP5	AB43	PCH_SSD_RESET_L	OUT 13 34
17 OUT PCH_SATAXPCIE6_SATAGP6	AC43	GPP_F3/SATAXPCIE6/SATAGP6	GPP_F8/DEVSLP6	AB36	HDD_PWR_EN	OUT 13 35
17 OUT PCH_SATAXPCIE7_SATAGP7	AB44	GPP_F4/SATAXPCIE7/SATAGP7	GPP_F9/DEVSLP7	AB39	SSD_PWR_EN	OUT 13 34
28 13 OUT TBT_POC_RESET	AB33	GPP_F10/SCLOCK	GPP_E8/SATALED*	AD44	PCH_SATALED_L	BI 13 17 35
40 OUT PCH_CAM_RESET_R	AB35	GPP_F11/SLOAD				
40 OUT PCH_CAM_EXT_BOOT_R_L	AA45	GPP_F12/SDATAOUT1				
37 13 OUT WOL_EN	AA44	GPP_F13/SDATAOUT0				

PLACE_NEAR=U1100.U36:10MM

R1320 33 NF-LF 1/16W 402 5% 13 PCH_CAM_RESET

R1321 33 NF-LF 1/16W 402 5% 13 PCH_CAM_EXT_BOOT_L

PLACE_NEAR=U1100.P44:10MM

R1324 NOSTUFF 33 NF-LF 402 5% 1/16W

PLACE_NEAR=U1100.AB41:15.0MM

PCIE_RCOMPP C17

PCIE_RCOMPN B18

R1315 100 1% 1/20W MF 201

PLACE_NEAR=U1100.C17:11.4MM

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

CPU & CHIPSET: PCH PCI-E/USB

Apple Inc.

DRAWING NUMBER 051-01506

SIZE D

REVISION 5.19.0

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PAGE 13 OF 121

SHEET 13 OF 102

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BUILD BITS[4:0]

POC:	11111
PROTO-0:	11110
PRE-PROTO-1:	11101
PROTO-1:	11100
PRE-EVT:	11011
EVT:	11010 <<

SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

U1100 SKL-PCH-H TBD FCBGA SYM 6 OF 12

CRITICAL

OMIT_TABLE

R1452 33 ENET_SD_RESET_L → ENET_SD_RESET_R_L
R1450 33 AP_RESET_L → AP_RESET_R_L
NOSTUFF PLACE_NEAR=U1100.BC19:15.0MM
NOSTUFF PLACE_NEAR=U1100.BB22:15.0MM

BC19	GPP_A17/ISH_GP7		
BB22	GPP_A18/ISH_GP0		
BD21	GPP_A19/ISH_GP1		
BD22	GPP_A20/ISH_GP2		
BE21	GPP_A21/ISH_GP3		
BD18	GPP_A22/ISH_GP4	GPPA/ INTEGRATED SENSOR	
BC22	GPP_A23/ISH_GP5		
AR27	GPP_B0		
AL27	GPP_B1		
AN24	GPP_B11	GPPB	GPPC/SMLINK/I2C/UART
BD26	GPP_B14/SPKR		
AL39	GPP_D0		
AN36	GPP_D1		
AN38	GPP_D2		
AN41	GPP_D3		
AL44	GPP_D9		
AL36	GPP_D10		
AL35	GPP_D11		
AJ39	GPP_D12		
AK45	GPP_D13/ISH_UART0_RXD/SML0BDATA/I2C2_SDA		
AK44	GPP_D14/ISH_UART0_TXD/SML0BCLK/I2C2_SCL		
AL43	GPP_D15/ISH_UART0_RTS*		
AJ43	GPP_D16/ISH_UART0_CTS*		
AG44	GPP_D21		
AH43	GPP_D22		
AM44	GPP_D4/ISH_I2C2_SDA/I2C3_SDA		
AJ44	GPP_D23/ISH_I2C2_SCL/I2C3_SCL	GPPD/INTEGRATED SENSOR/UART/I2C	

AW44	GPP_C0/SMBCLK	SMBUS_PCH_CLK	OUT 48
BB43	GPP_C1/SMBDATA	SMBUS_PCH_DATA	BI 48
BB41	GPP_C2/SMBALERT*	PCH_SMBALERT_L	14
AY44	GPP_C3/SML0CLK	SML_PCH_0_CLK	OUT 48
BB39	GPP_C4/SML0DATA	SML_PCH_0_DATA	BI 48
BA40	GPP_C5/SML0ALERT*	PCH_STRP_LPC_L	14
AW42	GPP_C6/SML1CLK	SML_PCH_1_CLK	IN 48
AW45	GPP_C7/SML1DATA	SML_PCH_1_DATA	BI 48
BA41	GPP_C8/UART0_RXD	NC	
AV44	GPP_C9/UART0_TXD	NC	
AV43	GPP_C10/UART0_RTS*	NC	
AU44	GPP_C11/UART0_CTS*	NC	
AU43	GPP_C12/UART1_RXD/ISH_UART1_RXD	NC	
AT43	GPP_C13/UART1_TXD/ISH_UART1_TXD	NC	
AT44	GPP_C14/UART1_RTS*/ISH_UART1_RTS*	NC	
AU41	GPP_C15/UART1_CTS*/ISH_UART1_CTS*	NC	
AT42	GPP_C16/I2C0_SDA	NC	
AR38	GPP_C17/I2C0_SCL	NC	
AR44	GPP_C18/I2C1_SDA	PCH_ALS_SDA	BI 14 39
AR41	GPP_C19/I2C1_SCL	PCH_ALS_SCL	OUT 14 39
AR45	GPP_C20/UART2_RXD	NC	
AR39	GPP_C21/UART2_TXD	NC	
AN44	GPP_C22/UART2_RTS*	NC	
AN43	GPP_C23/UART2_CTS*	NC	

MEM_OK, PCH_GPP_B1, PCH_STRP_TOPBLK_SWP_L, PCH_GPU_RESET_L, USB_EXTA_HILIM_EN, USB_EXTB_HILIM_EN, GPU_ID, PCH_BOARD_REV<0>, PCH_BOARD_REV<1>, PCH_BOARD_REV<2>, PCH_BOARD_REV<3>, PCH_BOARD_REV<4>

R1448 NOSTUFF, R1444 NOSTUFF, R1445, R1446 NOSTUFF, R1447 — 1K 5% 201 1/20W MF

GPP_D23 → TP1400

=PP3V3_S0_PCH_VCC

R1430 2.2K PCH_ALS_SDA
R1431 2.2K PCH_ALS_SCL
R1421 10K SDCONN_OC_L
R1411 10K JTAG_TBT_TMS_PCH
R1425 10K TCON_RESET_R_L
R1415 10K JTAG_ISP_TDO
R1416 NOSTUFF 10K JTAG_ISP_TDI
R1454 100K

=PP3V3_S4_PCH_VCC

R1432 10K SDCONN_STATE_CHANGE_L

=PP3V3_S5_PCH_VCC

R1404 100K PCH_SMBALERT_L

R1434 100K MEM_OK
R1433 100K PCH_GPP_B1
R1453 NOSTUFF 10K PCH_STRP_LPC_L
R1490 100K AUD_IPHS_SWITCH_EN_PCH
R1414 10K JTAG_ISP_TCK
R1417 10K ENET_LOW_PWR_PCH
R1451 10K ENET_MEDIA_SENSE
R1424 10K PCH_GPP_H16
R1435 249K SSD_BFH_L

SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

U1100 SKL-PCH-H TBD CRITICAL FCBGA SYM 9 OF 12

OMIT_TABLE

ENET_LOW_PWR_PCH
R1405 33 PLACE_NEAR=U1100.W39:10MM TCON_RESET_L → TCON_RESET_R_L
SPIROM_USE_MLB

Y44	GPP_F14		
W42	GPP_F19/EDP_VDDEN		
W35	GPP_F20/EDP_BKLTEN		
W36	GPP_F21/EDP_BKLTCTL	GPPF/ BACKLIGHT	
W39	GPP_F22		
V44	GPP_F23		
AW4	GPP_I0/DDPB_HPD0		
AY2	GPP_I1/DDPC_HPD1		
AV4	GPP_I2/DDPD_HPD2		
BA4	GPP_I3/DDPE_HPD3		
BD7	GPP_I4/EDP_HPD	GPPI/DISPLAY	GPPG
BA5	GPP_I5/DDPB_CTRLCLK		
BC4	GPP_I6/DDPB_CTRLDATA		
BB3	GPP_I7/DDPC_CTRLCLK		
BD6	GPP_I8/DDPC_CTRLDATA		
BE5	GPP_I9/DDPD_CTRLCLK		
BE6	GPP_I10/DDPD_CTRLDATA		
BD34	GPP_H10/SML2CLK		
AW35	GPP_H11/SML2DATA		
BD35	GPP_H12/SML2ALERT*		
BC35	GPP_H13/SML3CLK		
BA35	GPP_H14/SML3DATA		
BB36	GPP_H15/SML3ALERT*		
BD39	GPP_H16/SML4CLK		
BE34	GPP_H17/SML4DATA		
BC36	GPP_H18/SML4ALERT*		
BB38	GPP_H19/ISH_I2C0_SDA	GPPH/I2C/INTEGRATED SENSOR /SMLINK	
BC38	GPP_H20/ISH_I2C0_SCL		
BE39	GPP_H21/ISH_I2C1_SDA		
BD38	GPP_H22/ISH_I2C1_SCL		
BD36	GPP_H23		

U43	GPP_G0/FAN_TACH_0	SSD_BFH_L	OUT 14 34
U42	GPP_G1/FAN_TACH_1	NC	
U41	GPP_G2/FAN_TACH_2	NC	
M44	GPP_G3/FAN_TACH_3	NC	
U36	GPP_G4/FAN_TACH_4	NC	
P44	GPP_G5/FAN_TACH_5	NC	
T45	GPP_G6/FAN_TACH_6	NC	
T44	GPP_G7/FAN_TACH_7	NC	
R44	GPP_G8/FAN_PWM_0	NC	
R43	GPP_G9/FAN_PWM_1	PCH_BT_ROM_BOOT	OUT 33
U39	GPP_G10/FAN_PWM_2	NC	
N42	GPP_G11/FAN_PWM_3	NC	
R39	GPP_G12/GSXDOUT	NC	
R36	GPP_G13/GSXSLOAD	NC	
R42	GPP_G14/GSXDIN	NC	
R41	GPP_G15/GSXSRESET*	NC	
P43	GPP_G16/GSXCLK	NC	
N44	GPP_G17/ADR_COMPLETE	NC	
N43	GPP_G18/NMI*	NC	
M45	GPP_G19/SMI*	NC	
R35	GPP_G20	NC	
U35	GPP_G21	AUD_IPHS_SWITCH_EN_PCH	14
L44	GPP_G22	TP_AUD_IP_PERIPHERAL_DET	
L43	GPP_G23	TP_AUD_I2C_INT_L	

JTAG_ISP_TDO, JTAG_ISP_TDI, JTAG_ISP_TCK, JTAG_TBT_TMS_PCH

R1426 100K PCH_EDP_HPD_PD

GPPI IS ON VCC3V3_PRIM

R1427 NOSTUFF 33 SMC_EXT_VIDEO_SEL, ENET_MEDIA_SENSE, BKLT_FAULT_INT_L, SDCONN_STATE_CHANGE_L, SDCONN_OC_L, TBT_CIO_PLUG_EVENT_R_L

PCH_GPP_H16

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE: CPU & CHIPSET: PCH GPIO/Misc

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D
REVISION 5.19.0
BRANCH
PAGE 14 OF 121
SHEET 14 OF 102

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NEED TO ADD 47UF BULK CAPS TO 1.0V AND 3.3V RAIL

SNS_1V0_S5_P
SNS_1V0_S5_N
NO_XNET_CONNECTION=1
PLACE_NEAR=XW1501.2:1MM
XW1500
NO_XNET_CONNECTION=1
XW1501

=PP1V0_S5_PCH_VCC
C1520 0.1UF 20% 10V CERM 402
C1521 0.1UF 20% 10V CERM 402
C1522 0.1UF 20% 10V CERM 402
C1523 0.1UF 20% 10V CERM 402
C1524 0.1UF 20% 10V CERM 402
C1525 22UF 20% 25V X5R 0805
C1526 22UF 20% 25V X5R 0805
C1527 1UF 10% 6.3V CERM 402
C1528 1UF 10% 6.3V CERM 402
C1529 1UF 10% 6.3V CERM 402

U1100
SKL-PCH-H
TBD
FCBGA
SYM 10 OF 12
POWER
CRITICAL
OMIT_TABLE

PRIMARY WELL
Y23 VCCPRIM_1P0
Y25 VCCPRIM_1P0
AA23 VCCPRIM_1P0
AA26 VCCPRIM_1P0
AA28 VCCPRIM_1P0
AC17 VCCPRIM_1P0
AC23 VCCPRIM_1P0
AC26 VCCPRIM_1P0
AC28 VCCPRIM_1P0
AD15 VCCPRIM_1P0
AE23 VCCPRIM_1P0
AE26 VCCPRIM_1P0
AJ20 VCCPRIM_1P0
AJ21 VCCPRIM_1P0
AJ23 VCCPRIM_1P0
AJ25 VCCPRIM_1P0
AL22 VCCPRIM_1P0
VCC: 6010mA Max, 64mA Idle

AUDIO PLL
AN19 VCCHDAPLL_1P0 — PP1V0_S5_PCH_VCCUSB2HDA_PLL
VCCHDAPLL: 8mA Max, 0mA Idle
C1533 0.1UF 20% 10V CERM 402

MOD PHY PRIMARY
U21 VCCMPHY_1P0
U23 VCCMPHY_1P0
U25 VCCMPHY_1P0
U26 VCCMPHY_1P0
V26 VCCMPHY_1P0 — =PP1V0_S5_PCH_VCC
VCCMPHY: 4090mA Max, 0mA Idle
C1511 1UF 10% 6.3V CERM 402

DEEP SX WELL
W15 VCCDSW_3P3
BA24 VCCDSW_3P3 — =PP3V3_S5_PCH_VCC
VCCDSW: 502mA Max, 2.00mA Idle
C1512 1UF 10% 6.3V CERM 402

ANALOG PLL USB3/ PCIE2/SATA2/PCIE3
A43 VCCAMPHYPLL_1P0
B43 VCCAMPHYPLL_1P0 — PP1V0_S5_PCH_VCC_MPHYPCIE3_PLL
VCCMPHYPLL: 80mA Max, <1mA Idle
C1513 1UF 10% 6.3V CERM 402

MIPI PLL
C44 VCCMIPIPLL_1P0
C45 VCCMIPIPLL_1P0
VCCMIPIPLL: 37mA Max, <1mA Idle

GPPA PRIMARY WELL
BA31 VCCPGPPA — =PP3V3_S5_PCH_VCC
VCCPGPPA: 82mA Max, <1mA Idle
C1534 0.1UF 20% 10V CERM 402

GPPB/GPPC/GPPH PRIMARY WELL
BC42 VCCPGPPBCH
BD40 VCCPGPPBCH — =PP3V3_S5_PCH_VCC
VCCPGPPBCH: 229mA Max, <1mA Idle
C1514 0.1UF 20% 10V CERM 402

GPPD PRIMARY WELL
BA45 VCCPGPPD
BB45 VCCPGPPD
BC44 VCCPGPPD
BC45 VCCPGPPD — =PP3V3_S5_PCH_VCC
VCCPGPPD: 78mA Max, <1mA Idle
C1535 0.1UF 20% 10V CERM 402
C1536 1UF 10% 6.3V CERM 402

GPPE/GPPEF PRIMARY WELL
AJ41 VCCPGPPEF
AL41 VCCPGPPEF — =PP3V3_S5_PCH_VCC
VCCPGPPEF: 114mA Max, <1mA Idle
C1515 0.1UF 20% 10V CERM 402

GPPG PRIMARY WELL
AD41 VCCPGPPG — =PP3V3_S5_PCH_VCC
VCCPGPPG: 65mA Max, <1mA Idle
C1516 0.1UF 20% 10V CERM 402

DCPRTC
BA26 — PPV0UT_PCH_DCPRTC
C1517 0.1UF 20% 10V CERM 402

RTC WELL SUPPLY
BA22 VCCRTC — =PP3V3_G3_PCH_RTC
VCCRTC: 1mA Max, 1mA Idle
C1537 0.1UF 20% 10V CERM 402
C1518 1UF 10% 6.3V CERM 402

RTC LOGIC PW/VRM
BA20 VCCRTCPRIM_3P3 — =PP3V3_S5_PCH_VCC
VCCRTCPRIM: 1mA Max, 1mA Idle
C1538 0.1UF 20% 10V CERM 402
C1519 1UF 10% 6.3V CERM 402

SPI
BE41 VCCSPI
BE42 VCCSPI
BE43 VCCSPI — =PP3V3_S5_PCH_VCC
VCCSPI: 29mA Max, <1mA Idle

ANALOG PLL USB2/VRM
AJ5 VCCUSB2PLL_1P0 — PP1V0_S5_PCH_VCCUSB2HDA_PLL
AL5 VCCUSB2PLL_1P0
VCCUSB2PLL: 12mA Max, <1mA Idle
C1539 0.1UF 20% 10V CERM 402

PRIMARY WELL HVCMOS
AN5 VCCPRIM_3P3
BD3 VCCPRIM_3P3
BE3 VCCPRIM_3P3
BE4 VCCPRIM_3P3 — =PP3V3_S5_PCH_VCC
VCCPRIM_3P3: 171mA Max, <1mA Idle
C1501 0.1UF 20% 10V CERM 402
C1504 1UF 10% 6.3V CERM 402
PPV0UT_PCH_DCPDSW_1P0
BA29 DCPDSW_1P0

PCIE PLL EBB PRIMARY
V28 VCCAPLLEBB_1P0 — =PP1V0_S5_PCH_VCC
VCCAPLLEBB: 30mA Max, <1mA Idle
C1530 0.1UF 20% 10V CERM 402

THERMAL SENSOR PW
AD13 VCCATS — =PP3V3_S0_PCH_VCC
C1505 1UF 10% 6.3V CERM 402

CLOCK BUFFERS PRIMARY 1.0 V
N17 VCCCLK1 — =PP1V0_S5_PCH_VCC
VCCCLK1: 35mA Max, <1mA Idle
R17 VCCCLK2
V17 VCCCLK2
VCCCLK2: 204mA Max, <1mA Idle
C1532 1UF 10% 6.3V CERM 402
C1531 0.1UF 20% 10V CERM 402
R19 VCCCLK3
VCCCLK3: 57mA Max, <1mA Idle
U20 VCCCLK4
VCCCLK4: 36mA Max, <1mA Idle
K2 VCCCLK5 — PP1V0_S5_PCH_VCC_CLK_F
K3 VCCCLK5
VCCCLK5: 7mA Max, <1mA Idle
C1510 1UF 10% 6.3V CERM 402

HD AUDIO POWER
BA15 VCCHDA — =PP1V8_S5_PCH_VCCHDA
VCCHDA: 75mA Max, <1mA Idle
C1540 0.1UF 20% 10V CERM 402

placeholder for 2.2uH, .33ohm, 100mA inductor
L1501 2.2UH-20%-0.19A-0.221OHM 0603
=PP1V0_S5_PCH_VCC — PP1V0_S5_PCH_VCC_MPHYPCIE3_PLL
C1502 22UF 20% 25V X5R 0805
C1503 22UF 20% 25V X5R 0805

placeholder for 2.2uH, .33ohm, 100mA inductor
L1502 2.2UH-20%-0.19A-0.221OHM 0603
=PP1V0_S5_PCH_VCC — PP1V0_S5_PCH_VCC_CLK_F
C1506 22UF 20% 25V X5R 0805
C1507 22UF 20% 25V X5R 0805

placeholder for 2.2uH, .33ohm, 100mA inductor
L1503 2.2UH-20%-0.19A-0.221OHM 0603
=PP1V0_S5_PCH_VCC — PP1V0_S5_PCH_VCCUSB2HDA_PLL
C1508 22UF 20% 25V X5R 0805
C1509 22UF 20% 25V X5R 0805

Current data from EDS (doc #546717, Rev 0.7).

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=ANDREW
SYNC_DATE=02/19/2016
PAGE TITLE
CPU & CHIPSET: PCH Power/Decoupling

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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
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PAGE 15 OF 121
SHEET 15 OF 102

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SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

U1100
SKL-PCH-H
TBD
FCBGA
SYM 11 OF 12

CRITICAL

OMIT_TABLE

A2	VSS	VSS	M35
A3	VSS	VSS	M42
A4	VSS	VSS	N4
A18	VSS	VSS	N5
A25	VSS	VSS	N10
A32	VSS	VSS	N15
A37	VSS	VSS	N19
A42	VSS	VSS	N22
A44	VSS	VSS	N24
B1	VSS	VSS	N35
B2	VSS	VSS	N36
B3	VSS	VSS	N41
B6	VSS	VSS	P17
B25	VSS	VSS	P19
B37	VSS	VSS	P22
B40	VSS	VSS	P45
B44	VSS	VSS	R5
B45	VSS	VSS	R10
C2	VSS	VSS	R14
C4	VSS	VSS	R22
C10	VSS	VSS	R29
C28	VSS	VSS	R33
C37	VSS	VSS	R38
C42	VSS	VSS	T1
D10	VSS	VSS	T2
D12	VSS	VSS	T4
D15	VSS	VSS	T42
D16	VSS	VSS	U4
D17	VSS	VSS	U8
D19	VSS	VSS	U10
D21	VSS	VSS	U11
D24	VSS	VSS	U14
D25	VSS	VSS	U15
D27	VSS	VSS	U17
D29	VSS	VSS	U18
D30	VSS	VSS	U28
D31	VSS	VSS	U29
D33	VSS	VSS	U31
D35	VSS	VSS	U32
D36	VSS	VSS	U33
D45	VSS	VSS	U38
E13	VSS	VSS	V3
E15	VSS	VSS	V18
E31	VSS	VSS	V20
E33	VSS	VSS	V21
F8	VSS	VSS	V23
F44	VSS	VSS	V25
G9	VSS	VSS	V29
G42	VSS	VSS	V45
H3	VSS	VSS	W4
H17	VSS	VSS	W8
H19	VSS	VSS	W14
H22	VSS	VSS	W31
H24	VSS	VSS	W32
H27	VSS	VSS	W33
H29	VSS	VSS	W38
H35	VSS	VSS	Y17
J3	VSS	VSS	Y18
J5	VSS	VSS	Y20
J7	VSS	VSS	Y21
J10	VSS	VSS	Y26
J11	VSS	VSS	Y28
J39	VSS	VSS	Y29
K4	VSS	VSS	AA4
K10	VSS	VSS	AA17
K27	VSS	VSS	AA18
K33	VSS	VSS	AA20
K36	VSS	VSS	AA21
K42	VSS	VSS	AA25
K43	VSS	VSS	AA29
L4	VSS	VSS	AA42
L8	VSS	VSS	AB4
L12	VSS	VSS	AB5
L13	VSS	VSS	AB7
L15	VSS	VSS	AB8
L41	VSS	VSS	AB10

SKYLAKE_PCH_FCBGA837_FCBGA_SKL_PCH_H_TBD

U1100
SKL-PCH-H
TBD
FCBGA
SYM 12 OF 12

CRITICAL

OMIT_TABLE

AB11	VSS	VSS	AL32
AB14	VSS	VSS	AL33
AB15	VSS	VSS	AL38
AB31	VSS	VSS	AM15
AB32	VSS	VSS	AM17
AB38	VSS	VSS	AM19
AC1	VSS	VSS	AM22
AC18	VSS	VSS	AM24
AC20	VSS	VSS	AM27
AC21	VSS	VSS	AM29
AC25	VSS	VSS	AM45
AC29	VSS	VSS	AN4
AC45	VSS	VSS	AN7
AD4	VSS	VSS	AN8
AD8	VSS	VSS	AN10
AD11	VSS	VSS	AN11
AD14	VSS	VSS	AN22
AD32	VSS	VSS	AN27
AD33	VSS	VSS	AN31
AD36	VSS	VSS	AN39
AE4	VSS	VSS	AP4
AE18	VSS	VSS	AP11
AE20	VSS	VSS	AR5
AE21	VSS	VSS	AR7
AE25	VSS	VSS	AR9
AE28	VSS	VSS	AR33
AE29	VSS	VSS	AR34
AE42	VSS	VSS	AR42
AF18	VSS	VSS	AT9
AF20	VSS	VSS	AT10
AF21	VSS	VSS	AT15
AF23	VSS	VSS	AT36
AF25	VSS	VSS	AU1
AF26	VSS	VSS	AU7
AF28	VSS	VSS	AU35
AF29	VSS	VSS	AU36
AG4	VSS	VSS	AU39
AG11	VSS	VSS	AU45
AG13	VSS	VSS	AV6
AG31	VSS	VSS	AV17
AG32	VSS	VSS	AV24
AG33	VSS	VSS	AV27
AG38	VSS	VSS	AV31
AH1	VSS	VSS	AV33
AH17	VSS	VSS	AW9
AH18	VSS	VSS	AW13
AH20	VSS	VSS	AW19
AH21	VSS	VSS	AW29
AH23	VSS	VSS	AW37
AH25	VSS	VSS	AY38
AH26	VSS	VSS	AY45
AH28	VSS	VSS	BA1
AH29	VSS	VSS	BB1
AH45	VSS	VSS	BB11
AJ10	VSS	VSS	BB16
AJ14	VSS	VSS	BB21
AJ15	VSS	VSS	BB25
AJ17	VSS	VSS	BB30
AJ18	VSS	VSS	BB34
AJ26	VSS	VSS	BC1
AJ28	VSS	VSS	BC2
AJ29	VSS	VSS	BD2
AJ31	VSS	VSS	BD43
AJ32	VSS	VSS	BD44
AJ36	VSS	VSS	BD45
AK4	VSS	VSS	BE9
AK42	VSS	VSS	BE14
AL4	VSS	VSS	BE18
AL10	VSS	VSS	BE23
AL11	VSS	VSS	BE28
AL13	VSS	VSS	BE32
AL17	VSS	VSS	BE37
AL19	VSS	VSS	BE40
AL24	VSS	VSS	BE44
AL29	VSS		

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=kchooMLB
SYNC_DATE=10/30/2015

PAGE TITLE

CPU & CHIPSET: PCH Grounds

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 16 OF 121
SHEET 16 OF 102

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Primary Micro2-XDP

Extra BPM Testpoints

XDP_BPM_L<2> → TP1802
XDP_BPM_L<3> → TP1803

CRB has no-stuff S3 pull-up on PREQ_L
PDG allows any value from 51 to 3K

Connect Hook2 to CFG<0>
and pull-up to S0
(as in CRB)

WF: SB DPDG says HOOK1 is BP_PWRGD_RST#

NOTE: This is a MERGED XDP: = Primary XDP with CPU & PCH JTAGs in a Dual-Scan-Chain

NOTE: XDP_DBRESET_L pulled-up to 3.3V on PCH Support Page

NOTE: This is not the standard XDP pinout,
there is a horizontal mirroring for
use with 921-0133 Adapter Flex to
support chipset debug.

ICT CPU & PCH JTAG Test-Points:

TP1840 = TMS
TP1841 = TDI
TP1842 = TDO
TP1843 = TCK0
(TP1844 = TCK1)
TP1847 = TRST_L

NOTE: TDI and TMS are terminated in CPU.

PLACE_NEAR=U1100.AN3:28mm
PLACE_NEAR=U0500.F11:50.0mm

PLACE_NEAR=U0500.B9:50.0mm
PLACE_NEAR=U1100.AT3:150.0mm
PLACE_NEAR=R1850.1:10mm
PLACE_NEAR=U0500.B10:50.0mm
PLACE_NEAR=U1100.AT4:150.0mm
PLACE_NEAR=R1852.1:10mm

Secondary (PCH) Micro2-XDP

NOTE: This is not the standard XDP pinout,
there is a horizontal mirroring for
use with 921-0133 Adapter Flex to
support chipset debug.

PCH SIGNALS / XDP SIGNALS

PCH SIGNALS		XDP SIGNALS
PCH_PREQ_L	R1890	XDP_PCH_PREQ_L
PCH_PRDY_L	R1893	XDP_PCH_PRDY_L
PCH_SATAXPCIE0_SATAGP0	R1894	XDP_PCH_SATAXPCIE0_SATAGP0
PCH_SATAXPCIE1_SATAGP1	R1895	XDP_PCH_SATAXPCIE1_SATAGP1
PCH_SATAXPCIE2_SATAGP2	R1880	XDP_PCH_SATAXPCIE2_SATAGP2
PCH_SATAXPCIE3_SATAGP3	R1881	XDP_PCH_SATAXPCIE3_SATAGP3
PCH_SATAXPCIE4_SATAGP4	R1896	XDP_PCH_SATAXPCIE4_SATAGP4
PCH_SATAXPCIE5_SATAGP5	R1897	XDP_PCH_SATAXPCIE5_SATAGP5
PCH_SATAXPCIE6_SATAGP6	R1872	XDP_PCH_SATAXPCIE6_SATAGP6
PCH_SATAXPCIE7_SATAGP7	R1873	XDP_PCH_SATAXPCIE7_SATAGP7
XDP_PCH_OBSDATA_A2	R1874	XDP_PCH_CPU_GP0
PCH_DEVSLP0	R1875	XDP_PCH_DEVSLP0
PCH_ITP_PMODE	R1878	XDP_PCH_ITP_PMODE
PCH_DEVSLP1	R1879	XDP_PCH_DEVSLP1
PCH_DEVSLP2	R1882	XDP_PCH_DEVSLP2
XDP_PCH_OBSDATA_B2	R1883	XDP_PCH_CPU_GP1
PCH_SATALED_L	R1886	XDP_PCH_SATALED_L
PM_RSMRST_PCH_L	R1887	XDP_PCH_RSMRST_L
PM_SYSRST_L	R1884	XDP_PM_SYSRST_L
USB_EXTA_OC_L	R1833	XDP_PCH_USB2_OC0_L
USB_EXTB_OC_L	R1835	XDP_PCH_USB2_OC1_L
USB_EXTC_OC_L	R1832	XDP_PCH_USB2_OC2_L
USB_EXTD_OC_L	R1834	XDP_PCH_USB2_OC3_L

PCH/XDP Signal Isolation Notes:

R187x and R189x should be placed where
signal path needs to split between route
from PCH to J1850 and path to non-XDP
signal destination (to minimize stub).

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE	CPU & CHIPSET: XDP
Apple Inc.	DRAWING NUMBER 051-01506 · SIZE D
	REVISION 5.19.0
	BRANCH
	PAGE 18 OF 121
	SHEET 17 OF 102

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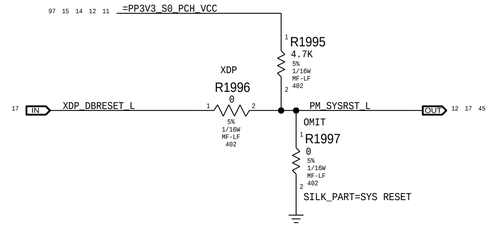
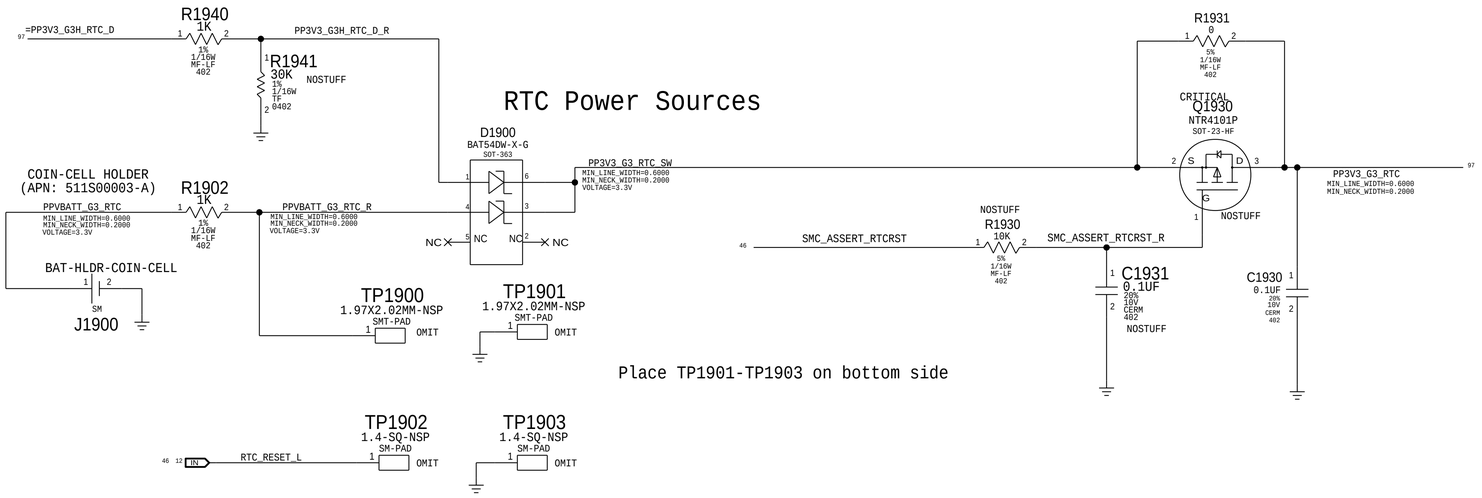
PCH Reset Button

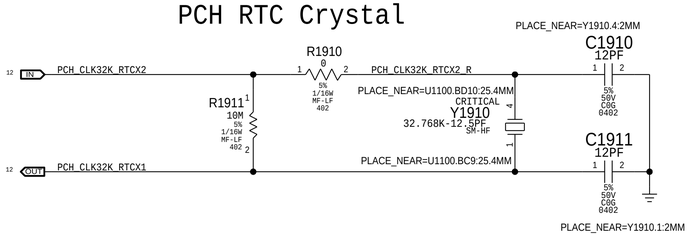


RTC Power Sources



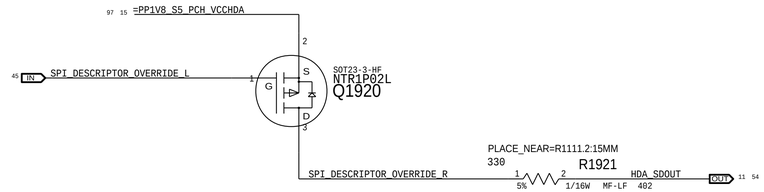
Place TP1901-TP1903 on bottom side

PCH RTC Crystal



PCH ME Disable Strap

PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.



BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE
CPU & CHIPSET: Chipset Support

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 19 OF 121
SHEET 18 OF 102

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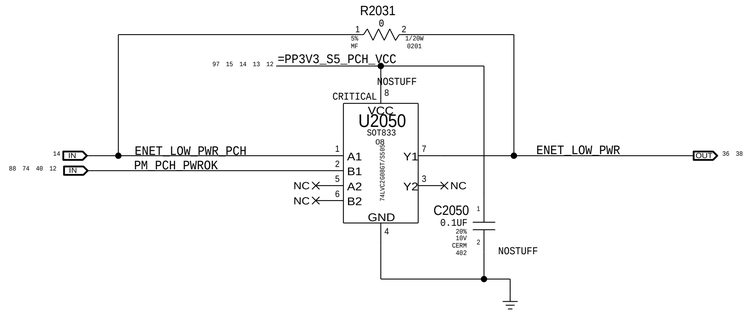
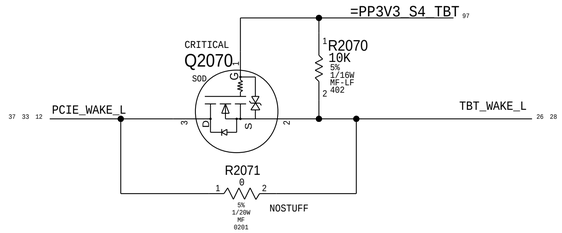
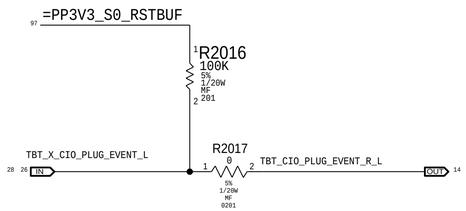
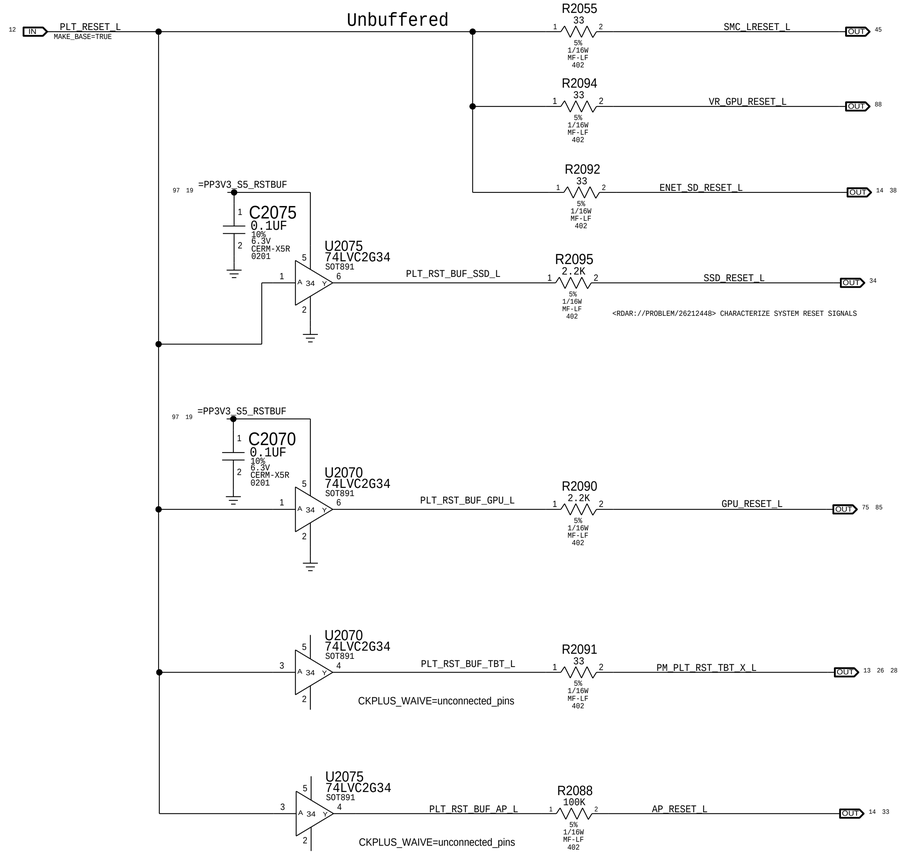
GPIO Glitch Prevention

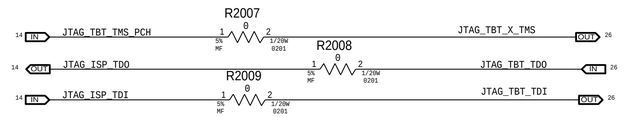




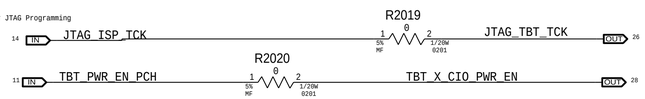


Platform Reset Connections





TBT_PWR_EN goes high for JTAG Programming



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=HARRIS
SYNC_DATE=06/22/2012

PAGE TITLE

CPU & CHIPSET: Project Chipset Support

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PAGE 20 OF 121
SHEET 19 OF 102

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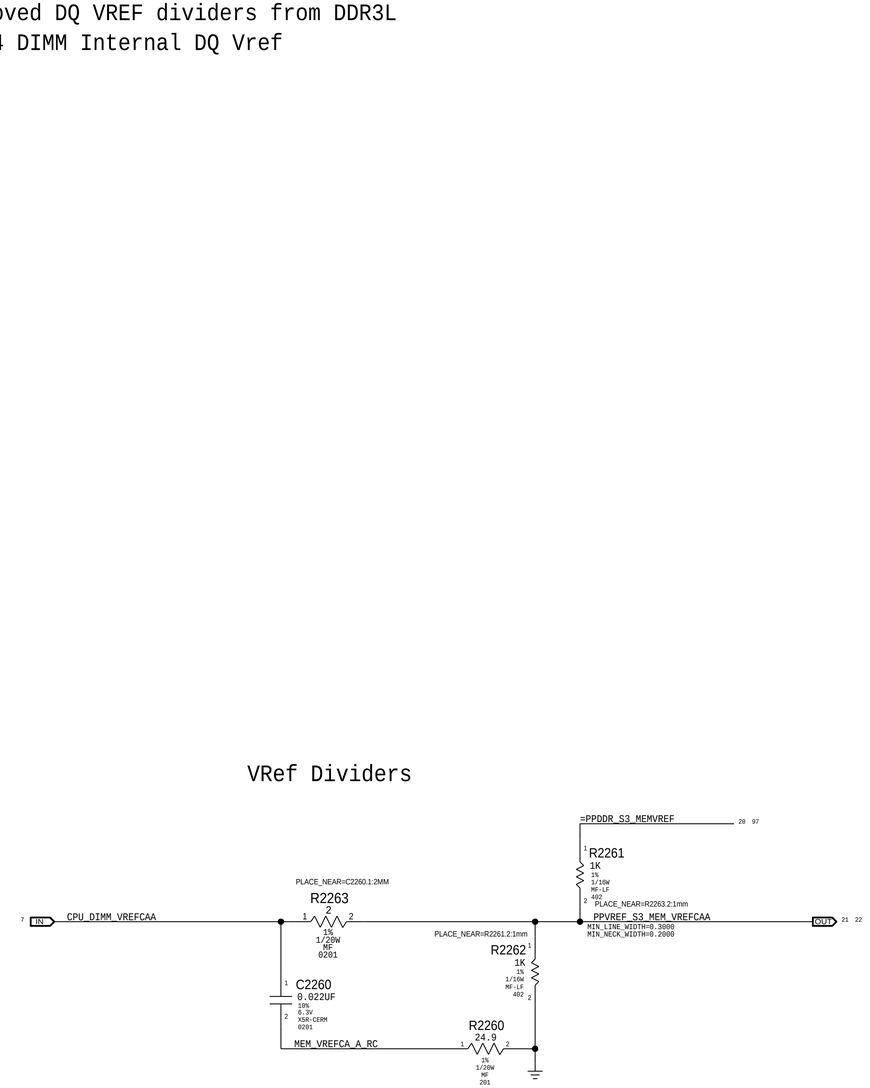
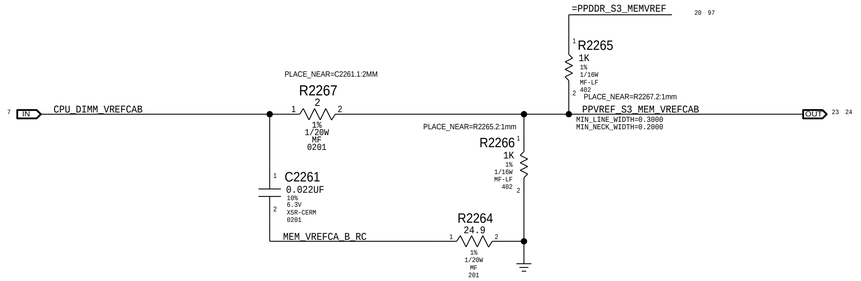
**Removed DQ VREF dividers from DDR3L
**DDR4 DIMM Internal DQ Vref

VRef Dividers



VRef Dividers



BOM_COST_GROUP=DRAM

SYNC_MASTER=DDRSSLER SYNC_DATE=02/03/2015

PAGE TITLE

DRAM:VREF/VTT EN

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

BRANCH

PAGE 22 OF 121

SHEET 20 OF 102

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Page Notes

Power aliases required by this page:
- =PPDDRVPP_S3_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PP3V3_S0_MEM_A_SPD (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

97 22 21 =PPDDRVTT_S0_MEM_A

C2350 1UF 10V 10% X5R 402
C2351 1UF 10V 10% X5R 402
C2352 1UF 10V 10% X5R 402
C2353 10UF 20% 6.3V CERM-X6S 0402

=PPDDRVPP_S3_MEM_A 21 22 97

C2360 10UF 20% 6.3V CERM-X6S 0402
C2361 1UF 20% 6.3V X6S-CERM 0201

DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

97 22 21 =PPVDDQ_S3_MEM_A

C2310 1UF 20% 6.3V X6S-CERM 0201
C2311 1UF 20% 6.3V X6S-CERM 0201
C2312 1UF 20% 6.3V X6S-CERM 0201
C2313 1UF 20% 6.3V X6S-CERM 0201
C2314 1UF 20% 6.3V X6S-CERM 0201
C2315 1UF 20% 6.3V X6S-CERM 0201
C2316 1UF 20% 6.3V X6S-CERM 0201
C2317 1UF 20% 6.3V X6S-CERM 0201

C2300 10UF 20% 6.3V CERM-X6S 0402
C2301 10UF 20% 6.3V CERM-X6S 0402
C2318 10UF 20% 6.3V CERM-X6S 0402
C2319 10UF 20% 6.3V CERM-X6S 0402
C2320 10UF 20% 6.3V CERM-X6S 0402
C2321 10UF 20% 6.3V CERM-X6S 0402
C2322 10UF 20% 6.3V CERM-X6S 0402
C2323 10UF 20% 6.3V CERM-X6S 0402

PPVREF_S3_MEM_VREFCAA 20 21 22

C2335 2.2UF 20% 6.3V CERM 402-LF
C2336 0.1UF 10% 10V CERM 402

=PP3V3_S0_MEM_A_SPD 21 22 97

C2337 2.2UF 20% 6.3V CERM 402-LF
C2338 0.1UF 10% 10V CERM 402

21 MEM_DIMM0_SA<2>
21 MEM_DIMM0_SA<1>
21 MEM_DIMM0_SA<0>
97 22 21 =PP3V3_S0_MEM_A_SPD

C2340 2.2UF 20% 6.3V CERM 402-LF
R2340 10K 5% 1/16W MF-LF 402
R2341 10K 5% 1/16W MF-LF 402
R2342 10K 5% 1/16W MF-LF 402

J2300
DDR4-SODIMM-X655
F-ANG-SM
SYM 3 OF 3

=PPVDDQ_S3_MEM_A 21 22 97

251	VSS	VDD	163
247	VSS	VDD	159
243	VSS	VDD	153
239	VSS	VDD	147
235	VSS	VDD	141
231	VSS	VDD	135
227	VSS	VDD	129
223	VSS	VDD	123
217	VSS	VDD	117
213	VSS	VDD	111
209	VSS	VDD	112
205	VSS	VDD	118
201	VSS	VDD	124
197	VSS	VDD	130
193	VSS	VDD	136
189	VSS	VDD	142
185	VSS	VDD	148
181	VSS	VDD	154
175	VSS	VDD	160
171	VSS		
167	VSS	VTT	258
107	VSS		
103	VSS	IO261	261
99	VSS	IO262	262
93	VSS	IO263	263
89	VSS	IO264	264
85	VSS	IO265	265
81	VSS		
77	VSS	VSS	60
73	VSS	VSS	64
69	VSS	VSS	68
65	VSS	VSS	72
61	VSS	VSS	78
57	VSS	VSS	82
51	VSS	VSS	86
47	VSS	VSS	90
43	VSS	VSS	94
39	VSS	VSS	98
35	VSS	VSS	102
31	VSS	VSS	106
27	VSS	VSS	168
23	VSS	VSS	172
19	VSS	VSS	176
15	VSS	VSS	180
9	VSS	VSS	184
5	VSS	VSS	188
1	VSS	VSS	192
2	VSS	VSS	196
6	VSS	VSS	202
10	VSS	VSS	206
14	VSS	VSS	210
18	VSS	VSS	214
22	VSS	VSS	218
26	VSS	VSS	222
30	VSS	VSS	226
36	VSS	VSS	230
40	VSS	VSS	234
44	VSS	VSS	238
48	VSS	VSS	244
52	VSS	VSS	248
56	VSS	VSS	252

=PPDDRVTT_S0_MEM_A 21 22 97

J2300
DDR4-SODIMM-X655
F-ANG-SM
SYM 2 OF 3

=PPVDDQ_S3_MEM_A 97 21 22

DM8*/DBI8*/NC	96		
DM7*/DBI7*/NC	241		
DM6*/DBI6*/NC	220		
DM5*/DBI5*/NC	199		
DM4*/DBI4*/NC	178		
DM3*/DBI3*/NC	75		
DM2*/DBI2*/NC	54		
DM1*/DBI1*/NC	33		
DM0*/DBI0*/NC	12		
DQS8_T	97	NC	
DQS8_C	95	NC	
DQS7_T	242	=MEM_A_DQS_P<7>	BI 22 25
DQS7_C	240	=MEM_A_DQS_N<7>	BI 22 25
DQS6_T	221	=MEM_A_DQS_P<6>	BI 22 25
DQS6_C	219	=MEM_A_DQS_N<6>	BI 22 25
DQS5_T	200	=MEM_A_DQS_P<5>	BI 22 25
DQS5_C	198	=MEM_A_DQS_N<5>	BI 22 25
DQS4_T	179	=MEM_A_DQS_P<4>	BI 22 25
DQS4_C	177	=MEM_A_DQS_N<4>	BI 22 25
DQS3_T	76	=MEM_A_DQS_P<3>	BI 22 25
DQS3_C	74	=MEM_A_DQS_N<3>	BI 22 25
DQS2_T	55	=MEM_A_DQS_P<2>	BI 22 25
DQS2_C	53	=MEM_A_DQS_N<2>	BI 22 25
DQS1_T	34	=MEM_A_DQS_P<1>	BI 22 25
DQS1_C	32	=MEM_A_DQS_N<1>	BI 22 25
DQS0_T	13	=MEM_A_DQS_P<0>	BI 22 25
DQS0_C	11	=MEM_A_DQS_N<0>	BI 22 25

SKL/KBL DOES NOT SUPPORT DM (DATAMASK)
SIGNAL IS ACTIVE LOW FOR DDR4
RECOMMENDED TO PULL TO VDDQ TO DISABLE FUNCITON

P/N: 516S00204

J2300
DDR4-SODIMM-X655
F-ANG-SM
SYM 1 OF 3

22 7 IN	MEM_A_A<16>	152	RAS*/A16
22 7 IN	MEM_A_A<15>	156	CAS*/A15
22 7 IN	MEM_A_A<14>	151	WE*/A14
22 7 IN	MEM_A_A<13>	158	A13
22 7 IN	MEM_A_A<12>	119	A12
22 7 IN	MEM_A_A<11>	120	A11
22 7 IN	MEM_A_A<10>	146	A10/AP
22 7 IN	MEM_A_A<9>	121	A9
22 7 IN	MEM_A_A<8>	125	A8
22 7 IN	MEM_A_A<7>	122	A7
22 7 IN	MEM_A_A<6>	127	A6
22 7 IN	MEM_A_A<5>	126	A5
22 7 IN	MEM_A_A<4>	128	A4
22 7 IN	MEM_A_A<3>	131	A3
22 7 IN	MEM_A_A<2>	132	A2
22 7 IN	MEM_A_A<1>	133	A1
22 7 IN	MEM_A_A<0>	144	A0
22 7 IN	MEM_A_BA<1>	145	BA1
22 7 IN	MEM_A_BA<0>	150	BA0
7 IN	MEM_A_CLK_P<1>	138	CK1_T/NF
7 IN	MEM_A_CLK_N<1>	140	CK1_C/NF
7 IN	MEM_A_CLK_P<0>	137	CK0_T
7 IN	MEM_A_CLK_N<0>	139	CK0_C
NC		165	C1/CS3*/NC
NC		162	C0/CS2*/NC
7 IN	MEM_A_CS_L<1>	157	CS1*/NC
7 IN	MEM_A_CS_L<0>	149	CS0*
7 IN	MEM_A_CKE<1>	110	CKE1/NC
7 IN	MEM_A_CKE<0>	109	CKE0
22 7 IN	MEM_A_BG<1>	113	BG1
22 7 IN	MEM_A_BG<0>	115	BG0
7 IN	MEM_A_ODT<1>	161	ODT1/NC
7 IN	MEM_A_ODT<0>	155	ODT0
22 7 IN	MEM_A_PARITY	143	PARITY
NC		104	CB7/NC
NC		100	CB6/NC
NC		87	CB5/NC
NC		88	CB4/NC
NC		105	CB3/NC
NC		101	CB2/NC
NC		91	CB1/NC
NC		92	CB0/NC
24 23 22 12 IN	MEM_RESET_L	108	RESET*
97 22 21	=PP3V3_S0_MEM_A_SPD	255	VDDSPD
21	MEM_DIMM0_SA<2>	166	SA2
21	MEM_DIMM0_SA<1>	260	SA1
21	MEM_DIMM0_SA<0>	256	SA0
48 22 BI	=I2C_SODIMMA_SDA	254	SDA
48 22 IN	=I2C_SODIMMA_SCL	253	SCL
22 21 20 IN	PPVREF_S3_MEM_VREFCA	164	VREFCA
97 22 21	=PPDDRVPP_S3_MEM_A	257	VPP
		259	VPP
22 7 IN	MEM_A_ACT_L	114	ACT*
22 7 OUT	MEM_A_ALERT_L	116	ALERT*

DQ63	246	=MEM_A_DQ<63>	BI 22 25
DQ62	245	=MEM_A_DQ<62>	BI 22 25
DQ61	233	=MEM_A_DQ<61>	BI 22 25
DQ60	232	=MEM_A_DQ<60>	BI 22 25
DQ59	250	=MEM_A_DQ<59>	BI 22 25
DQ58	249	=MEM_A_DQ<58>	BI 22 25
DQ57	236	=MEM_A_DQ<57>	BI 22 25
DQ56	237	=MEM_A_DQ<56>	BI 22 25
DQ55	225	=MEM_A_DQ<55>	BI 22 25
DQ54	224	=MEM_A_DQ<54>	BI 22 25
DQ53	212	=MEM_A_DQ<53>	BI 22 25
DQ52	211	=MEM_A_DQ<52>	BI 22 25
DQ51	229	=MEM_A_DQ<51>	BI 22 25
DQ50	228	=MEM_A_DQ<50>	BI 22 25
DQ49	215	=MEM_A_DQ<49>	BI 22 25
DQ48	216	=MEM_A_DQ<48>	BI 22 25
DQ47	204	=MEM_A_DQ<47>	BI 22 25
DQ46	203	=MEM_A_DQ<46>	BI 22 25
DQ45	190	=MEM_A_DQ<45>	BI 22 25
DQ44	191	=MEM_A_DQ<44>	BI 22 25
DQ43	208	=MEM_A_DQ<43>	BI 22 25
DQ42	207	=MEM_A_DQ<42>	BI 22 25
DQ41	194	=MEM_A_DQ<41>	BI 22 25
DQ40	195	=MEM_A_DQ<40>	BI 22 25
DQ39	182	=MEM_A_DQ<39>	BI 22 25
DQ38	183	=MEM_A_DQ<38>	BI 22 25
DQ37	169	=MEM_A_DQ<37>	BI 22 25
DQ36	170	=MEM_A_DQ<36>	BI 22 25
DQ35	186	=MEM_A_DQ<35>	BI 22 25
DQ34	187	=MEM_A_DQ<34>	BI 22 25
DQ33	173	=MEM_A_DQ<33>	BI 22 25
DQ32	174	=MEM_A_DQ<32>	BI 22 25
DQ31	80	=MEM_A_DQ<31>	BI 22 25
DQ30	79	=MEM_A_DQ<30>	BI 22 25
DQ29	67	=MEM_A_DQ<29>	BI 22 25
DQ28	66	=MEM_A_DQ<28>	BI 22 25
DQ27	84	=MEM_A_DQ<27>	BI 22 25
DQ26	83	=MEM_A_DQ<26>	BI 22 25
DQ25	71	=MEM_A_DQ<25>	BI 22 25
DQ24	70	=MEM_A_DQ<24>	BI 22 25
DQ23	59	=MEM_A_DQ<23>	BI 22 25
DQ22	58	=MEM_A_DQ<22>	BI 22 25
DQ21	45	=MEM_A_DQ<21>	BI 22 25
DQ20	46	=MEM_A_DQ<20>	BI 22 25
DQ19	63	=MEM_A_DQ<19>	BI 22 25
DQ18	62	=MEM_A_DQ<18>	BI 22 25
DQ17	49	=MEM_A_DQ<17>	BI 22 25
DQ16	50	=MEM_A_DQ<16>	BI 22 25
DQ15	37	=MEM_A_DQ<15>	BI 22 25
DQ14	38	=MEM_A_DQ<14>	BI 22 25
DQ13	25	=MEM_A_DQ<13>	BI 22 25
DQ12	24	=MEM_A_DQ<12>	BI 22 25
DQ11	42	=MEM_A_DQ<11>	BI 22 25
DQ10	41	=MEM_A_DQ<10>	BI 22 25
DQ9	29	=MEM_A_DQ<9>	BI 22 25
DQ8	28	=MEM_A_DQ<8>	BI 22 25
DQ7	17	=MEM_A_DQ<7>	BI 22 25
DQ6	16	=MEM_A_DQ<6>	BI 22 25
DQ5	3	=MEM_A_DQ<5>	BI 22 25
DQ4	4	=MEM_A_DQ<4>	BI 22 25
DQ3	21	=MEM_A_DQ<3>	BI 22 25
DQ2	20	=MEM_A_DQ<2>	BI 22 25
DQ1	7	=MEM_A_DQ<1>	BI 22 25
DQ0	8	=MEM_A_DQ<0>	BI 22 25
EVENT*/NF	134	MEM_EVENT_L	OUT 22 23 24 45 46

BOM_COST_GROUP=DRAM

PAGE TITLE
DRAM: SO-DIMM Connector A Slot0

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 23 OF 121
SHEET 21 OF 102

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Page Notes

Power aliases required by this page:
- =PPDDRVPP_S3_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PP3V3_S0_MEM_A_SPD (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

97 22 21 =PPDDRVTT_S0_MEM_A

C2450 1UF 10% 10V X5S 402
C2451 1UF 10% 10V X5S 402
C2452 1UF 10% 10V X5S 402
C2453 10UF 20% 6.3V CERM-X6S 0402

=PPDDRVPP_S3_MEM_A 21 22 97

C2460 10UF 20% 6.3V CERM-X6S 0402
C2461 1UF 20% 6.3V X6S-CERM 0201

DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

97 22 21 =PPVDDQ_S3_MEM_A

C2410 C2411 C2412 C2413 C2414 C2415 C2416 C2417 1UF 20% 6.3V X6S-CERM 0201

C2400 C2401 C2418 C2419 C2420 C2421 C2422 C2423 10UF 20% 6.3V CERM-X6S 0402

PPVREF_S3_MEM_VREFCAA 20 21 22

C2435 2.2UF 20% 6.3V CERM 402-LF
C2436 0.1UF 10% 10V CERM 402

=PP3V3_S0_MEM_A_SPD 97 22 21

C2437 2.2UF 20% 6.3V CERM 402-LF
C2438 0.1UF 20% 10V CERM 402

97 22 21 =PP3V3_S0_MEM_A_SPD

R2440 10K 5% 1/16W MF-LF 402

22 MEM_DIMM1_SA<0>

22 MEM_DIMM1_SA<2>
22 MEM_DIMM1_SA<1>
97 22 21 =PP3V3_S0_MEM_A_SPD

C2440 2.2UF 20% 6.3V CERM 402-LF
R2441 10K 5% 1/16W MF-LF 402
R2442 10K 5% 1/16W MF-LF 402

J2400
DDR4-SODIMM-X655
F-ANG-SM
SYM 2 OF 3

=PPVDDQ_S3_MEM_A 21 22 97

DM8*/DBI8*/NC 96
DM7*/DBI7*/NC 241
DM6*/DBI6*/NC 220
DM5*/DBI5*/NC 199
DM4*/DBI4*/NC 178
DM3*/DBI3*/NC 75
DM2*/DBI2*/NC 54
DM1*/DBI1*/NC 33
DM0*/DBI0*/NC 12

DQS8_T 97 NC
DQS8_C 95 NC
DQS7_T 242 =MEM_A_DQS_P<7>
DQS7_C 240 =MEM_A_DQS_N<7>
DQS6_T 221 =MEM_A_DQS_P<6>
DQS6_C 219 =MEM_A_DQS_N<6>
DQS5_T 200 =MEM_A_DQS_P<5>
DQS5_C 198 =MEM_A_DQS_N<5>
DQS4_T 179 =MEM_A_DQS_P<4>
DQS4_C 177 =MEM_A_DQS_N<4>
DQS3_T 76 =MEM_A_DQS_P<3>
DQS3_C 74 =MEM_A_DQS_N<3>
DQS2_T 55 =MEM_A_DQS_P<2>
DQS2_C 53 =MEM_A_DQS_N<2>
DQS1_T 34 =MEM_A_DQS_P<1>
DQS1_C 32 =MEM_A_DQS_N<1>
DQS0_T 13 =MEM_A_DQS_P<0>
DQS0_C 11 =MEM_A_DQS_N<0>

SKL/KBL DOES NOT SUPPORT DM (DATAMASK)
SIGNAL IS ACTIVE LOW FOR DDR4
RECOMMENDED TO PULL TO VDDQ TO DISABLE FUNCITON

J2400
DDR4-SODIMM-X655
F-ANG-SM
SYM 3 OF 3

VSS: 251 247 243 239 235 231 227 223 217 213 209 205 201 197 193 189 185 181 175 171 167 107 103 99 93 89 85 81 77 73 69 65 61 57 51 47 43 39 35 31 27 23 19 15 9 5 1 2 6 10 14 18 22 26 30 36 40 44 48 52 56

VDD: 163 159 153 147 141 135 129 123 117 111 112 118 124 130 136 142 148 154 160

VTT 258 =PPDDRVTT_S0_MEM_A 21 22 97

IO261 261
IO262 262
IO263 263
IO264 264
IO265 265

VSS: 60 64 68 72 78 82 86 90 94 98 102 106 168 172 176 180 184 188 192 196 202 206 210 214 218 222 226 230 234 238 244 248 252

P/N: 516S00204

J2400
DDR4-SODIMM-X655
F-ANG-SM
SYM 1 OF 3

21 7 IN	MEM_A_A<16>	152	RAS*/A16
21 7 IN	MEM_A_A<15>	156	CAS*/A15
21 7 IN	MEM_A_A<14>	151	WE*/A14
21 7 IN	MEM_A_A<13>	158	A13
21 7 IN	MEM_A_A<12>	119	A12
21 7 IN	MEM_A_A<11>	120	A11
21 7 IN	MEM_A_A<10>	146	A10/AP
21 7 IN	MEM_A_A<9>	121	A9
21 7 IN	MEM_A_A<8>	125	A8
21 7 IN	MEM_A_A<7>	122	A7
21 7 IN	MEM_A_A<6>	127	A6
21 7 IN	MEM_A_A<5>	126	A5
21 7 IN	MEM_A_A<4>	128	A4
21 7 IN	MEM_A_A<3>	131	A3
21 7 IN	MEM_A_A<2>	132	A2
21 7 IN	MEM_A_A<1>	133	A1
21 7 IN	MEM_A_A<0>	144	A0
21 7 IN	MEM_A_BA<1>	145	BA1
21 7 IN	MEM_A_BA<0>	150	BA0
7 IN	MEM_A_CLK_P<3>	138	CK1_T/NF
7 IN	MEM_A_CLK_N<3>	140	CK1_C/NF
7 IN	MEM_A_CLK_P<2>	137	CK0_T
7 IN	MEM_A_CLK_N<2>	139	CK0_C
	NC	165	C1/CS3*/NC
	NC	162	C0/CS2*/NC
7 IN	MEM_A_CS_L<3>	157	CS1*/NC
7 IN	MEM_A_CS_L<2>	149	CS0*
7 IN	MEM_A_CKE<3>	110	CKE1/NC
7 IN	MEM_A_CKE<2>	109	CKE0
21 7 IN	MEM_A_BG<1>	113	BG1
21 7 IN	MEM_A_BG<0>	115	BG0
7 IN	MEM_A_ODT<3>	161	ODT1/NC
7 IN	MEM_A_ODT<2>	155	ODT0
21 7 IN	MEM_A_PARITY	143	PARITY
	NC	104	CB7/NC
	NC	100	CB6/NC
	NC	87	CB5/NC
	NC	88	CB4/NC
	NC	105	CB3/NC
	NC	101	CB2/NC
	NC	91	CB1/NC
	NC	92	CB0/NC
24 23 21 12 IN	MEM_RESET_L	108	RESET*
97 22 21	=PP3V3_S0_MEM_A_SPD	255	VDDSPD
22	MEM_DIMM1_SA<2>	166	SA2
22	MEM_DIMM1_SA<1>	260	SA1
22	MEM_DIMM1_SA<0>	256	SA0
48 21 BI	=I2C_SODIMMA_SDA	254	SDA
48 21 IN	=I2C_SODIMMA_SCL	253	SCL
22 21 20 IN	PPVREF_S3_MEM_VREFCAA	164	VREFCA
97 22 21	=PPDDRVPP_S3_MEM_A	257	VPP
		259	VPP
21 7 IN	MEM_A_ACT_L	114	ACT*
21 7 OUT	MEM_A_ALERT_L	116	ALERT*

DQ63	246	=MEM_A_DQ<63>	BI 21 25
DQ62	245	=MEM_A_DQ<62>	BI 21 25
DQ61	233	=MEM_A_DQ<61>	BI 21 25
DQ60	232	=MEM_A_DQ<60>	BI 21 25
DQ59	250	=MEM_A_DQ<59>	BI 21 25
DQ58	249	=MEM_A_DQ<58>	BI 21 25
DQ57	236	=MEM_A_DQ<57>	BI 21 25
DQ56	237	=MEM_A_DQ<56>	BI 21 25
DQ55	225	=MEM_A_DQ<55>	BI 21 25
DQ54	224	=MEM_A_DQ<54>	BI 21 25
DQ53	212	=MEM_A_DQ<53>	BI 21 25
DQ52	211	=MEM_A_DQ<52>	BI 21 25
DQ51	229	=MEM_A_DQ<51>	BI 21 25
DQ50	228	=MEM_A_DQ<50>	BI 21 25
DQ49	215	=MEM_A_DQ<49>	BI 21 25
DQ48	216	=MEM_A_DQ<48>	BI 21 25
DQ47	204	=MEM_A_DQ<47>	BI 21 25
DQ46	203	=MEM_A_DQ<46>	BI 21 25
DQ45	190	=MEM_A_DQ<45>	BI 21 25
DQ44	191	=MEM_A_DQ<44>	BI 21 25
DQ43	208	=MEM_A_DQ<43>	BI 21 25
DQ42	207	=MEM_A_DQ<42>	BI 21 25
DQ41	194	=MEM_A_DQ<41>	BI 21 25
DQ40	195	=MEM_A_DQ<40>	BI 21 25
DQ39	182	=MEM_A_DQ<39>	BI 21 25
DQ38	183	=MEM_A_DQ<38>	BI 21 25
DQ37	169	=MEM_A_DQ<37>	BI 21 25
DQ36	170	=MEM_A_DQ<36>	BI 21 25
DQ35	186	=MEM_A_DQ<35>	BI 21 25
DQ34	187	=MEM_A_DQ<34>	BI 21 25
DQ33	173	=MEM_A_DQ<33>	BI 21 25
DQ32	174	=MEM_A_DQ<32>	BI 21 25
DQ31	80	=MEM_A_DQ<31>	BI 21 25
DQ30	79	=MEM_A_DQ<30>	BI 21 25
DQ29	67	=MEM_A_DQ<29>	BI 21 25
DQ28	66	=MEM_A_DQ<28>	BI 21 25
DQ27	84	=MEM_A_DQ<27>	BI 21 25
DQ26	83	=MEM_A_DQ<26>	BI 21 25
DQ25	71	=MEM_A_DQ<25>	BI 21 25
DQ24	70	=MEM_A_DQ<24>	BI 21 25
DQ23	59	=MEM_A_DQ<23>	BI 21 25
DQ22	58	=MEM_A_DQ<22>	BI 21 25
DQ21	45	=MEM_A_DQ<21>	BI 21 25
DQ20	46	=MEM_A_DQ<20>	BI 21 25
DQ19	63	=MEM_A_DQ<19>	BI 21 25
DQ18	62	=MEM_A_DQ<18>	BI 21 25
DQ17	49	=MEM_A_DQ<17>	BI 21 25
DQ16	50	=MEM_A_DQ<16>	BI 21 25
DQ15	37	=MEM_A_DQ<15>	BI 21 25
DQ14	38	=MEM_A_DQ<14>	BI 21 25
DQ13	25	=MEM_A_DQ<13>	BI 21 25
DQ12	24	=MEM_A_DQ<12>	BI 21 25
DQ11	42	=MEM_A_DQ<11>	BI 21 25
DQ10	41	=MEM_A_DQ<10>	BI 21 25
DQ9	29	=MEM_A_DQ<9>	BI 21 25
DQ8	28	=MEM_A_DQ<8>	BI 21 25
DQ7	17	=MEM_A_DQ<7>	BI 21 25
DQ6	16	=MEM_A_DQ<6>	BI 21 25
DQ5	3	=MEM_A_DQ<5>	BI 21 25
DQ4	4	=MEM_A_DQ<4>	BI 21 25
DQ3	21	=MEM_A_DQ<3>	BI 21 25
DQ2	20	=MEM_A_DQ<2>	BI 21 25
DQ1	7	=MEM_A_DQ<1>	BI 21 25
DQ0	8	=MEM_A_DQ<0>	BI 21 25
EVENT*/NF	134	MEM_EVENT_L	OUT 21 23 24 45 46

BOM_COST_GROUP=DRAM

PAGE TITLE
DRAM: SO-DIMM Connector A Slot1

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 24 OF 121
SHEET 22 OF 102

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Page Notes

Power aliases required by this page:
- =PPDDRVPP_S3_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PP3V3_S0_MEM_B_SPD (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

P/N: 516S00204

=PPDDRVTT_S0_MEM_B

C2550 1UF 20% 10V X5R 402
C2551 1UF 20% 10V X5R 402
C2552 1UF 20% 10V X5R 402
C2553 10UF 20% 6.3V CERM-X6S 0402

=PPDDRVPP_S3_MEM_B

C2560 10UF 20% 6.3V CERM-X6S 0402
C2561 1UF 20% 6.3V X6S-CERM 0201

DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

=PPVDDQ_S3_MEM_B

C2510 1UF 20% 6.3V X6S-CERM 0201
C2511 1UF 20% 6.3V X6S-CERM 0201
C2512 1UF 20% 6.3V X6S-CERM 0201
C2513 1UF 20% 6.3V X6S-CERM 0201
C2514 1UF 20% 6.3V X6S-CERM 0201
C2515 1UF 20% 6.3V X6S-CERM 0201
C2516 1UF 20% 6.3V X6S-CERM 0201
C2517 1UF 20% 6.3V X6S-CERM 0201

C2500 10UF 20% 6.3V CERM-X6S 0402
C2501 10UF 20% 6.3V CERM-X6S 0402
C2518 10UF 20% 6.3V CERM-X6S 0402
C2519 10UF 20% 6.3V CERM-X6S 0402
C2520 10UF 20% 6.3V CERM-X6S 0402
C2521 10UF 20% 6.3V CERM-X6S 0402
C2522 10UF 20% 6.3V CERM-X6S 0402
C2523 10UF 20% 6.3V CERM-X6S 0402

PPVREF_S3_MEM_VREFCAB

C2535 2.2UF 20% 6.3V CERM 402-LF
C2536 0.1UF 20% 10V CERM 402

=PP3V3_S0_MEM_B_SPD

C2537 2.2UF 20% 6.3V CERM 402-LF
C2538 0.1UF 20% 10V CERM 402

=PP3V3_S0_MEM_B_SPD

R2541 10K 5% 1/16W MF-LF 402

MEM_DIMM2_SA<1>
MEM_DIMM2_SA<2>
MEM_DIMM2_SA<0>
=PP3V3_S0_MEM_B_SPD

C2540 2.2UF 20% 6.3V CERM 402-LF
R2540 10K 5% 1/16W MF-LF 402
R2542 10K 5% 1/16W MF-LF 402

J2500
DDR4-SODIMM-X655
F-ANG-SM
SYM 3 OF 3

251	VSS	VDD	163
247	VSS	VDD	159
243	VSS	VDD	153
239	VSS	VDD	147
235	VSS	VDD	141
231	VSS	VDD	135
227	VSS	VDD	129
223	VSS	VDD	123
217	VSS	VDD	117
213	VSS	VDD	111
209	VSS	VDD	112
205	VSS	VDD	118
201	VSS	VDD	124
197	VSS	VDD	130
193	VSS	VDD	136
189	VSS	VDD	142
185	VSS	VDD	148
181	VSS	VDD	154
175	VSS	VDD	160
171	VSS		
167	VSS	VTT	258
107	VSS		
103	VSS	IO261	261
99	VSS	IO262	262
93	VSS	IO263	263
89	VSS	IO264	264
85	VSS	IO265	265
81	VSS		
77	VSS	VSS	60
73	VSS	VSS	64
69	VSS	VSS	68
65	VSS	VSS	72
61	VSS	VSS	78
57	VSS	VSS	82
51	VSS	VSS	86
47	VSS	VSS	90
43	VSS	VSS	94
39	VSS	VSS	98
35	VSS	VSS	102
31	VSS	VSS	106
27	VSS	VSS	168
23	VSS	VSS	172
19	VSS	VSS	176
15	VSS	VSS	180
9	VSS	VSS	184
5	VSS	VSS	188
1	VSS	VSS	192
2	VSS	VSS	196
6	VSS	VSS	202
10	VSS	VSS	206
14	VSS	VSS	210
18	VSS	VSS	214
22	VSS	VSS	218
26	VSS	VSS	222
30	VSS	VSS	226
36	VSS	VSS	230
40	VSS	VSS	234
44	VSS	VSS	238
48	VSS	VSS	244
52	VSS	VSS	248
56	VSS	VSS	252

J2500
DDR4-SODIMM-X655
F-ANG-SM
SYM 2 OF 3

=PPVDDQ_S3_MEM_B

DM8*/DBI8*/NC	96	DQS8_T	97
DM7*/DBI7*/NC	241	DQS8_C	95
DM6*/DBI6*/NC	220	DQS7_T	242 =MEM_B_DQS_P<7>
DM5*/DBI5*/NC	199	DQS7_C	240 =MEM_B_DQS_N<7>
DM4*/DBI4*/NC	178	DQS6_T	221 =MEM_B_DQS_P<6>
DM3*/DBI3*/NC	75	DQS6_C	219 =MEM_B_DQS_N<6>
DM2*/DBI2*/NC	54	DQS5_T	200 =MEM_B_DQS_P<5>
DM1*/DBI1*/NC	33	DQS5_C	198 =MEM_B_DQS_N<5>
DM0*/DBI0*/NC	12	DQS4_T	179 =MEM_B_DQS_P<4>
		DQS4_C	177 =MEM_B_DQS_N<4>
		DQS3_T	76 =MEM_B_DQS_P<3>
		DQS3_C	74 =MEM_B_DQS_N<3>
		DQS2_T	55 =MEM_B_DQS_P<2>
		DQS2_C	53 =MEM_B_DQS_N<2>
		DQS1_T	34 =MEM_B_DQS_P<1>
		DQS1_C	32 =MEM_B_DQS_N<1>
		DQS0_T	13 =MEM_B_DQS_P<0>
		DQS0_C	11 =MEM_B_DQS_N<0>

SKL/KBL DOES NOT SUPPORT DM (DATAMASK)
SIGNAL IS ACTIVE LOW FOR DDR4
RECOMMENDED TO PULL TO VDDQ TO DISABLE FUNCITON

J2500
DDR4-SODIMM-X655
F-ANG-SM
SYM 1 OF 3

MEM_B_A<16>	152	RAS*/A16	
MEM_B_A<15>	156	CAS*/A15	
MEM_B_A<14>	151	WE*/A14	
MEM_B_A<13>	158	A13	
MEM_B_A<12>	119	A12	
MEM_B_A<11>	120	A11	
MEM_B_A<10>	146	A10/AP	
MEM_B_A<9>	121	A9	
MEM_B_A<8>	125	A8	
MEM_B_A<7>	122	A7	
MEM_B_A<6>	127	A6	
MEM_B_A<5>	126	A5	
MEM_B_A<4>	128	A4	
MEM_B_A<3>	131	A3	
MEM_B_A<2>	132	A2	
MEM_B_A<1>	133	A1	
MEM_B_A<0>	144	A0	
MEM_B_BA<1>	145	BA1	
MEM_B_BA<0>	150	BA0	
MEM_B_CLK_P<1>	138	CK1_T/NF	
MEM_B_CLK_N<1>	140	CK1_C/NF	
MEM_B_CLK_P<0>	137	CK0_T	
MEM_B_CLK_N<0>	139	CK0_C	
NC	165	C1/CS3*/NC	
NC	162	C0/CS2*/NC	
MEM_B_CS_L<1>	157	CS1*/NC	
MEM_B_CS_L<0>	149	CS0*	
MEM_B_CKE<1>	110	CKE1/NC	
MEM_B_CKE<0>	109	CKE0	
MEM_B_BG<1>	113	BG1	
MEM_B_BG<0>	115	BG0	
MEM_B_ODT<1>	161	ODT1/NC	
MEM_B_ODT<0>	155	ODT0	
MEM_B_PARITY	143	PARITY	
NC	104	CB7/NC	
NC	100	CB6/NC	
NC	87	CB5/NC	
NC	88	CB4/NC	
NC	105	CB3/NC	
NC	101	CB2/NC	
NC	91	CB1/NC	
NC	92	CB0/NC	
MEM_RESET_L	108	RESET*	
=PP3V3_S0_MEM_B_SPD	255	VDDSPD	
MEM_DIMM2_SA<2>	166	SA2	
MEM_DIMM2_SA<1>	260	SA1	
MEM_DIMM2_SA<0>	256	SA0	
=I2C_SODIMMB_SDA	254	SDA	
=I2C_SODIMMB_SCL	253	SCL	
PPVREF_S3_MEM_VREFCAB	164	VREFCA	
=PPDDRVPP_S3_MEM_B	257	VPP	
	259	VPP	
MEM_B_ACT_L	114	ACT*	
MEM_B_ALERT_L	116	ALERT*	

DQ63	246	=MEM_B_DQ<63>	
DQ62	245	=MEM_B_DQ<62>	
DQ61	233	=MEM_B_DQ<61>	
DQ60	232	=MEM_B_DQ<60>	
DQ59	250	=MEM_B_DQ<59>	
DQ58	249	=MEM_B_DQ<58>	
DQ57	236	=MEM_B_DQ<57>	
DQ56	237	=MEM_B_DQ<56>	
DQ55	225	=MEM_B_DQ<55>	
DQ54	224	=MEM_B_DQ<54>	
DQ53	212	=MEM_B_DQ<53>	
DQ52	211	=MEM_B_DQ<52>	
DQ51	229	=MEM_B_DQ<51>	
DQ50	228	=MEM_B_DQ<50>	
DQ49	215	=MEM_B_DQ<49>	
DQ48	216	=MEM_B_DQ<48>	
DQ47	204	=MEM_B_DQ<47>	
DQ46	203	=MEM_B_DQ<46>	
DQ45	190	=MEM_B_DQ<45>	
DQ44	191	=MEM_B_DQ<44>	
DQ43	208	=MEM_B_DQ<43>	
DQ42	207	=MEM_B_DQ<42>	
DQ41	194	=MEM_B_DQ<41>	
DQ40	195	=MEM_B_DQ<40>	
DQ39	182	=MEM_B_DQ<39>	
DQ38	183	=MEM_B_DQ<38>	
DQ37	169	=MEM_B_DQ<37>	
DQ36	170	=MEM_B_DQ<36>	
DQ35	186	=MEM_B_DQ<35>	
DQ34	187	=MEM_B_DQ<34>	
DQ33	173	=MEM_B_DQ<33>	
DQ32	174	=MEM_B_DQ<32>	
DQ31	80	=MEM_B_DQ<31>	
DQ30	79	=MEM_B_DQ<30>	
DQ29	67	=MEM_B_DQ<29>	
DQ28	66	=MEM_B_DQ<28>	
DQ27	84	=MEM_B_DQ<27>	
DQ26	83	=MEM_B_DQ<26>	
DQ25	71	=MEM_B_DQ<25>	
DQ24	70	=MEM_B_DQ<24>	
DQ23	59	=MEM_B_DQ<23>	
DQ22	58	=MEM_B_DQ<22>	
DQ21	45	=MEM_B_DQ<21>	
DQ20	46	=MEM_B_DQ<20>	
DQ19	63	=MEM_B_DQ<19>	
DQ18	62	=MEM_B_DQ<18>	
DQ17	49	=MEM_B_DQ<17>	
DQ16	50	=MEM_B_DQ<16>	
DQ15	37	=MEM_B_DQ<15>	
DQ14	38	=MEM_B_DQ<14>	
DQ13	25	=MEM_B_DQ<13>	
DQ12	24	=MEM_B_DQ<12>	
DQ11	42	=MEM_B_DQ<11>	
DQ10	41	=MEM_B_DQ<10>	
DQ9	29	=MEM_B_DQ<9>	
DQ8	28	=MEM_B_DQ<8>	
DQ7	17	=MEM_B_DQ<7>	
DQ6	16	=MEM_B_DQ<6>	
DQ5	3	=MEM_B_DQ<5>	
DQ4	4	=MEM_B_DQ<4>	
DQ3	21	=MEM_B_DQ<3>	
DQ2	20	=MEM_B_DQ<2>	
DQ1	7	=MEM_B_DQ<1>	
DQ0	8	=MEM_B_DQ<0>	
EVENT*/NF	134	MEM_EVENT_L	

BOM_COST_GROUP=DRAM

SYNC_MASTER=DDRESSLER
SYNC_DATE=02/03/2016

PAGE TITLE
DRAM: SO-DIMM CONNECTOR B SLOT0

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 25 OF 121
SHEET 23 OF 102

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Page Notes

Power aliases required by this page:
- =PPDDRVPP_S3_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PP3V3_S0_MEM_B_SPD (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

P/N: 516S00204

=PPDDRVTT_S0_MEM_B

=PPDDRVPP_S3_MEM_B

DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

=PPVDDQ_S3_MEM_B

PPVREF_S3_MEM_VREFCAB

=PP3V3_S0_MEM_B_SPD

=PP3V3_S0_MEM_B_SPD

MEM_DIMM3_SA<1>

MEM_DIMM3_SA<0>

MEM_DIMM3_SA<2>

=PP3V3_S0_MEM_B_SPD

J2600
DDR4-SODIMM-X655
F-ANG-SM
SYM 1 OF 3

J2600
DDR4-SODIMM-X655
F-ANG-SM
SYM 2 OF 3

J2600
DDR4-SODIMM-X655
F-ANG-SM
SYM 3 OF 3

SKL/KBL DOES NOT SUPPORT DM (DATAMASK)
SIGNAL IS ACTIVE LOW FOR DDR4
RECOMMENDED TO PULL TO VDDQ TO DISABLE FUNCITON

BOM_COST_GROUP=DRAM

SYNC_MASTER=DRESSLER SYNC_DATE=02/03/2016

PAGE TITLE
DRAM: SO-DIMM CONNECTOR B SLOT1

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 26 OF 121
SHEET 24 OF 102

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THERE ARE DQ PIN SWAPS WITHIN A BYTE

MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	21 22	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	=MEM_B_DQS_N<0>	23 24
MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	21 22	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	=MEM_B_DQS_P<0>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<7>	MAKE_BASE=TRUE	=MEM_A_DQ<7>	21 22	MEM_B_DQ<7>	MAKE_BASE=TRUE	=MEM_B_DQ<2>	23 24
MEM_A_DQ<6>	MAKE_BASE=TRUE	=MEM_A_DQ<6>	21 22	MEM_B_DQ<6>	MAKE_BASE=TRUE	=MEM_B_DQ<3>	23 24
MEM_A_DQ<5>	MAKE_BASE=TRUE	=MEM_A_DQ<0>	21 22	MEM_B_DQ<5>	MAKE_BASE=TRUE	=MEM_B_DQ<0>	23 24
MEM_A_DQ<4>	MAKE_BASE=TRUE	=MEM_A_DQ<1>	21 22	MEM_B_DQ<4>	MAKE_BASE=TRUE	=MEM_B_DQ<1>	23 24
MEM_A_DQ<3>	MAKE_BASE=TRUE	=MEM_A_DQ<2>	21 22	MEM_B_DQ<3>	MAKE_BASE=TRUE	=MEM_B_DQ<7>	23 24
MEM_A_DQ<2>	MAKE_BASE=TRUE	=MEM_A_DQ<3>	21 22	MEM_B_DQ<2>	MAKE_BASE=TRUE	=MEM_B_DQ<6>	23 24
MEM_A_DQ<1>	MAKE_BASE=TRUE	=MEM_A_DQ<4>	21 22	MEM_B_DQ<1>	MAKE_BASE=TRUE	=MEM_B_DQ<5>	23 24
MEM_A_DQ<0>	MAKE_BASE=TRUE	=MEM_A_DQ<5>	21 22	MEM_B_DQ<0>	MAKE_BASE=TRUE	=MEM_B_DQ<4>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<1>	MAKE_BASE=TRUE	=MEM_A_DQS_N<1>	21 22	MEM_B_DQS_N<1>	MAKE_BASE=TRUE	=MEM_B_DQS_N<1>	23 24
MEM_A_DQS_P<1>	MAKE_BASE=TRUE	=MEM_A_DQS_P<1>	21 22	MEM_B_DQS_P<1>	MAKE_BASE=TRUE	=MEM_B_DQS_P<1>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<15>	MAKE_BASE=TRUE	=MEM_A_DQ<14>	21 22	MEM_B_DQ<15>	MAKE_BASE=TRUE	=MEM_B_DQ<11>	23 24
MEM_A_DQ<14>	MAKE_BASE=TRUE	=MEM_A_DQ<15>	21 22	MEM_B_DQ<14>	MAKE_BASE=TRUE	=MEM_B_DQ<10>	23 24
MEM_A_DQ<13>	MAKE_BASE=TRUE	=MEM_A_DQ<13>	21 22	MEM_B_DQ<13>	MAKE_BASE=TRUE	=MEM_B_DQ<8>	23 24
MEM_A_DQ<12>	MAKE_BASE=TRUE	=MEM_A_DQ<9>	21 22	MEM_B_DQ<12>	MAKE_BASE=TRUE	=MEM_B_DQ<13>	23 24
MEM_A_DQ<11>	MAKE_BASE=TRUE	=MEM_A_DQ<10>	21 22	MEM_B_DQ<11>	MAKE_BASE=TRUE	=MEM_B_DQ<14>	23 24
MEM_A_DQ<10>	MAKE_BASE=TRUE	=MEM_A_DQ<11>	21 22	MEM_B_DQ<10>	MAKE_BASE=TRUE	=MEM_B_DQ<15>	23 24
MEM_A_DQ<9>	MAKE_BASE=TRUE	=MEM_A_DQ<12>	21 22	MEM_B_DQ<9>	MAKE_BASE=TRUE	=MEM_B_DQ<9>	23 24
MEM_A_DQ<8>	MAKE_BASE=TRUE	=MEM_A_DQ<8>	21 22	MEM_B_DQ<8>	MAKE_BASE=TRUE	=MEM_B_DQ<12>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<2>	MAKE_BASE=TRUE	=MEM_A_DQS_N<2>	21 22	MEM_B_DQS_N<2>	MAKE_BASE=TRUE	=MEM_B_DQS_N<2>	23 24
MEM_A_DQS_P<2>	MAKE_BASE=TRUE	=MEM_A_DQS_P<2>	21 22	MEM_B_DQS_P<2>	MAKE_BASE=TRUE	=MEM_B_DQS_P<2>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<23>	MAKE_BASE=TRUE	=MEM_A_DQ<22>	21 22	MEM_B_DQ<23>	MAKE_BASE=TRUE	=MEM_B_DQ<18>	23 24
MEM_A_DQ<22>	MAKE_BASE=TRUE	=MEM_A_DQ<23>	21 22	MEM_B_DQ<22>	MAKE_BASE=TRUE	=MEM_B_DQ<19>	23 24
MEM_A_DQ<21>	MAKE_BASE=TRUE	=MEM_A_DQ<20>	21 22	MEM_B_DQ<21>	MAKE_BASE=TRUE	=MEM_B_DQ<16>	23 24
MEM_A_DQ<20>	MAKE_BASE=TRUE	=MEM_A_DQ<16>	21 22	MEM_B_DQ<20>	MAKE_BASE=TRUE	=MEM_B_DQ<20>	23 24
MEM_A_DQ<19>	MAKE_BASE=TRUE	=MEM_A_DQ<19>	21 22	MEM_B_DQ<19>	MAKE_BASE=TRUE	=MEM_B_DQ<23>	23 24
MEM_A_DQ<18>	MAKE_BASE=TRUE	=MEM_A_DQ<18>	21 22	MEM_B_DQ<18>	MAKE_BASE=TRUE	=MEM_B_DQ<22>	23 24
MEM_A_DQ<17>	MAKE_BASE=TRUE	=MEM_A_DQ<17>	21 22	MEM_B_DQ<17>	MAKE_BASE=TRUE	=MEM_B_DQ<21>	23 24
MEM_A_DQ<16>	MAKE_BASE=TRUE	=MEM_A_DQ<21>	21 22	MEM_B_DQ<16>	MAKE_BASE=TRUE	=MEM_B_DQ<17>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<3>	MAKE_BASE=TRUE	=MEM_A_DQS_N<3>	21 22	MEM_B_DQS_N<3>	MAKE_BASE=TRUE	=MEM_B_DQS_N<3>	23 24
MEM_A_DQS_P<3>	MAKE_BASE=TRUE	=MEM_A_DQS_P<3>	21 22	MEM_B_DQS_P<3>	MAKE_BASE=TRUE	=MEM_B_DQS_P<3>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<31>	MAKE_BASE=TRUE	=MEM_A_DQ<27>	21 22	MEM_B_DQ<31>	MAKE_BASE=TRUE	=MEM_B_DQ<26>	23 24
MEM_A_DQ<30>	MAKE_BASE=TRUE	=MEM_A_DQ<30>	21 22	MEM_B_DQ<30>	MAKE_BASE=TRUE	=MEM_B_DQ<27>	23 24
MEM_A_DQ<29>	MAKE_BASE=TRUE	=MEM_A_DQ<25>	21 22	MEM_B_DQ<29>	MAKE_BASE=TRUE	=MEM_B_DQ<24>	23 24
MEM_A_DQ<28>	MAKE_BASE=TRUE	=MEM_A_DQ<29>	21 22	MEM_B_DQ<28>	MAKE_BASE=TRUE	=MEM_B_DQ<25>	23 24
MEM_A_DQ<27>	MAKE_BASE=TRUE	=MEM_A_DQ<31>	21 22	MEM_B_DQ<27>	MAKE_BASE=TRUE	=MEM_B_DQ<30>	23 24
MEM_A_DQ<26>	MAKE_BASE=TRUE	=MEM_A_DQ<26>	21 22	MEM_B_DQ<26>	MAKE_BASE=TRUE	=MEM_B_DQ<31>	23 24
MEM_A_DQ<25>	MAKE_BASE=TRUE	=MEM_A_DQ<28>	21 22	MEM_B_DQ<25>	MAKE_BASE=TRUE	=MEM_B_DQ<29>	23 24
MEM_A_DQ<24>	MAKE_BASE=TRUE	=MEM_A_DQ<24>	21 22	MEM_B_DQ<24>	MAKE_BASE=TRUE	=MEM_B_DQ<28>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<4>	MAKE_BASE=TRUE	=MEM_A_DQS_N<4>	21 22	MEM_B_DQS_N<4>	MAKE_BASE=TRUE	=MEM_B_DQS_N<4>	23 24
MEM_A_DQS_P<4>	MAKE_BASE=TRUE	=MEM_A_DQS_P<4>	21 22	MEM_B_DQS_P<4>	MAKE_BASE=TRUE	=MEM_B_DQS_P<4>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<39>	MAKE_BASE=TRUE	=MEM_A_DQ<39>	21 22	MEM_B_DQ<39>	MAKE_BASE=TRUE	=MEM_B_DQ<35>	23 24
MEM_A_DQ<38>	MAKE_BASE=TRUE	=MEM_A_DQ<38>	21 22	MEM_B_DQ<38>	MAKE_BASE=TRUE	=MEM_B_DQ<39>	23 24
MEM_A_DQ<37>	MAKE_BASE=TRUE	=MEM_A_DQ<36>	21 22	MEM_B_DQ<37>	MAKE_BASE=TRUE	=MEM_B_DQ<32>	23 24
MEM_A_DQ<36>	MAKE_BASE=TRUE	=MEM_A_DQ<37>	21 22	MEM_B_DQ<36>	MAKE_BASE=TRUE	=MEM_B_DQ<37>	23 24
MEM_A_DQ<35>	MAKE_BASE=TRUE	=MEM_A_DQ<34>	21 22	MEM_B_DQ<35>	MAKE_BASE=TRUE	=MEM_B_DQ<34>	23 24
MEM_A_DQ<34>	MAKE_BASE=TRUE	=MEM_A_DQ<35>	21 22	MEM_B_DQ<34>	MAKE_BASE=TRUE	=MEM_B_DQ<38>	23 24
MEM_A_DQ<33>	MAKE_BASE=TRUE	=MEM_A_DQ<33>	21 22	MEM_B_DQ<33>	MAKE_BASE=TRUE	=MEM_B_DQ<33>	23 24
MEM_A_DQ<32>	MAKE_BASE=TRUE	=MEM_A_DQ<32>	21 22	MEM_B_DQ<32>	MAKE_BASE=TRUE	=MEM_B_DQ<36>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<5>	MAKE_BASE=TRUE	=MEM_A_DQS_N<5>	21 22	MEM_B_DQS_N<5>	MAKE_BASE=TRUE	=MEM_B_DQS_N<5>	23 24
MEM_A_DQS_P<5>	MAKE_BASE=TRUE	=MEM_A_DQS_P<5>	21 22	MEM_B_DQS_P<5>	MAKE_BASE=TRUE	=MEM_B_DQS_P<5>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<47>	MAKE_BASE=TRUE	=MEM_A_DQ<43>	21 22	MEM_B_DQ<47>	MAKE_BASE=TRUE	=MEM_B_DQ<43>	23 24
MEM_A_DQ<46>	MAKE_BASE=TRUE	=MEM_A_DQ<46>	21 22	MEM_B_DQ<46>	MAKE_BASE=TRUE	=MEM_B_DQ<42>	23 24
MEM_A_DQ<45>	MAKE_BASE=TRUE	=MEM_A_DQ<44>	21 22	MEM_B_DQ<45>	MAKE_BASE=TRUE	=MEM_B_DQ<44>	23 24
MEM_A_DQ<44>	MAKE_BASE=TRUE	=MEM_A_DQ<40>	21 22	MEM_B_DQ<44>	MAKE_BASE=TRUE	=MEM_B_DQ<40>	23 24
MEM_A_DQ<43>	MAKE_BASE=TRUE	=MEM_A_DQ<47>	21 22	MEM_B_DQ<43>	MAKE_BASE=TRUE	=MEM_B_DQ<46>	23 24
MEM_A_DQ<42>	MAKE_BASE=TRUE	=MEM_A_DQ<42>	21 22	MEM_B_DQ<42>	MAKE_BASE=TRUE	=MEM_B_DQ<47>	23 24
MEM_A_DQ<41>	MAKE_BASE=TRUE	=MEM_A_DQ<45>	21 22	MEM_B_DQ<41>	MAKE_BASE=TRUE	=MEM_B_DQ<41>	23 24
MEM_A_DQ<40>	MAKE_BASE=TRUE	=MEM_A_DQ<41>	21 22	MEM_B_DQ<40>	MAKE_BASE=TRUE	=MEM_B_DQ<45>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<6>	MAKE_BASE=TRUE	=MEM_A_DQS_N<6>	21 22	MEM_B_DQS_N<6>	MAKE_BASE=TRUE	=MEM_B_DQS_N<6>	23 24
MEM_A_DQS_P<6>	MAKE_BASE=TRUE	=MEM_A_DQS_P<6>	21 22	MEM_B_DQS_P<6>	MAKE_BASE=TRUE	=MEM_B_DQS_P<6>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQ<55>	MAKE_BASE=TRUE	=MEM_A_DQ<54>	21 22	MEM_B_DQ<55>	MAKE_BASE=TRUE	=MEM_B_DQ<50>	23 24
MEM_A_DQ<54>	MAKE_BASE=TRUE	=MEM_A_DQ<49>	21 22	MEM_B_DQ<54>	MAKE_BASE=TRUE	=MEM_B_DQ<54>	23 24
MEM_A_DQ<53>	MAKE_BASE=TRUE	=MEM_A_DQ<55>	21 22	MEM_B_DQ<53>	MAKE_BASE=TRUE	=MEM_B_DQ<48>	23 24
MEM_A_DQ<52>	MAKE_BASE=TRUE	=MEM_A_DQ<53>	21 22	MEM_B_DQ<52>	MAKE_BASE=TRUE	=MEM_B_DQ<49>	23 24
MEM_A_DQ<51>	MAKE_BASE=TRUE	=MEM_A_DQ<51>	21 22	MEM_B_DQ<51>	MAKE_BASE=TRUE	=MEM_B_DQ<51>	23 24
MEM_A_DQ<50>	MAKE_BASE=TRUE	=MEM_A_DQ<52>	21 22	MEM_B_DQ<50>	MAKE_BASE=TRUE	=MEM_B_DQ<55>	23 24
MEM_A_DQ<49>	MAKE_BASE=TRUE	=MEM_A_DQ<50>	21 22	MEM_B_DQ<49>	MAKE_BASE=TRUE	=MEM_B_DQ<52>	23 24
MEM_A_DQ<48>	MAKE_BASE=TRUE	=MEM_A_DQ<48>	21 22	MEM_B_DQ<48>	MAKE_BASE=TRUE	=MEM_B_DQ<53>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		
MEM_A_DQS_N<7>	MAKE_BASE=TRUE	=MEM_A_DQS_N<7>	21 22	MEM_B_DQS_N<7>	MAKE_BASE=TRUE	=MEM_B_DQS_N<7>	23 24
MEM_A_DQS_P<7>	MAKE_BASE=TRUE	=MEM_A_DQS_P<7>	21 22	MEM_B_DQS_P<7>	MAKE_BASE=TRUE	=MEM_B_DQS_P<7>	23 24
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MEM_A_DQ<63>	MAKE_BASE=TRUE	=MEM_A_DQ<57>	21 22	MEM_B_DQ<63>	MAKE_BASE=TRUE	=MEM_B_DQ<59>	23 24
MEM_A_DQ<62>	MAKE_BASE=TRUE	=MEM_A_DQ<59>	21 22	MEM_B_DQ<62>	MAKE_BASE=TRUE	=MEM_B_DQ<63>	23 24
MEM_A_DQ<61>	MAKE_BASE=TRUE	=MEM_A_DQ<56>	21 22	MEM_B_DQ<61>	MAKE_BASE=TRUE	=MEM_B_DQ<56>	23 24
MEM_A_DQ<60>	MAKE_BASE=TRUE	=MEM_A_DQ<62>	21 22	MEM_B_DQ<60>	MAKE_BASE=TRUE	=MEM_B_DQ<60>	23 24
MEM_A_DQ<59>	MAKE_BASE=TRUE	=MEM_A_DQ<58>	21 22	MEM_B_DQ<59>	MAKE_BASE=TRUE	=MEM_B_DQ<62>	23 24
MEM_A_DQ<58>	MAKE_BASE=TRUE	=MEM_A_DQ<60>	21 22	MEM_B_DQ<58>	MAKE_BASE=TRUE	=MEM_B_DQ<58>	23 24
MEM_A_DQ<57>	MAKE_BASE=TRUE	=MEM_A_DQ<63>	21 22	MEM_B_DQ<57>	MAKE_BASE=TRUE	=MEM_B_DQ<61>	23 24
MEM_A_DQ<56>	MAKE_BASE=TRUE	=MEM_A_DQ<61>	21 22	MEM_B_DQ<56>	MAKE_BASE=TRUE	=MEM_B_DQ<57>	23 24
	MAKE_BASE=TRUE				MAKE_BASE=TRUE		

BOM_COST_GROUP=DRAM

SYNC_MASTER=DRESSLER SYNC_DATE=02/03/2016

PAGE TITLE

DRAM: ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

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PAGE 27 OF 121

SHEET 25 OF 102

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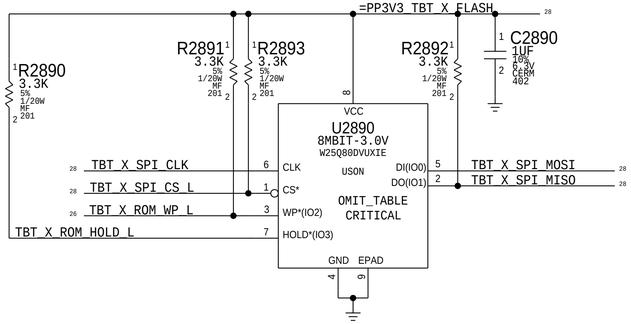


SNK0 AC Coupling

DP_X_SNK0_ML_C_P<0>	C2820	0.1UF	DP_X_SNK0_ML_P<0>
DP_X_SNK0_ML_C_N<0>	C2821	0.1UF	DP_X_SNK0_ML_N<0>
DP_X_SNK0_ML_C_P<1>	C2822	0.1UF	DP_X_SNK0_ML_P<1>
DP_X_SNK0_ML_C_N<1>	C2823	0.1UF	DP_X_SNK0_ML_N<1>
DP_X_SNK0_ML_C_P<2>	C2824	0.1UF	DP_X_SNK0_ML_P<2>
DP_X_SNK0_ML_C_N<2>	C2825	0.1UF	DP_X_SNK0_ML_N<2>
DP_X_SNK0_ML_C_P<3>	C2826	0.1UF	DP_X_SNK0_ML_P<3>
DP_X_SNK0_ML_C_N<3>	C2827	0.1UF	DP_X_SNK0_ML_N<3>
DP_X_SNK0_AUXCH_C_P	C2828	0.1UF	DP_X_SNK0_AUXCH_P
DP_X_SNK0_AUXCH_C_N	C2829	0.1UF	DP_X_SNK0_AUXCH_N

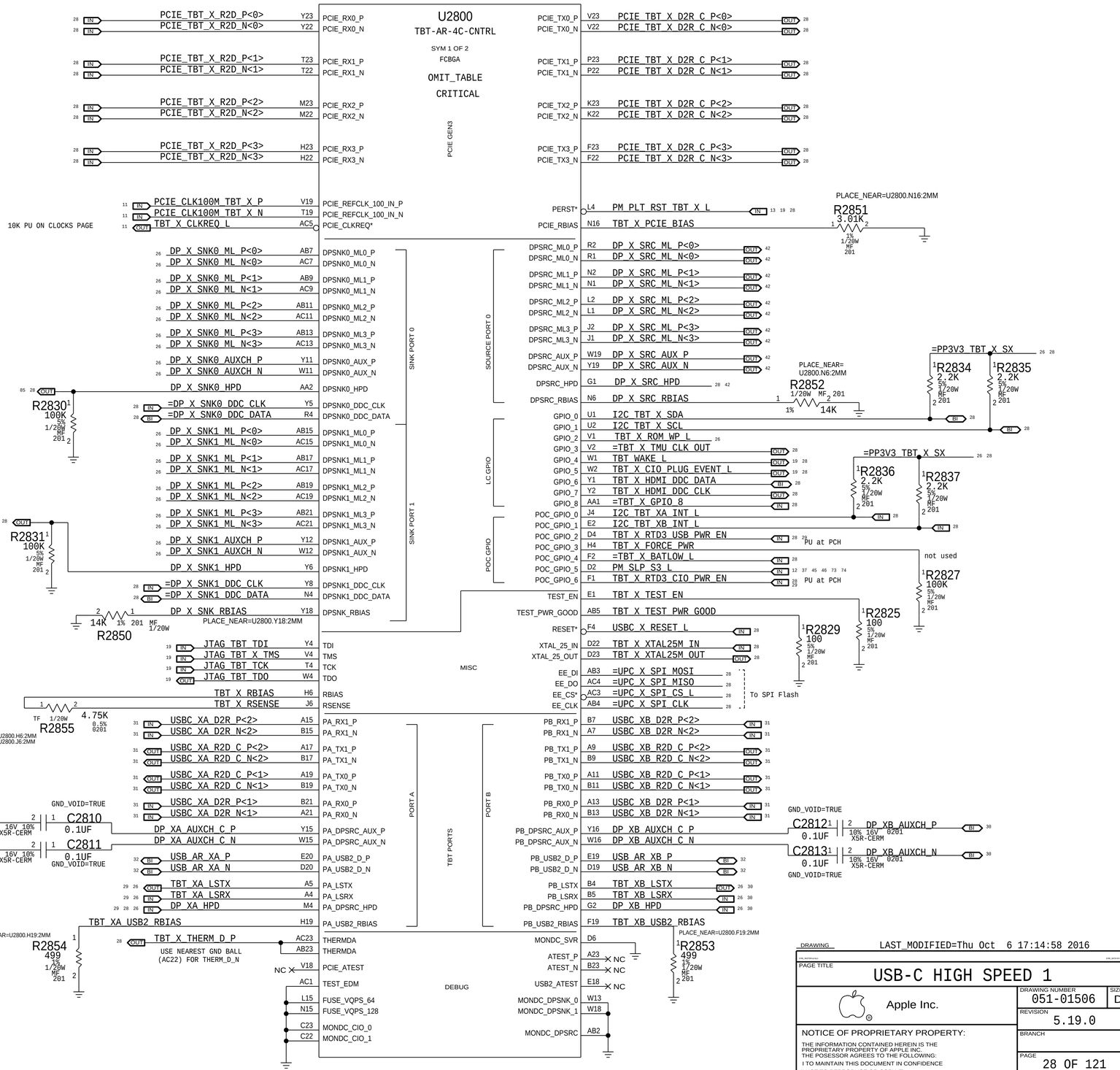
All: GND_VOID=TRUE; 0.1UF 10% 16V X5R-CERM 0201

SNK1 AC Coupling

DP_X_SNK1_ML_C_P<0>	C2830	0.1UF	DP_X_SNK1_ML_P<0>
DP_X_SNK1_ML_C_N<0>	C2831	0.1UF	DP_X_SNK1_ML_N<0>
DP_X_SNK1_ML_C_P<1>	C2832	0.1UF	DP_X_SNK1_ML_P<1>
DP_X_SNK1_ML_C_N<1>	C2833	0.1UF	DP_X_SNK1_ML_N<1>
DP_X_SNK1_ML_C_P<2>	C2834	0.1UF	DP_X_SNK1_ML_P<2>
DP_X_SNK1_ML_C_N<2>	C2835	0.1UF	DP_X_SNK1_ML_N<2>
DP_X_SNK1_ML_C_P<3>	C2836	0.1UF	DP_X_SNK1_ML_P<3>
DP_X_SNK1_ML_C_N<3>	C2837	0.1UF	DP_X_SNK1_ML_N<3>
DP_X_SNK1_AUXCH_C_P	C2838	0.1UF	DP_X_SNK1_AUXCH_P
DP_X_SNK1_AUXCH_C_N	C2839	0.1UF	DP_X_SNK1_AUXCH_N

All: GND_VOID=TRUE; 0.1UF 10% 16V X5R-CERM 0201

R2862 100K 5% 1/20W MF 201	DP_XA_HPD
R2872 100K 5% 1/20W MF 201	DP_XB_HPD
R2860 1M 5% 1/20W MF 201	TBT_XA_LSTX
R2861 1M 5% 1/20W MF 201	TBT_XA_LSRX
R2870 1M 5% 1/20W MF 201	TBT_XB_LSTX
R2871 1M 5% 1/20W MF 201	TBT_XB_LSRX



BOM_COST_GROUP=TBT

DRAWING LAST_MODIFIED=Thu Oct 6 17:14:58 2016

PAGE TITLE: USB-C HIGH SPEED 1

Apple Inc.

DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	28 OF 121
SHEET	26 OF 102

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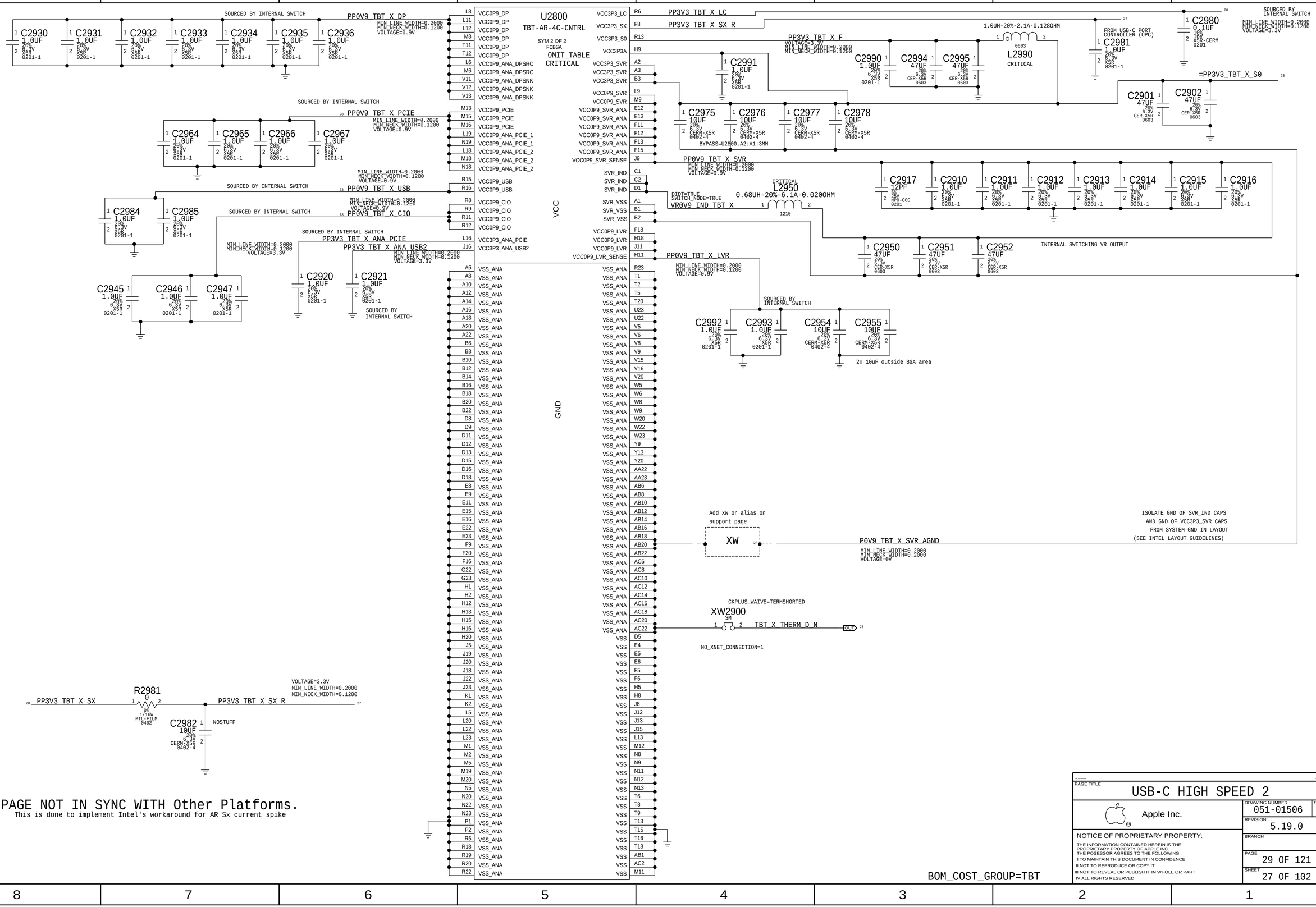
U2800
TBT-AR-4C-CNTRL
SYM 2 OF 2
FCBGA
OMIT_TABLE
CRITICAL
VCC
GND
SOURCED BY INTERNAL SWITCH
PP0V9_TBT_X_DP
PP0V9_TBT_X_PCIE
PP0V9_TBT_X_USB
PP0V9_TBT_X_CIO
PP3V3_TBT_X_ANA_PCIE
PP3V3_TBT_X_ANA_USB2
PP3V3_TBT_X_LC
PP3V3_TBT_X_SX_R
PP3V3_TBT_X_F
PP3V3_TBT_X_S0
PP0V9_TBT_X_SVR
PP0V9_TBT_X_LVR
VR0V9_IND_TBT_X
0.68UH-20%-6.1A-0.020OHM
L2950
1.0UH-20%-2.1A-0.128OHM
L2990
FROM USB-C PORT CONTROLLER (UPC)
INTERNAL SWITCHING VR OUTPUT
2x 10uF outside BGA area
BYPASS=U2800.A2:A1:3MM
Add XW or alias on support page
XW
P0V9_TBT_X_SVR_AGND
ISOLATE GND OF SVR_IND CAPS AND GND OF VCC3P3_SVR CAPS FROM SYSTEM GND IN LAYOUT (SEE INTEL LAYOUT GUIDELINES)
CKPLUS_WAIVE=TERMSHORTED
XW2900
TBT_X_THERM_D_N
NO_XNET_CONNECTION=1
R2981
PP3V3_TBT_X_SX
PP3V3_TBT_X_SX_R
C2982
NOSTUFF
PAGE NOT IN SYNC WITH Other Platforms.
This is done to implement Intel's workaround for AR Sx current spike
BOM_COST_GROUP=TBT
PAGE TITLE
USB-C HIGH SPEED 2
Apple Inc.
DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 29 OF 121
SHEET 27 OF 102
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Ridge PUs

=TBT_X_BATLOW_L — MAKE_BASE=TRUE PM_BATLOW_L — R3075 100K 1/20W 201 5% MF — PP3V3_TBT_X_SX

TBT_X_HDMI_DDC_CLK — R3074 100K 1/20W 201 MF 5% — PP3V3_TBT_X_LC

USBC 5V EN PD

UPC_XA_5V_EN — R3032 100K MF 1/20W 5% 201

UPC_XB_5V_EN — R3012 100K MF 1/20W 5% 201

ACE Debug Support

TP_UPC_XA_DBG_UART_TX — R3089 NOSTUFF NONE NONE NONE 402 — TP_UPC_XA_DBG_UART_RX

TP_UPC_XB_DBG_UART_TX — R3088 NOSTUFF NONE NONE NONE 402 — TP_UPC_XB_DBG_UART_RX

Ridge PDs

=DP_X_SNK0_DDC_CLK (MAKE_BASE=TRUE) DP_X_SNK0_DDC_CLK	R3067 100K 5% 1/20W MF 201	
=DP_X_SNK0_DDC_DATA (MAKE_BASE=TRUE) DP_X_SNK0_DDC_DATA	R3068 100K 5% 1/20W MF 201	
=DP_X_SNK1_DDC_CLK (MAKE_BASE=TRUE) DP_X_SNK1_DDC_CLK	R3069 100K 5% 1/20W MF 201	
=DP_X_SNK1_DDC_DATA (MAKE_BASE=TRUE) DP_X_SNK1_DDC_DATA	R3070 100K 5% 1/20W MF 201	
=TBT_X_GPIO_8 (MAKE_BASE=TRUE) PD_TBT_X_GPIO_8	R3071 100K 5% 1/20W MF 201	
=TBT_X_TMU_CLK_OUT (MAKE_BASE=TRUE) PD_TBT_X_TMU_CLK_OUT	R3072 100K 5% 1/20W MF 201	
DP_X_SRC_HPD	R3073 100K 1/20W 201 MF 5%	
TBT_X_HDMI_DDC_DATA	R3076 100K 1/20W 201 MF 5%	

SMC RESET Connection

UPC_XA_HPD_RX — R3033 100K 5% 1/20W MF 201 — R3024 MF 0201 NOSTUFF 0 5% 1/20W

UPC_XB_HPD_RX — R3034 MF 0201 NOSTUFF 0 5% 1/20W — SMC_RESET_L

R3025 100K 5% 1/20W MF 201

GND ALIASES

MAKE_BASE=TRUE

USBC_XB_RPD_G2
USBC_XB_RPD_G1
USBC_XA_RPD_G2
USBC_XA_RPD_G1
UPC_XB_SPI_CLK
UPC_XB_SPI_MOSI
UPC_XB_SPI_MISO
UPC_XB_SPI_CS_L
UPC_XA_RESET
UPC_XB_RESET
TBT_X_THERM_D_P
TBT_X_THERM_D_N

POWER ALIASES

=PP5V_USBC_XA
=PP5V_USBC_XB
=PP5V_S4_USBC

=PP3V3_TBT_X_S0 — MIN_LINE_WIDTH=0.2000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V — PP3V3_TBT_X_S0 (MAKE_BASE=TRUE) — XW3001 SM — =PP3V3_TBT_X_S0_R

=PPHV_TBT_XA
=PPHV_TBT_XB
=PPHV_SW_TBT_X

MIN_LINE_WIDTH=0.2000 MIN_NECK_WIDTH=0.1200 VOLTAGE=3.3V

=PP3V3_TBT_X_SX
=PP3V3_UPC_XA_SX
=PP3V3_UPC_XB_SX
MAKE_BASE=TRUE PP3V3_TBT_X_SX

=PP3V3R1V8_UPC_XB
=PP3V3_UPC_XB_LDO
MAKE_BASE=TRUE PP3V3_UPC_XB_LDO — MIN_LINE_WIDTH=0.2000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V

=PP3V3_TBT_X_FLASH
=PP3V3R1V8_UPC_XA
=PP3V3_UPC_XA_LDO
MAKE_BASE=TRUE PP3V3_UPC_XA_LDO — MIN_LINE_WIDTH=0.2000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V

SIGNAL ALIASES

TBT_X_RTD3_CIO_PWR_EN — MAKE_BASE=TRUE TBT_X_CIO_PWR_EN
UPC_XB_GPIO2

TBT_X_RTD3_USB_PWR_EN — MAKE_BASE=TRUE TBT_X_USB_PWR_EN
UPC_XB_GPIO3

=UPC_WAKE_L — MAKE_BASE=TRUE TBT_WAKE_L

=UPC_X_SPI_CLK	MAKE_BASE=TRUE UPC_X_SPI_CLK
=UPC_X_SPI_CS_L	MAKE_BASE=TRUE UPC_X_SPI_CS_L
=UPC_X_SPI_MOSI	MAKE_BASE=TRUE UPC_X_SPI_MOSI
=UPC_X_SPI_MISO	MAKE_BASE=TRUE UPC_X_SPI_MISO

USBC_X_POC_RESET — MAKE_BASE=TRUE TBT_POC_RESET

I2C SERIES R'S

PLACE_NEAR=U5000:5mm

=I2C_UPC_XA_SDA2 — R3041 33 5% 1/20W MF 201 — =I2C_UPC_XA_SDA2_R
=I2C_UPC_XB_SDA2 (MAKE_BASE=TRUE) — =I2C_UPC_XB_SDA2_R
I2C_UPC_X_SDA2

PLACE_NEAR=U5000:5mm

=I2C_UPC_XA_SCL2 — R3042 33 5% 1/20W MF 201 — =I2C_UPC_XA_SCL2_R
=I2C_UPC_XB_SCL2 (MAKE_BASE=TRUE) — =I2C_UPC_XB_SCL2_R
I2C_UPC_X_SCL2

RIDGE 0.9V SVR XW

XW3000 SM

P0V9_TBT_X_SVR_AGND

NO_XNET_CONNECTION=1

TBT to ACE

TBT
Alpine Ridge U2800
(MASTER)

Pri ACE
U3100
(Write: 0x70 Read: 0x71)

PLACE_NEAR=U2800:10MM

I2C_TBT_X_SCL — R3047 33 5% 1/20W MF 201 — I2C_TBT_SCL_R (MAKE_BASE=TRUE) — =I2C_UPC_XA_SCL1
I2C_TBT_X_SDA — R3048 33 5% 1/20W MF 201 — I2C_TBT_SDA_R (MAKE_BASE=TRUE) — =I2C_UPC_XA_SDA1

PLACE_NEAR=U2800:10MM
I2C_TBT_XA_INT_L (MAKE_BASE=TRUE) — =I2C_UPC_XA_INT1_L
I2C_TBT_XB_INT_L (MAKE_BASE=TRUE)

Sec ACE
U3200
(Write: 0x7E Read: 0x7F)

=I2C_UPC_XB_SCL1
=I2C_UPC_XB_SDA1
=I2C_UPC_XB_INT1_L

Ridge PCIE Caps

PCIE_TBT_X_R2D_C_P<0>	C3040 0.22UF	PCIE_TBT_X_R2D_P<0>
PCIE_TBT_X_R2D_C_N<0>	C3041 0.22UF	PCIE_TBT_X_R2D_N<0>
PCIE_TBT_X_R2D_C_P<1>	C3042 0.22UF	PCIE_TBT_X_R2D_P<1>
PCIE_TBT_X_R2D_C_N<1>	C3043 0.22UF	PCIE_TBT_X_R2D_N<1>
PCIE_TBT_X_R2D_C_P<2>	C3044 0.22UF	PCIE_TBT_X_R2D_P<2>
PCIE_TBT_X_R2D_C_N<2>	C3045 0.22UF	PCIE_TBT_X_R2D_N<2>
PCIE_TBT_X_R2D_C_P<3>	C3046 0.22UF	PCIE_TBT_X_R2D_P<3>
PCIE_TBT_X_R2D_C_N<3>	C3047 0.22UF	PCIE_TBT_X_R2D_N<3>
PCIE_TBT_X_D2R_C_P<0>	C3050 0.22UF	PCIE_TBT_X_D2R_P<0>
PCIE_TBT_X_D2R_C_N<0>	C3051 0.22UF	PCIE_TBT_X_D2R_N<0>
PCIE_TBT_X_D2R_C_P<1>	C3052 0.22UF	PCIE_TBT_X_D2R_P<1>
PCIE_TBT_X_D2R_C_N<1>	C3053 0.22UF	PCIE_TBT_X_D2R_N<1>
PCIE_TBT_X_D2R_C_P<2>	C3054 0.22UF	PCIE_TBT_X_D2R_P<2>
PCIE_TBT_X_D2R_C_N<2>	C3055 0.22UF	PCIE_TBT_X_D2R_N<2>
PCIE_TBT_X_D2R_C_P<3>	C3056 0.22UF	PCIE_TBT_X_D2R_P<3>
PCIE_TBT_X_D2R_C_N<3>	C3057 0.22UF	PCIE_TBT_X_D2R_N<3>

All: 0201 X5R 6.3V 20%; GND_VOID=TRUE

RIDGE DEBUG CONN

Place on bottom

BOM OPTION Removed
No Stuff at PVT

J3099 505070-1220 M-ST-SM DEVELOPMENT

TBT_X_CIO_PLUG_EVENT_L	1 2	PM_PLT_RST_TBT_X_L
DP_X_SNK0_HPD	3 4	USBC_X_RESET_L
ARKANOID_P5	5 6	PP3V3_TBT_X_SX
ARKANOID_P7	7 8	PP0V9_TBT_X_PCIE
TBT_POC_RESET	9 10	PP0V9_TBT_X_USB
PP3V3_TBT_X_LC	11 12	PP0V9_TBT_X_CIO

13 14 15 16 — 516S00115

ACE DEBUG CONN

Place on bottom

BOM OPTION Removed
No Stuff at PVT

J3000 505070-1220 M-ST-SM DEVELOPMENT

I2C_TBT_XB_INT_L	1 2	I2C_TBT_XA_INT_L
I2C_UPC_X_SCL2	3 4	I2C_TBT_X_SDA
I2C_UPC_X_SDA2	5 6	I2C_TBT_X_SCL
SMC_USBC_INT_L	7 8	I2C_UPC_XB_DBG_CTL_SDA
TBT_X_SPI_CLK_DBG	9 10	I2C_UPC_XB_DBG_CTL_SCL
UPC_XA_UART_TX	11 12	UPC_XA_UART_RX

13 14 15 16

AR/ACE SPI BUS SERIES R'S

USBC DEBUG CONN

TBT_X_SPI_CLK	R3094 100	TBT_X_SPI_CLK_DBG	
TBT_X_SPI_CS_L	R3095 15	UPC_XA_SPI_CLK	IN
TBT_X_SPI_MOSI	R3096 15	UPC_XA_SPI_CS_L	IN
TBT_X_SPI_MISO	R3097 15	UPC_XA_SPI_MOSI	IN
ROM	R3098 15	UPC_XA_SPI_MISO	OUT
	R3090 15	=UPC_X_SPI_CLK	IN
	R3091 15	=UPC_X_SPI_CS_L	IN
	R3092 15	=UPC_X_SPI_MOSI	IN
	R3093 15	=UPC_X_SPI_MISO	OUT

All: 5% 1/20W MF 201

Ace

AR

=PP5V_S4_USBC

C3089 220UF-35MOHM 20% 35V POLY-TANT CASE-B2-SM1

ACE XA DEBUG PORTS

SMC SWD

MAKE_BASE=TRUE	SMC_TCK	=	UPC_XA_DBG1
MAKE_BASE=TRUE	SMC_TMS	=	UPC_XA_DBG2
MAKE_BASE=TRUE	SMC_TX_L	=	UPC_XA_DBG3
MAKE_BASE=TRUE	SMC_RX_L	=	UPC_XA_DBG4

ARKANOID_P5 — R3082 MF 0201 0 5% 1/20W — DP_X_SNK1_HPD
R3083 NOSTUFF MF 0201 0 5% 1/20W — SMC_DEBUGPRT_RX_L

ARKANOID_P7 — R3084 MF 0201 0 5% 1/20W — DP_XA_HPD
R3085 NOSTUFF MF 0201 0 5% 1/20W — SMC_DEBUGPRT_TX_L

ACE XB DEBUG PORTS

R3062 100K 1/20W 201 MF 5% — UPC_XB_DBG1
R3063 100K 1/20W 201 MF 5% — UPC_XB_DBG2

SMC_DEBUGPRT_TX_L — R3080 MF 0201 NOSTUFF 0 5% 1/20W — UPC_XB_DBG3

R3015 100K MF 201 5% 1/20W
R3016 100K MF 201 5% 1/20W

SMC MOJO — SMC_DEBUGPRT_RX_L — R3081 MF 0201 NOSTUFF 0 5% 1/20W — UPC_XB_DBG4

Place Resistors close to USB-A Mojo Mux resistors.

PPVBUS_USBC_XA_CONN (MAKE_BASE=TRUE)	=	TP_PPVBUS_USBC_XA — TP3025
PPVBUS_USBC_XB_CONN (MAKE_BASE=TRUE)	=	TP_PPVBUS_USBC_XB — TP3026
TP_UPC_XA_SWD_DATA		TP3001
TP_UPC_XA_SWD_CLK		TP3002
TP_UPC_XB_SWD_DATA		TP3003
TP_UPC_XB_SWD_CLK		TP3004
=USBC_XB_RESET_L	=	TP_USBC_XB_RESET_L (MAKE_BASE=TRUE) — TP3014

All: TP-P5

RIDGE 0.9V SVR XW

XW3000 SM

P0V9_TBT_X_SVR_AGND

NO_XNET_CONNECTION=1

TBT to ACE

TBT
Alpine Ridge U2800
(MASTER)

Pri ACE
U3100
(Write: 0x70 Read: 0x71)

PLACE_NEAR=U2800:10MM

I2C_TBT_X_SCL — R3047 33 5% 1/20W MF 201 — I2C_TBT_SCL_R (MAKE_BASE=TRUE) — =I2C_UPC_XA_SCL1
I2C_TBT_X_SDA — R3048 33 5% 1/20W MF 201 — I2C_TBT_SDA_R (MAKE_BASE=TRUE) — =I2C_UPC_XA_SDA1

PLACE_NEAR=U2800:10MM
I2C_TBT_XA_INT_L (MAKE_BASE=TRUE) — =I2C_UPC_XA_INT1_L
I2C_TBT_XB_INT_L (MAKE_BASE=TRUE)

Sec ACE
U3200
(Write: 0x7E Read: 0x7F)

=I2C_UPC_XB_SCL1
=I2C_UPC_XB_SDA1
=I2C_UPC_XB_INT1_L

Ridge PCIE Caps

PCIE_TBT_X_R2D_C_P<0>	C3040 0.22UF	PCIE_TBT_X_R2D_P<0>
PCIE_TBT_X_R2D_C_N<0>	C3041 0.22UF	PCIE_TBT_X_R2D_N<0>
PCIE_TBT_X_R2D_C_P<1>	C3042 0.22UF	PCIE_TBT_X_R2D_P<1>
PCIE_TBT_X_R2D_C_N<1>	C3043 0.22UF	PCIE_TBT_X_R2D_N<1>
PCIE_TBT_X_R2D_C_P<2>	C3044 0.22UF	PCIE_TBT_X_R2D_P<2>
PCIE_TBT_X_R2D_C_N<2>	C3045 0.22UF	PCIE_TBT_X_R2D_N<2>
PCIE_TBT_X_R2D_C_P<3>	C3046 0.22UF	PCIE_TBT_X_R2D_P<3>
PCIE_TBT_X_R2D_C_N<3>	C3047 0.22UF	PCIE_TBT_X_R2D_N<3>
PCIE_TBT_X_D2R_C_P<0>	C3050 0.22UF	PCIE_TBT_X_D2R_P<0>
PCIE_TBT_X_D2R_C_N<0>	C3051 0.22UF	PCIE_TBT_X_D2R_N<0>
PCIE_TBT_X_D2R_C_P<1>	C3052 0.22UF	PCIE_TBT_X_D2R_P<1>
PCIE_TBT_X_D2R_C_N<1>	C3053 0.22UF	PCIE_TBT_X_D2R_N<1>
PCIE_TBT_X_D2R_C_P<2>	C3054 0.22UF	PCIE_TBT_X_D2R_P<2>
PCIE_TBT_X_D2R_C_N<2>	C3055 0.22UF	PCIE_TBT_X_D2R_N<2>
PCIE_TBT_X_D2R_C_P<3>	C3056 0.22UF	PCIE_TBT_X_D2R_P<3>
PCIE_TBT_X_D2R_C_N<3>	C3057 0.22UF	PCIE_TBT_X_D2R_N<3>

All: 0201 X5R 6.3V 20%; GND_VOID=TRUE

RIDGE DEBUG CONN

Place on bottom

BOM OPTION Removed
No Stuff at PVT

J3099 505070-1220 M-ST-SM DEVELOPMENT

TBT_X_CIO_PLUG_EVENT_L	1 2	PM_PLT_RST_TBT_X_L
DP_X_SNK0_HPD	3 4	USBC_X_RESET_L
ARKANOID_P5	5 6	PP3V3_TBT_X_SX
ARKANOID_P7	7 8	PP0V9_TBT_X_PCIE
TBT_POC_RESET	9 10	PP0V9_TBT_X_USB
PP3V3_TBT_X_LC	11 12	PP0V9_TBT_X_CIO

13 14 15 16 — 516S00115

ACE DEBUG CONN

Place on bottom

BOM OPTION Removed
No Stuff at PVT

J3000 505070-1220 M-ST-SM DEVELOPMENT

I2C_TBT_XB_INT_L	1 2	I2C_TBT_XA_INT_L
I2C_UPC_X_SCL2	3 4	I2C_TBT_X_SDA
I2C_UPC_X_SDA2	5 6	I2C_TBT_X_SCL
SMC_USBC_INT_L	7 8	I2C_UPC_XB_DBG_CTL_SDA
TBT_X_SPI_CLK_DBG	9 10	I2C_UPC_XB_DBG_CTL_SCL
UPC_XA_UART_TX	11 12	UPC_XA_UART_RX

13 14 15 16

AR/ACE SPI BUS SERIES R'S

USBC DEBUG CONN

TBT_X_SPI_CLK	R3094 100	TBT_X_SPI_CLK_DBG	
TBT_X_SPI_CS_L	R3095 15	UPC_XA_SPI_CLK	IN
TBT_X_SPI_MOSI	R3096 15	UPC_XA_SPI_CS_L	IN
TBT_X_SPI_MISO	R3097 15	UPC_XA_SPI_MOSI	IN
ROM	R3098 15	UPC_XA_SPI_MISO	OUT
	R3090 15	=UPC_X_SPI_CLK	IN
	R3091 15	=UPC_X_SPI_CS_L	IN
	R3092 15	=UPC_X_SPI_MOSI	IN
	R3093 15	=UPC_X_SPI_MISO	OUT

All: 5% 1/20W MF 201

Ace

AR

=PP5V_S4_USBC

C3089 220UF-35MOHM 20% 35V POLY-TANT CASE-B2-SM1

ACE XA DEBUG PORTS

SMC SWD

MAKE_BASE=TRUE	SMC_TCK	=	UPC_XA_DBG1
MAKE_BASE=TRUE	SMC_TMS	=	UPC_XA_DBG2
MAKE_BASE=TRUE	SMC_TX_L	=	UPC_XA_DBG3
MAKE_BASE=TRUE	SMC_RX_L	=	UPC_XA_DBG4

ARKANOID_P5 — R3082 MF 0201 0 5% 1/20W — DP_X_SNK1_HPD
R3083 NOSTUFF MF 0201 0 5% 1/20W — SMC_DEBUGPRT_RX_L

ARKANOID_P7 — R3084 MF 0201 0 5% 1/20W — DP_XA_HPD
R3085 NOSTUFF MF 0201 0 5% 1/20W — SMC_DEBUGPRT_TX_L

ACE XB DEBUG PORTS

R3062 100K 1/20W 201 MF 5% — UPC_XB_DBG1
R3063 100K 1/20W 201 MF 5% — UPC_XB_DBG2

SMC_DEBUGPRT_TX_L — R3080 MF 0201 NOSTUFF 0 5% 1/20W — UPC_XB_DBG3

R3015 100K MF 201 5% 1/20W
R3016 100K MF 201 5% 1/20W

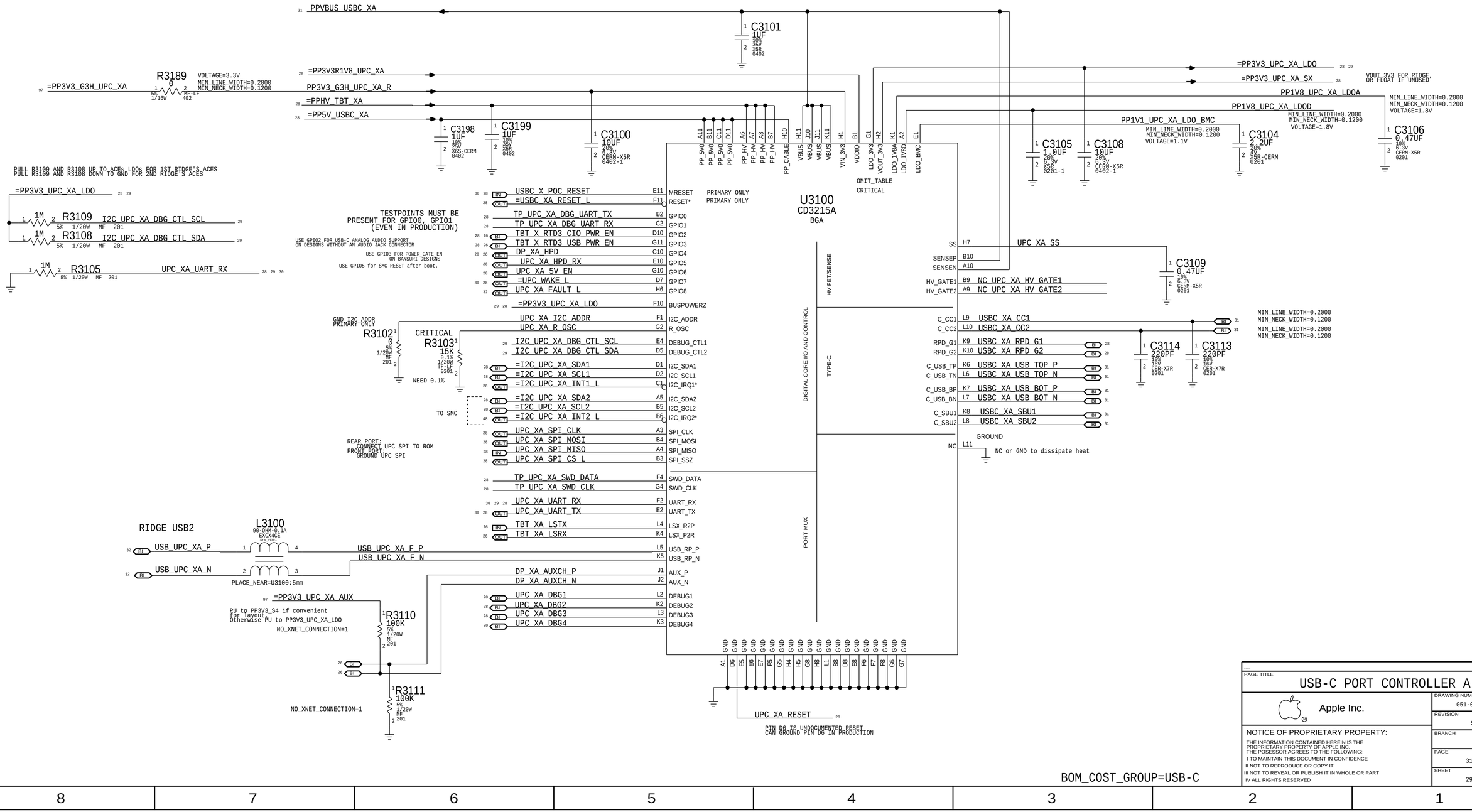
SMC MOJO — SMC_DEBUGPRT_RX_L — R3081 MF 0201 NOSTUFF 0 5% 1/20W — UPC_XB_DBG4

Place Resistors close to USB-A Mojo Mux resistors.

PPVBUS_USBC_XA_CONN (MAKE_BASE=TRUE)	=	TP_PPVBUS_USBC_XA — TP3025
PPVBUS_USBC_XB_CONN (MAKE_BASE=TRUE)	=	TP_PPVBUS_USBC_XB — TP3026
TP_UPC_XA_SWD_DATA		TP3001
TP_UPC_XA_SWD_CLK		TP3002
TP_UPC_XB_SWD_DATA		TP3003
TP_UPC_XB_SWD_CLK		TP3004
=USBC_XB_RESET_L	=	TP_USBC_XB_RESET_L (MAKE_BASE=TRUE) — TP3014

All: TP-P5

PRIMARY ACE USB-C PORT CONTROLLER (UPC)



PAGE TITLE
USB-C PORT CONTROLLER A

Apple Inc.

DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	31 OF 121
SHEET	29 OF 102

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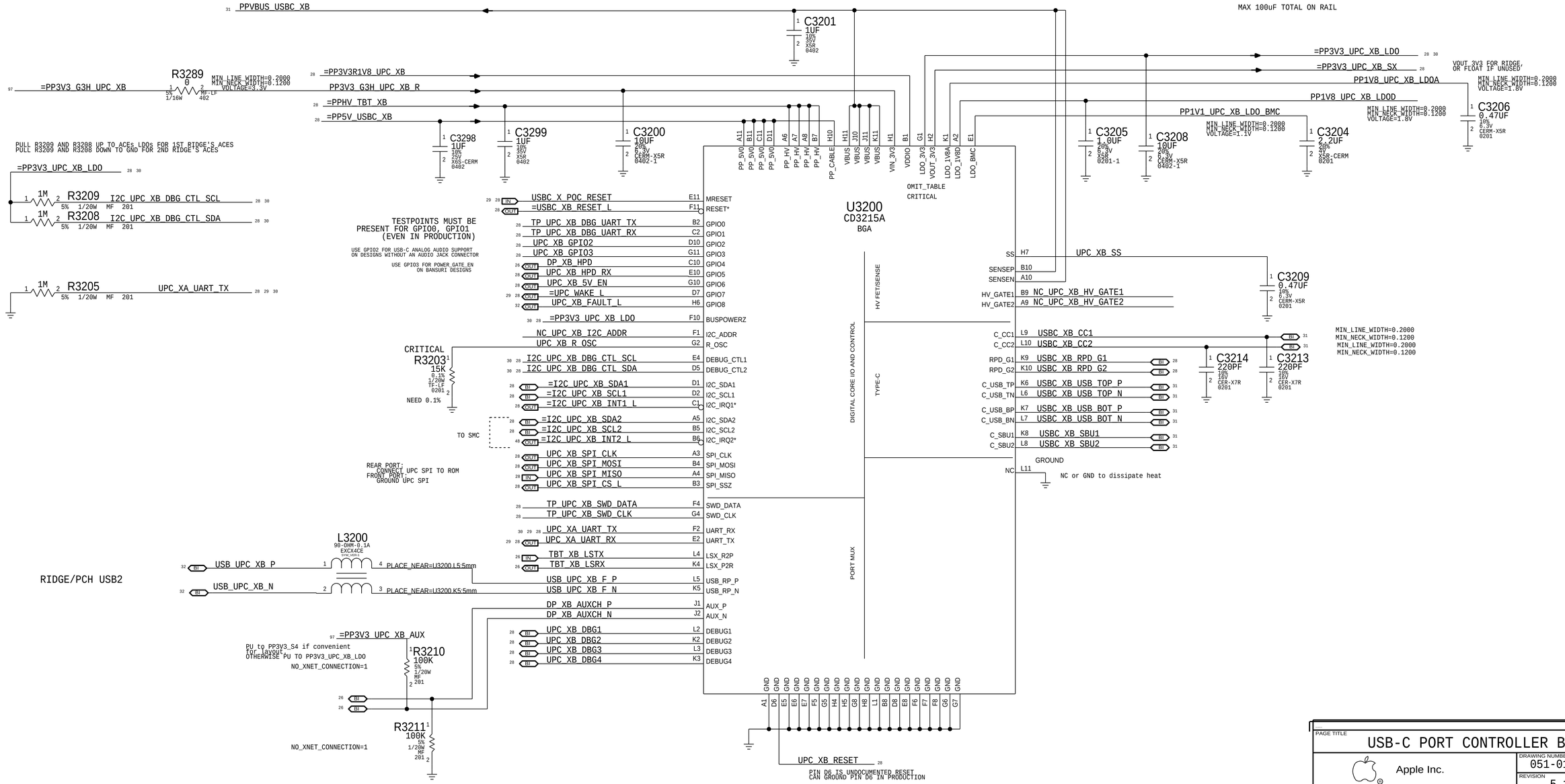
SECONDARY ACE USB-C PORT CONTROLLER (UPC)



PAGE TITLE	USB-C PORT CONTROLLER B
	Apple Inc.
DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	32 OF 121
SHEET	30 OF 102

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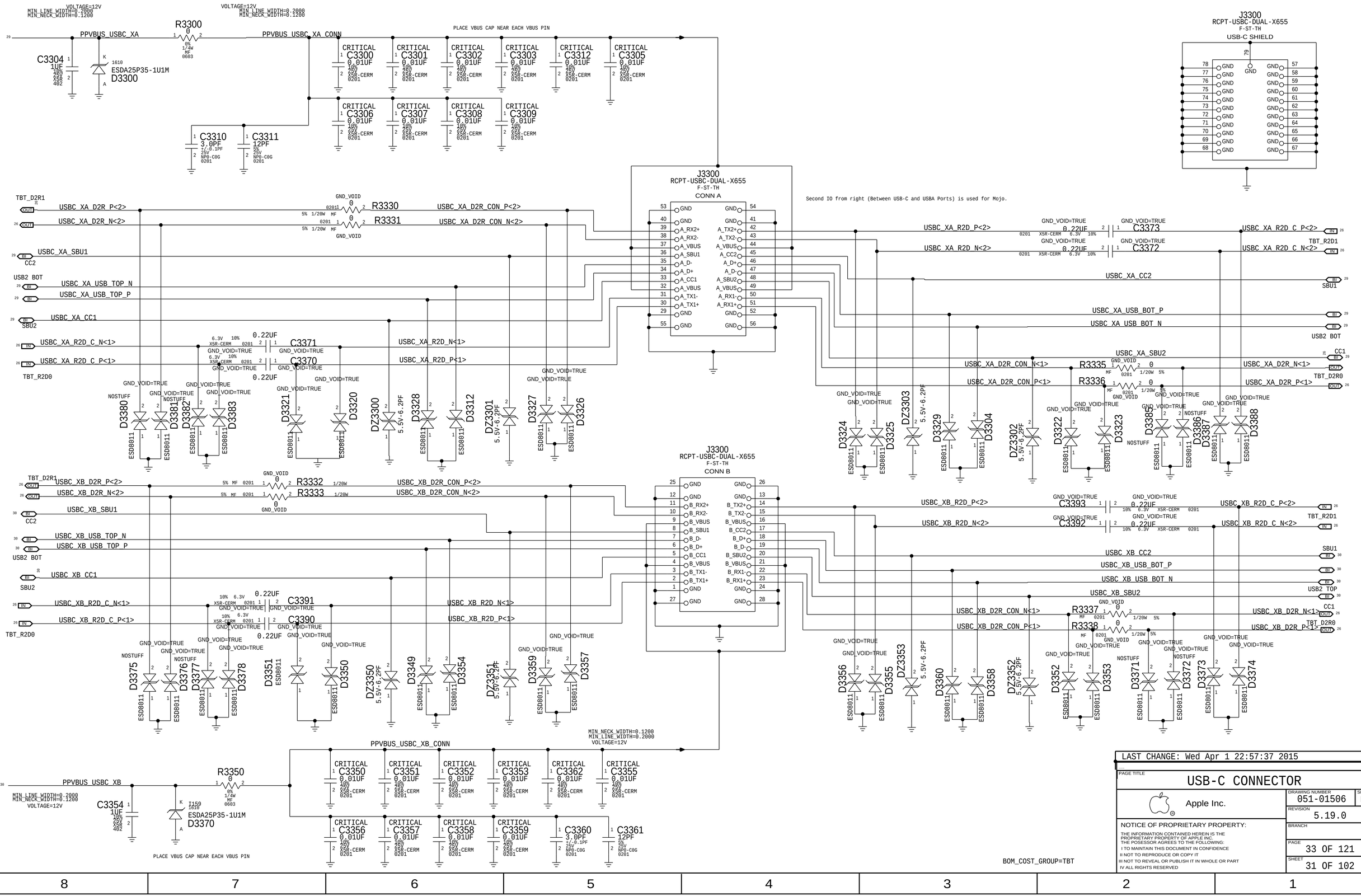
VOLTAGE=12V
MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1200
R3300
0
0% 1/4W MF 0603
PPVBUS_USBC_XA
PPVBUS_USBC_XA_CONN
PLACE VBUS CAP NEAR EACH VBUS PIN
C3304 1UF 10% 25V X5S 402
D3300 ESDA25P35-1U1M
C3300 C3301 C3302 C3303 C3312 C3305 C3306 C3307 C3308 C3309 0.01UF 10% 25V X5R-CERM 0201
C3310 3.0PF +/-0.1PF 25V NP0-C0G 0201
C3311 12PF 5% 25V NP0-C0G 0201
J3300 RCPT-USBC-DUAL-X655 F-ST-TH
USB-C SHIELD
CONN A
CONN B
Second IO from right (Between USB-C and USBA Ports) is used for Mojo.
TBT_D2R1
TBT_D2R0
USB2 BOT
USBC_XA_D2R_P<2>
USBC_XA_D2R_N<2>
USBC_XA_SBU1
USBC_XA_USB_TOP_N
USBC_XA_USB_TOP_P
USBC_XA_CC1
USBC_XA_R2D_C_N<1>
USBC_XA_R2D_C_P<1>
R3330
R3331
USBC_XA_D2R_CON_P<2>
USBC_XA_D2R_CON_N<2>
C3371
C3370
USBC_XA_R2D_N<1>
USBC_XA_R2D_P<1>
USBC_XA_R2D_P<2>
USBC_XA_R2D_N<2>
C3373
C3372
USBC_XA_R2D_C_P<2>
USBC_XA_R2D_C_N<2>
USBC_XA_CC2
USBC_XA_USB_BOT_P
USBC_XA_USB_BOT_N
USBC_XA_SBU2
R3335
R3336
USBC_XA_D2R_CON_N<1>
USBC_XA_D2R_CON_P<1>
USBC_XA_D2R_N<1>
USBC_XA_D2R_P<1>
D3380 D3381 D3382 D3383 D3321 D3320 DZ3300 D3328 D3312 DZ3301 D3327 D3326
D3324 D3325 DZ3303 D3329 D3304 DZ3302 D3322 D3323 D3385 D3386 D3387 D3388
TBT_D2R1
TBT_D2R0
USB2 BOT
USBC_XB_D2R_P<2>
USBC_XB_D2R_N<2>
USBC_XB_SBU1
USBC_XB_USB_TOP_N
USBC_XB_USB_TOP_P
USBC_XB_CC1
USBC_XB_R2D_C_N<1>
USBC_XB_R2D_C_P<1>
R3332
R3333
USBC_XB_D2R_CON_P<2>
USBC_XB_D2R_CON_N<2>
C3391
C3390
USBC_XB_R2D_N<1>
USBC_XB_R2D_P<1>
USBC_XB_R2D_P<2>
USBC_XB_R2D_N<2>
C3393
C3392
USBC_XB_R2D_C_P<2>
USBC_XB_R2D_C_N<2>
USBC_XB_CC2
USBC_XB_USB_BOT_P
USBC_XB_USB_BOT_N
USBC_XB_SBU2
R3337
R3338
USBC_XB_D2R_CON_N<1>
USBC_XB_D2R_CON_P<1>
USBC_XB_D2R_N<1>
USBC_XB_D2R_P<1>
D3375 D3376 D3377 D3378 D3351 D3350 DZ3350 D3349 D3354 DZ3351 D3359 D3357
D3356 D3355 DZ3353 D3360 D3358 DZ3352 D3352 D3353 D3371 D3372 D3373 D3374
PPVBUS_USBC_XB
PPVBUS_USBC_XB_CONN
MIN_NECK_WIDTH=0.1200
MIN_LINE_WIDTH=0.2000
VOLTAGE=12V
R3350
C3354 1UF 10% 25V X5S 402
D3370 ESDA25P35-1U1M
C3350 C3351 C3352 C3353 C3362 C3355 C3356 C3357 C3358 C3359 0.01UF 10% 25V X5R-CERT 0201
C3360 3.0PF +/-0.1PF 25V NP0-C0G 0201
C3361 12PF 5% 25V NP0-C0G 0201
PLACE VBUS CAP NEAR EACH VBUS PIN
BOM_COST_GROUP=TBT
LAST CHANGE: Wed Apr 1 22:57:37 2015
PAGE TITLE
USB-C CONNECTOR
Apple Inc.
DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 33 OF 121
SHEET 31 OF 102
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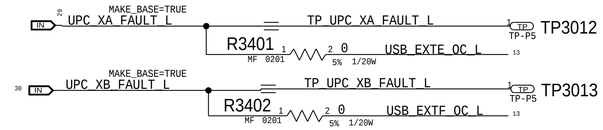

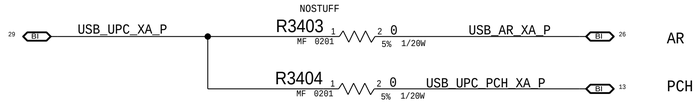

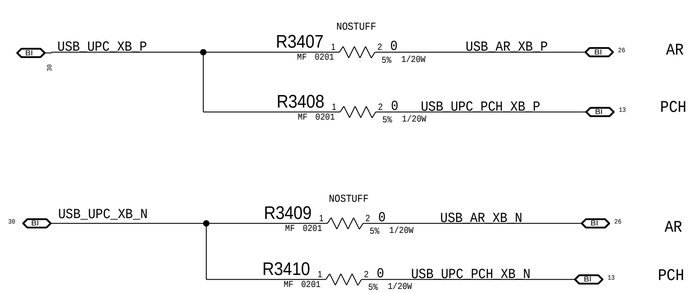

BOM_COST_GROUP=USB-C

PAGE TITLE

USB-C Support 2

Apple Inc.

DRAWING NUMBER 051-01506

SIZE D

REVISION 5.19.0

BRANCH

PAGE 34 OF 121

SHEET 32 OF 102

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WLAN
BLUETOOTH

AP & BT Load Switch

SWITCH	TPS22924B
CHANNEL	N-TYPE
RDS(ON)	18.4 MOHM @3.3V
LOADING	2 A (EDP)

CRITICAL L3502 FERR-220-OHM-2.5A 0603
PP3V3_S4_AP_FET — MIN_LINE_WIDTH=0.6000 MIN_NECK_WIDTH=0.2000
PP3V3_S4_AP_FLT — MIN_LINE_WIDTH=0.6000 MIN_NECK_WIDTH=0.1800 VOLTAGE=3.3V

C3501 3PF ±/-0.1PF 25V NP0-C0G 0201
C3502 12PF 5% 25V NP0-C0G 0201
C3503 0.1UF 10% 16V X5R-CERM 0201
C3504 10UF 20% 6.3V X5R 0402

514S0335 CRITICAL
J3500 SSD-K99 F-RT-SM1

PCIE_AP_R2D_C_N — C3505 0.1UF 10% 16V X5R-CERM 0201
PCIE_AP_R2D_C_P — C3506 0.1UF 10% 16V X5R-CERM 0201

PCH_BT_ROM_BOOT_L_R
PCIE_AP_R2D_N
PCIE_AP_R2D_P
PCIE_CLK100M_AP_N
PCIE_CLK100M_AP_P
PCIE_AP_D2R_N
PCIE_AP_D2R_P
AP_WAKE_L
AP_RESET_CONN_L
AP_CLKREQ_Q_L
USB_BT_MUX_N
USB_BT_MUX_P

U3510 TPS22966 DPU

=PP3V3_G3H_BT	1	VIN1	VOUT1 13
SMC_BT_PWR_EN	3	ON1	
BT_PWR_SS	12	CT1	
=PP3V3_G3H_BT	4	VBIAS	
=PP3V3_S4_AP	6	VIN2	VOUT2 8
AP_PWR_EN	5	ON2	
WL_PWR_SS	10	CT2	
	11	GND	
	15	THRM PAD	

C3511 0.001UF 10% 50V X7R-CERM 0402
C3510 0.1UF 10% 16V X7R-CERM 0402
C3512 0.001UF 10% 50V X7R-CERM 0402

CRITICAL L3501 FERR-220-OHM-2.5A 0603
PP3V3_G3H_BT_FET — VOLTAGE=3.3V MIN_LINE_WIDTH=0.5000 MIN_NECK_WIDTH=0.2000
PP3V3_G3H_BT_FLT — VOLTAGE=3.3V MIN_LINE_WIDTH=0.5000 MIN_NECK_WIDTH=0.2000

C3509 3PF ±/-0.1PF 25V NP0-C0G 0201
C3513 12PF 5% 25V NP0-C0G 0201
C3507 0.1UF 10% 16V X5R-CERM 0201
C3508 10UF 20% 6.3V X5R 0402

=PP3V3_G3H_BT
R3502 10K 5% 1/20W MF 201
R3504 0 5% 1/10W MF-LF 603
R3503 10K 5% 1/20W MF 201 NOSTUFF
CRITICAL Q3502 SOD
PCH_BT_ROM_BOOT
R3505 100K 5% 1/20W MF 201

Supervisor & CLKREQ# Isolation
Delay = 130 ms +/- 20%

PP3V3_S4_AP_FLT — =PP3V3_S4_AP
R3530 100K 1% 1/10W MF-LF 402
R3531 232K 1% 1/10W MF-LF 402
P3V3AP_VMON
CRITICAL U3530 SLG4AP041V TDFN
C3530 0.1uF 20% 10V CERM 402
AP_RESET_CONN_R_L
R3533 33 5% 1/20W MF 201
R3532 100K 1% 1/10W MF-LF 402

AP_RESET_CONN_L	4	RESET*	MR* 3 AP_RESET_L
	6	EN	AP_PWR_EN
AP_CLKREQ_Q_L	7	IN	OUT 8 AP_CLKREQ_L
	2	SENSE	
	1	VREF	
	5	DLY	
		THRM PAD	GND

Wake from BT in G3H circuit

SMC_PME_S4_WAKE_L
CRITICAL Q3501 SOD
R3501 15K 1% 1/20W MF 201
=PP3V3_G3H_BT
C3500 0.1UF 10% 10V X5R-CERM 0201

U3501 PI3USB102EZLE TQFN

USB_BT_WAKEN	NC		
	5	M+	Y+ 1 USB_BT_MUX_P
	4	M-	Y- 2 USB_BT_MUX_N
USB_BT_P	7	D+	
USB_BT_N	6	D-	
	8	OE*	SEL 10 PM_SLP_S5_L
		VCC	GND

PP3V3_S4_AP_FLT
CRITICAL Q3570 SOD
R3570 10K 5% 1/10W MF-LF 402
PCIE_WAKE_L — AP_WAKE_L

SYNC_MASTER=EIYIN	SYNC_DATE=09/15/2010
PAGE TITLE	WIRELESS: Airport/Bluetooth
Apple Inc.	DRAWING NUMBER 051-01506
	SIZE D
	REVISION 5.19.0
	BRANCH
	PAGE 35 OF 121
	SHEET 33 OF 102

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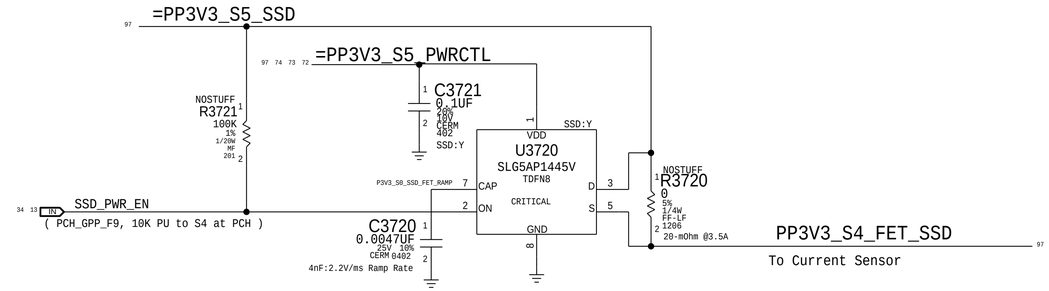

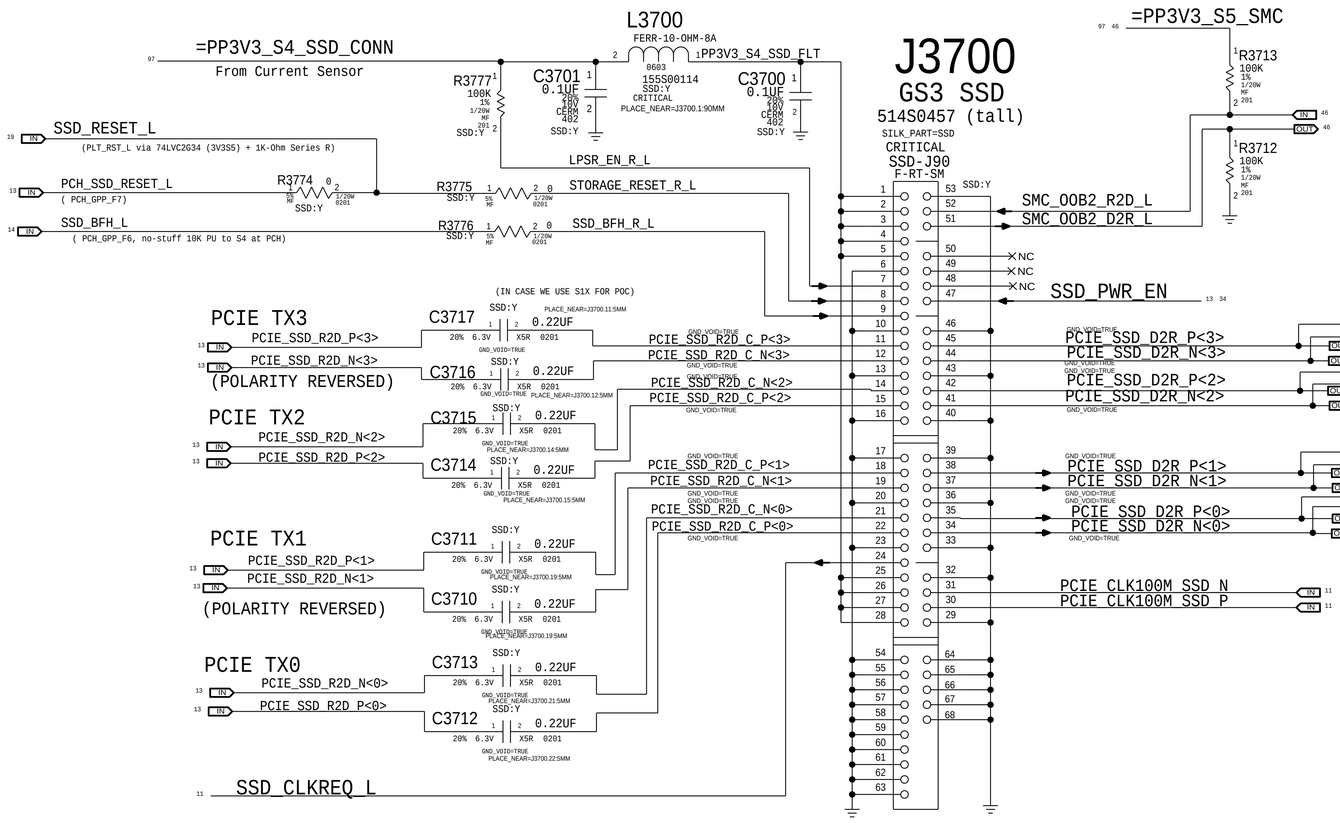

BOM_COST_GROUP=SSD

SYNC_MASTER=JERRY SYNC_DATE=02/13/2010

PAGE TITLE

SSD:CONNECTOR

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

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PAGE 37 OF 121

SHEET 34 OF 102

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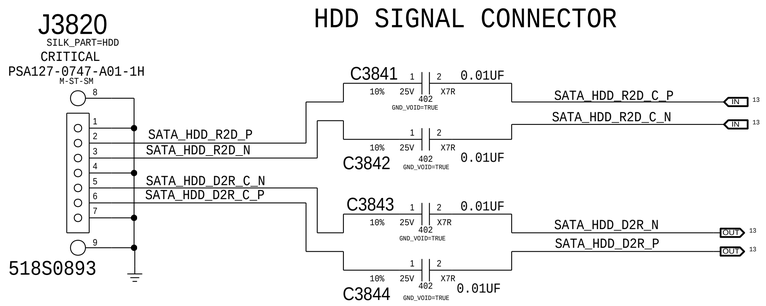
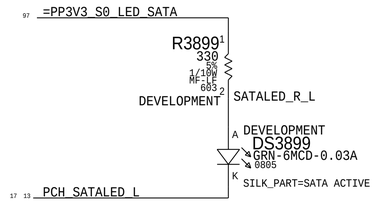
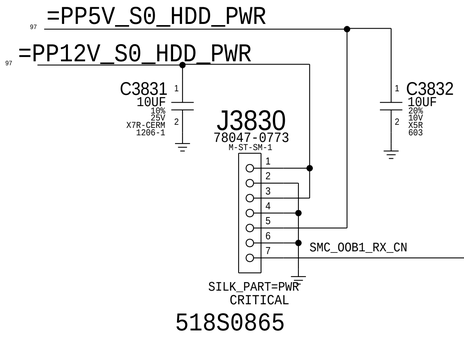
HDD SIGNAL CONNECTOR



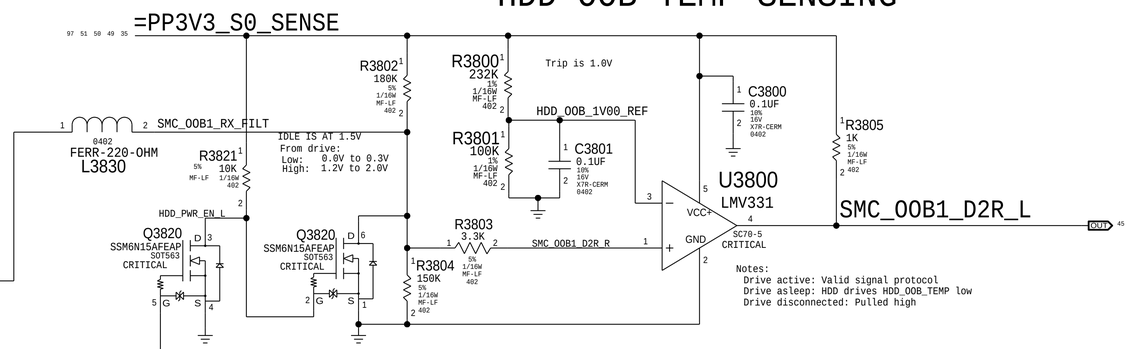
SATA Activity LED



HDD Power Connector

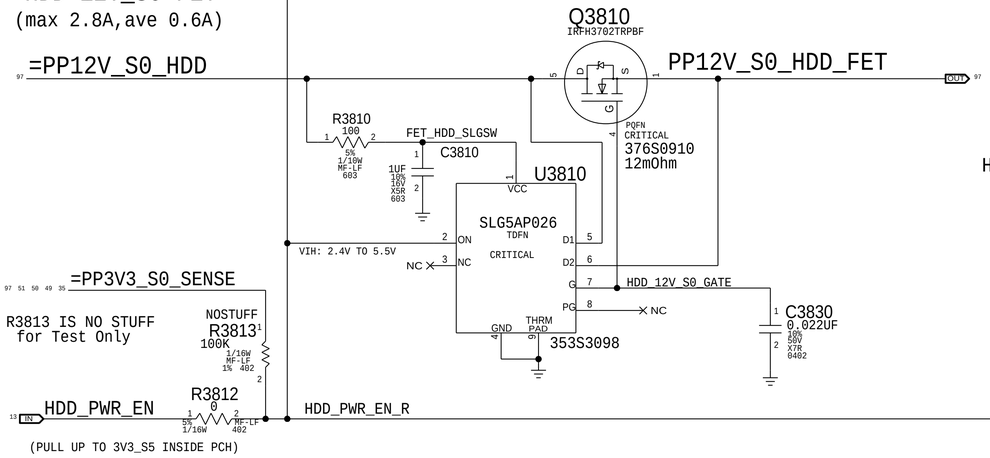


HDD OOB TEMP SENSING



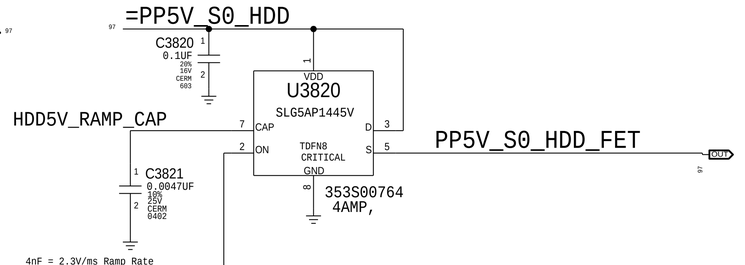
HDD 12V_S0 FET

(max 2.8A,ave 0.6A)



HDD 5V_S0 FET

(max 0.7A,ave 0.3A)



BOM_COST_GROUP=HDD

SYNC_MASTER=JERRY SYNC_DATE=02/01/2016

PAGE TITLE

HDD: CONN & Temp Sense

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

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PAGE 38 OF 121

SHEET 35 OF 102

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BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.

97 37 36 =PP3V3_ENET_PHY

281mA (1000base-T max power, Caesar IV)

VDD for Card Reader I/O
37 =PP3V3R1V8_CR_VDDIO

ENET_SR_LX 37

=ENET_SR_VFB 37

Internal 1.2V Switching Regulator pins.

=PP1V2_ENET_PHY 37

396mA (1000base-T, Caesar II)

CRITICAL L3900 FERR-600-OHM-300MA-0.85OHM 0402
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V
C3900 0.1UF 10% 16V X5R-CERM 0201
ENET_XTALVDDH
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V

CRITICAL L3905 FERR-600-OHM-300MA-0.85OHM 0402
PP3V3_ENET_PHY_BIASVDDH
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V
C3905 0.1UF 10% 16V X5R-CERM 0201

CRITICAL L3910 FERR-600-OHM-300MA-0.85OHM 0402
PP3V3_ENET_PHY_AVDDH
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000 VOLTAGE=3.3V
R3910 4.7K 5% 1/10W MF-LF 402
C3910 0.1UF 10% 16V X5R-CERM 0201
C3911 0.1UF 10% 16V X5R-CERM 0201

R3940 4.7K 5% 1/10W MF-LF 402
R3941 4.7K 5% 1/10W MF-LF 402
C3915 4.7UF 20% 6.3V X5R-CERM 402
C3916 0.1UF 10% 16V X5R-CERM 0201

42 AVDDH, 48 BIASVDDH, 37 XTALVDDH, 17 VDDIO, 7 20 56 62 VDDO, 14 SR_VDD, 15 SR_VDDP, 16 SR_LX, 13 SR_VFB, 39 AVDDL, 45 51 AVDDL, 29 PCIE_PLLVDDL, 32 GPHY_PLLVDDL, 36 VDDC, 35 61 VDDC

CRITICAL L3920 FERR-600-OHM-300MA-0.85OHM 0402
PP1V2_ENET_PHY_AVDDL
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.1500 VOLTAGE=1.2V
C3921 0.1UF 10% 16V X5R-CERM 0201
C3920 4.7UF 20% 6.3V X5R-CERM1 402

CRITICAL L3925 FERR-600-OHM-0.5A SM
PP1V2_ENET_PHY_PCIEPLL
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000 VOLTAGE=1.2V
C3926 0.1UF 10% 16V X5R-CERM 0201
C3925 4.7UF 20% 6.3V X5R-CERM1 402

CRITICAL L3930 FERR-600-OHM-300MA-0.85OHM 0402
PP1V2_ENET_PHY_GPHYPLL
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.1500 VOLTAGE=1.2V
C3931 0.1UF 10% 16V X5R-CERM 0201
C3930 4.7UF 20% 6.3V X5R-CERM1 402

C3936 0.1UF 10% 16V X5R-CERM 0201
C3935 10UF 20% 6.3V X5R 603

LR_OUT/GPIO1 is used as a 3.3V/1.8V internal LDO out for the card reader on-chip I/O.
Connect only to U3900 pin 20.

97 =PP3V3_S0_ENET

R3942 1K 5% 1/10W MF-LF 402 Current Limiting Resistor

C3950 0.1UF 10% 16V X5R-CERM 0201
C3951 0.1UF 10% 16V X5R-CERM 0201
C3955 0.1UF 10% 16V X5R-CERM 0201
C3956 0.1UF 10% 16V X5R-CERM 0201

13 OUT PCIE_ENET_D2R_N
13 OUT PCIE_ENET_D2R_P
13 IN PCIE_ENET_R2D_C_P
13 IN PCIE_ENET_R2D_C_N
37 OUT =ENET_WAKE_L (See note)

WAKE#
Must isolate from PCIe WAKE# if PHY is powered-down in S3/S5. Standard N-channel FET isolation suggested.
If PHY is always powered then alias =ENET_WAKE_L to PCIE_WAKE_L.

OMIT_TABLE
U3900
BCM57766C0KMLG
QFN-8X8

ENET_VMAIN_PRSNT	58	VMAIN_PRSNT	(IPD-ENET)
PCIE_ENET_D2R_C_N	27	PCIE_TXD_N	
PCIE_ENET_D2R_C_P	28	PCIE_TXD_P	
PCIE_ENET_R2D_P	33	PCIE_RXD_P	
PCIE_ENET_R2D_N	34	PCIE_RXD_N	
11 IN PCIE_CLK100M_ENET_P	31	PCIE_REFCLK_P	
11 IN PCIE_CLK100M_ENET_N	30	PCIE_REFCLK_N	
38 IN ENET_RESET_L	11	PERST*	(IPD)
11 OUT ENET_CLKREQ_L	12	CLKREQ*	(OD)
	3	WAKE*	(OD)
38 19 IN ENET_LOW_PWR	4	LOW_PWR	(IPD)
	6	SMB_CLK	
SMB_ENET_SDA	10	SMB_DATA	(IPD-ENETM)
36 ENET_SCLK MF R3975 33 5% 1/20W 201 ENET_SCLK_R	66	SCLK_SPD1000LED*	
36 ENET_MISO_R MF R3976 33 5% 1/20W 201 ENET_MISO	64	SI/EEDATA	
36 ENET_MOSI MF R3977 33 5% 1/20W 201 ENET_MOSI_R	65	SO_LINKLED*	(IPU)
36 ENET_CS_L MF R3978 33 5% 1/20W 201 ENET_CS_R_L	63	CS*/EECLK	
NC	2	SPD100LED*/SERIAL_DO	(OD)
37 OUT ENET_TRAFFICLED_L	67	TRAFFICLED*/SERIAL_DI	(OD)
36 ENET_XTAL_IN	18	XTALI	
36 ENET_XTAL_OUT	19	XTALO	
ENET_RDAC	38	RDAC	
	69	THRM_PAD	(NO IPU OR IPD-ENET)

NOTE: "IPx" == Programmable pull-up/down

SD_DETECT can only be used active low due to errata.

TRD0_P	40	ENETCONN_MDI_P<0>	BI 37
TRD0_N	41	ENETCONN_MDI_N<0>	BI 37
TRD1_P	44	ENETCONN_MDI_P<1>	BI 37
TRD1_N	43	ENETCONN_MDI_N<1>	BI 37
TRD2_P	46	ENETCONN_MDI_P<2>	BI 37
TRD2_N	47	ENETCONN_MDI_N<2>	BI 37
TRD3_P	50	ENETCONN_MDI_P<3>	BI 37
TRD3_N	49	ENETCONN_MDI_N<3>	BI 37
(IPD) GPIO_0/CR_ACT_LED*	5	NC	
GPIO_1/LR_OUT	8		
GPIO_2/MEDIA_SENSE	9	ENET_MEDIA_SENSE	OUT 14
(IPx-ENET) SD_DETECT	1	ENET_SD_DETECT_L	IN 38
(IPU-ENET) CR_CMD	26	ENET_SD_CMD R3961 33 5% 1/20W MF 201	SDCONN_CMD IN 38
CR_CLK	21	ENET_SD_CLK R3979 33 5% 1/20W MF 201	SDCONN_CLK OUT 38
CR_DATA0	25	ENET_CR_DATA<0> R3971 33 5% 1/20W MF 201	SDCONN_DATA<0> BI 38
CR_DATA1	24	ENET_CR_DATA<1> R3972 33 5% 1/20W MF 201	SDCONN_DATA<1> BI 38
CR_DATA2	23	ENET_CR_DATA<2> R3973 33 5% 1/20W MF 201	SDCONN_DATA<2> BI 38
CR_DATA3	22	ENET_CR_DATA<3> R3974 33 5% 1/20W MF 201	SDCONN_DATA<3> BI 38
(IPU-ENET) CR_DATA4	52	NC	
CR_DATA5	53	NC	
CR_DATA6	54	NC	
CR_DATA7	55	NC	
(IPU-ENET) MS_INS*	59	NC	
CR_LED*/CR_BUS_PWR	60	ENET_CR_PWREN	OUT 38
CR_WP*	57		SDCONN_WP IN 38
SR_DISABLE	68	ENET_SR_DISABLE R3981 1K 5% 1/16W MF-LF 402 (See note)	

No MS (Memory Stick) Insert feature needed.
Control signal to light LED or control SD bus power.

ENET 1.2V SR IS ENABLED IF FLOATING.

ENET supports both active-levels for WP.

PP3V3R1V8_ENET_LR_OUT 37
MIN_LINE_WIDTH=0.3000 MIN_NECK_WIDTH=0.1500 VOLTAGE=3.3V

PHY Non-Volatile Memory

ROM contains MAC address, PCIe config info as well as code for Bonjour proxy.
Avoids need for EFI to program at startup.
(Required ROM size 1 Mbit)

97 37 36 =PP3V3_ENET_PHY

C3990 0.1UF 10% 16V X5R-CERM 0201

U3990 AT45DB011D SOIC-8S1 OMIT_TABLE

36 ENET_SCLK	2	SCK	SI 1 ENET_MOSI 36
36 ENET_CS_L	4	CS*	
	5	WP*	SO 8 ENET_MISO_R 36
	3	RESET*	
	6	VCC	
	7	GND	

NOSTUFF R3990 4.7K 5% 1/16W MF-LF 402
R3997 4.7K 5% 1/16W MF-LF 402

NOTE: Pull-down on SO plus internal pull-ups on other 3 SPI pins configures ENET for the Atmel AT45DB011D (1Mbit) ROM. If a different ROM is used then the straps must change.
NOTE: ENETM requires SI pull-down instead of SO.

Ethernet 25MHz Crystal

C3945 20PF 5% 50V C0GM 0402
ENET_XTAL_OUT_R
R3945 0 5% 1/16W MF-LF 402
ENET_XTAL_OUT 36
NOSTUFF R3946 1M 5% 1/16W MF-LF 402
CRITICAL Y3900 25.000MHZ-20PPM-12PF-85C 3.2X2.5MM-SM PLACE_NEAR=U3900.18:10MM
C3946 20PF 5% 50V C0GM 0402
NOTE: 30 PPM crystal required
ENET_XTAL_IN 36

ENET_CR Signals

BCM requests SD CR[0:7], CMD, CLK termination.

ENET_SR_DISABLE

If ENET switching regulator is used, this pin should have a 1K pull-down to GND

PLACEMENT_NOTE=PLACE R3961 NEAR U3900
PLACEMENT_NOTE=PLACE R3979 NEAR U3900
PLACEMENT_NOTE=PLACE R3971 NEAR U3900
PLACEMENT_NOTE=PLACE R3972 NEAR U3900
PLACEMENT_NOTE=PLACE R3973 NEAR U3900
PLACEMENT_NOTE=PLACE R3974 NEAR U3900

BOM_COST_GROUP=ETHERNET

SYNC_MASTER=FIY1N SYNC_DATE=01/19/2010

PAGE TITLE
ETHERNET: PHY (CAESAR IV)

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D
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PAGE 39 OF 121
SHEET 36 OF 102

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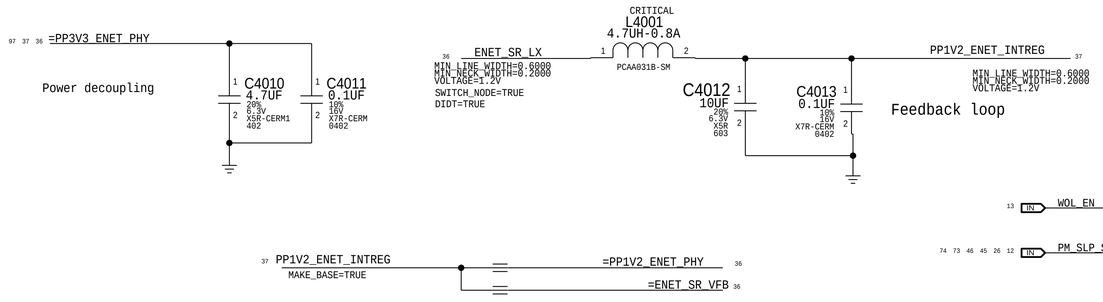
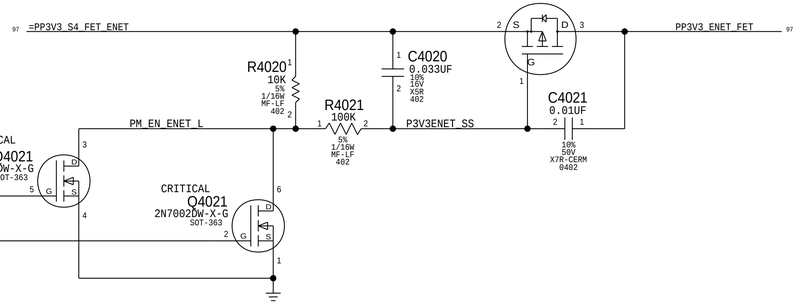
CAESAR IV 1.2V INT.VR CMPTS



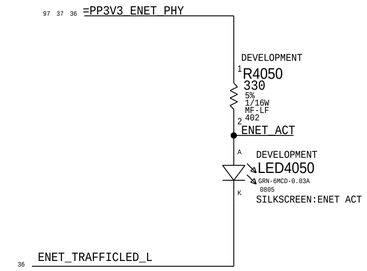
ENET Enable Generation

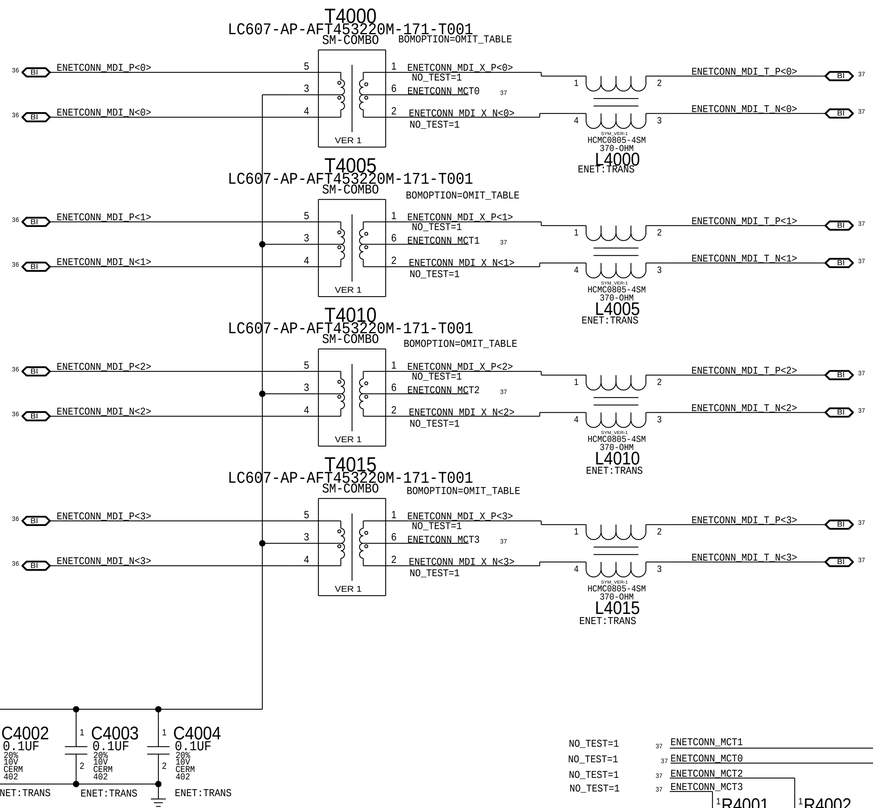
"ENET" = "S0" || ("S4" && "WOL_EN")

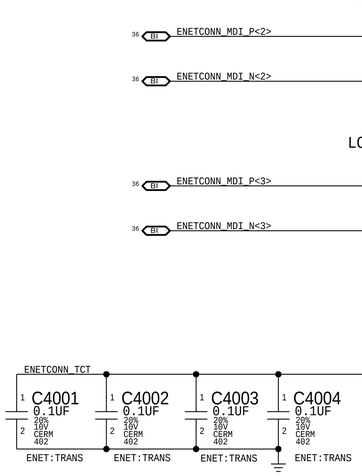
3.3V ENET FET

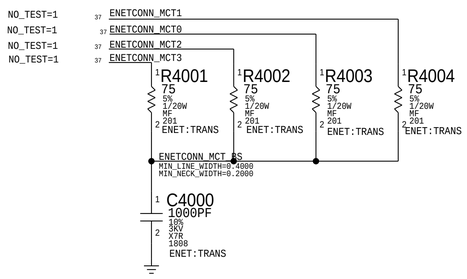


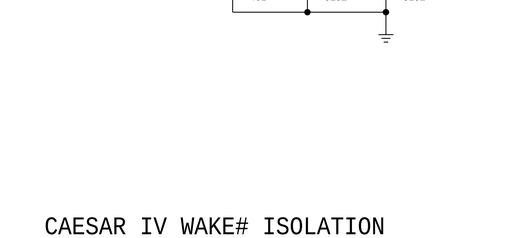
CAESAR IV ACTIVITY LED

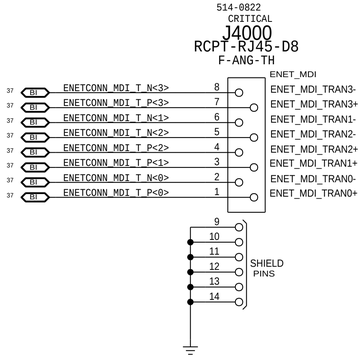




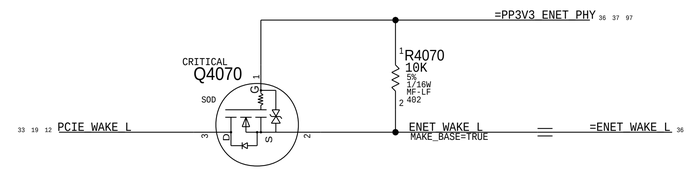








CAESAR IV WAKE# ISOLATION



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
157S00005	4	XFRMR,LAN CHIP,HALF-PORT,170UH,1000T,6P	T4000,T4005,T4010,T4015	CRITICAL	TRANS:DLT
157S00010	4	XFRMR,LAN,170UH,1000T,6P,4532	T4000,T4005,T4010,T4015	CRITICAL	TRANS:TDK

BOM_COST_GROUP=ETHERNET

PAGE TITLE
ETHERNET: Support & Connector

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SIZE D
REVISION 5.19.0
BRANCH
PAGE 40 OF 121
SHEET 37 OF 102

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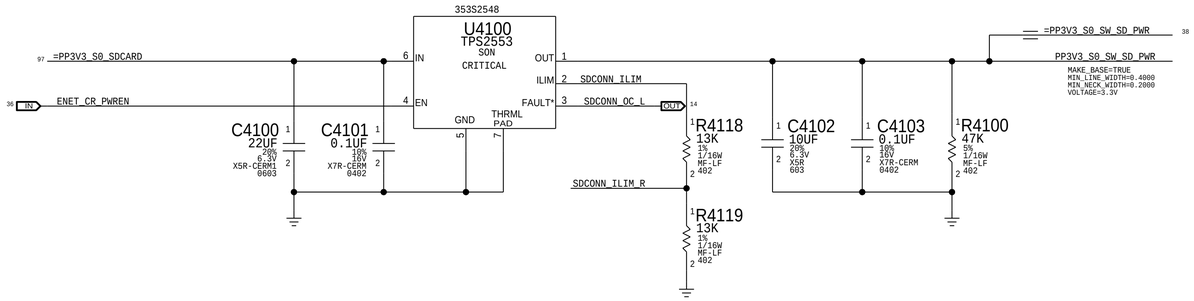
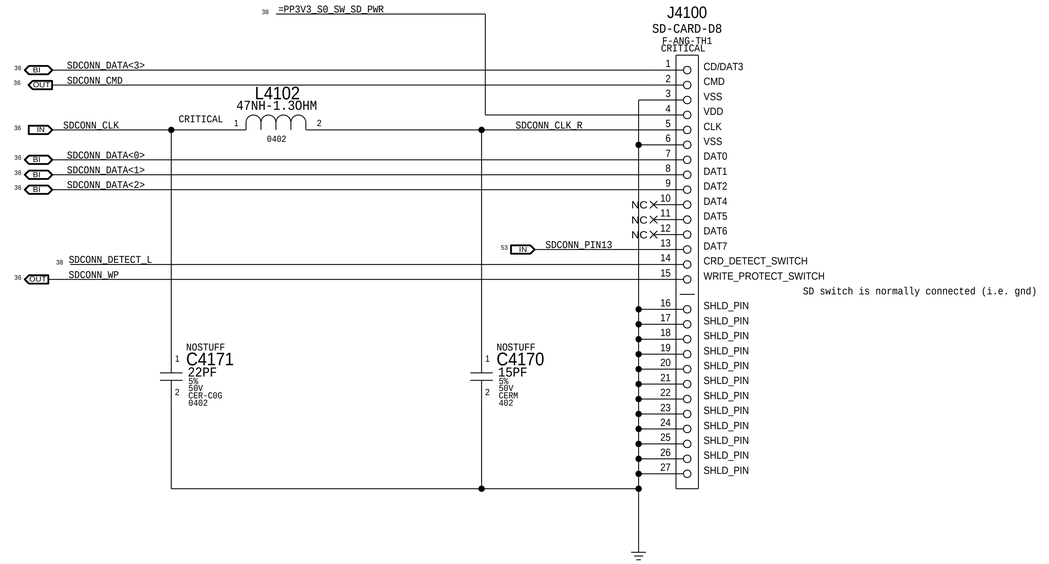
SD CARD 3.3V OVERCURRENT PROTECTION CHIP

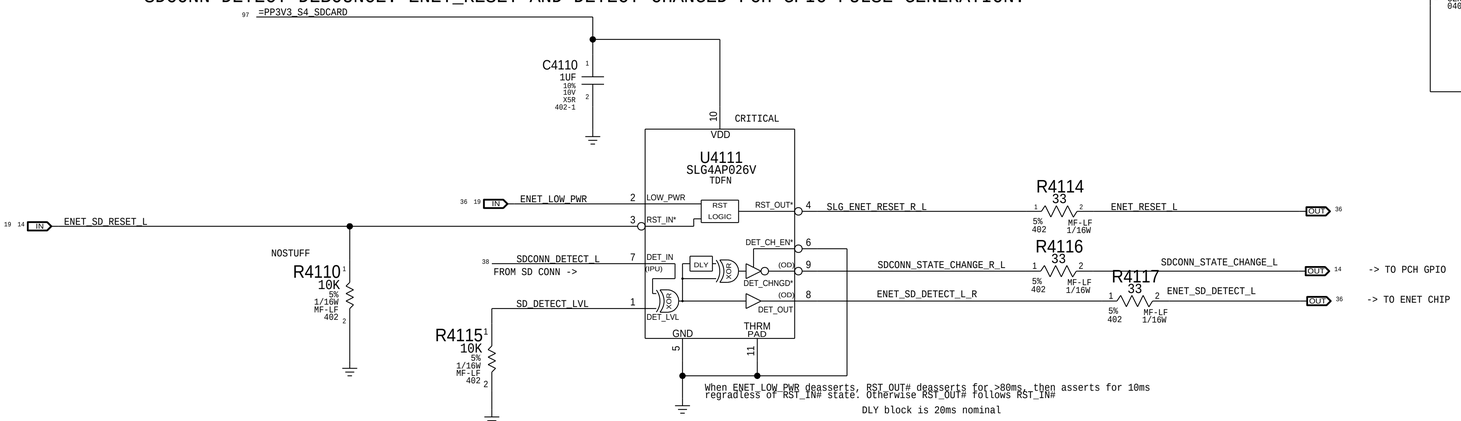


J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



BOM_COST_GROUP=SD CARD

SYNC_MASTER=FIYIN SYNC_DATE=01/19/2016

PAGE TITLE

SD CARD: Connector

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

BRANCH

PAGE 41 OF 121

SHEET 38 OF 102

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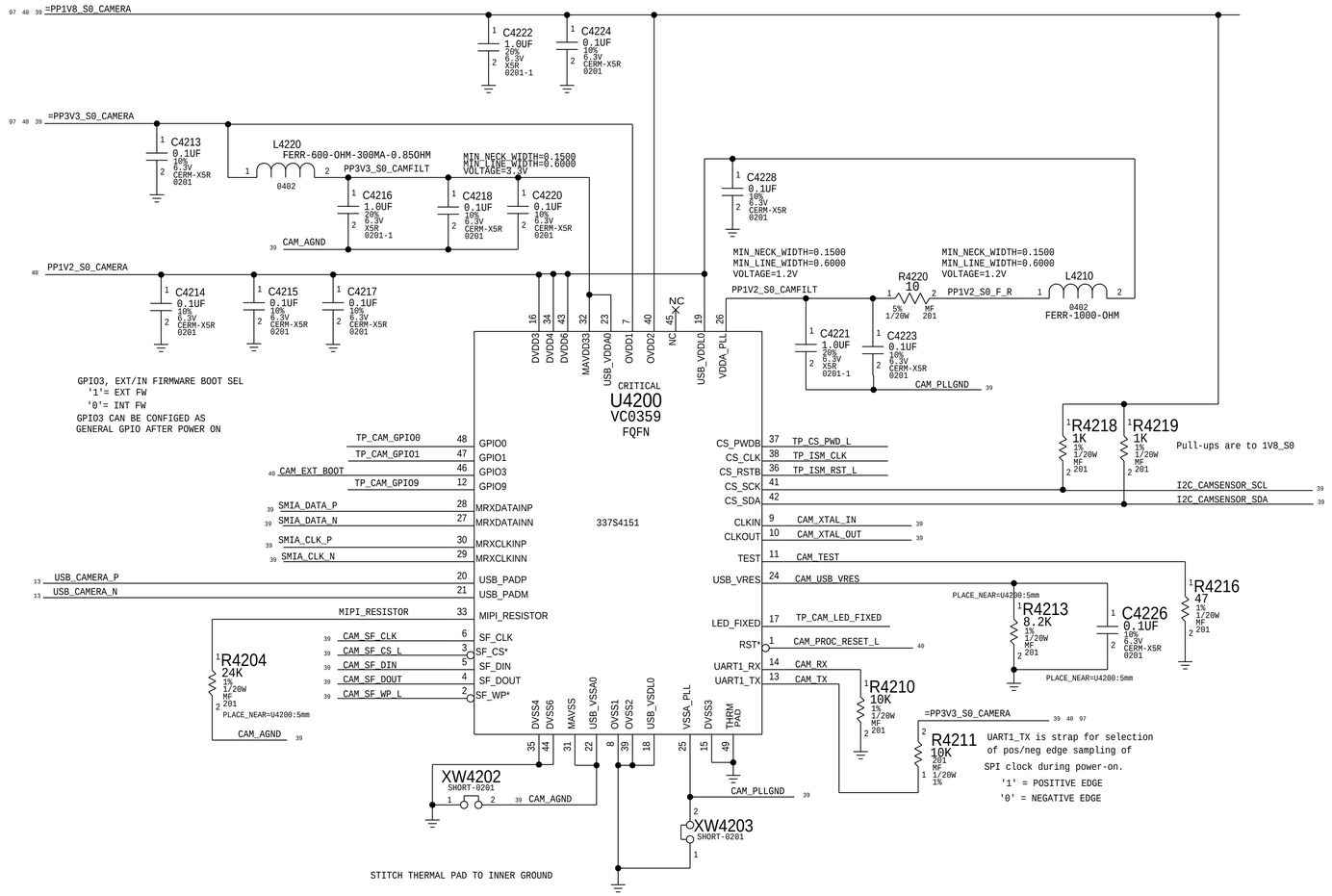
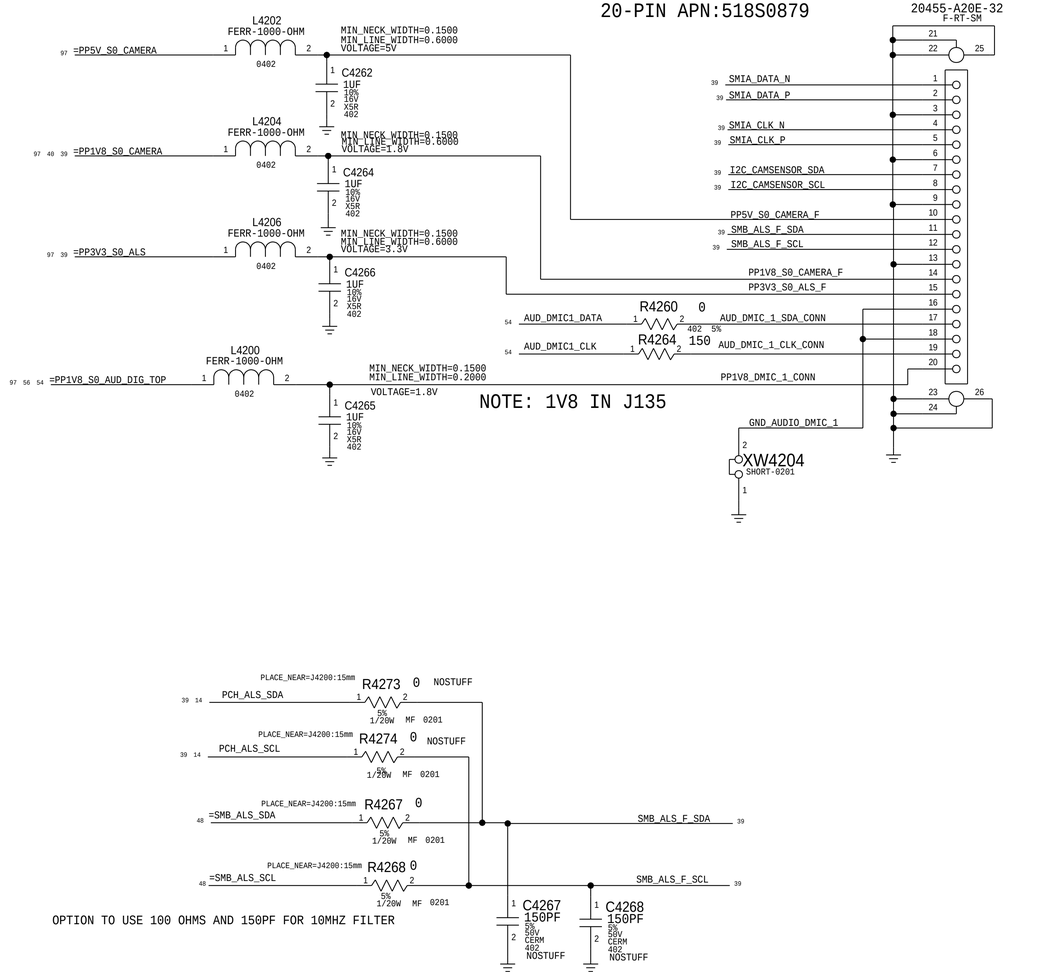
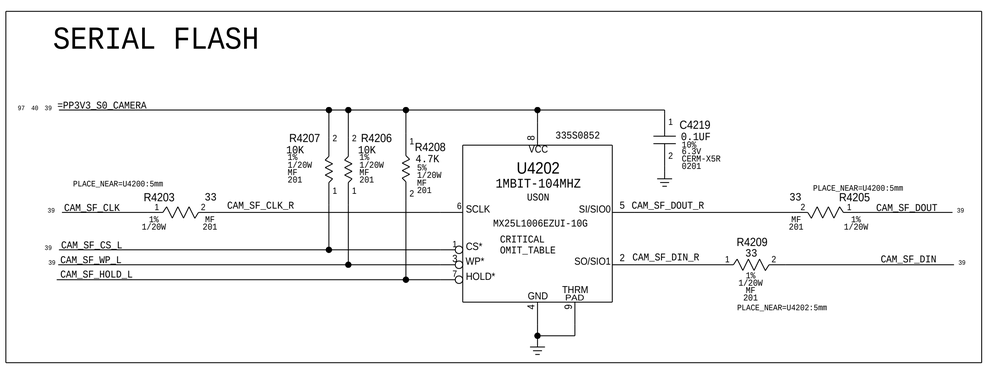
USB CAMERA CONTROLLER



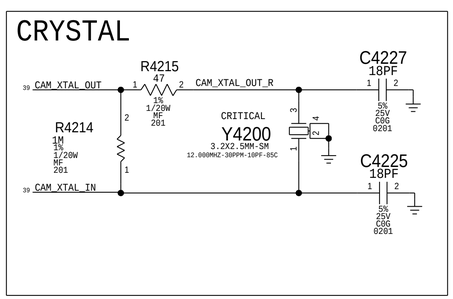
Camera/ALS/DMIC connector



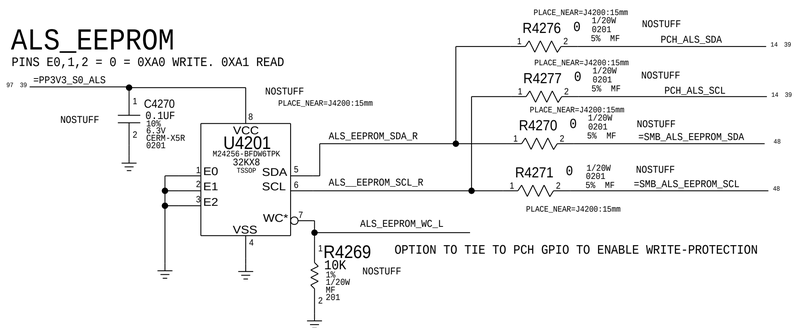
SERIAL FLASH



CRYSTAL



ALS_EEPROM



BOM_COST_GROUP=CAMERA

SYNC_MASTER=MBP8X SYNC_DATE=08/29/2012

PAGE TITLE

CAMERA: Controller

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

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PAGE 42 OF 121

SHEET 39 OF 102

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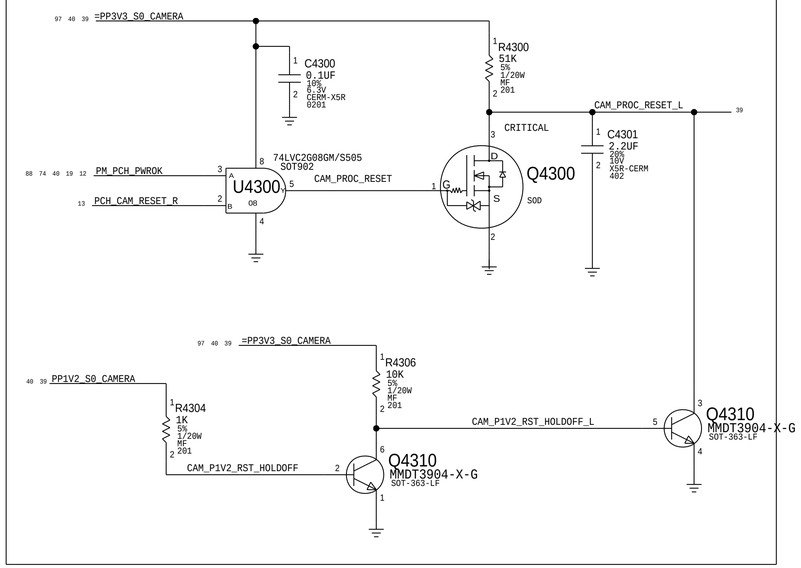
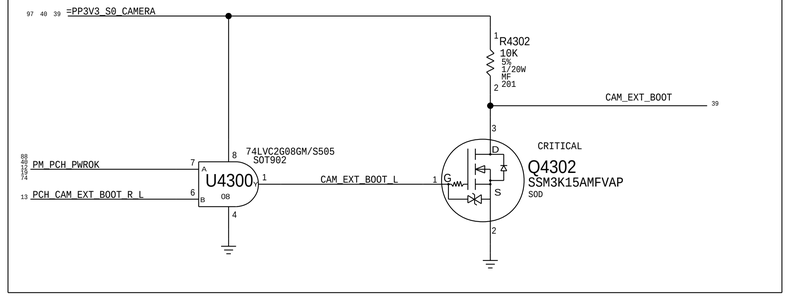
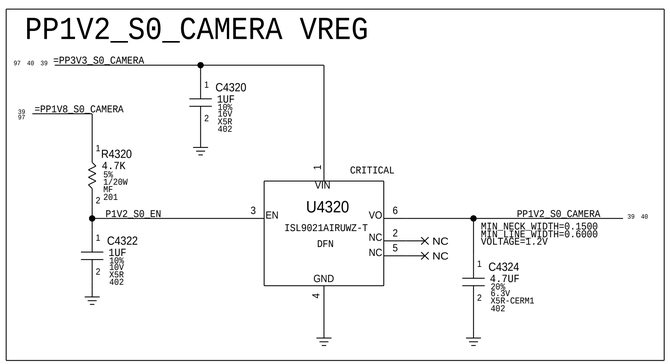
Camera Processor Reset



Camera Processor ExtBoot Cntl



PP1V2_S0_CAMERA VREG



BOM_COST_GROUP=CAMERA

SYNC_MASTER=DDRESSLER SYNC_DATE=08/27/2016

PAGE TITLE: CAMERA: Controller Support

Apple Inc.

DRAWING NUMBER: 051-01506 SIZE: D

REVISION: 5.19.0

BRANCH:

PAGE: 43 OF 121

SHEET: 40 OF 102

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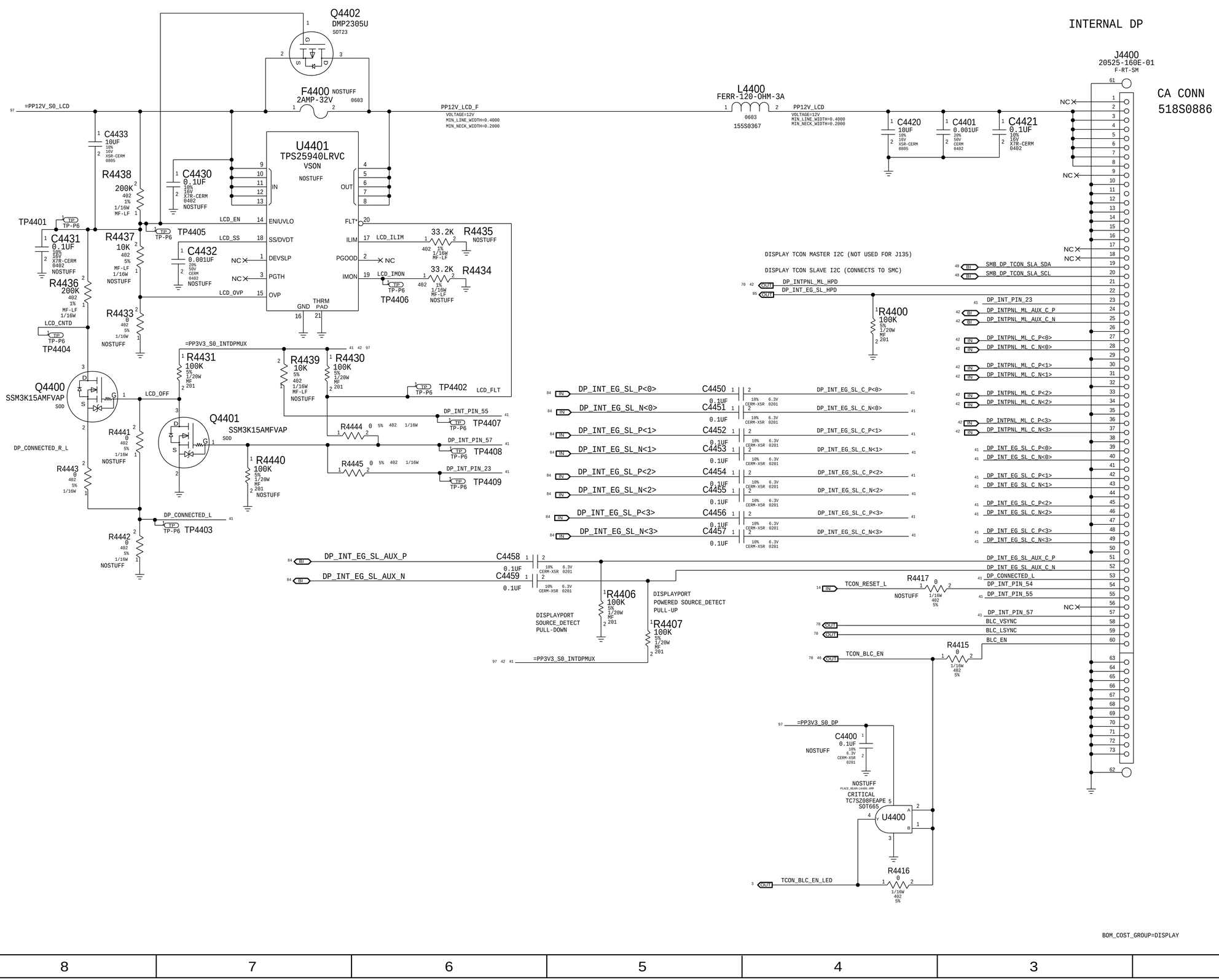
INTERNAL DP
J4400
20525-160E-01
F-RT-SM
CA CONN
518S0886
Q4402
DMP2305U
SOT23
F4400 NOSTUFF
2AMP-32V
0603
=PP12V_S0_LCD
PP12V_LCD_F
VOLTAGE=12V
MIN_LINE_WIDTH=0.4000
MIN_NECK_WIDTH=0.2000
L4400
FERR-120-OHM-3A
0603
155S0367
PP12V_LCD
U4401
TPS25940LRVC
VSON
NOSTUFF
IN
OUT
EN/UVLO
SS/DVDT
DEVSLP
PGTH
OVP
FLT*
ILIM
PGOOD
IMON
GND
THRM PAD
LCD_EN
LCD_SS
LCD_ILIM
LCD_IMON
LCD_OVP
C4420 10UF 10% 16V X5R-CERM 0805
C4401 0.001UF 20% 50V CERM 0402
C4421 0.1UF 10% 16V X7R-CERM 0402
C4430 0.1UF 10% 16V X7R-CERM 0402 NOSTUFF
C4433 10UF 10% 16V X5R-CERM 0805
C4431 0.1UF 10% 16V X7R-CERM 0402 NOSTUFF
C4432 0.001UF 20% 50V CERM 0402 NOSTUFF
R4438 200K 402 1% 1/16W MF-LF
R4437 10K 402 5% 1/16W MF-LF NOSTUFF
R4436 200K 402 1% 1/16W MF-LF
R4433 0 402 5% 1/16W NOSTUFF
R4435 33.2K 402 1% 1/16W MF-LF NOSTUFF
R4434 33.2K 402 1% 1/16W MF-LF NOSTUFF
TP4401 TP-P6
TP4405 TP-P6
TP4406 TP-P6
TP4404 TP-P6
LCD_CNTD
DISPLAY TCON MASTER I2C (NOT USED FOR J135)
DISPLAY TCON SLAVE I2C (CONNECTS TO SMC)
DP_INTPNL_ML_HPD
DP_INT_EG_SL_HPD
R4400 100K 5% 1/20W MF 201
SMB_DP_TCON_SLA_SDA
SMB_DP_TCON_SLA_SCL
DP_INT_PIN_23
DP_INTPNL_ML_AUX_C_P
DP_INTPNL_ML_AUX_C_N
DP_INTPNL_ML_C_P<0>
DP_INTPNL_ML_C_N<0>
DP_INTPNL_ML_C_P<1>
DP_INTPNL_ML_C_N<1>
DP_INTPNL_ML_C_P<2>
DP_INTPNL_ML_C_N<2>
DP_INTPNL_ML_C_P<3>
DP_INTPNL_ML_C_N<3>
DP_INT_EG_SL_C_P<0>
DP_INT_EG_SL_C_N<0>
DP_INT_EG_SL_C_P<1>
DP_INT_EG_SL_C_N<1>
DP_INT_EG_SL_C_P<2>
DP_INT_EG_SL_C_N<2>
DP_INT_EG_SL_C_P<3>
DP_INT_EG_SL_C_N<3>
DP_INT_EG_SL_AUX_C_P
DP_INT_EG_SL_AUX_C_N
DP_CONNECTED_L
DP_INT_PIN_54
DP_INT_PIN_55
DP_INT_PIN_57
BLC_VSYNC
BLC_LSYNC
BLC_EN
TCON_BLC_EN
TCON_RESET_L
R4417 0 1/16W 402 5% NOSTUFF
R4415 0 1/10W 402 5%
=PP3V3_S0_INTDPMUX
R4431 100K 5% 1/20W MF 201
R4439 10K 5% 402 1/16W MF-LF NOSTUFF
R4430 100K 5% 1/20W MF 201
TP4402
LCD_FLT
Q4400
SSM3K15AMFVAP
SOD
LCD_OFF
Q4401
SSM3K15AMFVAP
SOD
R4441 0 402 5% 1/16W NOSTUFF
R4443 0 402 5% 1/16W
R4440 100K 5% 1/20W MF 201 NOSTUFF
R4444 0 5% 402 1/16W
R4445 0 5% 402 1/16W
DP_INT_PIN_55
DP_INT_PIN_57
DP_INT_PIN_23
TP4407
TP4408
TP4409
DP_CONNECTED_R_L
DP_CONNECTED_L
TP4403
R4442 0 402 5% 1/16W NOSTUFF
DP_INT_EG_SL_P<0>
DP_INT_EG_SL_N<0>
DP_INT_EG_SL_P<1>
DP_INT_EG_SL_N<1>
DP_INT_EG_SL_P<2>
DP_INT_EG_SL_N<2>
DP_INT_EG_SL_P<3>
DP_INT_EG_SL_N<3>
C4450 0.1UF
C4451 0.1UF
C4452 0.1UF
C4453 0.1UF
C4454 0.1UF
C4455 0.1UF
C4456 0.1UF
C4457 0.1UF
10% 6.3V CERM-XSR 0201
DP_INT_EG_SL_AUX_P
DP_INT_EG_SL_AUX_N
C4458 0.1UF
C4459 0.1UF
R4406 100K 5% 1/20W MF 201
R4407 100K 5% 1/20W MF 201
DISPLAYPORT SOURCE_DETECT PULL-DOWN
DISPLAYPORT POWERED SOURCE_DETECT PULL-UP
=PP3V3_S0_INTDPMUX
=PP3V3_S0_DP
C4400 0.1UF 10% 6.3V CERM-XSR 0201 NOSTUFF
NOSTUFF PLACE_SOMEWHERE_ELSE
CRITICAL
TC7SZ08FEAPE
SOT665
U4400
R4416 0 1/10W 402 5%
TCON_BLC_EN_LED
BOM_COST_GROUP=DISPLAY
SYNC_MASTER=ANDREW
SYNC_DATE=10/04/2016
PAGE TITLE
DISPLAY: Support
Apple Inc.
DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 44 OF 121
SHEET 41 OF 102
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TP to DP aliases

26	IN	DP_X_SRC_ML_P<0>	DP_TBTSRC_ML_P<0>	42
26	IN	DP_X_SRC_ML_N<0>	DP_TBTSRC_ML_N<0>	42
26	IN	DP_X_SRC_ML_P<1>	DP_TBTSRC_ML_P<1>	42
26	IN	DP_X_SRC_ML_N<1>	DP_TBTSRC_ML_N<1>	42
26	IN	DP_X_SRC_ML_P<2>	DP_TBTSRC_ML_P<2>	42
26	IN	DP_X_SRC_ML_N<2>	DP_TBTSRC_ML_N<2>	42
26	IN	DP_X_SRC_ML_P<3>	DP_TBTSRC_ML_P<3>	42
26	IN	DP_X_SRC_ML_N<3>	DP_TBTSRC_ML_N<3>	42
26	BI	DP_X_SRC_AUX_P	DP_TBTSRC_AUXCH_P	42
26	BI	DP_X_SRC_AUX_N	DP_TBTSRC_AUXCH_N	42

All: MAKE_BASE=TRUE

97 42 =PP1V8_S0_DP

R4510 470K 5% 1/20W MF 201

R4511 470K 5% 1/20W MF 201

84 BI DP_INT_EG_ML_AUX_P — C4508 0.1UF — DP_INT_EG_ML_AUX_C_P

84 BI DP_INT_EG_ML_AUX_N — C4509 0.1UF — DP_INT_EG_ML_AUX_C_N

85 OUT DP_INT_EG_ML_HPD — R4500 200K 1% 1/20W MF 201

97 42 =PP1V8_S0_DP

R4512 470K 5% 1/20W MF 201

R4513 470K 5% 1/20W MF 201

42 DP_TBTSRC_AUXCH_P — C4510 0.1UF — DP_TBTSRC_AUX_C_P

42 DP_TBTSRC_AUXCH_N — C4511 0.1UF — DP_TBTSRC_AUX_C_N

28 26 OUT DP_X_SRC_HPD — PD is on the LR page

46 14 IN SMC_EXT_VIDEO_SEL

97 42 41 =PP3V3_S0_INTDPMUX

R4504 10K 5% 1/20W MF 201

DP_GPU_MUX_EN

R4501 10K 5% 1/20W MF 201

U4500.A1:8MM

U4500 CBTL06142EEE TFBGA CRITICAL

84 IN	DP_INT_EG_ML_P<0>	B4	DIN1_0+
84 IN	DP_INT_EG_ML_N<0>	A4	DIN1_0-
84 IN	DP_INT_EG_ML_P<1>	B5	DIN1_1+
84 IN	DP_INT_EG_ML_N<1>	A5	DIN1_1-
84 IN	DP_INT_EG_ML_P<2>	B6	DIN1_2+
84 IN	DP_INT_EG_ML_N<2>	A6	DIN1_2-
84 IN	DP_INT_EG_ML_P<3>	A8	DIN1_3+
84 IN	DP_INT_EG_ML_N<3>	A9	DIN1_3-
	DP_INT_EG_ML_AUX_C_P	H9	DAUX1+
	DP_INT_EG_ML_AUX_C_N	J9	DAUX1-
	NC	H8	DDC_CLK1
	NC	J8	DDC_DAT1
		J2	HPD_1
42 IN	DP_TBTSRC_ML_C_P<0>	B8	DIN2_0+
42 IN	DP_TBTSRC_ML_C_N<0>	B9	DIN2_0-
42 IN	DP_TBTSRC_ML_C_P<1>	D8	DIN2_1+
42 IN	DP_TBTSRC_ML_C_N<1>	D9	DIN2_1-
42 IN	DP_TBTSRC_ML_C_P<2>	E8	DIN2_2+
42 IN	DP_TBTSRC_ML_C_N<2>	E9	DIN2_2-
42 IN	DP_TBTSRC_ML_C_P<3>	F8	DIN2_3+
42 IN	DP_TBTSRC_ML_C_N<3>	F9	DIN2_3-
		H6	DAUX2+
		J6	DAUX2-
	NC	H5	DDC_CLK2
	NC	J5	DDC_DAT2
		H3	HPD_2
		A1	GPU_SEL
		B7	XSD*
		C2	DDC_AUX_SEL
		B3	GND
		C8	GND
		G8	GND
		H4	GND
		H7	GND
		G2	GND

VDD A2, VDD J4

=PP3V3_S0_INTDPMUX 41 42 97

C4568 0.1UF 10% 6.3V CERM-X5R 0201

C4569 0.1UF 10% 6.3V CERM-X5R 0201

DOUT_0+	B2	DP_INTPNL_ML_P<0>	C4500 0.1UF	DP_INTPNL_ML_C_P<0> OUT 41
DOUT_0-	B1	DP_INTPNL_ML_N<0>	C4501 0.1UF	DP_INTPNL_ML_C_N<0> OUT 41
DOUT_1+	D2	DP_INTPNL_ML_P<1>	C4502 0.1UF	DP_INTPNL_ML_C_P<1> OUT 41
DOUT_1-	D1	DP_INTPNL_ML_N<1>	C4503 0.1UF	DP_INTPNL_ML_C_N<1> OUT 41
DOUT_2+	E2	DP_INTPNL_ML_P<2>	C4504 0.1UF	DP_INTPNL_ML_C_P<2> OUT 41
DOUT_2-	E1	DP_INTPNL_ML_N<2>	C4505 0.1UF	DP_INTPNL_ML_C_N<2> OUT 41
DOUT_3+	F2	DP_INTPNL_ML_P<3>	C4506 0.1UF	DP_INTPNL_ML_C_P<3> OUT 41
DOUT_3-	F1	DP_INTPNL_ML_N<3>	C4507 0.1UF	DP_INTPNL_ML_C_N<3> OUT 41

All: 0201 10% 6.3V CERM-X5R 6.3V

AUX+ H2 DP_INTPNL_ML_AUX_P — R4558 0 5% 1/20W MF 0201 — DP_INTPNL_ML_AUX_C_P BI 41

AUX- H1 DP_INTPNL_ML_AUX_N — R4559 0 5% 1/20W MF 0201 — DP_INTPNL_ML_AUX_C_N BI 41

HPDIN J1 DP_INTPNL_ML_HPD IN 41 70

R4502 100K 5% 1/20W MF 201

R4503 100K 5% 1/20W MF 201

=PP3V3_S0_INTDPMUX 41 42 97

42 IN	DP_TBTSRC_ML_P<0>	DP_TBTSRC_ML_C_P<0>	42
42 IN	DP_TBTSRC_ML_N<0>	DP_TBTSRC_ML_C_N<0>	42
42 IN	DP_TBTSRC_ML_P<1>	DP_TBTSRC_ML_C_P<1>	42
42 IN	DP_TBTSRC_ML_N<1>	DP_TBTSRC_ML_C_N<1>	42
42 IN	DP_TBTSRC_ML_P<2>	DP_TBTSRC_ML_C_P<2>	42
42 IN	DP_TBTSRC_ML_N<2>	DP_TBTSRC_ML_C_N<2>	42
42 IN	DP_TBTSRC_ML_P<3>	DP_TBTSRC_ML_C_P<3>	42
42 IN	DP_TBTSRC_ML_N<3>	DP_TBTSRC_ML_C_N<3>	42

All: MAKE_BASE=TRUE

BOM_COST_GROUP=DISPLAY

SYNC_MASTER=ANDREW SYNC_DATE=02/17/2016

PAGE TITLE: DISPLAY: MUXing

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

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PAGE 45 OF 121

SHEET 42 OF 102

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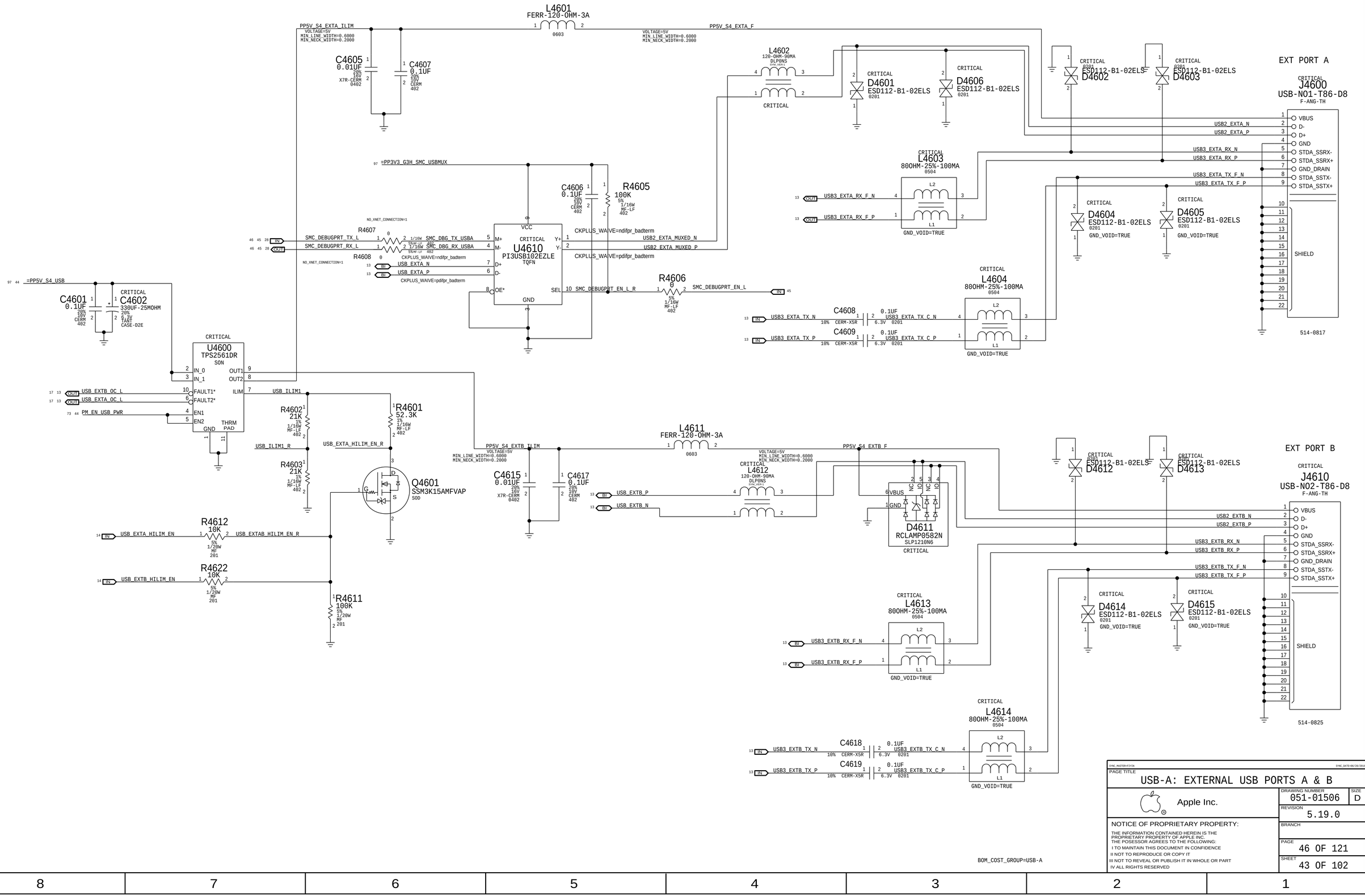

PAGE TITLE

USB-A: EXTERNAL USB PORTS A & B

Apple Inc.

DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	46 OF 121
SHEET	43 OF 102

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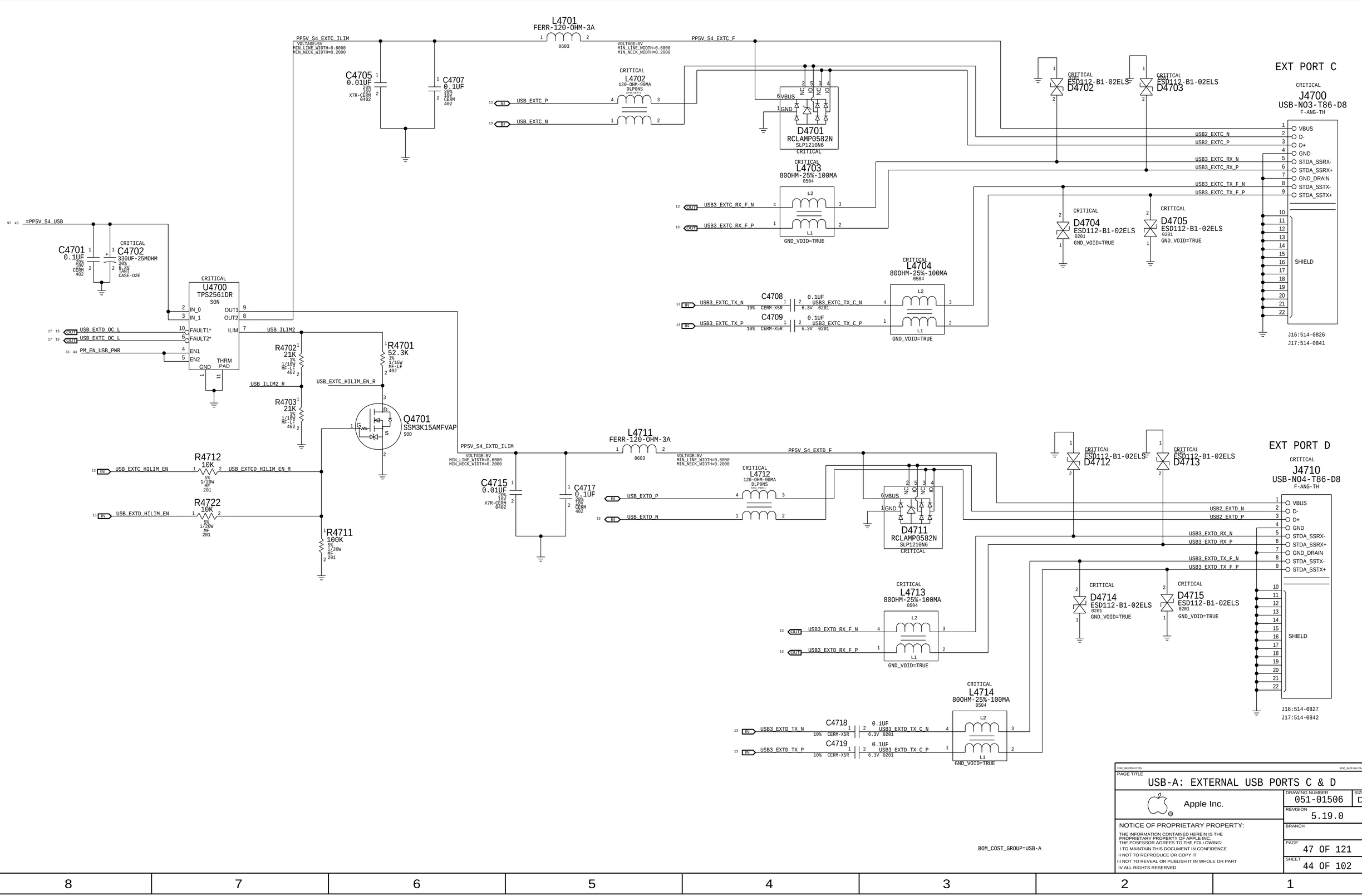
L4701
FERR-120-OHM-3A
0603
PP5V_S4_EXTC_ILIM
VOLTAGE=5V
MIN_LINE_WIDTH=0.6000
MIN_NECK_WIDTH=0.2000
PP5V_S4_EXTC_F
C4705
0.01UF
20%
16V
X7R-CERM
0402
C4707
0.1UF
20%
16V
CERM
402
CRITICAL
L4702
120-OHM-30MA
DLP0NS
USB_EXTC_P
USB_EXTC_N
D4701
RCLAMP0582N
SLP1210N6
CRITICAL
D4702
ESD112-B1-02ELS
0201
D4703
ESD112-B1-02ELS
0201
EXT PORT C
J4700
USB-NO3-T86-D8
F-ANG-TH
VBUS
D-
D+
GND
STDA_SSRX-
STDA_SSRX+
GND_DRAIN
STDA_SSTX-
STDA_SSTX+
SHIELD
USB2_EXTC_N
USB2_EXTC_P
USB3_EXTC_RX_N
USB3_EXTC_RX_P
USB3_EXTC_TX_F_N
USB3_EXTC_TX_F_P
J16:514-0826
J17:514-0841
L4703
80OHM-25%-100MA
0504
USB3_EXTC_RX_F_N
USB3_EXTC_RX_F_P
GND_VOID=TRUE
D4704
ESD112-B1-02ELS
D4705
ESD112-B1-02ELS
L4704
80OHM-25%-100MA
USB3_EXTC_TX_N
USB3_EXTC_TX_P
C4708
C4709
0.1UF
10%
CERM-X5R
6.3V
USB3_EXTC_TX_C_N
USB3_EXTC_TX_C_P
=PP5V_S4_USB
C4701
0.1UF
C4702
330UF-25MOHM
20%
6.3V
TANT
CASE-D2E
U4700
TPS2561DR
SON
IN_0
IN_1
OUT1
OUT2
FAULT1*
FAULT2*
ILIM
EN1
EN2
GND
THRM PAD
USB_EXTD_OC_L
USB_EXTC_OC_L
PM_EN_USB_PWR
USB_ILIM2
R4702
21K
1%
1/10W
MF-LF
402
R4701
52.3K
1%
1/16W
MF-LF
402
USB_ILIM2_R
USB_EXTC_HILIM_EN_R
R4703
21K
Q4701
SSM3K15AMFVAP
SOD
R4712
10K
5%
1/20W
MF
201
R4722
10K
USB_EXTC_HILIM_EN
USB_EXTCD_HILIM_EN_R
USB_EXTD_HILIM_EN
R4711
100K
L4711
FERR-120-OHM-3A
PP5V_S4_EXTD_ILIM
PP5V_S4_EXTD_F
C4715
0.01UF
C4717
0.1UF
L4712
120-OHM-30MA
DLP0NS
USB_EXTD_P
USB_EXTD_N
D4711
RCLAMP0582N
SLP1210N6
CRITICAL
D4712
ESD112-B1-02ELS
D4713
ESD112-B1-02ELS
EXT PORT D
J4710
USB-NO4-T86-D8
F-ANG-TH
VBUS
D-
D+
GND
STDA_SSRX-
STDA_SSRX+
GND_DRAIN
STDA_SSTX-
STDA_SSTX+
SHIELD
USB2_EXTD_N
USB2_EXTD_P
USB3_EXTD_RX_N
USB3_EXTD_RX_P
USB3_EXTD_TX_F_N
USB3_EXTD_TX_F_P
J16:514-0827
J17:514-0842
L4713
80OHM-25%-100MA
0504
USB3_EXTD_RX_F_N
USB3_EXTD_RX_F_P
GND_VOID=TRUE
D4714
ESD112-B1-02ELS
D4715
ESD112-B1-02ELS
L4714
80OHM-25%-100MA
USB3_EXTD_TX_N
USB3_EXTD_TX_P
C4718
C4719
0.1UF
10%
CERM-X5R
6.3V
USB3_EXTD_TX_C_N
USB3_EXTD_TX_C_P
BOM_COST_GROUP=USB-A
PAGE TITLE
USB-A: EXTERNAL USB PORTS C & D
Apple Inc.
DRAWING NUMBER
051-01506
SIZE
D
REVISION
5.19.0
BRANCH
PAGE
47 OF 121
SHEET
44 OF 102
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NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

APN IN BOM PAGE:
338S1214-A-TI (LM4FS1EH5BB)

U5000
BGA
(1 OF 2)
OMIT_TABLE

12	BI	LPC_AD<0>	arch		B13	LPC0AD0	AIN00	E2	proj	analog	SMC_ADC0	IN	46
12	BI	LPC_AD<1>	arch		A13	LPC0AD1	AIN01	E1	proj	analog	SMC_ADC1	IN	46
12	BI	LPC_AD<2>	arch		C12	LPC0AD2	AIN02	F2	proj	analog	SMC_ADC2	IN	46
12	BI	LPC_AD<3>	arch		D11	LPC0AD3	AIN03	F1	proj	analog	SMC_ADC3	IN	46
11	IN	LPC_CLK24M_SMC	arch		H12	LPC0CLK	AIN04	B3	proj	analog	SMC_ADC4	IN	46
12	IN	LPC_FRAME_L	arch		D12	LPC0FRAME*	AIN05	A3	proj	analog	SMC_ADC5	IN	46
19	IN	SMC_LRESET_L	arch		C13	LPC0RESET*	AIN06	B4	proj	analog	SMC_ADC6	IN	46
12	BI	LPC_SERIRQ	arch	od	H13	LPC0SERIRQ	AIN07	A4	proj	analog	SMC_ADC7	IN	46
46	OUT	PM_CLKRUN_L	arch	od	G11	LPC0CLKRUN*	AIN08	B5	proj	analog	SMC_ADC8	IN	46
12	IN	LPC_PWRDWN_L	arch		F13	LPC0PD*	AIN09	A5	proj	analog	SMC_ADC9	IN	46
12	OUT	SMC_RUNTIME_SCI_L	arch		F12	LPC0SCI*	AIN10	B6	proj	analog	SMC_ADC10	IN	46
12	OUT	SMC_WAKE_SCI_L	arch		B12	PK5	AIN11	A6	proj	analog	SMC_ADC11	IN	46
							AIN12	C1	proj	analog	SMC_ADC12	IN	46
48	BI	SMBUS_SMC_0_S0_SCL	arch	od	E10	I2C0SCL	AIN13	C2	proj	analog	SMC_ADC13	IN	46
48	BI	SMBUS_SMC_0_S0_SDA	arch	od	D13	I2C0SDA	AIN14	B1	proj	analog	SMC_ADC14	IN	46
48	BI	SMBUS_SMC_1_S0_SCL	arch	od	M4	I2C1SCL	AIN15	B2	proj	analog	SMC_ADC15	IN	46
48	BI	SMBUS_SMC_1_S0_SDA	arch	od	N2	I2C1SDA	AIN16	G2	proj	analog	SMC_ADC16	IN	46
48	BI	SMBUS_SMC_2_S0_SCL	arch	od	N8	I2C2SCL	AIN17	G1	proj	analog	SMC_ADC17	IN	46
48	BI	SMBUS_SMC_2_S0_SDA	arch	od	M8	I2C2SDA	AIN18	H1	proj	analog	SMC_ADC18	IN	46
48	BI	SMBUS_SMC_3_SCL	arch	od	L8	I2C3SCL	AIN19	H2	proj	analog	SMC_ADC19	IN	46
48	BI	SMBUS_SMC_3_SDA	arch	od	K8	I2C3SDA	AIN20	B7	proj	analog	SMC_ADC20	IN	46
NEW	48 BI	SMBUS_SMC_4_G3_SCL	arch	od	N7	I2C4SCL	AIN21	A7	proj	analog	SMC_ADC21	IN	46
NEW	48 BI	SMBUS_SMC_4_G3_SDA	arch	od	M7	I2C4SDA	AIN22	B8	proj	analog	SMC_ADC22	IN	46
NEW	48 BI	SMBUS_SMC_5_S0_SCL	arch	od	N4	I2C5SCL	AIN23	A8	proj	analog	SMC_ADC23	IN	46
NEW	48 BI	SMBUS_SMC_5_S0_SDA	arch	od	N3	I2C5SDA							
							C0-	K2	arch	analog	CPU_PROCHOT_L	IN	6 46 52 62
	53 OUT	SMC_FAN_0_CTL	arch		H11	PM6/FAN0PWM0	C0+	K1	arch	analog	SMC_VCCIO_CPU_DIV2	IN	46
	53 IN	SMC_FAN_0_TACH	arch		L13	PM7/FAN0TACH0	C1-	L2	arch	analog	SMC_PC4	IN	(MEM_OK)
	53 OUT	SMC_FAN_1_CTL	arch		C11	PK6/FAN0PWM1	PC5/C1+	L1	arch		SPI_DESCRIPTOR_OVERRIDE_L	OUT	18
	53 IN	SMC_FAN_1_TACH	arch		A12	PK7/FAN0TACH1	T3CCP1/PJ5/C2-	C5	arch		SMC_CPU_CATERR_L	IN	46
	46 OUT	SMC_TOPBLK_SWP_L	arch		G3	PN2/FAN0PWM2	T3CCP0/PJ4/C2+	D5	arch	analog	CPU_THRMTRIP_3V3	IN	46
TCON_OK	46 IN	SMC_PN3	proj		D10	PN3/FAN0TACH2							
	85 IN	GFX_SMC_ALERT_R	proj		L11	PN4/FAN0PWM3	SSI0CLK/PA2	M2	arch		SMC_PM_G2_EN	OUT	46 72
	46 OUT	SMC_PN5	proj		N12	PN5/FAN0TACH3	SSI0FSS/PA3	M3	arch		PM_DSW_PWRGD	IN	46 74
	46 OUT	SMC_PN6	arch		N11	PN6/FAN0PWM4	SSI0RX/PA4	L4	arch		SMC_DELAYED_PWRGD	IN	46 74
LINK_OK	46 OUT	SMC_PN7	proj		M11	PN7/FAN0TACH4	SSI0TX/PA5	N1	arch		SMC_PROCHOT	OUT	46
	46 IN	SMC_PH2	proj		J4	PH2/FAN0PWM5							
	46 OUT	SMC_PH3	proj		J2	PH3/FAN0TACH5	U1RX/B0	F11	arch		SMC_DEBUGPRT_RX_L	IN	28 43 46 (to USB-A Mojo)
							U1TX/PB1	E11	arch		SMC_DEBUGPRT_TX_L	OUT	28 43 46 (to USB-A Mojo)
	46 12 6 IN	CPU_PECI	arch	analog	C4	PECI0RX	T0CCP0/PB6	F4	arch	pwm	SMC_PB6	OUT	46 (new, Ext_Video_Sel)
	46 OUT	SMC_PECI_L	arch		C6	PECI0TX	T0CCP1/PB7	F3	arch		SMC_GFX_THROTTLE_L	OUT	52 85
							SSI1RX/PF0	M9	arch		SMC_PF0	IN	46
	46 IN	SMC_PP0	proj	int	M13	PP0/IRQ116	SSI1TX/PF1	N9	arch		SMC_PF1	OUT	46
	46 IN	SMC_PP1	proj	int	L12	PP1/IRQ117	SSI1CLK/PF2	L10	arch		SMC_PF2	OUT	46
	46 33 IN	SMC_PME_S4_WAKE_L	proj	int	M5	PP2/IRQ118	SSI1FSS/PF3	K10	arch		SMC_PF3	OUT	46
	46 IN	SMC_PP3	proj	int	J12	PP3/IRQ119	PF4	L9	arch		S5_PWRGD	IN	74
	46 OUT	SMC_PP4	proj	int	J13	PP4/IRQ120	PF5	K9	arch		SMC_PM_PCH_SYS_PWROK	OUT	46
	46 IN	SMC_PP5	proj	int	L5	PP5/IRQ121							
	46 IN	SMC_PP6	proj	int	D8	PP6/IRQ122	WT0CCP0/PG4	K7	arch		SMC_DEBUGPRT_EN_L	OUT	43
	46 IN	SMC_PP7	proj	int	K6	PP7/IRQ123	WT0CCP1/PG5	L7	arch		GFX_SMC_CTF_3V3	IN	46 GPU's GPIO-19 = CTF
	46 OUT	ENET_ASF_GPIO	arch	od	D4	PQ0/IRQ124	WT2CCP0/PH0	K3	arch		ALL_SYS_PWRGD	IN	3 74
10K PU at I2C page	28 IN	SMC_USBC_INT_L	arch	int	E4	PQ1/IRQ125	WT2CCP1/PH1	K4	arch		SMC_THRMTRIP	OUT	46
	46 IN	SMC_BC_ACOK	arch	int	F5	PQ2/IRQ126							
	46 IN	SMC_PQ3	arch	int	N5	PQ3/IRQ127	WT3CCP0/PH4	J3	arch	od	PM_PWRBTN_L	OUT	12 17
	74 73 46 37 26 12 IN	PM_SLP_S3_L	arch	int	N6	PQ4/IRQ128	WT3CCP1/PH5	H4	arch	od	PM_SYSRST_L	OUT	12 17 18
	73 12 IN	PM_SLP_S4_L	arch	int	K5	PQ5/IRQ129	WT4CCP0/PH6	H3	arch	od	MEM_EVENT_L	OUT	21 22 23 24 46 Throttle DIMM
	73 33 12 IN	PM_SLP_S5_L	arch	int	M6	PQ6/IRQ130	WT4CCP1/PH7	G4	proj		SMC_PH7	IN	46 GFX_OK
	46 IN	SMC_ONOFF_L	arch	int	L6	PQ7/IRQ131							
							T1CCP0/PJ0	C9	arch		SMC_OOB1_D2R_L	IN	35
(to ACE)	46 28 IN	SMC_RX_L (SMC_DEBUGPRT2_RX_L, PA0)	arch		L3	U0RX	T1CCP1/PJ1	B9	arch		SMC_OOB1_R2D_L	OUT	46
(to ACE)	46 28 OUT	SMC_TX_L (SMC_DEBUGPRT2_TX_L, PA1)	arch		M1	U0TX	T2CCP0/PJ2	A9	proj		SMC_PJ2	IN	46
							T2CCP1/PJ3	C8	proj		SMC_PJ3	IN	46
	46 BI	SMC_PL7 (SMC_BT_PWR_EN)	proj		E13	USB0DM							
	46 BI	SMC_WIFI_PWR_EN (PL6)	arch		E12	USB0DP	WT5CCP1/PM3	H10	arch		SMC_PM3	OUT	46 NEW PCC_ALERT

FILTER_2P_0402_30_OHM_1_7A
L5001
30-OHM-1.7A
0402
MIN_LINE_WIDTH=0.2500
MIN_NECK_WIDTH=0.1000
VOLTAGE=3.3V

97 53 46 =PP3V3_G3H_SMC
PP3V3_G3H_SMC_VDDA

C5002 1UF 20% 10V X5R-CERM 0603-1
C5003 0.1UF 10% 6.3V CERM-X5R 0201
C5004 0.1UF 10% 6.3V CERM-X5R 0201
C5005 0.1UF 10% 6.3V CERM-X5R 0201
C5006 0.1UF 10% 6.3V CERM-X5R 0201
C5001 0.1UF 10% 6.3V CERM-X5R 0201

R5002 1M 5% 1/20W MF 201

U5000
BGA
(2 OF 2)
OMIT_TABLE

47 46 28 IN SMC_RESET_L → G10 RST*
46 BI AP_EVENT_L → B11 PK4/RTCCLK
(OD) SMC_WAKE_L → N13 WAKE*
NC_SMC_HIB_L → M12 HIB*
12 IN SMC_CLK32K → M10 XOSC0
NC_SMC_XOSC1 → N10 XOSC1
46 SMC_EXTAL → G12 OSC0
46 SMC_XTAL → G13 OSC1

C10 SWCLK/TCK → SMC_TCK 28 46 47
A10 SWDIO/TMS → SMC_TMS 28 46 47
A11 SWO/TDO → SMC_TDO 46
B10 TDI ← SMC_TDI 46
A2 NC
D3 VDDA

(J135 only)
PP3V0_G3H_AVREF_SMC 46
D2 VREFA+
D1 VREFA- SHORT_SM_BASE
XWS000 SM
C3 GNDA
E3 PLACE_NEAR=U5000.A1:4.5MM
C5020 1.0UF 20% 6.3V X5R 0201-1
C5021 0.01UF 10% 16V X5R-CERM 0201
PLACE_NEAR=U5000.D1:4mm
PLACE_NEAR=U5000.D1:6mm

A1 C7 D9 E5 F9 H5 H9 J5 J8 J11 K11 → GND
GND_SMC_AVSS 46 49 50

K12 VBAT
D7 E6 E8 E9 F10 J7 J9 J10 → VDD
J1 J6 K13 D6 → VDDC

C5007 0.1UF 10% 6.3V CERM-X5R 0201
C5008 0.1UF 10% 6.3V CERM-X5R 0201
C5009 0.1UF 10% 6.3V CERM-X5R 0201

PP1V2_G3H_SMC_VDDC
C5010 1UF 10% 6.3V CERM 402
C5011 1UF 10% 6.3V CERM 402
C5012 1UF 10% 6.3V CERM 402

MIN_LINE_WIDTH=0.2500
MIN_NECK_WIDTH=0.1000
VOLTAGE=1.2V
C5013 0.1UF 10% 6.3V CERM-X5R 0201
C5014 0.1UF 10% 6.3V CERM-X5R 0201
C5015 0.1UF 10% 6.3V CERM-X5R 0201
C5016 0.1UF 10% 6.3V CERM-X5R 0201
C5017 0.1UF 10% 6.3V CERM-X5R 0201

BOM_COST_GROUP=SMC

PAGE TITLE
SMC: Controller

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 50 OF 121
SHEET 45 OF 102

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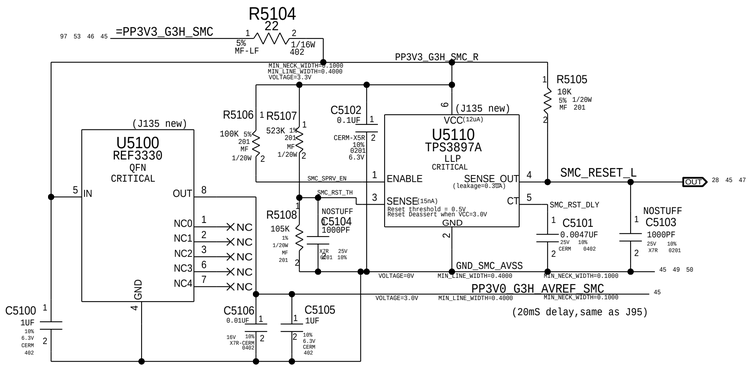
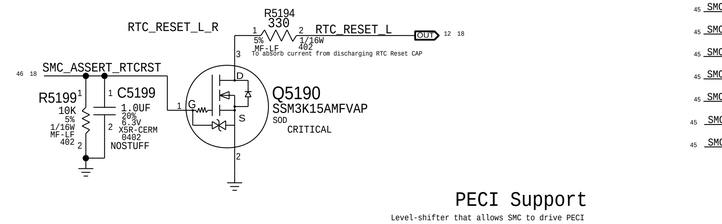
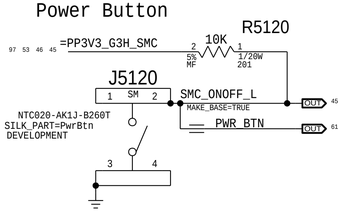
SMC Supervisor and AVREF Supply



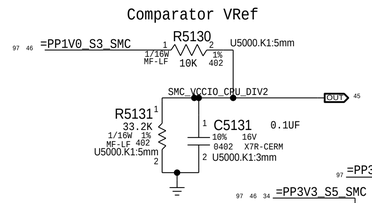
SMC Controlled RTC Reset



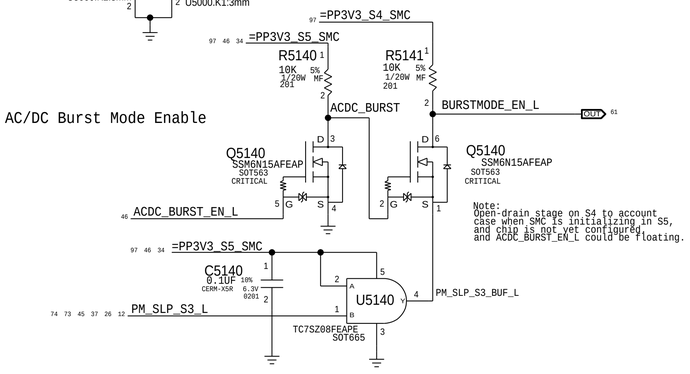
Power Button



Comparator VRef

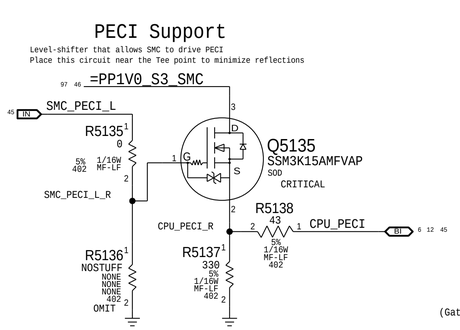


AC/DC Burst Mode Enable



PECI Support

Level-shifter that allows SMC to drive PECI
Place this circuit near the Tee point to minimize reflections



ADC Channel Aliases

SMC_ADC0	VSNS_P12VG3H	VD2R
SMC_ADC1	ISNS_P12VG3H	ID2R
SMC_ADC2	ISNS_P5V_S4	IR54
SMC_ADC3	ISNS_P12VS0_CPUCORE	ID20
SMC_ADC4	VSNS_CPUVCC	VC0C
SMC_ADC5	ISNS_CPUVCC	IC0C
SMC_ADC6	VSNS_CPUVCC_GT	VC0G
SMC_ADC7	ISNS_CPUVCC_GT	IC0G
SMC_ADC8	ISNS_CPUVCC_IO	IC0I
SMC_ADC9	ISNS_P1V2S0	IC0M
SMC_ADC10	ISNS_CPUVCC_SA	IC0S
SMC_ADC11	ISNS_P12VS0_FBVDDQ	IG1F
SMC_ADC12	VSNS_GPUCORE_ALT	VG0C
SMC_ADC13	ISNS_GPUCORE_ALT	IG0C
SMC_ADC14	VSNS_GPU_VDDCI	VG0I
SMC_ADC15	ISNS_GPU_VDDCI	IG0I
SMC_ADC16	ISNS_P12VS0_GPU_AUX	IG1A
SMC_ADC17	ISNS_P12VS0_GPUCORE	IG1C
SMC_ADC18	ISNS_P12VS0_HDD	IH02
SMC_ADC19	ISNS_HDDS0	IH05
SMC_ADC20	ISNS_P3V3_S5	IR35
SMC_ADC21	ISNS_SSD_S4	IH1R
SMC_ADC22	VSNS_VDDQS3_DDR	VM0R
SMC_ADC23	ISNS_VDDQS3_DDR	IM0R

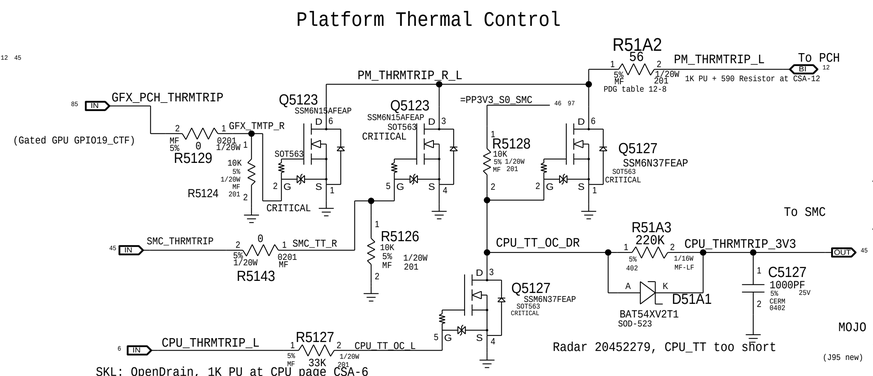
Project-specific Aliases

SMC_PN5	ACDC_BURST_EN_L	
SMC_PJ3	SMC_OOB2_R2D_L	
SMC_PJ2	SMC_OOB2_D2R_L	
SMC_PP0	SMC_ACDC_ID	
SMC_PH2	SMC_ASSERT_RTCRST	
SMC_PL7	SMC_BT_PWR_EN	
SMC_PC4	MEM_OK (NEW, LED-7)	(100K PD AT PCH)
SMC_PH7	GFX_OK (LED-3)	(10K PD AT GPU)
SMC_PN7	DP_LINK_OK (LED-3)	(10K PD AT GPU)
SMC_PN3	TCON_BLC_EN (LED-4)	(100K PD AT BLC)
SMC_PM3	PSU_OCP_ACTIVE (NEW)	
SMC_PP1	PM_EPO_SMC_L (NEW)	
SMC_PH3	SMC_HI_PWR_GFX (NEW)	
SMC_PB6	SMC_EXT_VIDEO_SEL (NEW)	

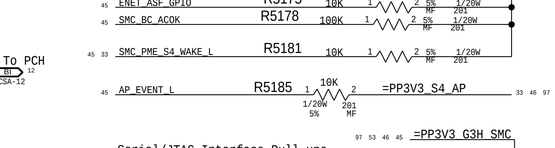
Unused Project-specific

SMC_PF0	NC_SMC_PF0	
SMC_PF1	NC_SMC_PF1	
SMC_PF2	NC_SMC_PF2	NO_TEST=1
SMC_PF3	NC_SMC_PF3	NO_TEST=1
SMC_PN6	NC_SMC_PN6	NO_TEST=1
SMC_PP3	NC_SMC_PME_S4_DARK_L	NO_TEST=1
SMC_PP4	NC_SMC_S4_WAKESRC_EN	
SMC_PP5	NC_SMC_PP5	NO_TEST=1
SMC_PP6	NC_SMC_PP6	NO_TEST=1
SMC_PP7	NC_SMC_PP7	NO_TEST=1
SMC_PQ3	PM_SLP_S0_L	12
SMC_OOB1_R2D_L	NC_SMC_OOB1_R2D_L	NO_TEST=1
SMC_PH3	NC_SMC_PH3	NO_TEST=1

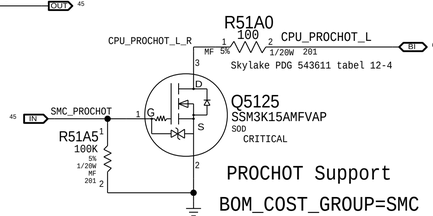
Platform Thermal Control



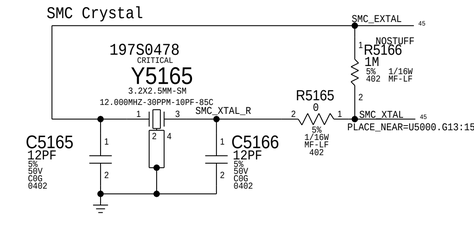
GPU BOM Option



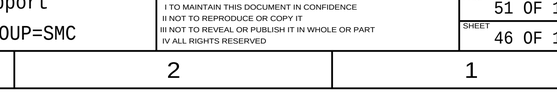
GPU_COST_GROUP=SMC



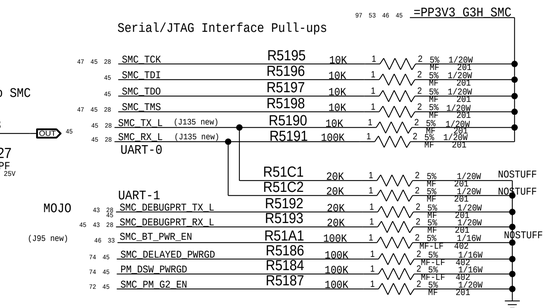
SMC Crystal



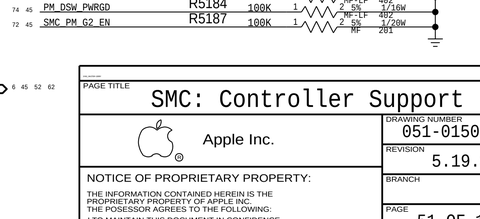
Arch Pull Up/Down



Serial/JTAG Interface Pull-ups



=PP3V3_S4_AP



PAGE TITLE

SMC: Controller Support

Apple Inc.

DRAWING NUMBER 051-01506 | SIZE D
REVISION 5.19.0
BRANCH
PAGE 51 OF 121
SHEET 46 OF 102

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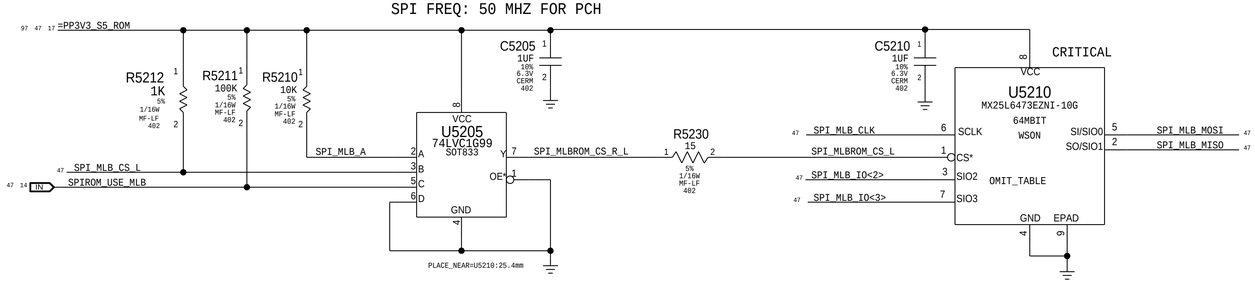
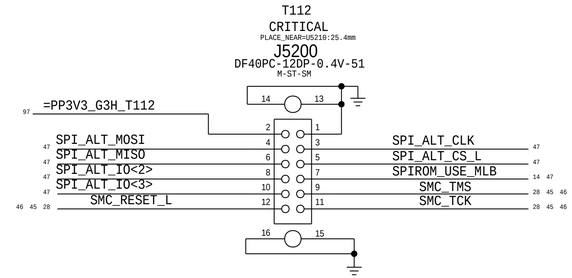
SPI BootROM

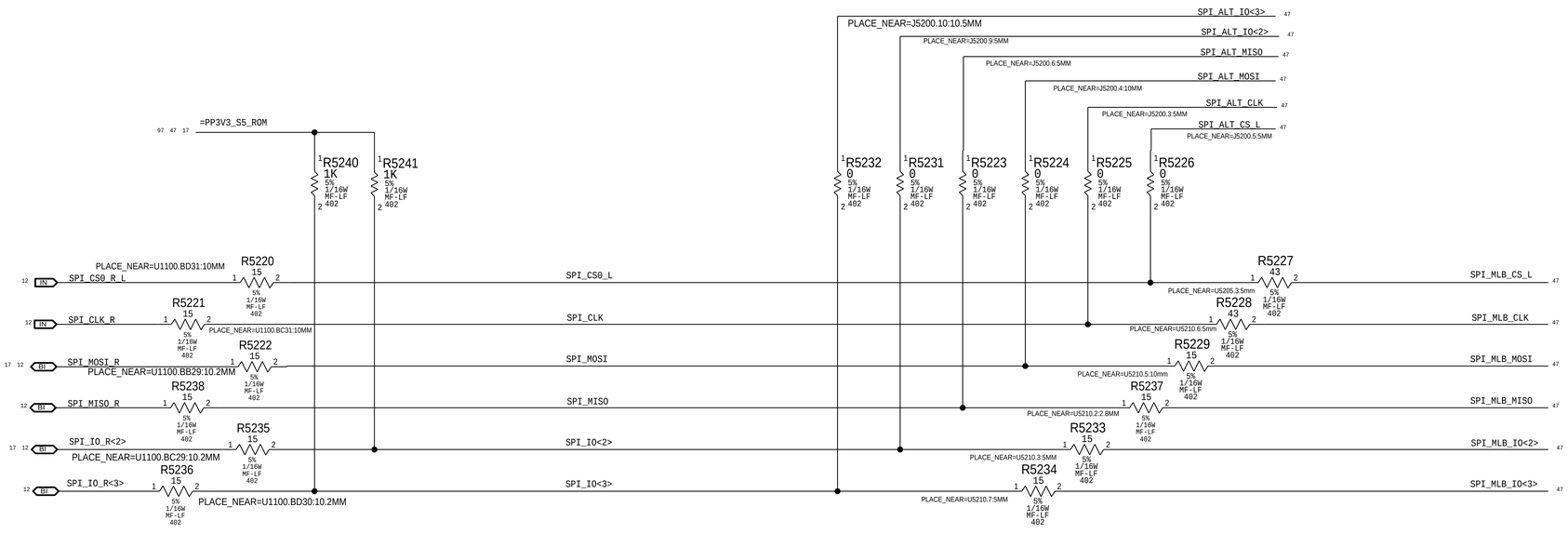
QUAD_IO MODE (MODE 0 & 3) SUPPORTED
SPI FREQ: 50 MHZ FOR PCH



T112 Connector



SPI Series Termination



SYNC_MASTER=ANDREW SYNC_DATE=02/23/2010

PAGE TITLE
CPU & CHIPSET: SPI and Debug Connector

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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 52 OF 121
SHEET 47 OF 102

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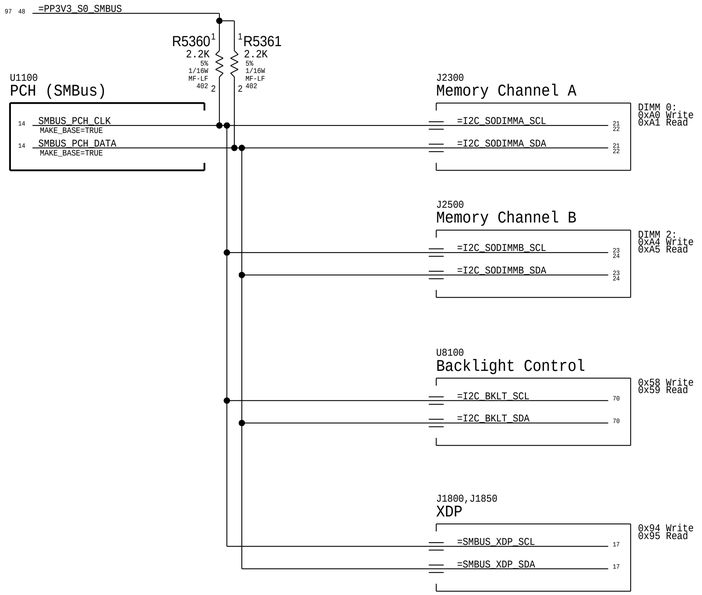

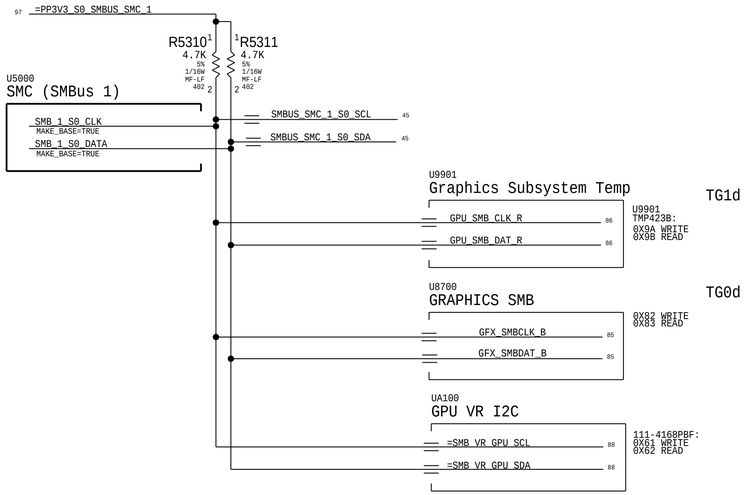

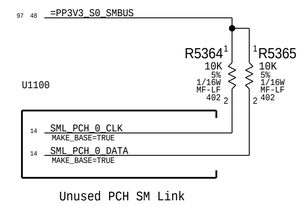

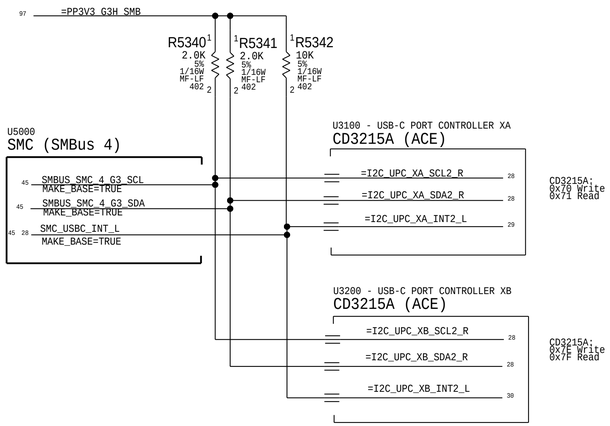

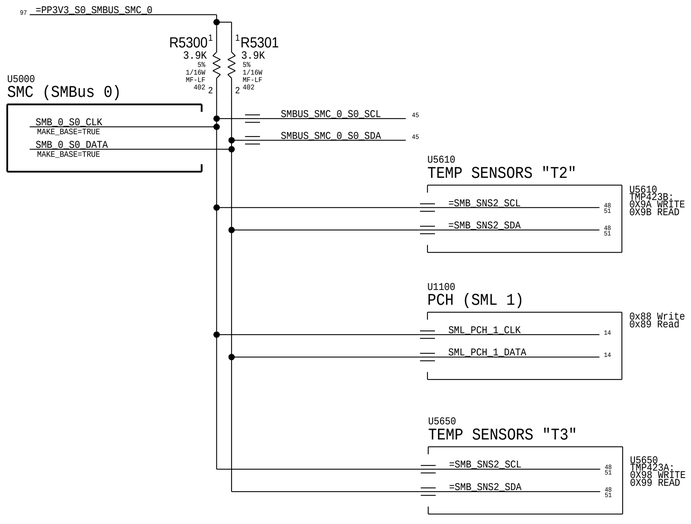

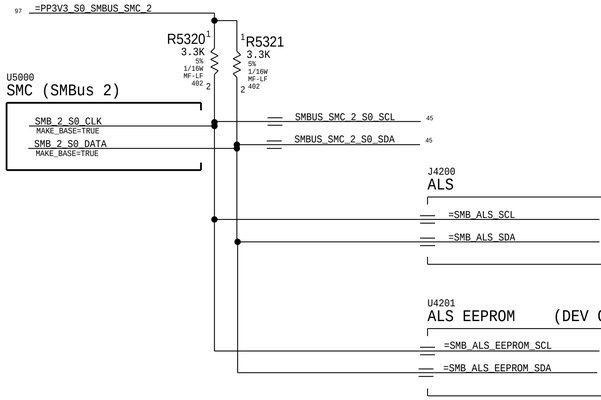

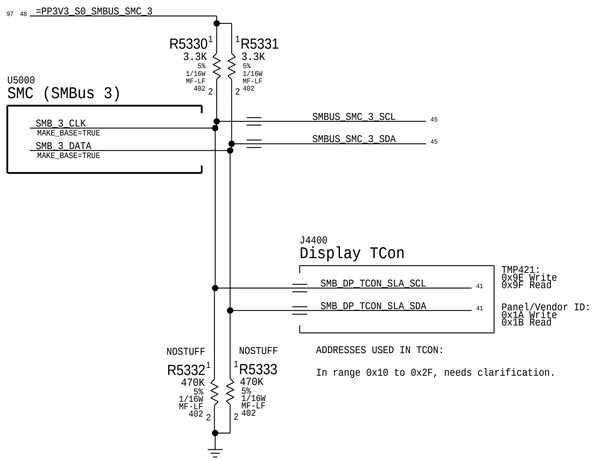

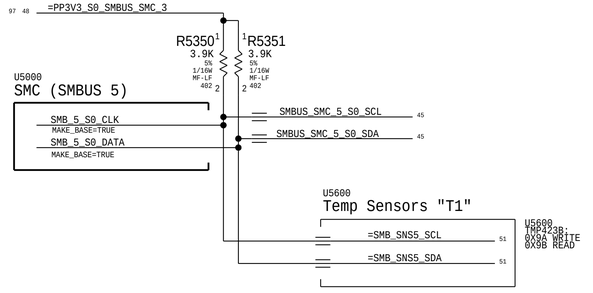

Line Legend
- Master
- Slave
- Mux

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SYNC_MASTER=KEVIN SYNC_DATE=02/04/2016

PAGE TITLE
SMC: SMBus Connections

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REVISION 5.19.0
BRANCH
PAGE 53 OF 121
SHEET 48 OF 102

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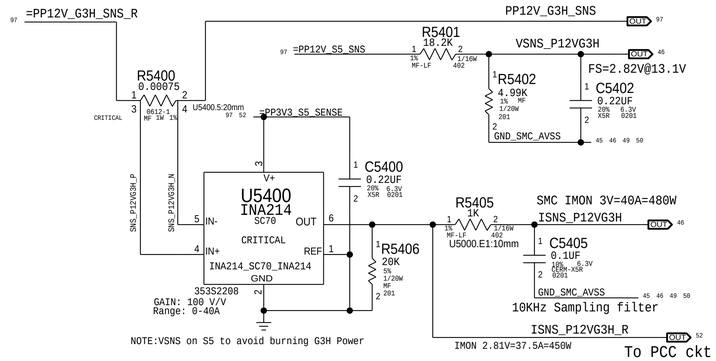
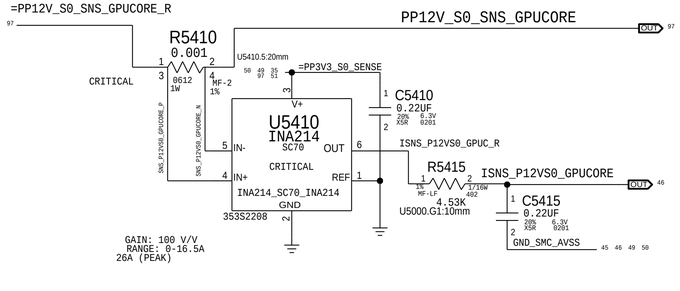
12V G3H (VD2R:ADC0/ID2R:ADC1)

AC/DC lowside sense (System total)



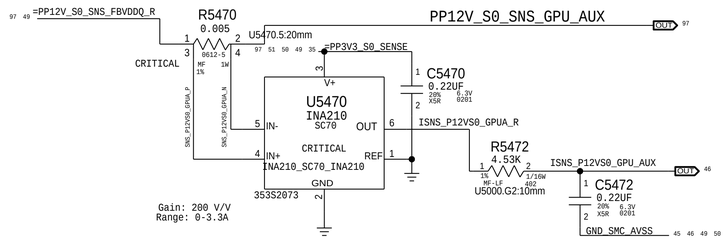
PP12V_S0_GPU (VG1C=VD20, IG1C:ADC17)

GPU highside sense for GPU Core Regulator



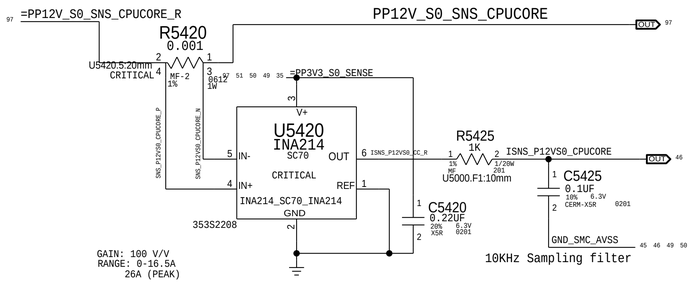
GPU AUX RAILS (VG1A=VD20, IG1A:ADC16)

GPU highside sense for GPU 0.9V & 1.8V VR



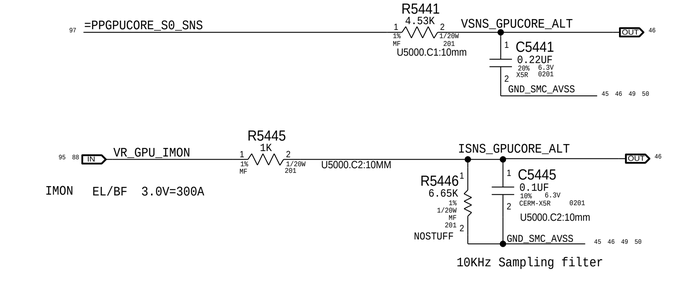
PP12V_S0_CPU (VD20:ADC2 /ID20:ADC3)

CPU highside sense for CPU Core Regulator



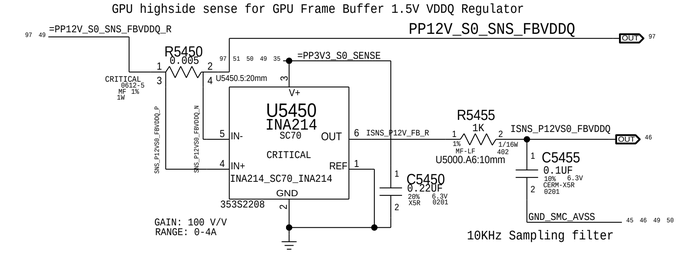
GPU Core - Alt (VG0C:ADC12/IG0C:ADC13)

Alternate low side V-sense and IMON amp

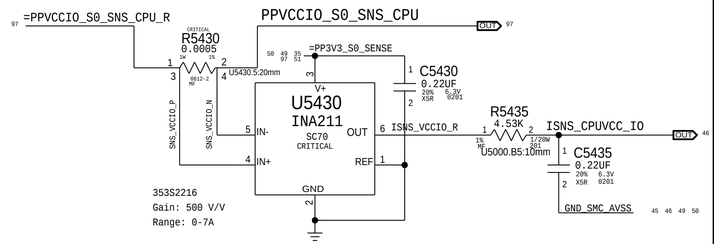


GPU FB (VG1F=VD20, IG1F:ADC11)

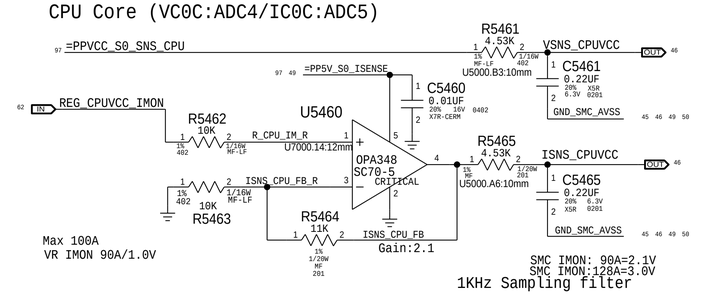
GPU highside sense for GPU Frame Buffer 1.5V VDDQ Regulator



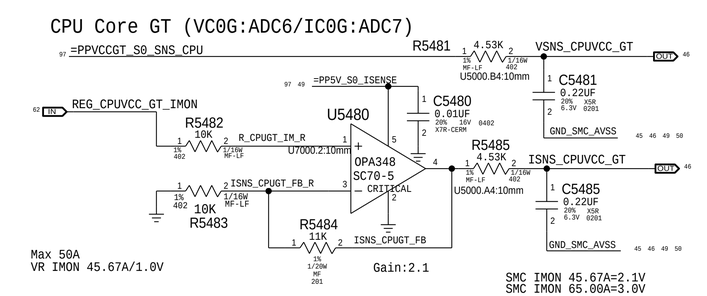
CPU VCCIO (VC0I=0.95V, IC0I:ADC8)



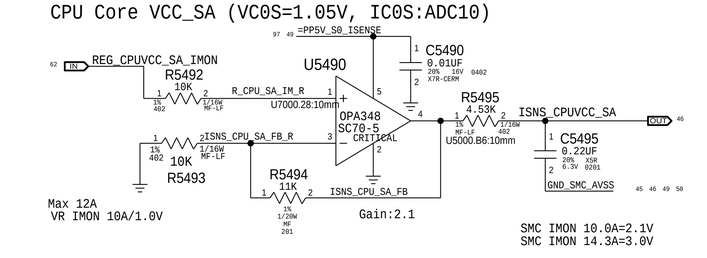
CPU Core (VC0C:ADC4/IC0C:ADC5)



CPU Core GT (VC0G:ADC6/IC0G:ADC7)



CPU Core VCC_SA (VC0S=1.05V, IC0S:ADC10)



BOM_COST_GROUP=Sensors

SYNC_MASTER=JERRY
SYNC_DATE=06/22/2016

PAGE TITLE

SENSORS: I and V Sense

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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 54 OF 121
SHEET 49 OF 102

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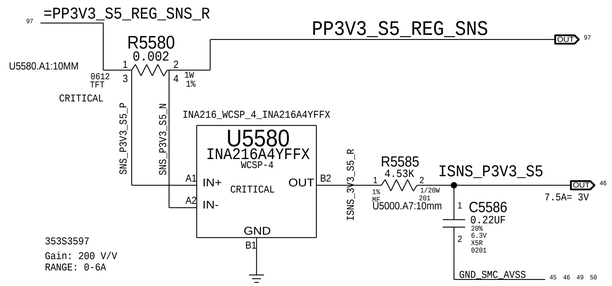
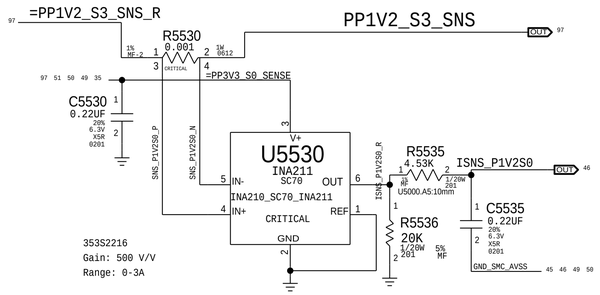
3V3_S5 (VC0M=3.3, IC35:ADC20)

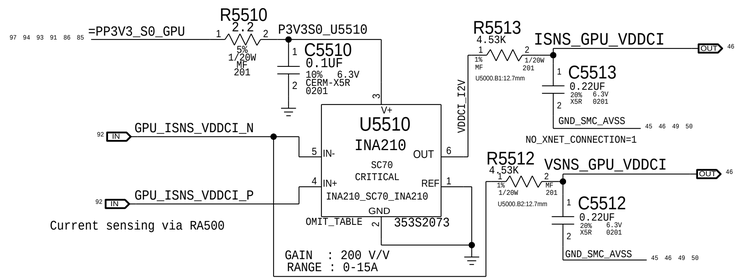


VDDQ S3 (VC0M=VM0R, IC0M:ADC9)

VDDQ lowside sense for S0-DIMM modules



GPU_VDDCI S0 (VG0I:ADC14 /IG0I: ADC15)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S2073	1	IC,INA210, 200V/V SC70-6	U5510	GPU:EL
353S2208	1	IC,INA214, 100V/V SC70-6	U5510	GPU:BF

RA500 = 0.5_mOHM

RA500 = 1.0_mOHM

5V_S4 (VC0M=5.0, IC54:ADC2)

VDDQ S3 (VM0R:ADC22/ IM0R:ADC23)

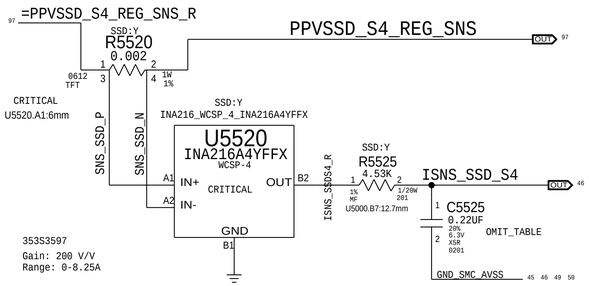
VDDQ lowside sense for S0-DIMM modules

HDD S0 (VH05=5V, IH05:ADC19)

I/V-sense for HDD 5V

SSD S0 (VR1R:ADC20 / IH1R:ADC21)

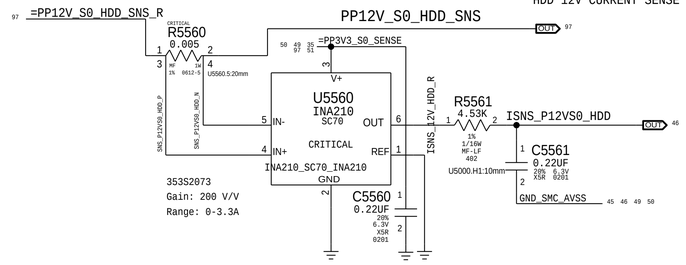
I-SENSE FOR SSD / V-SENSE FOR PPSSD_S4)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	1	CAP,0.22UF,201	C5525	SSD:Y
117S0201	1	RES,0 OHM,201	C5525	SSD:N

PP12V_S0_HDD (VH02=VD20, IH02:ADC18)

HDD 12V CURRENT SENSE



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SYNC_MASTER=JERRY SYNC_DATE=04/15/2019

PAGE TITLE

SENSORS: I and V Sense(Continued)

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REVISION 5.19.0

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PAGE 55 OF 121

SHEET 50 OF 102

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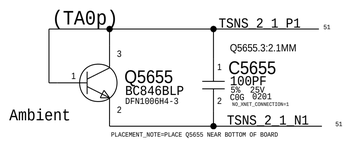

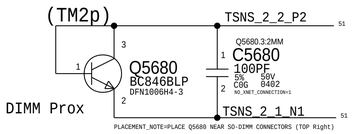

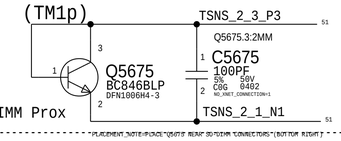

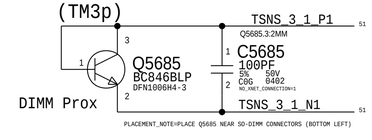

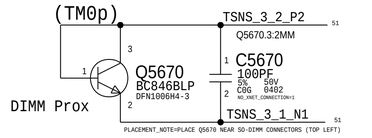

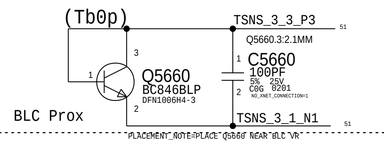

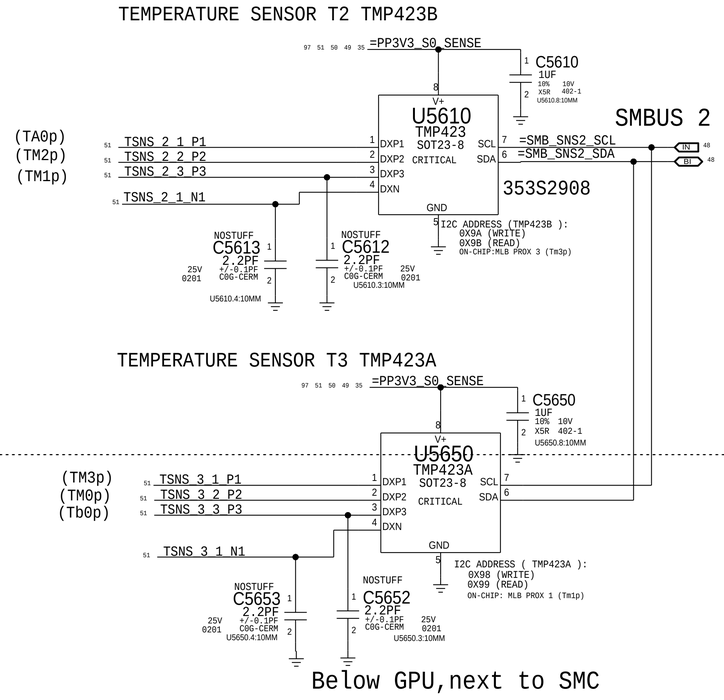

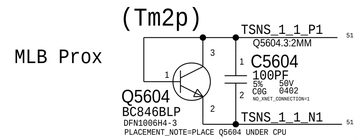

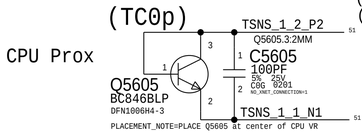

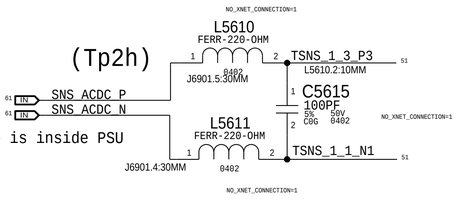

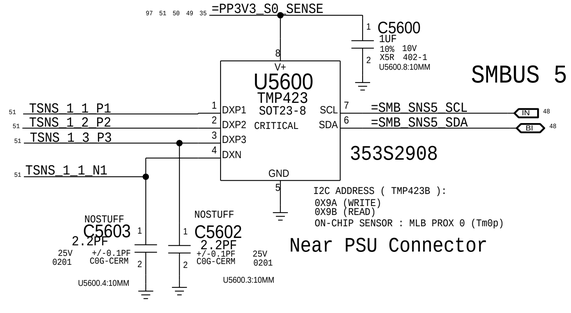

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PAGE TITLE

SENSORS: Temperature Sensors

Apple Inc.

DRAWING NUMBER 051-01506

SIZE D

REVISION 5.19.0

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PAGE 56 OF 121

SHEET 51 OF 102

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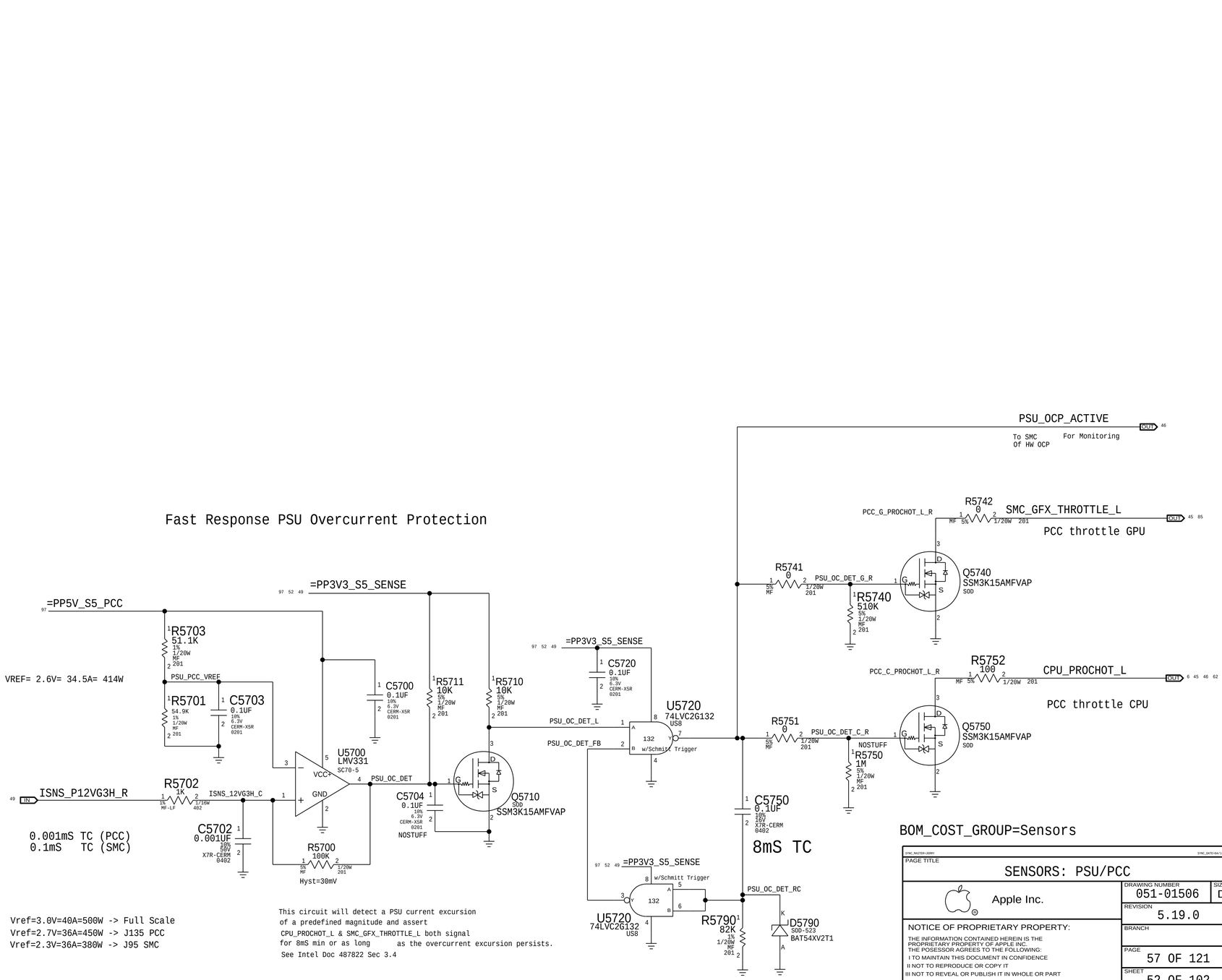
Fast Response PSU Overcurrent Protection



PAGE TITLE	SENSORS: PSU/PCC
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	REVISION 5.19.0
	BRANCH
	PAGE 57 OF 121
	SHEET 52 OF 102

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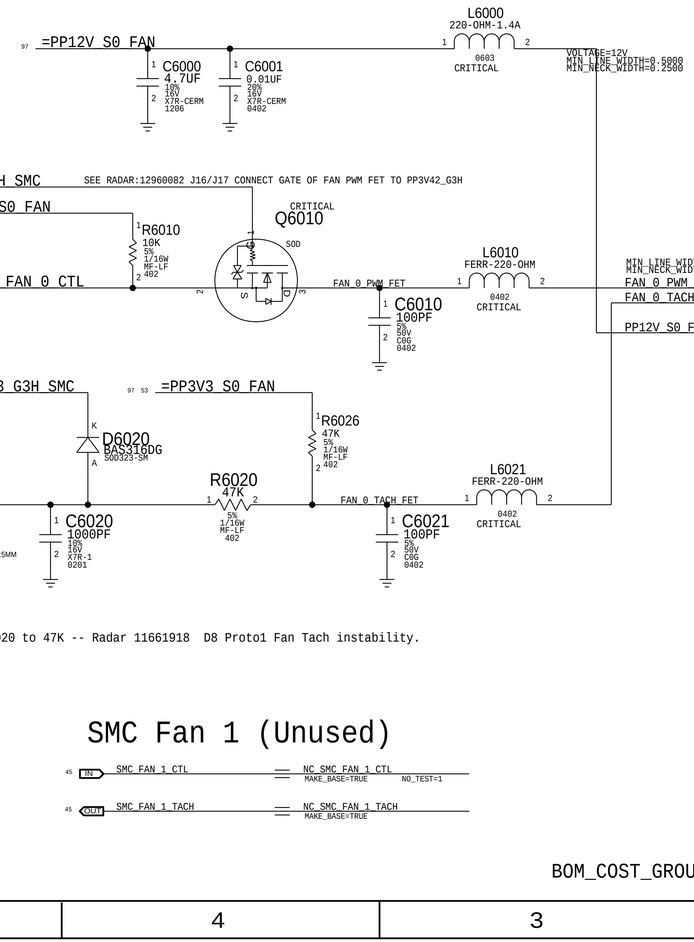
SMC Fan 0 (System)

Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)

BOM_COST_GROUP=FAN

SYNC_MASTER=ANDREW
SYNC_DATE=04/06/2016

PAGE TITLE
FAN: System Fan

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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 60 OF 121
SHEET 53 OF 102

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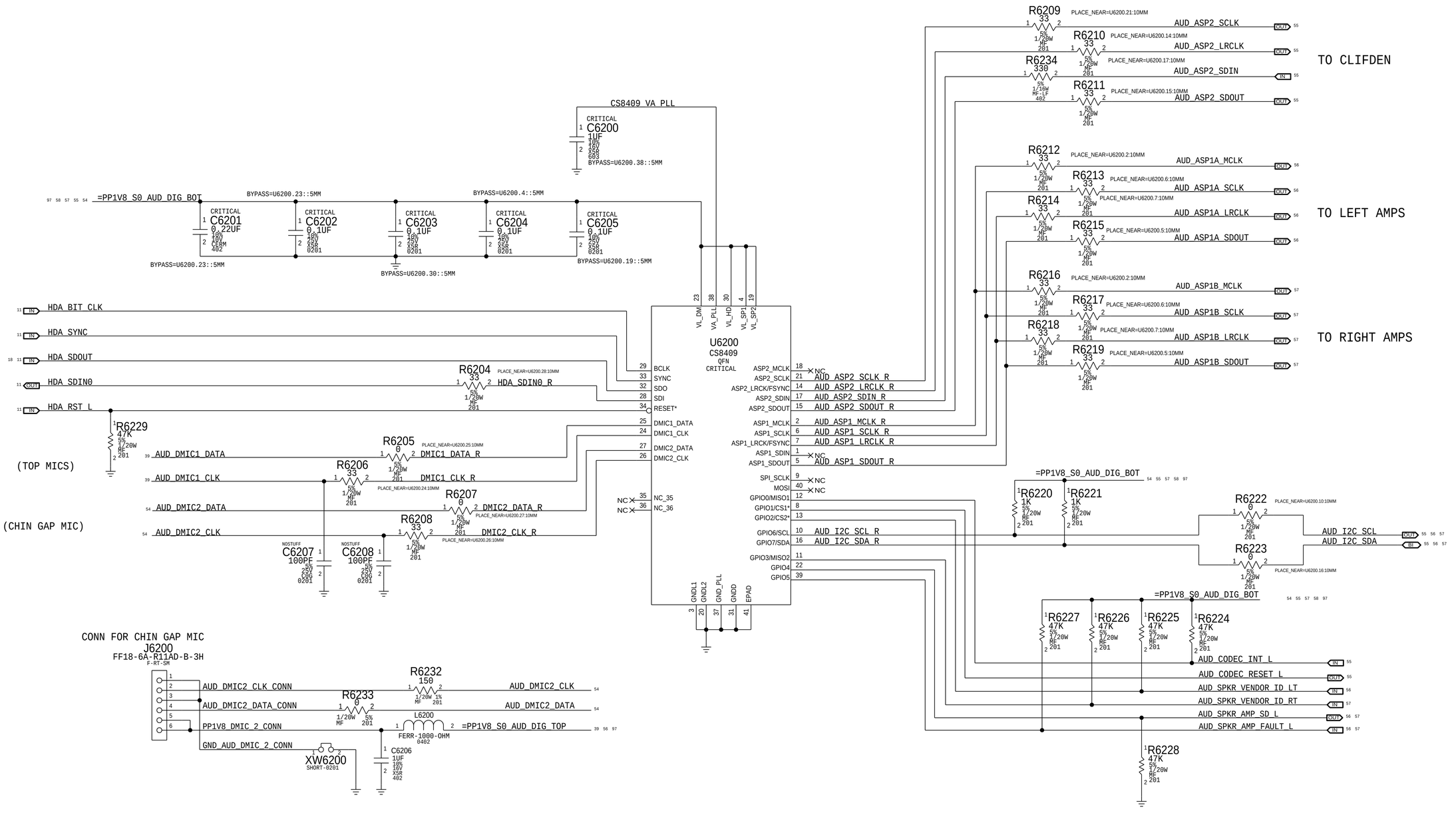

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LAST CHANGE:	Tue Aug 18 12:55:20 2015
PAGE TITLE	AUDIO: HDA BRIDGE
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DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	62 OF 121
SHEET	54 OF 102

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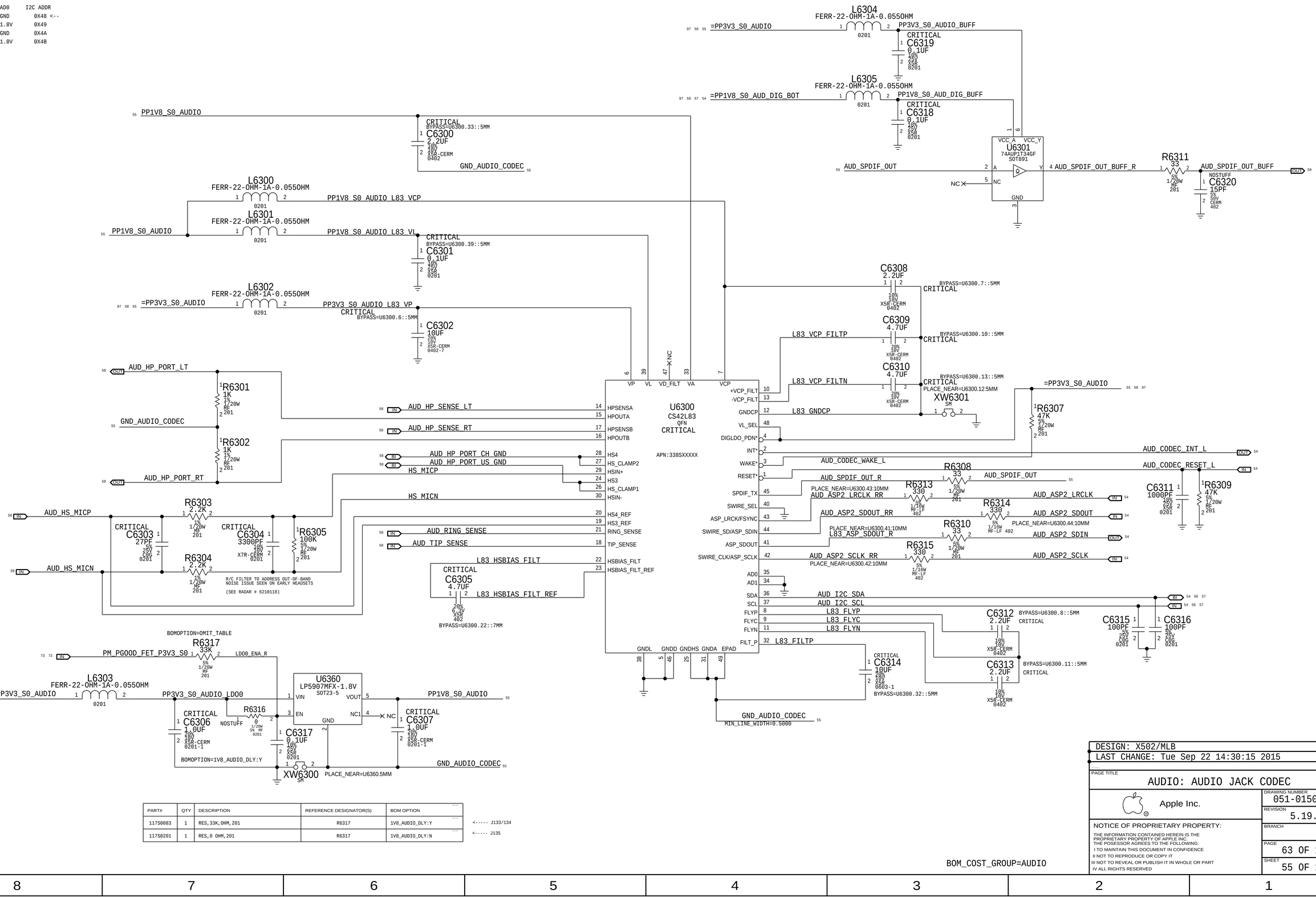
AD1	AD0	I2C ADDR
GND	GND	0X48 <--
GND	1.8V	0X49
1.8V	GND	0X4A
1.8V	1.8V	0X4B



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION	
117S0083	1	RES,33K,OHM,201	R6317	1V8_AUDIO_DLY:Y	<----- J133/134
117S0201	1	RES,0 OHM,201	R6317	1V8_AUDIO_DLY:N	<----- J135

BOM_COST_GROUP=AUDIO

DESIGN: X502/MLB

LAST CHANGE: Tue Sep 22 14:30:15 2015

PAGE TITLE

AUDIO: AUDIO JACK CODEC

Apple Inc.

DRAWING NUMBER 051-01506

SIZE D

REVISION 5.19.0

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PAGE 63 OF 121

SHEET 55 OF 102

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2X MONO SPEAKER AMPLIFIERS (TAS5764L)

SPEAKER AMPLIFIERS - LEFT CHANNEL

=PP1V8_S0_AUD_DIG_TOP

R6407 22K 5% 1/16W MF-LF 402 (NOSTUFF)

ADDRESS 0X6C

R6408 0 5% 1/20W MF 0201

C6426 1UF 10% 10V X5R 402-1

C6428 0.1UF 20% 10V X7R-CERM 0402

=PP12V_S0_AUDIO_SPKRAMP

C6430 1.0UF 20% 10V X5R-CERM 0201-1

C6436 0.1UF 10% 25V X5R 402

C6438 1UF 10% 25V X5R 603-1

C6440 220UF 20% 16V ELEC SM-CASE-C1-HF

AUD_LAMP_LT_GVDD

NOTE: AUD_LAMP_BOOTLT<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6408 AND C6409 ARE PLACED CLOSE TO U6403

=PP12V_S0_AUDIO_SPKRAMP

C6448 10UF 10% 25V X5R 805

C6449 10UF 10% 25V X5R 805

U6403
TAS5764L
QFN
CRITICAL
APN:353S00875
GAIN = TBD

AVDD	28	
DVDD	11	
GVDD	30	
PVDD	14	
PVDD	15	
PVDD	26	
PVDD	27	
SDZ*	3	
ADR0	13	
ADR1	12	
SCL	8	
SDA	9	
FAULTZ*	2	
SDI	7	
FSYNC	4	
SBCLK	6	
MCLK	5	
BST_P	23	
OUT_P	24	
OUT_P	25	
OUT_N	16	
OUT_N	17	
BST_N	18	
VCOM	32	AUD_LAMP_LT_VCOM
VREF_N	1	AUD_LAMP_LT_VREFN
VREG	31	AUD_LAMP_LT_VREG
GND	10	
GND	29	
PGND	19	
PGND	20	
PGND	21	
PGND	22	
EPAD	33	

AUD_SPKR_AMP_SD_L

AUD_LAMP_LT_ADR

AUD_SPKR_AMP_FAULT_L

AUD_I2C_SCL

AUD_I2C_SDA

PLACE_NEAR=U6403.7:15MM

PLACE_NEAR=U6403.4:15MM

AUD_ASP1A_SDOUT

AUD_ASP1A_LRCLK

AUD_ASP1A_SCLK

AUD_ASP1A_MCLK

R6409 100 1/20W 5% MF 201

R6410 100 1/20W 5% MF 201

R6411 100 1/20W 5% MF 201

R6412 100 1/20W 5% MF 201

AUD_ASP1A_SDOUT_R

AUD_ASP1A_LRCLK_R

AUD_ASP1A_SCLK_R

AUD_ASP1A_MCLK_R

PLACE_NEAR=U6403.5:15MM

PLACE_NEAR=U6403.6:15MM

C6432 1UF 10% X5R 0402

C6434 1UF 10% X5R 0402

NO_XNET_CONNECTION=1

C6441 0.22UF 20% 25V X5R 603

AUD_LAMP_BOOTLTP

MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

AUD_SPKR_LTWT_OUT_L_P

CKPLUS_WAIVE=MISS_P_DIFFPAIR

CRITICAL FL6405 180OHM-3.4A 0806

AUD_SPKR_LTWT_OUT_P

CKPLUS_WAIVE=MISS_P_DIFFPAIR

C6443 0.22UF 20% 25V X5R 603

AUD_LAMP_BOOTLTN

MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

AUD_SPKR_LTWT_OUT_L_N

CKPLUS_WAIVE=MISS_N_DIFFPAIR

CRITICAL FL6406 180OHM-3.4A 0806

AUD_SPKR_LTWT_OUT_N

CKPLUS_WAIVE=MISS_N_DIFFPAIR

NO_XNET_CONNECTION=1

=PP1V8_S0_AUD_DIG_TOP

R6405 22K 5% 1/16W MF-LF 402 (NOSTUFF)

ADDRESS 0X6D

R6406 22K 5% 1/16W MF-LF 402

C6427 1UF 10% 10V X5R 402-1

C6429 0.1UF 20% 10V X7R-CERM 0402

=PP12V_S0_AUDIO_SPKRAMP

C6431 1.0UF 20% 10V X5R-CERM 0201-1

C6437 0.1UF 10% 25V X5R 402

C6439 1UF 10% 25V X5R 603-1

C6451 220UF 20% 16V ELEC SM-CASE-C1-HF

AUD_LAMP_LW_GVDD

NOTE: AUD_LAMP_BOOTLW<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6418 AND C6419 ARE PLACED CLOSE TO U6404

U6404
TAS5764L
QFN
CRITICAL
APN:353S00875
GAIN = TBD

AVDD	28	
DVDD	11	
GVDD	30	
PVDD	14	
PVDD	15	
PVDD	26	
PVDD	27	
SDZ*	3	
ADR0	13	
ADR1	12	
SCL	8	
SDA	9	
FAULTZ*	2	
SDI	7	
FSYNC	4	
SBCLK	6	
MCLK	5	
BST_P	23	
OUT_P	24	
OUT_P	25	
OUT_N	16	
OUT_N	17	
BST_N	18	
VCOM	32	AUD_LAMP_LW_VCOM
VREF_N	1	AUD_LAMP_LW_VREFN
VREG	31	AUD_LAMP_LW_VREG
GND	10	
GND	29	
PGND	19	
PGND	20	
PGND	21	
PGND	22	
EPAD	33	

AUD_SPKR_AMP_SD_L

AUD_LAMP_LW_ADR

AUD_SPKR_AMP_FAULT_L

AUD_I2C_SCL

AUD_I2C_SDA

AUD_ASP1A_SDOUT_R

AUD_ASP1A_LRCLK_R

AUD_ASP1A_SCLK_R

AUD_ASP1A_MCLK_R

C6433 1UF 10% X5R 0402

C6435 1UF 10% X5R 0402

NO_XNET_CONNECTION=1

C6442 0.22UF 20% 25V X5R 603

AUD_LAMP_BOOTLWP

MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

AUD_SPKR_LWFR_OUT_L_P

CKPLUS_WAIVE=MISS_P_DIFFPAIR

CRITICAL FL6407 180OHM-3.4A 0806

AUD_SPKR_LWFR_OUT_P

CKPLUS_WAIVE=MISS_P_DIFFPAIR

C6444 0.22UF 20% 25V X5R 603

AUD_LAMP_BOOTLWN

MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

AUD_SPKR_LWFR_OUT_L_N

CKPLUS_WAIVE=MISS_N_DIFFPAIR

CRITICAL FL6408 180OHM-3.4A 0806

AUD_SPKR_LWFR_OUT_N

CKPLUS_WAIVE=MISS_N_DIFFPAIR

NO_XNET_CONNECTION=1

CRITICAL C6445 330PF 10% 50V CERM 603

CRITICAL C6446 330PF 10% 50V CERM 603

CRITICAL C6447 330PF 10% 50V CERM 603

CRITICAL C6450 330PF 10% 50V CERM 603

SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862

CRITICAL J6402 504050-0691 M-RT-SM

WOOFER (BL)

AUD_SPKR_LWFR_OUT_N	56	1
AUD_SPKR_LWFR_OUT_P	56	2
AUD_SPKR_VENDOR_ID_LT	OUT	3
		4
AUD_SPKR_LTWT_OUT_N	56	5
AUD_SPKR_LTWT_OUT_P	56	6
		7
		8

TWEETER (FL)

BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U

LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE: AUDIO: LEFT SPKR AMP

Apple Inc.

DRAWING NUMBER: 051-01506

SIZE: D

REVISION: 5.19.0

BRANCH:

PAGE: 64 OF 121

SHEET: 56 OF 102

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2X MONO SPEAKER AMPLIFIERS (TAS5764L)

SPEAKER AMPLIFIERS - RIGHT CHANNEL

=PP1V8_S0_AUD_DIG_BOT

R6501 22K 5% 1/16W MF-LF 402

C6501 1UF 10% 10V X5R 402-1

C6503 0.1UF 20% 10V X7R-CERM 0402

ADDRESS 0X6E

NOSTUFF R6502 22K 5% 1/16W MF-LF 402

AUD_RAMP_RT_GVDD

=PP12V_S0_AUDIO_SPKRAMP

C6505 1.0UF 20% 10V X5R-CERM 0201-1

C6511 0.1UF 10% 25V X5R 402

C6513 1UF 10% 25V X5R 603-1

C6515 220UF 20% 16V ELEC SM-CASE-C1-HF

=PP12V_S0_AUDIO_SPKRAMP

C6523 10UF 10% 25V X5R 805

C6524 10UF 10% 25V X5R 805

NOTE: AUD_LAMP_BOOTRT<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6408 AND C6409 ARE PLACED CLOSE TO U6501

U6501 TAS5764L QFN CRITICAL APN:353S00875 GAIN = TBD

AVDD	28	
DVDD	11	
GVDD	30	
PVDD	14	
PVDD	15	
PVDD	26	
PVDD	27	
SDZ*	3	AUD_SPKR_AMP_SD_L
ADR0	13	AUD_RAMP_RT_ADR
ADR1	12	
SCL	8	AUD_I2C_SCL
SDA	9	AUD_I2C_SDA
FAULTZ*	2	AUD_SPKR_AMP_FAULT_L
SDI	7	AUD_ASP1B_SDOUT_R
FSYNC	4	AUD_ASP1B_LRCLK_R
SBCLK	6	AUD_ASP1B_SCLK_R
MCLK	5	AUD_ASP1B_MCLK_R
BST_P	23	
OUT_P	24	
OUT_P	25	
OUT_N	16	
OUT_N	17	
BST_N	18	
VCOM	32	AUD_RAMP_RT_VCOM
VREF_N	1	AUD_RAMP_RT_VREFN
VREG	31	AUD_RAMP_RT_VREG
GND	10	
GND	29	
PGND	19	
PGND	20	
PGND	21	
PGND	22	
EPAD	33	

PLACE_NEAR=U6501.7:15MM
PLACE_NEAR=U6501.4:15MM
PLACE_NEAR=U6501.5:15MM
PLACE_NEAR=U6501.6:15MM

R6509 100 1/20W 5% MF 201
R6510 100 1/20W 5% MF 201
R6511 100 1/20W 5% MF 201
R6512 100 1/20W 5% MF 201

C6507 1UF 10% X5R 0402 25V
C6509 1UF 10% X5R 0402 25V

NO_XNET_CONNECTION=1

C6516 0.22UF 20% 25V X5R 603

AUD_RAMP_BOOTRTP
MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

C6518 0.22UF 20% 25V X5R 603

AUD_RAMP_BOOTRTN
MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

CRITICAL FL6501 180OHM-3.4A 0806

CRITICAL FL6502 180OHM-3.4A 0806

AUD_SPKR_RTWT_OUT_L_P (CKPLUS_WAIVE=MISS_P_DIFFPAIR) → AUD_SPKR_RTWT_OUT_P (CKPLUS_WAIVE=MISS_P_DIFFPAIR) → 57

AUD_SPKR_RTWT_OUT_L_N (CKPLUS_WAIVE=MISS_N_DIFFPAIR) → AUD_SPKR_RTWT_OUT_N (CKPLUS_WAIVE=MISS_N_DIFFPAIR) → 57

=PP1V8_S0_AUD_DIG_BOT

R6503 0 5% 1/20W MF 0201

C6502 1UF 10% 10V X5R 402-1

C6504 0.1UF 20% 10V X7R-CERM 0402

ADDRESS 0X6F

NOSTUFF R6504 22K 5% 1/16W MF-LF 402

AUD_RAMP_RW_GVDD

=PP12V_S0_AUDIO_SPKRAMP

C6506 1.0UF 20% 10V X5R-CERM 0201-1

C6512 0.1UF 10% 25V X5R 402

C6514 1UF 10% 25V X5R 603-1

C6526 220UF 20% 16V ELEC SM-CASE-C1-HF

NOTE: AUD_LAMP_BOOTRW<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6418 AND C6419 ARE PLACED CLOSE TO U6502

U6502 TAS5764L QFN CRITICAL APN:353S00875 GAIN = TBD

AVDD	28	
DVDD	11	
GVDD	30	
PVDD	14	
PVDD	15	
PVDD	26	
PVDD	27	
SDZ*	3	AUD_SPKR_AMP_SD_L
ADR0	13	AUD_RAMP_RW_ADR
ADR1	12	
SCL	8	AUD_I2C_SCL
SDA	9	AUD_I2C_SDA
FAULTZ*	2	AUD_SPKR_AMP_FAULT_L
SDI	7	AUD_ASP1B_SDOUT_R
FSYNC	4	AUD_ASP1B_LRCLK_R
SBCLK	6	AUD_ASP1B_SCLK_R
MCLK	5	AUD_ASP1B_MCLK_R
BST_P	23	
OUT_P	24	
OUT_P	25	
OUT_N	16	
OUT_N	17	
BST_N	18	
VCOM	32	AUD_RAMP_RW_VCOM
VREF_N	1	AUD_RAMP_RW_VREFN
VREG	31	AUD_RAMP_RW_VREG
GND	10	
GND	29	
PGND	19	
PGND	20	
PGND	21	
PGND	22	
EPAD	33	

C6508 1UF 10% X5R 0402 25V
C6510 1UF 10% X5R 0402 25V

NO_XNET_CONNECTION=1

C6517 0.22UF 20% 25V X5R 603

AUD_RAMP_BOOTRWP
MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

C6519 0.22UF 20% 25V X5R 603

AUD_RAMP_BOOTRWN
MIN_LINE_WIDTH=0.2000
MIN_NECK_WIDTH=0.1500

CRITICAL FL6503 180OHM-3.4A 0806

CRITICAL FL6504 180OHM-3.4A 0806

AUD_SPKR_RWFR_OUT_L_P (CKPLUS_WAIVE=MISS_P_DIFFPAIR) → AUD_SPKR_RWFR_OUT_P (CKPLUS_WAIVE=MISS_P_DIFFPAIR) → 57

AUD_SPKR_RWFR_OUT_L_N (CKPLUS_WAIVE=MISS_N_DIFFPAIR) → AUD_SPKR_RWFR_OUT_N (CKPLUS_WAIVE=MISS_N_DIFFPAIR) → 57

CRITICAL C6520 330PF 10% 50V CERM 603

CRITICAL C6521 330PF 10% 50V CERM 603

CRITICAL C6522 330PF 10% 50V CERM 603

CRITICAL C6525 330PF 10% 50V CERM 603

SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862

CRITICAL J6502 504050-0691 M-RT-SM

WOOFER (BR)

57	AUD_SPKR_RWFR_OUT_N	1
57	AUD_SPKR_RWFR_OUT_P	2
54 OUT	AUD_SPKR_VENDOR_ID_RT	3
		4
57	AUD_SPKR_RTWT_OUT_N	5
57	AUD_SPKR_RTWT_OUT_P	6
		7
		8

TWEETER (FR)

BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV_MLB_U

LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE: AUDIO: RIGHT SPKR AMP

Apple Inc.

DRAWING NUMBER: 051-01506 SIZE: D

REVISION: 5.19.0

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PAGE: 65 OF 121

SHEET: 57 OF 102

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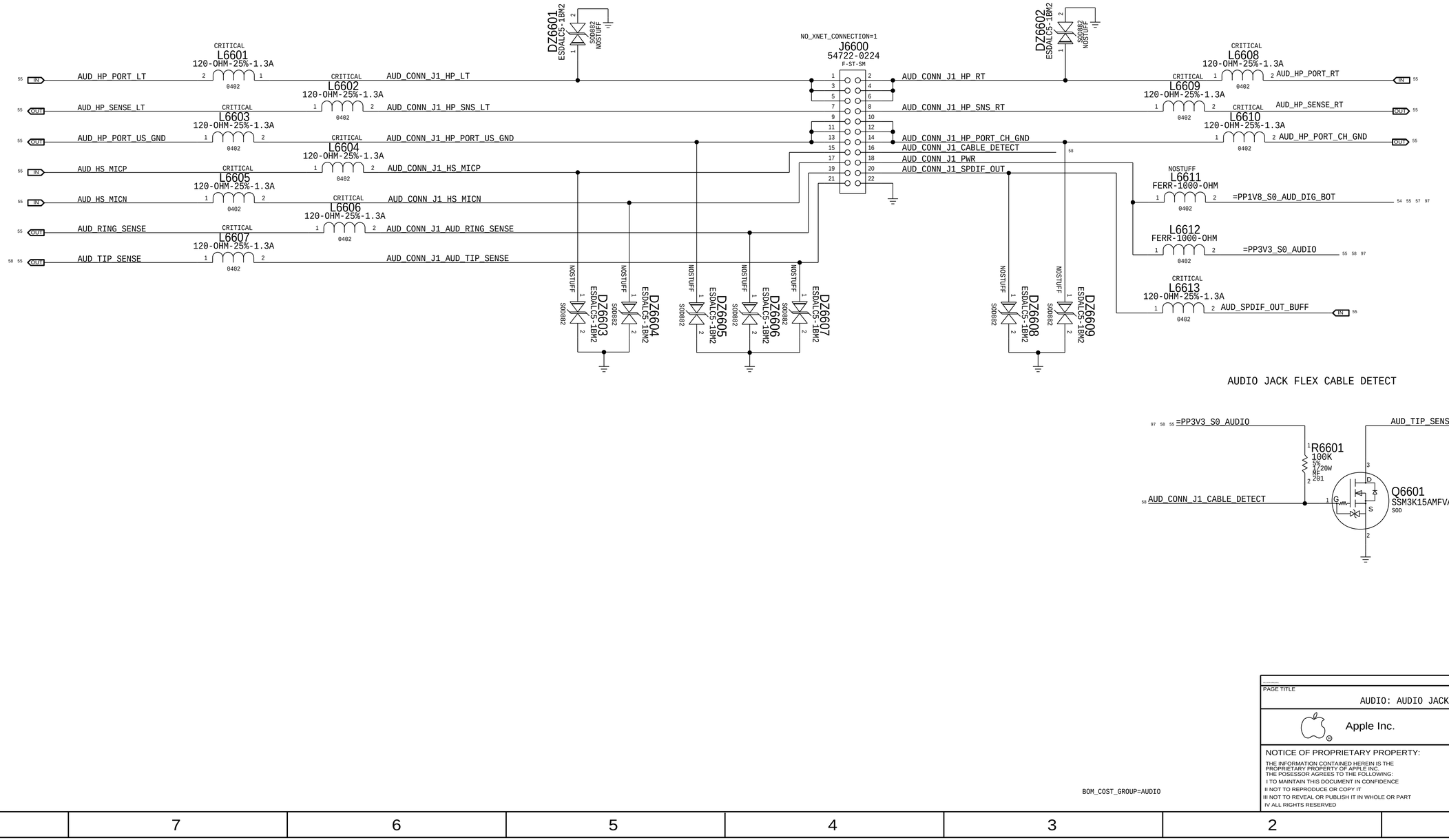

PAGE TITLE

AUDIO: AUDIO JACK CONN

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PAGE 66 OF 121

SHEET 58 OF 102

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PAGE 67 OF 121

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0x0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	Lynx POINT GPIO 16	Lynx POINT GPIO 5 (RCVR INT) Lynx POINT GPIO 3 (PERIPH DET)

OTHER DETECT

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

SYNC_MASTER=JAMES(AUDIO) SYNC_DATE=02/23/2016

PAGE TITLE

AUDIO: EMPTY

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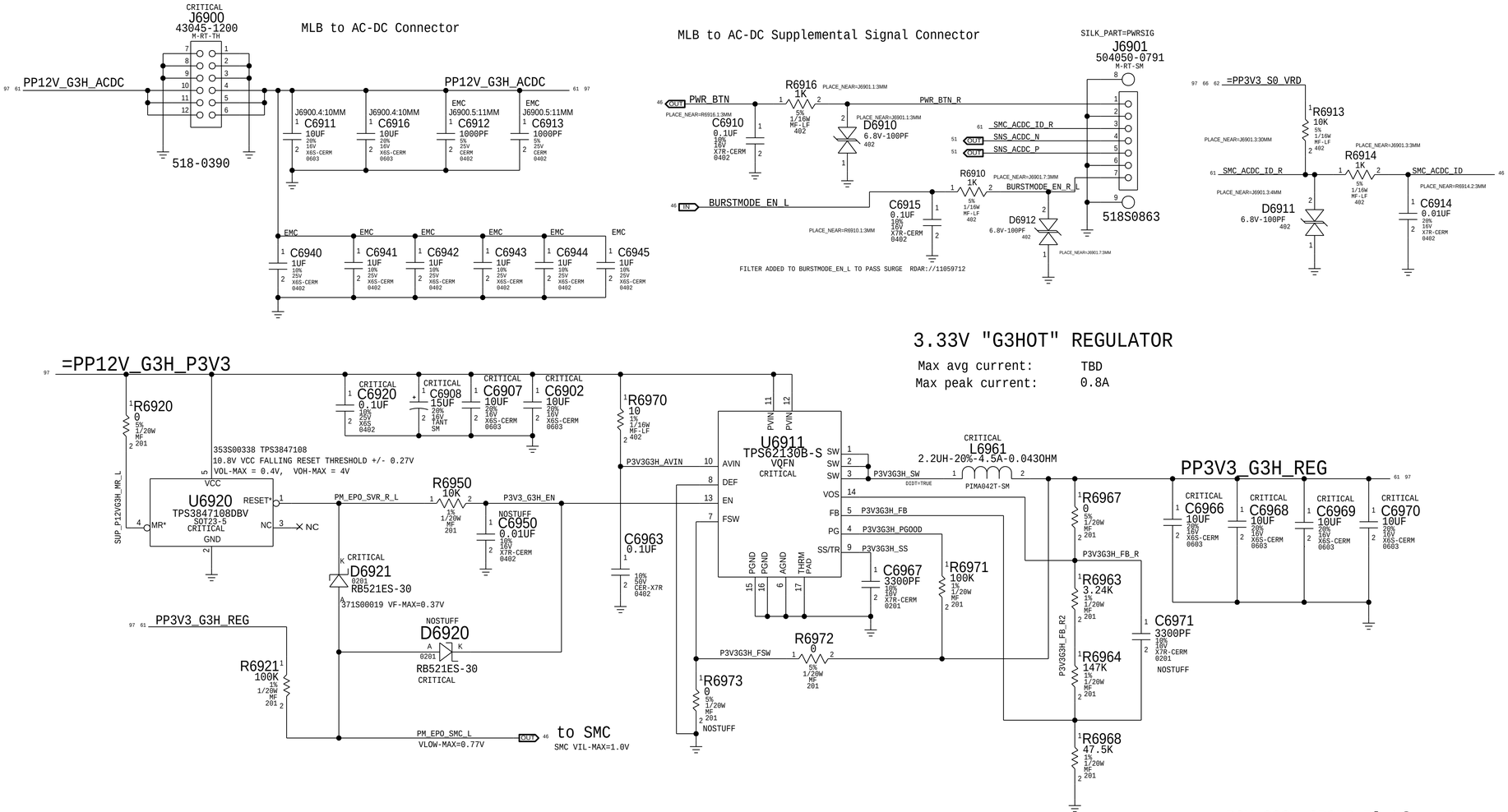
PAGE 68 OF 121
SHEET 60 OF 102



PAGE TITLE

PLATFORM POWER: Connectors / VReg G3Hot

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PAGE 69 OF 121

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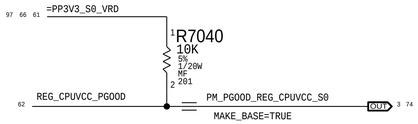

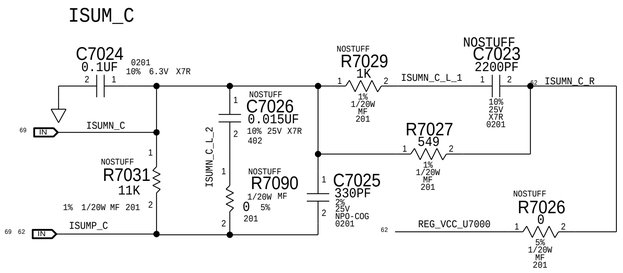

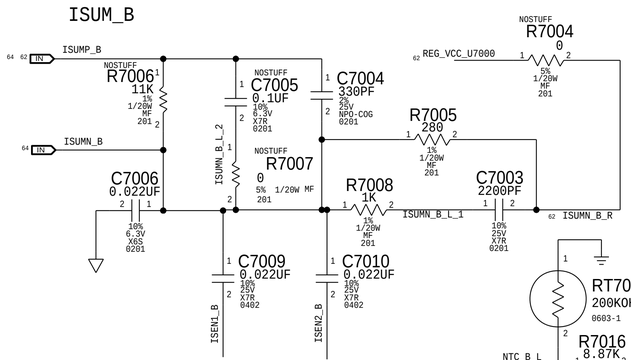

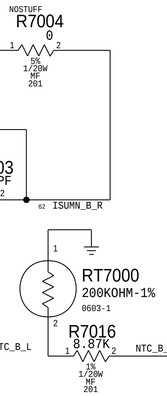

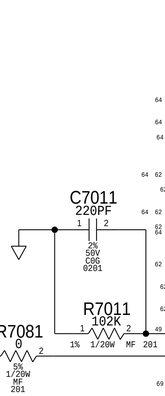

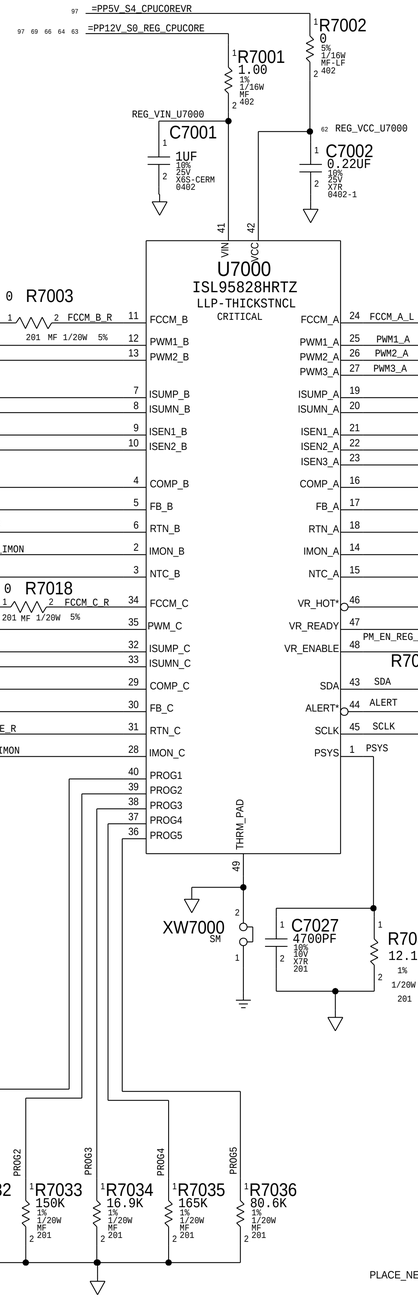

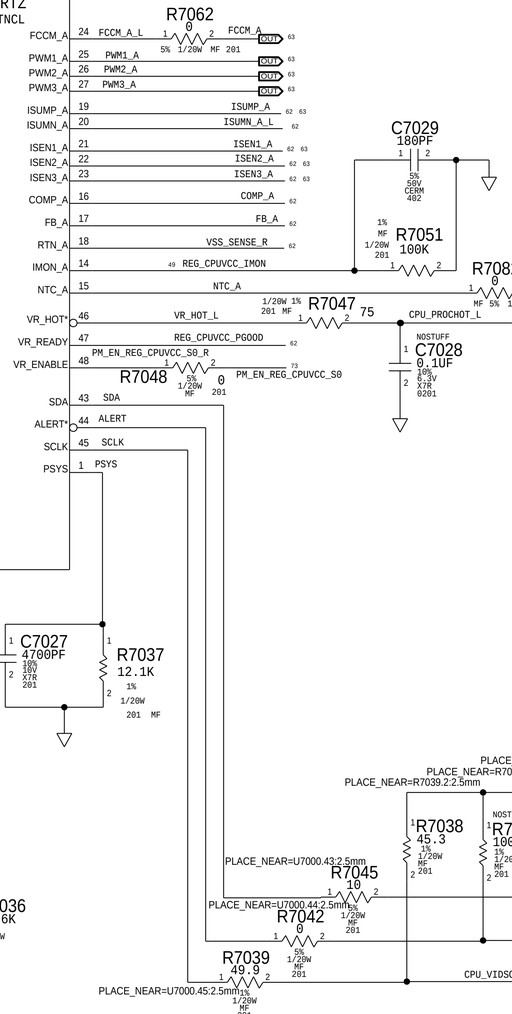

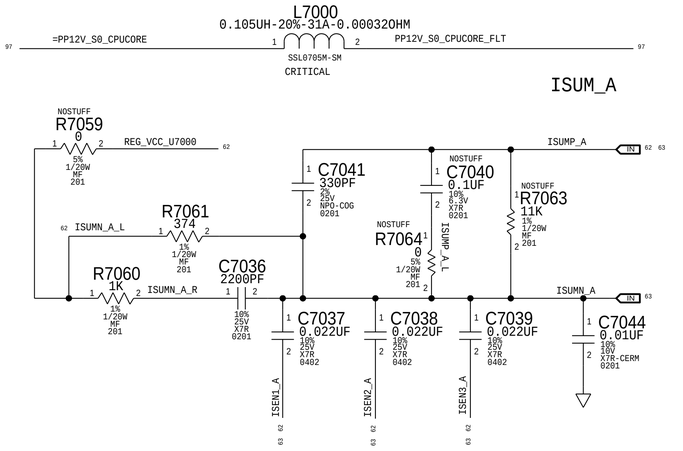

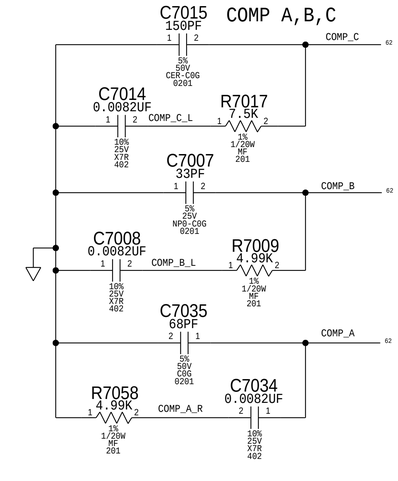

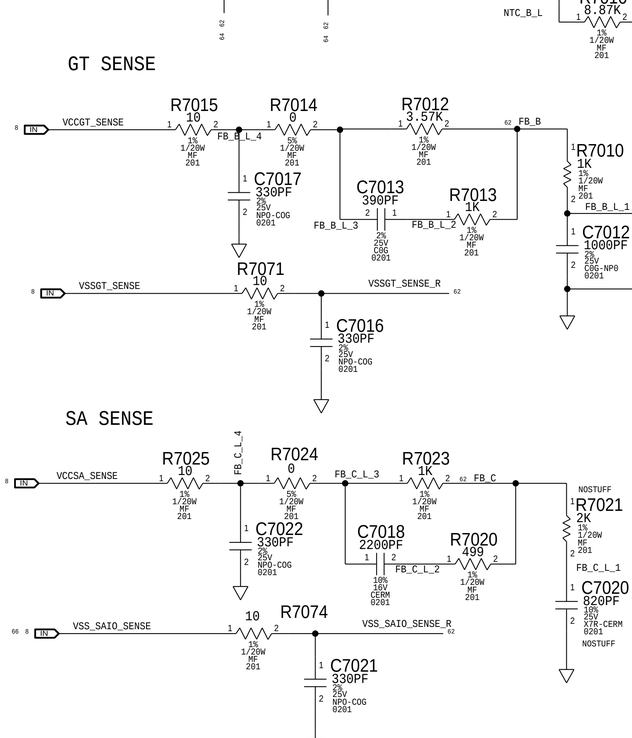

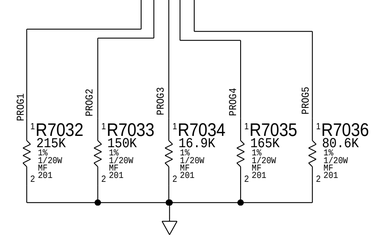

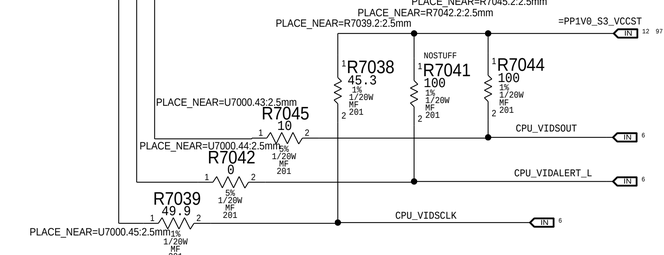

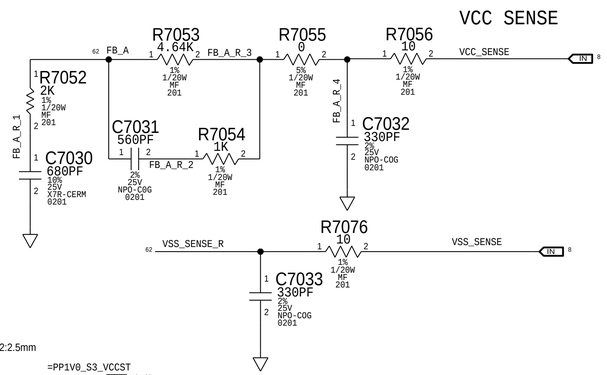

BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=SUNOL SYNC_DATE=09/21/2020

PAGE TITLE

CPU & CHIPSET: CPU CORE VR

Apple Inc.

DRAWING NUMBER 051-01506

SIZE D

REVISION 5.19.0

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PAGE 70 OF 121

SHEET 62 OF 102

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CPU VCC Regulator
EDC = 100A
TDC = 70A

CPU VCORE Phase 1

CPU VCORE Phase 2

CPU Phase 3

CPU VCORE OUTPUT DECOUPLING

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S00519	3	IC,SIC621,IMVP-8,60A,PQFN31	Q7110,Q7130,Q7150	CPU_DRMOS:FUTURE
353S00831	3	IC,FDMF5804,IMVP-8,50A,PQFN31	Q7110,Q7130,Q7150	CPU_DRMOS:CURRENT

BOM_COST_GROUP=CPU & CHIPSET

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:05 2016

PAGE TITLE
CPU & CHIPSET: CPU CORE VR (VCC)

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DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 71 OF 121
SHEET 63 OF 102

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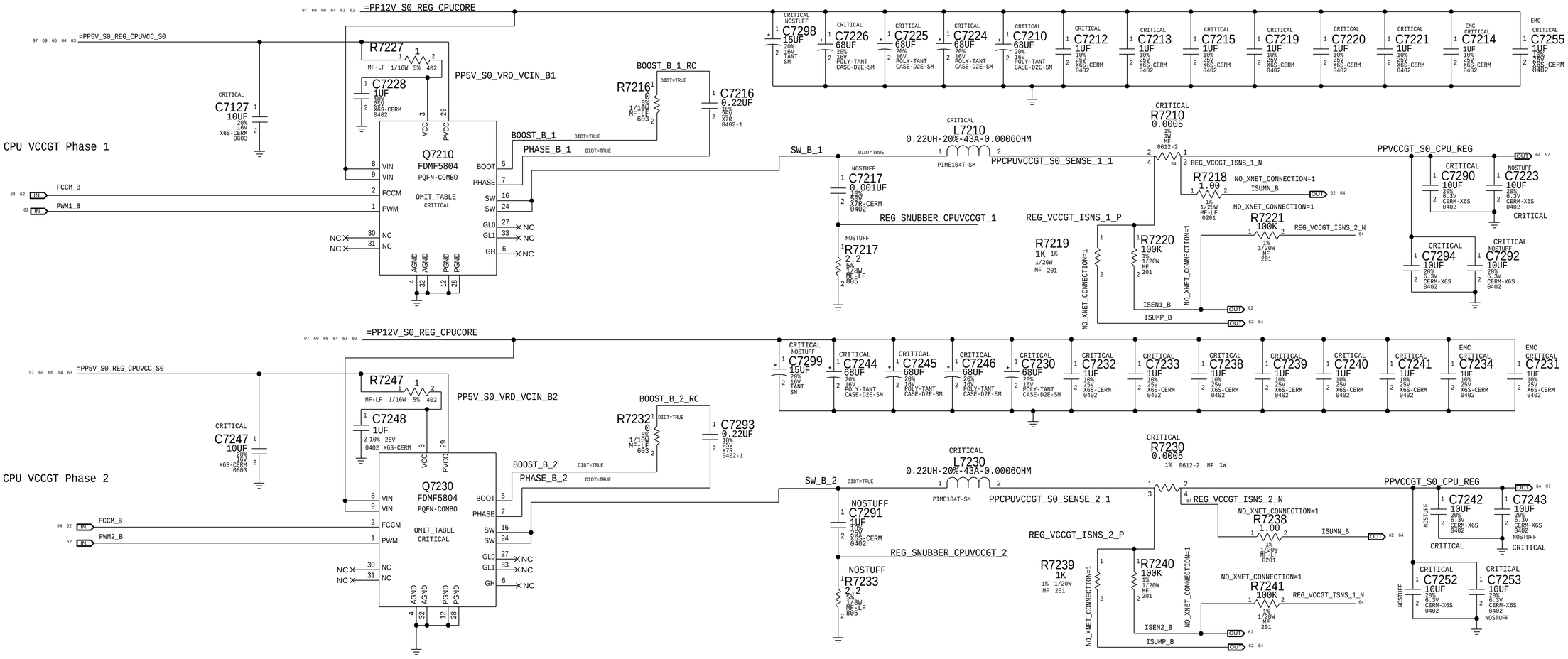
CPU VCCGT Regulator

EDC = 51A

TDC = 37A

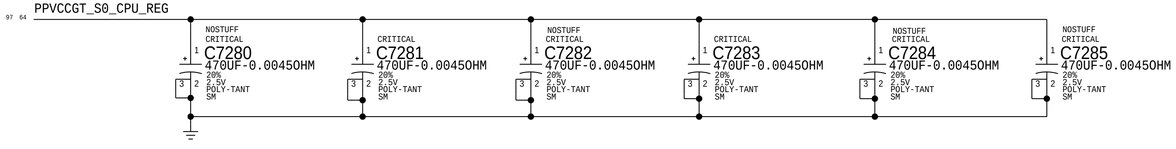
CPU VCCGT Phase 1



CPU VCCGT Phase 2

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S00519	2	IC,SIC621,IMVP-8,60A,PQFN31	Q7210,Q7230	CPU_DRMOS:FUTURE
353S00831	2	IC,FDMF5804,IMVP-8,50A,PQFN31	Q7210,Q7230	CPU_DRMOS:CURRENT

CPU VCCGT OUTPUT DECOUPLING



BOM_COST_GROUP=CPU & CHIPSET

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:05 2016

PAGE TITLE

CPU & CHIPSET: CPU CORE VR (VCCGT)

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SIZE D

REVISION 5.19.0

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PAGE 72 OF 121

SHEET 64 OF 102

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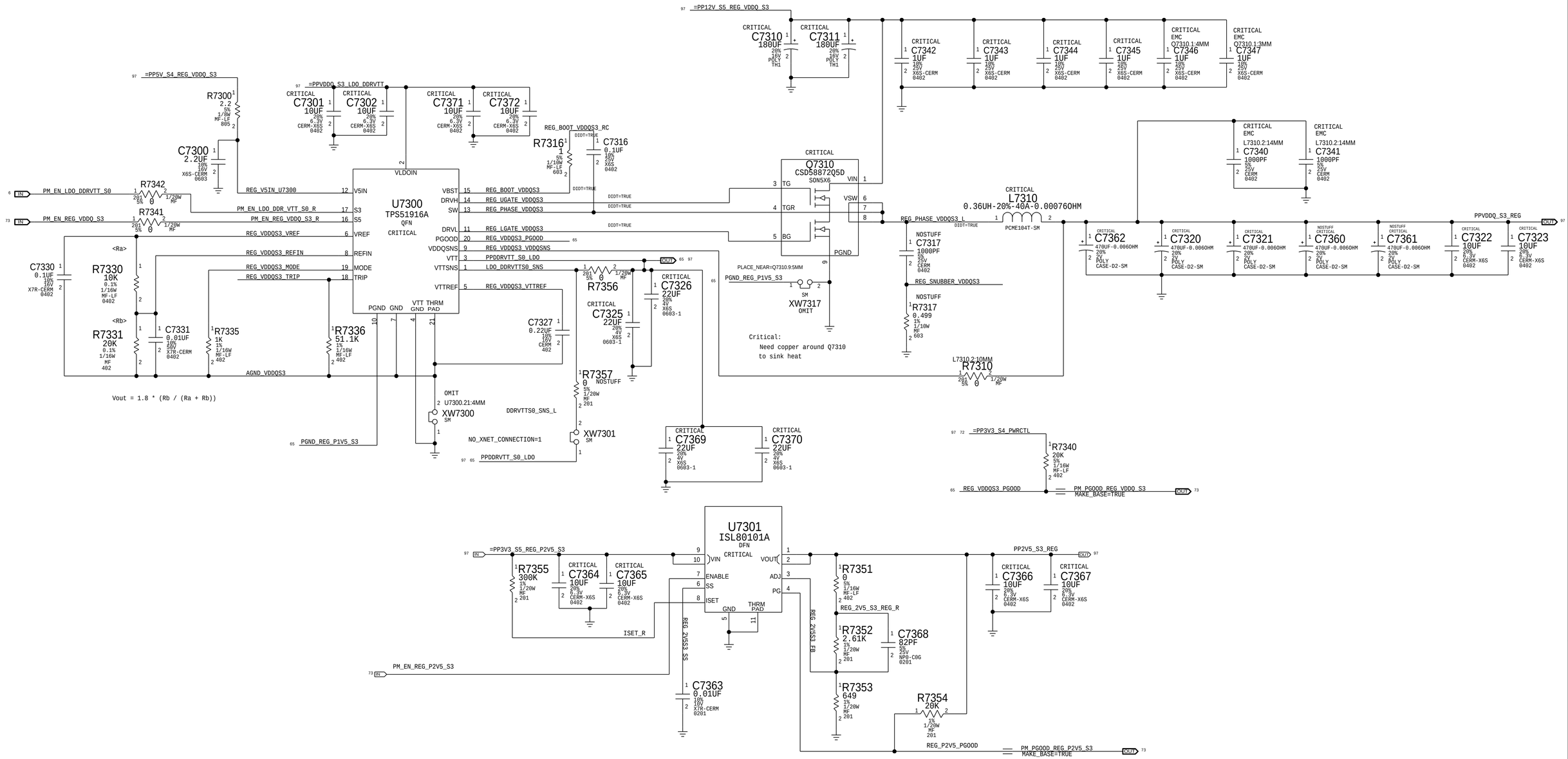
VDDQ S3 REGULATOR

VDDQ = 1.2V @ 12A (EDP)

VTT = 0.6V @ 3A (EDP)

VPP = 2.5V @ 1.1A (EDP)



BOM_COST_GROUP=DRAM

PAGE TITLE

CPU & CHIPSET: CPU VDDQ VR

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PAGE 73 OF 121

SHEET 65 OF 102

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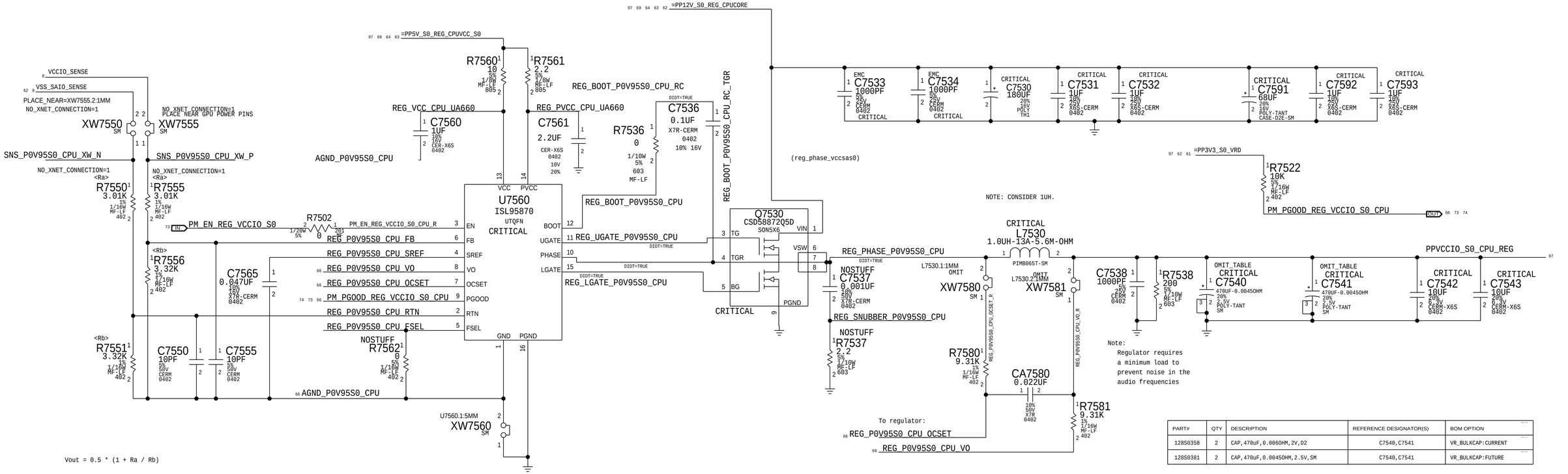
VCCIO S0 REGULATOR

0.95V @ 5.5A (EDP)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0358	2	CAP,470uF,0.0060HM,2V,D2	C7540,C7541	VR_BULKCAP:CURRENT
128S0381	2	CAP,470uF,0.00450HM,2.5V,SM	C7540,C7541	VR_BULKCAP:FUTURE

BOM_COST_GROUP=CPU & CHIPSET

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:05 2016

PAGE TITLE: CPU & CHIPSET: CPU VCCIO VR

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PAGE 75 OF 121

SHEET 66 OF 102

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3.3V S5 Regulator

3.3V @ 10A (EDP)

5V S4 Regulator @ 17Ak

5.1V @ 17A (EDP)

=PP12V_S5_FET_P3V3_S5_P5V_S4

C7610 180UF CRITICAL 20% 16V POLY TH1
C7690 180UF CRITICAL 20% 16V POLY TH1
C7650 180UF CRITICAL 20% 16V POLY TH1
C7651 180UF CRITICAL 20% 16V POLY TH1

R7603 1 5% 1/8W MF-LF 805

C76A0 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A1 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A2 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A3 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7644 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7691 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7642 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7643 1UF EMC CRITICAL 10% 25V X6S-CERM 0402

C7682 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7683 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7684 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C7685 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A4 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A5 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A6 1UF EMC CRITICAL 10% 25V X6S-CERM 0402
C76A7 1UF EMC CRITICAL 10% 25V X6S-CERM 0402

C7640 1000PF EMC CRITICAL 5% 25V CERM 0402
C7641 1000PF EMC CRITICAL 5% 25V CERM 0402

C7680 1000PF EMC L7650.2:4MM 5% 25V CERM 0402
C7681 1000PF CRITICAL EMC L7650.2:4MM 5% 25V CERM 0402

PP5V_S5_LDO

C7601 10UF 20% 10V X6S-CERM 0603

R7602 2.2 5% 1/8W MF-LF 805

C7602 1UF 10% 16V CER-X6S 0402

C7603 1UF 10% 25V X6S-CERM 0402

REG_VIN_U7600
REG_VCC2_U7600
REG_VCC1_U7600

C7600 1UF 10% 16V CER-X6S 0402

REG_BOOT_P5VS4_RC

R7656 0 5% 1/10W MF-LF 603

C7656 0.1UF 10% 16V X6S 0402

Q7650 IRF3575D PQFN CRITICAL
VIN
SW
GATEH
GATEL
PGND

L7610 1.0UH-20A-2.64M-OHM CRITICAL PIMB103E-SM-1

REG_PHASE_P3V3S5 DIDT=TRUE

Q7610 CSD5887Q2D SON5X6 CRITICAL
VIN
VSW
PGND
TG
TGR
BG

PP3V3_S5_REG

C7621 330UF NOSTUFF CRITICAL 20% 6.3V POLY-TANT CASE-D3L-SM
C7620 330UF NOSTUFF CRITICAL 20% 6.3V POLY-TANT CASE-D3L-SM

C7623 10UF CRITICAL 20% 6.3V CERM-X6S 0402
C7622 10UF CRITICAL 20% 6.3V CERM-X6S 0402

L7610.0.16MM

R7610 0 5% 1/20W MF 201

REG_XW_PP3V3_S5_XW

XW7610 OMIT
C7618 0.047UF 10% 25V X7R 0402
XW7618 OMIT
R7618 7.15K 1% 1/16W MF-LF 402 DIDT=TRUE
REG_PHASE_P3V3_S5_XW

C7617 0.001UF NOSTUFF 10% 50V CERM 402
C7616 0.1UF 10% 16V X6S 0402

REG_SNUBBER_P3V3S5

R7619 7.15K 1% 1/16W MF-LF 402

R7617 0.499 NOSTUFF 1% 1/10W MF 603

REG_BOOT_P3V3S5_RC DIDT=TRUE

R7616 2.2 5% 1/10W MF-LF 603

REG_P3V3S5_PGOOD
REG_UGATE_P3V3S5 DIDT=TRUE
REG_BOOT_P3V3S5 DIDT=TRUE
REG_PHASE_P3V3S5_TGR DIDT=TRUE
REG_LGATE_P3V3S5 DIDT=TRUE
REG_P3V3S5_ISEN
REG_P3V3S5_OCSET
REG_P3V3S5_VOUT
REG_P3V3S5_FB
REG_P3V3S5_FSET

U7600 ISL62383CRTZ QFN CRITICAL
LDO5
VCC1
VCC2
FCCM
NC
PGOOD1
UGATE1
BOOT1
PHASE1
LGATE1
ISEN1
OCSET1
VOUT1
FB1
FSET1
EN1
THRM PAD
PGND
PGOOD2
UGATE2
BOOT2
PHASE2
LGATE2
ISEN2
OCSET2
VOUT2
FB2
FSET2
EN2

REG_P5VS4_PGOOD
REG_UGATE_P5VS4 DIDT=TRUE
REG_BOOT_P5VS4 DIDT=TRUE
PHASE_2
REG_LGATE_P5VS4 DIDT=TRUE
REG_P5VS4_ISEN
REG_P5VS4_OCSET
REG_P5VS4_VOUT
REG_P5VS4_FB
REG_P5VS4_FSET

L7650 1.0UH-20A-2.64M-OHM CRITICAL PIMB103E-SM-1

REG_PHASE_P5VS4 DIDT=TRUE

C7657 0.001UF NOSTUFF 10% 50V X7R-CERM 0402

XW7658 OMIT
R7658 10K 1% 1/16W MF-LF 402 DIDT=TRUE
REG_PHASE_P5VS4_XW

XW7659 OMIT
C7658 0.033UF 10% 16V X5M 402

R7659 10K 1% 1/16W MF-LF 402

REG_SNUBBER_P5VS4

R7657 0.499 NOSTUFF 1% 1/10W MF 603

PP5V_S4_REG

L7650.23.8MM

R7650 9 5% 1/20W MF 201

C7662 10UF CRITICAL 20% 16V CERM-X6S 0603
C7663 10UF CRITICAL 20% 16V CERM-X6S 0603

C7670 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7671 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7674 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7676 150UF CRITICAL 20% 6.3V POLY B1A-SM-1

C7646 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7647 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7648 150UF CRITICAL 20% 6.3V POLY B1A-SM-1
C7649 150UF CRITICAL 20% 6.3V POLY B1A-SM-1

<Ra>
R7630 45.3K 1% 1/16W MF-LF 402
<Rb>
R7631 10.0K 0.5% 1/16W MF 402

R7632 976 1% 1/16W MF-LF 402
C7632 1000PF 5% 25V CERM 0402
REG_P3V3S5_VOUT_R

Vout = 0.6 * (1 + Ra / Rb)

C7633 0.01UF 10% 16V X7R-CERM 0402
R7633 11.8K 1% 1/16W MF-LF 402

C7673 0.01UF 10% 16V X7R-CERM 0402
R7673 11.8K 1% 1/16W MF-LF 402

REG_P5VS4_VOUT_R

R7672 976 1% 1/16W MF-LF 402
C7672 1000PF 5% 25V CERM 0402

<Ra>
R7670 75K 1% 1/16W MF-LF 402
<Rb>
R7671 10K 1% 1/16W MF-LF 402

Vout = 0.6 * (1 + Ra / Rb)

(reg_p3v3s4_isen)
(reg_p3v3s4_ocset)
(reg_p3v3s4_vout)

PM_EN_REG_P3V3_S5 → R7698 0 5% 1/16W MF-LF 402 → PM_EN_REG_P3V3_S5_R

PM_EN_REG_P5V_S4 → R7646 0 5% 1/20W MF 201 → PM_EN_REG_P5V_S4_R

=PP3V3_S5_VRD → R7680 20K 5% 1/16W MF-LF 402 → REG_P5VS4_PGOOD → PM_PGOOD_REG_P5V_S4 MAKE_BASE=TRUE

=PP3V3_S5_VRD → R7640 20K 5% 1/16W MF-LF 402 → REG_P3V3S5_PGOOD → PM_PGOOD_REG_P3V3_S5 MAKE_BASE=TRUE

BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE	PLATFORM POWER: 3.3V S5/5V S4 VR
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	REVISION 5.19.0
	BRANCH
	PAGE 76 OF 121
	SHEET 67 OF 102

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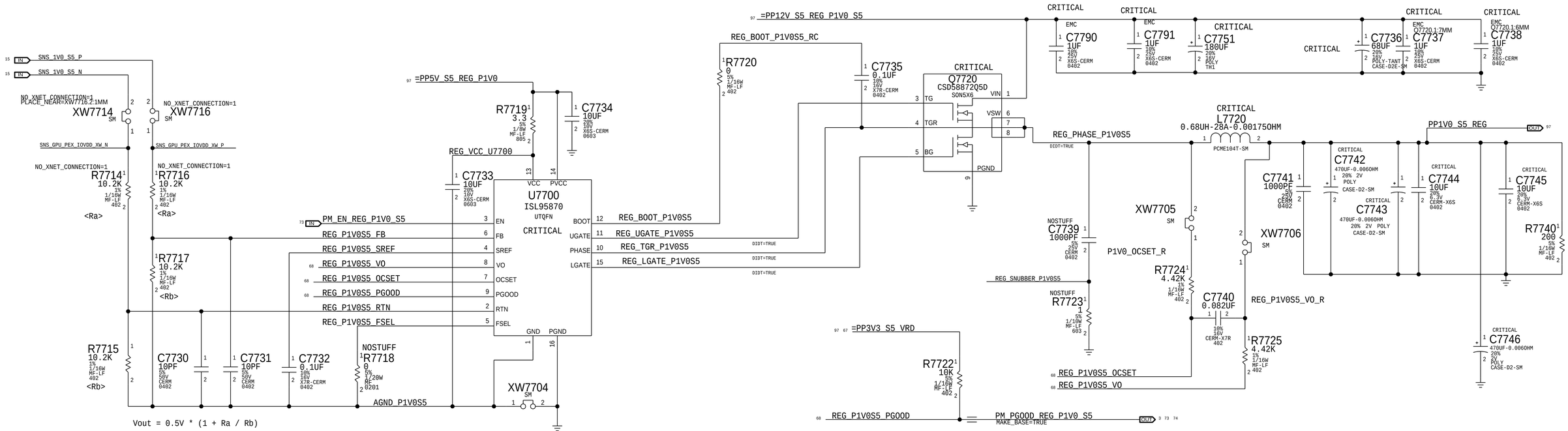
PCH 1V0 SUPPLY

1V @ 10A (EDP)



BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE

CPU & CHIPSET: PCH 1V0 & 1V8 VR

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SIZE D

REVISION 5.19.0

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PAGE 77 OF 121

SHEET 68 OF 102

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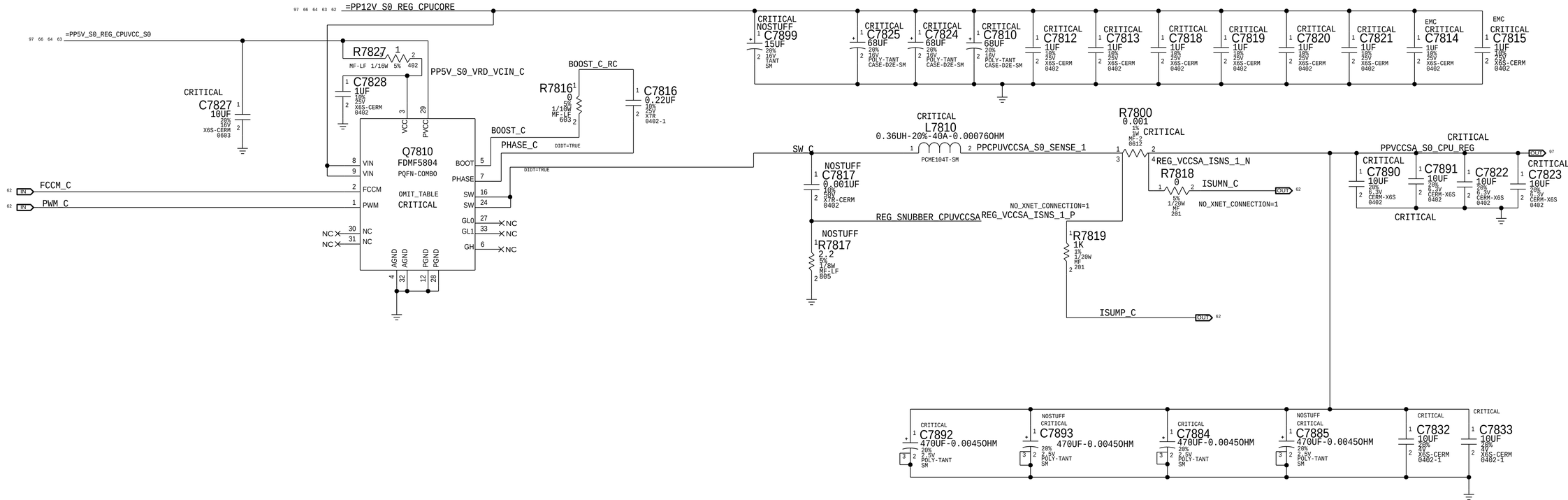
CPU VCCSA Regulator

EDC = 11.1A

TDC = 10A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S00519	1	IC,SIC621,IMVP-8,60A,PQFN31	Q7810	CPU_DRMOS:FUTURE
353S00831	1	IC,FDMF5804,IMVP-8,50A,PQFN31	Q7810	CPU_DRMOS:CURRENT

BOM_COST_GROUP=CPU & CHIPSET

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:06 2016

SYNC_MASTER=SUNIL SYNC_DATE=08/22/2016

PAGE TITLE

CPU & CHIPSET: CPU CORE VR (VCCSA)

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DRAWING NUMBER 051-01506 SIZE D

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PAGE 78 OF 121

SHEET 69 OF 102

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OMIT/table L8100 due to late sourcing change, did not want footprint change
Add 152s1668 (Cyntec) directly for next program

PP12V_S0_BKLT_PWR
D8103 SOD-323 SBR130S3
C8140 1000PF 5% 25V CERM 0402
C8141 1000PF 5% 25V CERM 0402
C8102 10UF 10% 25V X7R-CERM 1206-1 CRITICAL
C8103 10UF 10% 25V X7R-CERM 1206-1 CRITICAL
C8114 100UF 20% 35V ELEC 6.3X11-TH CRITICAL
PLACE C8114 NEAR HIGHER FREQUENCY CAPS
GND

L8100 OMIT_TABLE
33UH-20%-10A-0.03510HM
IHLP6767GZ-IHLP4040DZ11-SM
BKLT_PHASE
D8101 VS-8TQ100SPBF TO263
R8123 4.7 5% 1/10W MF-LF 603 NOSTUFF
BKLT_SNUBBER
C8113 100PF 5% 100V C0G-CERM 0603 NOSTUFF
GND

C8119 100PF 5% 100V C0G-CERM 0603
C8106 2.2UF 10% 100V X7S-CERM 1206
C8108 2.2UF 10% 100V X7S-CERM 1206
C8109 18UF 20% 100V AL-EL 10X12.5-TH CRITICAL OMIT_TABLE
C8118 18UF 20% 100V AL-EL 10X12.5-TH CRITICAL OMIT_TABLE
C8120 100PF 5% 100V C0G-CERM 0603
GND
BKLT_BOOST

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S1078	138S0810		C8106, C8108	BLC OUTPUT CAPS
371S0748	371S0731		D8103	INPUT DIODE
371S0782	371S0783		D8101	OUTPUT DIODE
376S0845	376S1067		Q8119	POWERLINE FET
376S1204	376S1106		Q8211 TO Q8219	LED SINK FET
740S0145	740S0146		F8100	FUSE FOR BLC
107S0359	107S0229		R8102	BOOST SENSE RESISTOR
376S1184	376S1183		Q8120	BLC BOOST FET

Q8120 FDMC86102LZ MLP-3.3X3.3-PWRPK-12128-COMBO

PP3V3_S0_BKLT_VDDIO
PP5V_S0_BKLT_VDDD
PP5V_S0_BKLT_VDDA
PP12V_S0_BKLT_FILT
BKLT_GATE_R
R8119 30.1 1% 1/8W MF-LF 805
BKLT_SNSP_R
R8102 0.03 1% 2W MF 2512-LF
BKLT_SNSN_R
R8114 20 5% 1/8W MF-LF 805
R8104 887K 1% 1/16W MF-LF 402
R8103 330K 1% 1/8W MF 0402
C8111 1000PF 10% 100V X7R-CERM 0603
BKLT_BST_FB_R
R8113 887K 1% 1/16W MF-LF 402
R8112 330K 1% 1/8W MF 0402
C8110 1000PF 10% 100V X7R-CERM 0603
XW8109 SHORT-1206
BKLT_SNSN_R

U8100 LP8565A13RWC WQFN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	1	IND,PWR,33UH,20%,10A,35MOHM	L8100	CRITICAL	
353S00079	1	IC,PLP8565A13,LED BCKLGHT CTRL,7X7 QFN56	U8100	CRITICAL	
126-00013	2	BOOST POLYMER OUTPUT CAP	C8109, C8118	CRITICAL	

U8100 pin		
4	VDDIO	
5	VDDD	
42	VDDA	
39	SD	BKLT_SHUTDOWN
40	UVLO_OVP	
9	SDA	=I2C_BKLT_SDA
8	SCL	=I2C_BKLT_SCL
2	HPD	DP_INTPNL_ML_HPD_R
3	BL_EN	TCON_BLC_EN
12	INT	BKLT_FAULT_INT_R_L
10	BRT_SYNC	BLC_VSYNC_R
7	PWM	BKLT_PWM_IN
11	SYNC	BLC_LSYNC_R
43	FILTER	BKLT_PLL_FILT
32	ANALOG_TEST	BKLT_ANG_TEST
1	DIGITAL_TEST	BKLT_DIG_TEST
38	VING	PP12V_S0_BKLT_PWR_R
37	BST_GD_RAIL	BKLT_GD_RAIL
13	NC	BKLT_NC_TEST
41	GNDA	BKLT_GNDA
6	GNDD	BKLT_GNDD
23	LED_GND1	BKLT_GND_BST_GD
50	LED_GND2	BKLT_GND_BST_GD
35	GND_BST_GD	BKLT_GND_BST_GD
57	THRM PAD	
36	BST_GD	BKLT_GATE
34	BST_ISENSE_P	BKLT_SNSP
33	BST_ISENSE_N	BKLT_SNSN
31	BST_FB	BKLT_BST_FB
30	BST_SHORT_FB	BKLT_BST_SHORT_FB
14	LED1_HDR	BKLT_VSEN1
15	LED1_GD	BKLT_GD1
16	LED1_R	BKLT_ISEN1_Q
17	LED2_HDR	BKLT_VSEN2
18	LED2_GD	BKLT_GD2
19	LED2_R	BKLT_ISEN2_Q
20	LED3_HDR	BKLT_VSEN3
21	LED3_GD	BKLT_GD3
22	LED3_R	BKLT_ISEN3_Q
24	LED4_HDR	BKLT_VSEN4
25	LED4_GD	BKLT_GD4
26	LED4_R	BKLT_ISEN4_Q
29	LED5_HDR	BKLT_VSEN5
27	LED5_GD	BKLT_GD5
28	LED5_R	BKLT_ISEN5_Q
44	LED6_HDR	BKLT_VSEN6
45	LED6_GD	BKLT_GD6
46	LED6_R	BKLT_ISEN6_Q
47	LED7_HDR	BKLT_VSEN7
48	LED7_GD	BKLT_GD7
49	LED7_R	BKLT_ISEN7_Q
51	LED8_HDR	BKLT_VSEN8
52	LED8_GD	BKLT_GD8
53	LED8_R	BKLT_ISEN8_Q
54	LED9_HDR	BKLT_VSEN9
55	LED9_GD	BKLT_GD9
56	LED9_R	BKLT_ISEN9_Q

BKLT_PLL_FILT
R8111 270K 5% 1/16W MF-LF 402
D8102 SOD-523 BAT54XV2T1 NOSTUFF
C8115 330PF 5% 50V NP0-C0G-CERM 0402
C8116 4700PF 10% 50V X7R-CERM 0402
BKLT_PLL_FILT_R
BKLT_GNDA

BKLT_GD_RAIL
C8145 0.1UF 10% 25V X6S 0402 NOSTUFF
C8117 0.1UF 10% 25V X6S 0402
BKLT_GND_BST_GD

R8140 0 MF-LF 1/16W 5% 402 NOSTUFF BKLT_GD_RAIL
R8138 1 5% 1/16W MF-LF 402 PP12V_S0_BKLT_PWR
PP12V_S0_BKLT_PWR_R
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000
C8142 1UF 10% 25V X7R 0603
C8143 0.01UF 10% 16V X7R-CERM 0402
C8144 0.1UF 10% 16V X7R-CERM 0402
BKLT_GND_BST_GD

XW8105 SMD BKLT_GNDA
XW8106 SMD BKLT_GNDD
XW8108 SMD BKLT_GND_BST_GD

R8137 1 5% 1/16W MF-LF 402 =PP5V_S0_BKLT
PP5V_S0_BKLT_VDDA
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000
C8137 22UF 20% 10V X5R-CERM 0603
C8138 0.01UF 10% 16V X7R-CERM 0402
C8139 0.1UF 10% 16V X7R-CERM 0402
BKLT_GNDA

R8131 1 5% 1/16W MF-LF 402 =PP5V_S0_BKLT
PP5V_S0_BKLT_VDDD
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000
C8131 22UF 20% 10V X5R-CERM 0603
C8132 0.01UF 10% 16V X7R-CERM 0402
C8133 0.1UF 10% 16V X7R-CERM 0402
BKLT_GNDD

R8134 1 5% 1/16W MF-LF 402 =PP3V3_S0_BKLT
PP3V3_S0_BKLT_VDDIO
MIN_LINE_WIDTH=0.4000 MIN_NECK_WIDTH=0.2000
C8134 1UF 10% 25V X7R 0603
C8135 0.01UF 10% 16V X7R-CERM 0402
C8136 0.1UF 10% 16V X7R-CERM 0402
BKLT_GNDD

=PP3V3_S0_BKLT
R8124 4.7K 5% MF 1/20W 201
BKLT_FAULT_INT_L
R8172 0 5% 1/16W MF-LF 402 BKLT_FAULT_INT_R_L

R8106 200K 1% MF 1/20W 201 DP_INTPNL_ML_HPD_R
R8105 100K 5% MF 1/20W 201 TCON_BLC_EN
R8107 10K 5% MF 1/20W 201 BKLT_PWM_IN
R8121 10K 5% MF 1/20W 201 NOSTUFF BLC_VSYNC_R
R8120 10K 5% MF 1/20W 201 NOSTUFF BLC_LSYNC_R
BKLT_GNDD

R8173 0 5% MF-LF 1/16W 402 DP_INTPNL_ML_HPD
R8170 0 5% MF-LF 1/16W 402 BLC_VSYNC
R8139 0 5% MF-LF 1/16W 402 NOSTUFF
R8171 0 5% MF-LF 1/16W 402 BLC_LSYNC

R8180 10K 5% MF 1/20W 201 BKLT_ANG_TEST
R8181 10K 5% MF 1/20W 201 BKLT_DIG_TEST
R8182 10K 5% MF 1/20W 201 BKLT_NC_TEST
BKLT_GNDA

BKLT_SNSP R8125 0 5% 1/10W MF-LF 402 BKLT_SNSP_R
C8130 1000PF 10% 100V X7R 603 NOSTUFF
BKLT_SNSN R8191 0 5% 1/10W MF-LF 402 BKLT_SNSN_R

LED_RETURN_1_FB
LED_RETURN_2_FB
LED_RETURN_3_FB
LED_RETURN_4_FB
LED_RETURN_5_FB
LED_RETURN_6_FB
LED_RETURN_7_FB
LED_RETURN_8_FB
LED_RETURN_9_FB

C8122 470PF 5% 100V CER-C0G 0603
C8121 470PF 5% 100V CER-C0G 0603
C8124 470PF 5% 100V CER-C0G 0603
C8123 470PF 5% 100V CER-C0G 0603
C8126 470PF 5% 100V CER-C0G 0603
C8125 470PF 5% 100V CER-C0G 0603
C8128 470PF 5% 100V CER-C0G 0603
C8127 470PF 5% 100V CER-C0G 0603
C8129 470PF 5% 100V CER-C0G 0603

=PP12V_S0_BKLT
F8100 6AMP-32V 0603-1
PP12V_BKLT_SNS
R8132 0.005 1% 1/4W MF 1206-1 CRITICAL
PP12V_BKLT_FUSED
FB8108 160-OHM-6A 1206 CRITICAL
PP12V_S0_BKLT_FILT
C8171 0.1UF 10% 25V X6S 0402 NOSTUFF
R8101 330K 1% 1/8W MF 0402 NOSTUFF
Q8119 FDMS6681Z POWER56
C8101 0.01UF 10% 16V X7R-CERM 0402
PP12V_S0_BKLT_PWR

BKLT_EN_L
R8133 147K 1% 1/16W MF-LF 402 NOSTUFF
BKLT_EN_DIV
R8122 0 5% 1/16W MF-LF 402
TCON_BLC_EN
Q8101 SSM3K15AMFVAP SOD CRITICAL NOSTUFF
BKLT_SHUTDOWN
GND

BOM_COST_GROUP=DISPLAY

SYNC_MASTER=kchooMLB SYNC_DATE=10/30/2015
PAGE TITLE
DISPLAY: LCD Backlight Driver (LP8565)
Apple Inc.
DRAWING NUMBER 051-01506 SIZE D
REVISION 5.19.0
BRANCH
PAGE 81 OF 121
SHEET 70 OF 102

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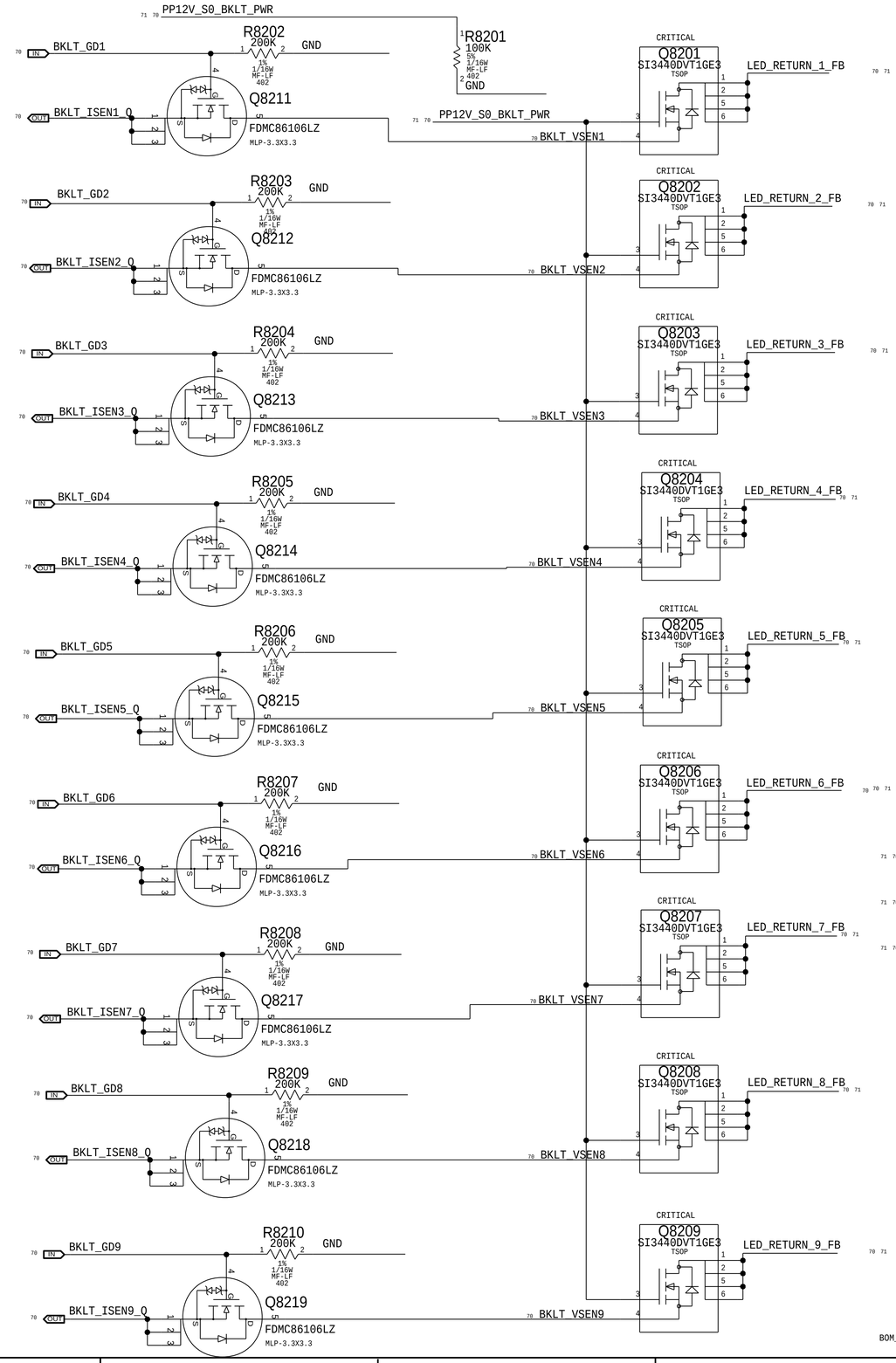
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1256	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8201 TO FB8211



BOM_COST_GROUP=DISPLAY

SYNC_MASTER=kchooMLB SYNC_DATE=10/30/2015

PAGE TITLE

DISPLAY: Backlight Driver 2

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D

REVISION 5.19.0

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PAGE 82 OF 121

SHEET 71 OF 102

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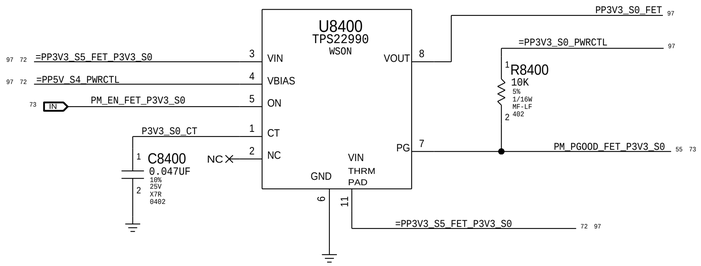
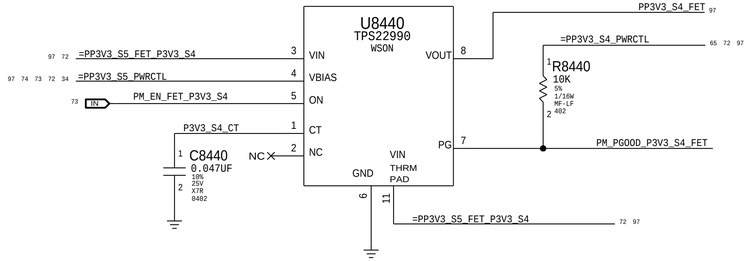
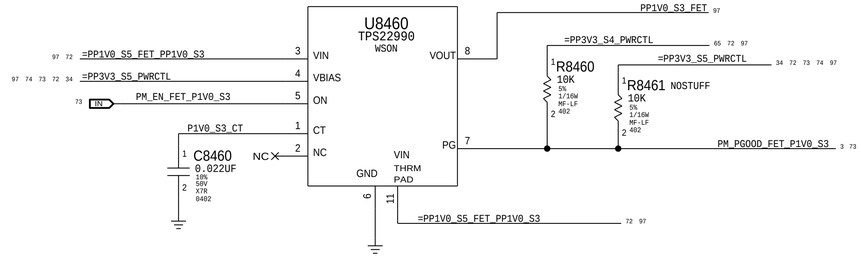
3.3V S0 FET (? PK / 2.0A AVG)
RAMP RATE = ~1.56V/MS



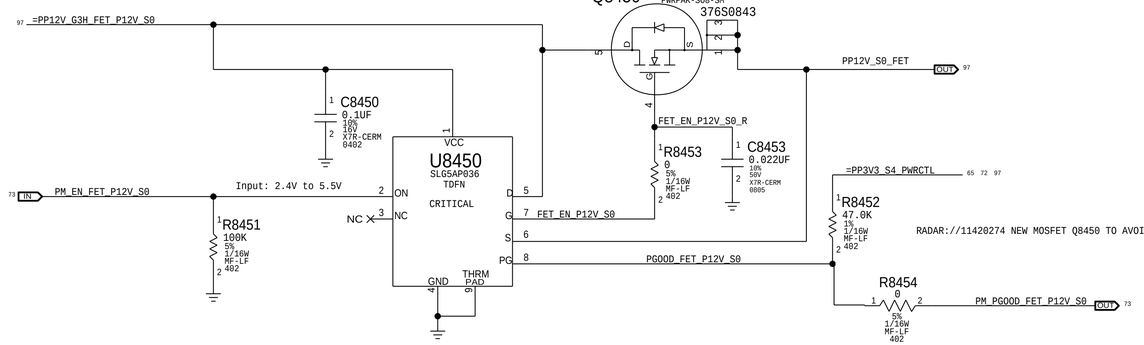
3.3V S4 FET (2.7A PK / 2.0A AVG)
RAMP RATE = ~1.74V/MS



1.0V S3 FET (??A PK / 0.27A AVG)
RAMP RATE = ~3.0V/MS

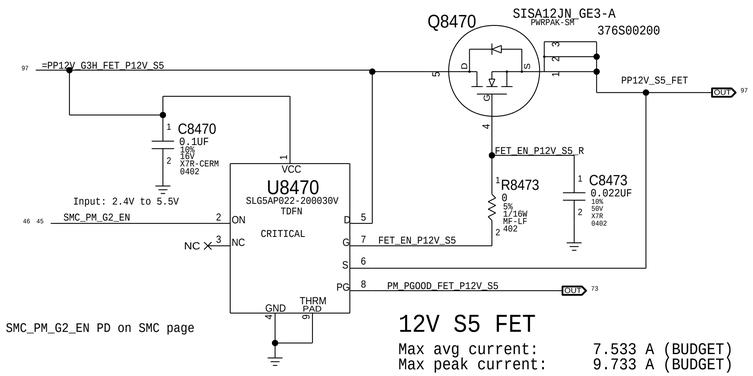


12V S0 FET
Max avg current: 20.4 A (BUDGET)
Max peak current: 24.1 A (BUDGET)



RADAR://11420274 NEW MOSFET Q8450 TO AVOID 50A 100US PULSE ON 12V S0.

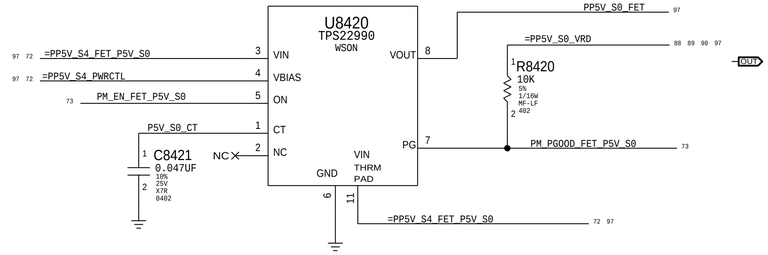
RADAR://10865139 U8470 ENSURES S5_PWRGD ISN'T ASSERTED AFTER HARD SHUTDOWN.



SMC_PM_G2_EN PD on SMC page

12V S5 FET
Max avg current: 7.533 A (BUDGET)
Max peak current: 9.733 A (BUDGET)

5V S0 FET
Max avg current: 2.7 A (BUDGET)
Max peak current: 3.9 A (BUDGET)
RAMP RATE = ~1.46V/MS



BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=kchooMLB SYNC_DATE=10/30/2015

PAGE TITLE
PLATFORM POWER: FET-Controlled S0 and S4

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D
REVISION 5.19.0
BRANCH
PAGE 84 OF 121
SHEET 72 OF 102

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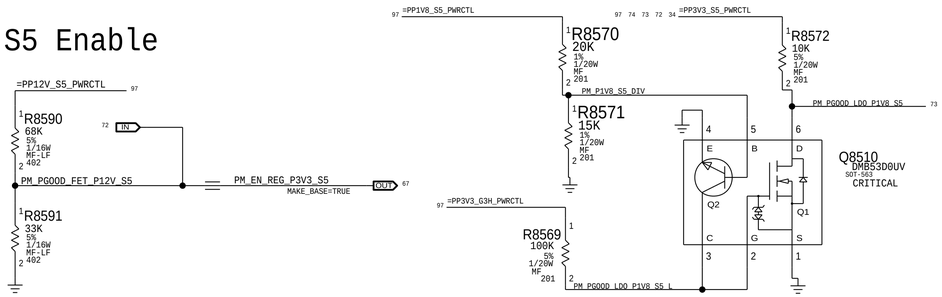
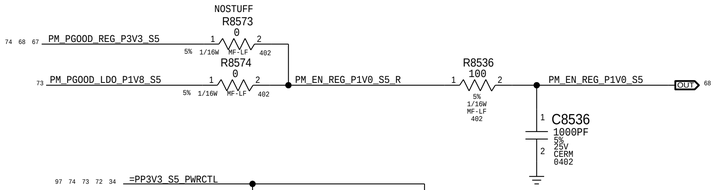
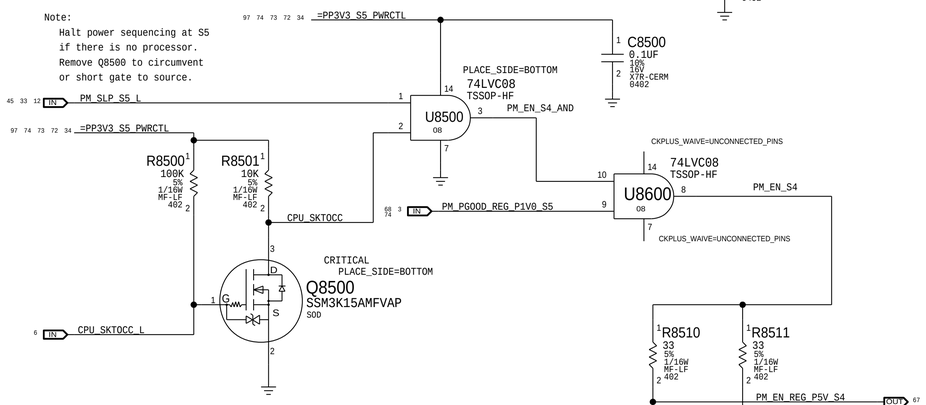
S5 Enable





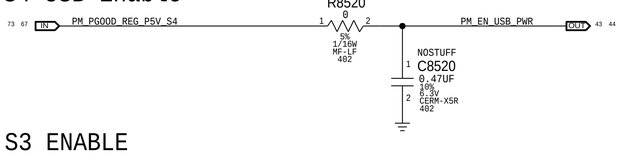
S4 Enables

Note:
Halt power sequencing at S5
if there is no processor.
Remove Q8500 to circumvent
or short gate to source.

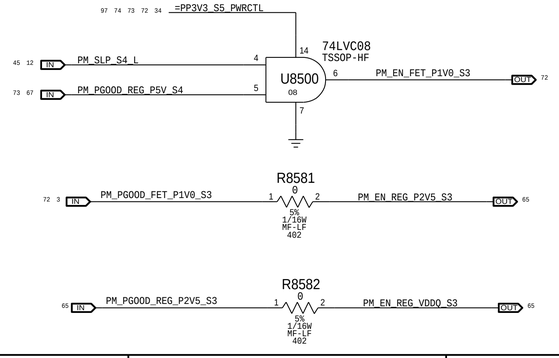


tau (RC delay, ms): 0.0 0.0

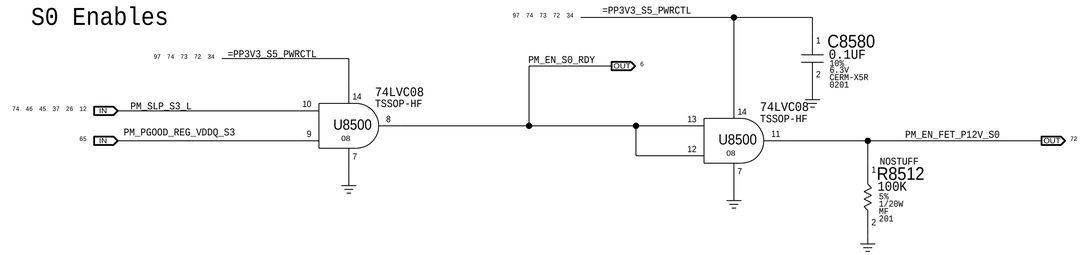
S4 USB Enable

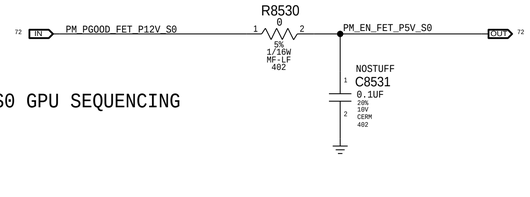


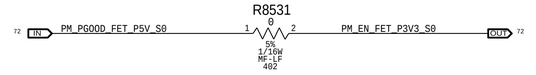
S3 ENABLE



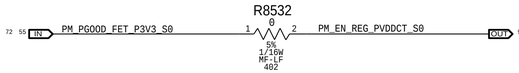
S0 Enables

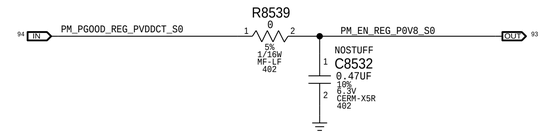


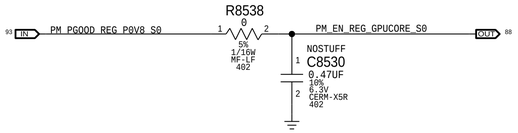


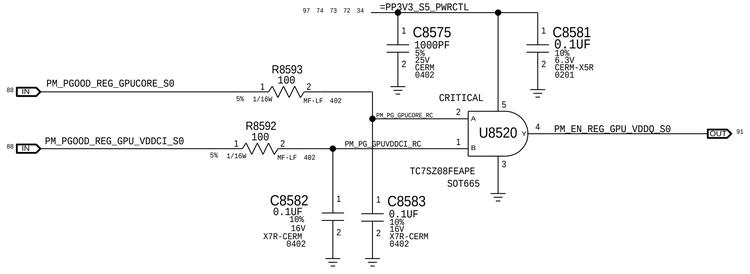


S0 GPU SEQUENCING

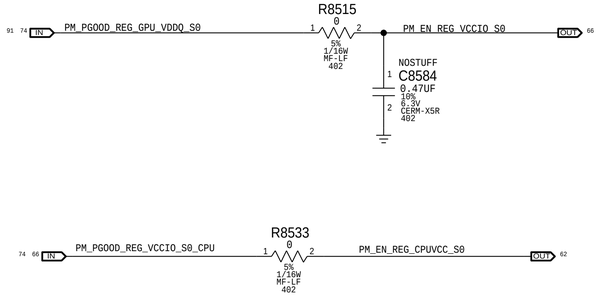








S0 CPU Sequencing



Power Sequencing requirements

System:

1. 12V_ACDC -> 3V42G3H -> 12V_S5/12V_S5_SSD -> 3V3_S5->1V8_S5 -> 1V0_S5 -> 5V_S4/3V3_S4 -> 1V0_S3

GPU RAILS (AMD):

2. 12V_S0 -> 5V_S0 ->3V3_S0 -> 1V8(VDD_CT) -> 0V8 -> GPUCORE (VDDC) -> VDDCI -> FBVDDQ

Intel CPU:

3. 1V0_S3 -> 2V5_S3 -> VDDQ_S3 -> 12V_S0 -> 5V_S0 -> 3V3_S0 -> GPU RAILS -> VCCIO_S0-> VCC_CPU_S0 (VCCGT & VCCSA)

BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=kchooMLB SYNC_DATE=10/30/2015

PAGE TITLE
PLATFORM POWER: Regulator Enables

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 85 OF 121
SHEET 73 OF 102

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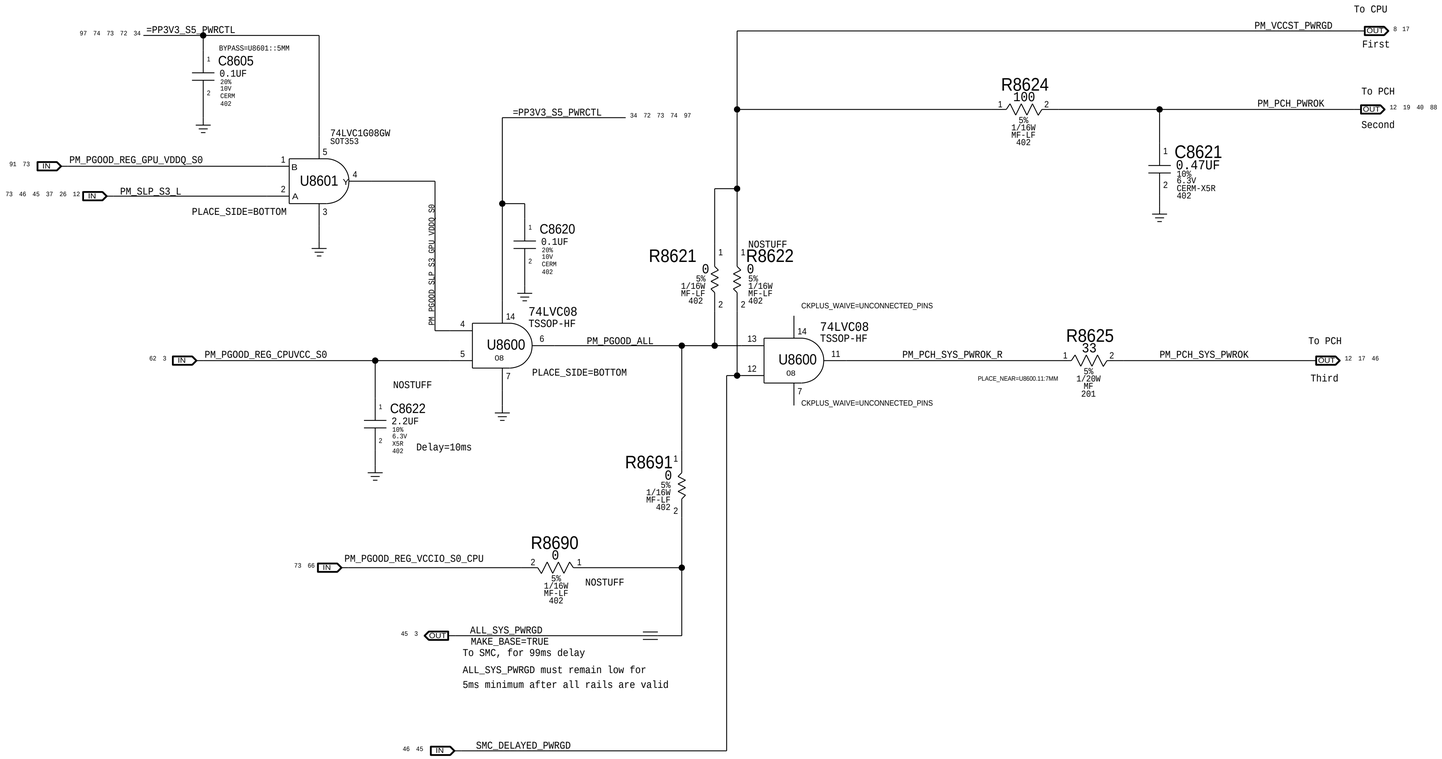
ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



Resume Reset

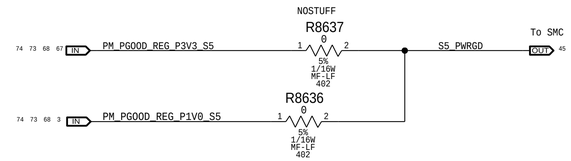
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

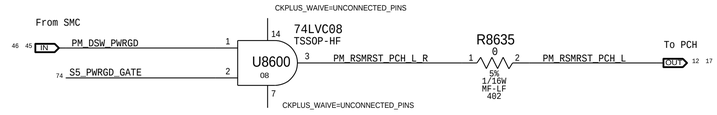
Note:
THE IMAC J135 DESIGNS DOES NOT SUPPORT DEEP SX MODES SO BOTH DPWROK AND RSMRST# signals are shorted together

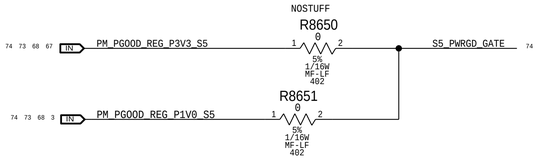
Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.







BOM_COST_GROUP=PLATFORM POWER

SYNC_MASTER=kchooMLB SYNC_DATE=10/30/2015

PAGE TITLE
PLATFORM POWER: PM Power Good

Apple Inc.

DRAWING NUMBER 051-01506 SIZE D
REVISION 5.19.0
BRANCH
PAGE 86 OF 121
SHEET 74 OF 102

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ALL LANES ARE ALSO REVERSED, SEE ALIASES ON CSA 112

SEE OMIT TABLE ON CSA 96 FOR STUFFING OPTIONS
NOTE: ELLESMERE'S PEG WIDTH IS X16, BAFFIN X8

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20% 6.3V X5R 0201

U8700
ELLSMERE-XTA
BGA
100-CK4788-ES
SYM 2 OF 13
PCIEXPRESS

OMIT_TABLE

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11 IN	PEG_CLK100M_N	AR43	PCIE_REFCLK-	WAKE*	AV43	GPU_WAKE_L	
85 19 IN	GPU_RESET_L	AV42	PERST*	PCIE_ZVSS	AK30	PCIE_ZVSS	
14 IN	PCH_GPU_RESET_L			VDD_08	AJ30	
		AM25	VDD_18	VDD_08	AA34	
		AN25	VDD_18	VDD_08	AA35	
		AN26	VDD_18	VDD_08	AD36	
		AP26	VDD_18	VDD_08	AE35	
		AP27	VDD_18	VDD_08	AF35	
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NOSTUFF
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PLACE_NEAR=U8700.AV42:6MM

R8700 200 1% 1/20W MF 201
R8701 10K 5% 1/20W MF 201

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L8700
0.24UH-20%-10A-0.012OHM
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C87F1 4.7UF 20% 6.3V X6S 0402
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C87F3 4.7UF 20% 6.3V X6S 0402
C87F4 1UF 10% 6.3V X7R 0402
C87F5 0.1UF 10% 4V CERM-X6S 0201
C87F6 0.01UF 10% 10V X7R-CERM 0201

PLACE IND & CAPS ALONG EDGE OF GPU

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C8791 4.7UF 20% 6.3V X6S 0402
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C87B0 1UF 10% 6.3V X7R 0402
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C87B3 0.1UF 10% 4V CERM-X6S 0201
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PLACE THESE CAPS UNDER GPU

97 75 =PP0V8_S0_GPU_PCIEVDDC

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C8793 10UF 20% 4V X6S 0402
C8794 4.7UF 20% 6.3V X6S 0402
C8795 4.7UF 20% 6.3V X6S 0402
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C87E5 0.1UF 10% 4V CERM-X6S 0201
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PLACE THESE CAPS UNDER GPU

BOM_COST_GROUP=GRAPHICS

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:07 2016

PAGE TITLE
GFX: GPU PCIE INTF

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 87 OF 121
SHEET 75 OF 102

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GPU VDDC DECOUPLING

PLACE ALL CAPS UNDER GPU
PAIR A 1-UF WITH A 20-UF

97 =PPGPUCORE_S0

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1UF 20% 4V CERM-X6S 0201

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12PF 5% 25V NP0-C0G 0201

CAPS FOR DESENSE
PLACED UNDERNEATH GPU

U8700
ELLSMERE-XTA
BGA
100-CK4788-ES
SYM 9 OF 13
POWER1
OMIT_TABLE

VDDC: AA16 AA17 AA20 AA21 AA24 AA25 AA28 AA29 AB16 AB17 AB20 AB21 AB24 AB25 AB28 AB29 AC16 AC17 AC20 AC21 AC24 AC25 AC28 AC29 AD16 AD17 AD20 AD21 AD24 AD25 AD28 AD29 AE16 AE17 AE20 AE21 AE24 AE25 AE28 AE29 AF16 AF17 AF20 AF21 AF24 AF25 AF28 AF29 AG17 AG20 AG21 AG24 AG25 AG28 AG29 AH18 AH19 AH20 AH21 AH22 AH23 AH24 AH25 AH26 AH27 AH28 AH29 AJ19 AJ20 AJ21

VDDC: AJ22 AJ23 AJ24 AJ25 AJ26 AJ27 AJ28 T16 T17 T20 T21 T24 T25 U16 U17 U20 U21 U24 U25 V16 V17 V20 V21 V24 V25 V28 W16 W17 W20 W21 W24 W25 W28 W29 Y16 Y17 Y20 Y21 Y24 Y25 Y28 Y29

FB_VDDC AL21
FB_VSS AK21

C8863 C8864 C8865 C8866 C8867 C8868 C8869 C8870 C8871 C8872
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C88D7 C88D8 C88D9 C88DA C88DB C88DC C88DD C88DE
20UF 20% 2.5V X6S-CERM 0402

R8800
100
5%
1/20W
MF
201

VR_VSNS_GPU_VDDC OUT 88
VR_VSNS_GPU_VSS OUT 88

R8801
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5%
1/20W
MF
201

BOM_COST_GROUP=GRAPHICS

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:07 2016

PAGE TITLE
GFX: GPU VDDC DECOUPLING

Apple Inc.

DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
BRANCH
PAGE 88 OF 121
SHEET 76 OF 102

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GPU VMEMIO/VDDCI DECOUPLING

PPFBVDDQ_S0_GPU = +1.5V

=PPFBVDDQ_S0_GPU

78 79 80 81 82 83 91 97

PLACE ALL CAPS UNDER GPU

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1UF 20% 4V CERM-X6S 0201

U8700
ELLSMERE-XTA
BGA
100-CK4788-ES
SYM 10 OF 13

OMIT_TABLE

AC9	VMEMIO
AD11	VMEMIO
AE13	VMEMIO
AE7	VMEMIO
AF9	VMEMIO
AG11	VMEMIO
AG5	VMEMIO
AJ13	VMEMIO
AJ7	VMEMIO
AK14	VMEMIO
AK9	VMEMIO
AL15	VMEMIO
AL19	VMEMIO
AM5	VMEMIO
AN17	VMEMIO
AN20	VMEMIO
AN7	VMEMIO
AR14	VMEMIO
AR18	VMEMIO
AU11	VMEMIO
AU15	VMEMIO
AW12	VMEMIO
AW17	VMEMIO
E12	VMEMIO
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U39	VMEMIO
U5	VMEMIO
V35	VMEMIO
V9	VMEMIO
W13	VMEMIO
W31	VMEMIO
W7	VMEMIO
Y11	VMEMIO
Y33	VMEMIO

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AC14	VDDCI
AD15	VDDCI
AE15	VDDCI
AF15	VDDCI
AG16	VDDCI
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R26	VDDCI
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T27	VDDCI
T15	VDDCI
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U28	VDDCI
V29	VDDCI
W15	VDDCI
Y15	VDDCI
AK23	FB_VDDCI

R8900 100 5% 1/20W MF 201

VR_VSNS_GPU_VDDCI → OUT 88

SHARING FEEDBACK VSS PIN WITH VDDC

PLACE ALL CAPS UNDER GPU
PAIR A 20-UF WITH A 1-UF

=PPVDDCI_S0_GPU 97

C8960 C8961 C8962 C8963 C8964 C8965 C8966 C8967 C8968 C8969
C8970 C8971 C8972 C8973 C8974 C8975 C8976 C8977 C8978 C8979

1UF 20% 4V CERM-X6S 0201

C89A0 C89A1 C89A2 C89A3 C89A4 C89A5 C89A6 C89A7 C89A8 C89A9

20UF 20% 2.5V X6S-CERM 0402

BOM_COST_GROUP=GRAPHICS

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:07 2016

SYNC_MASTER=PETE SYNC_DATE=07/17/2015

PAGE TITLE
GFX: GPU VMEMIO/VDDCI DECOUPLING

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PAGE 89 OF 121
SHEET 77 OF 102

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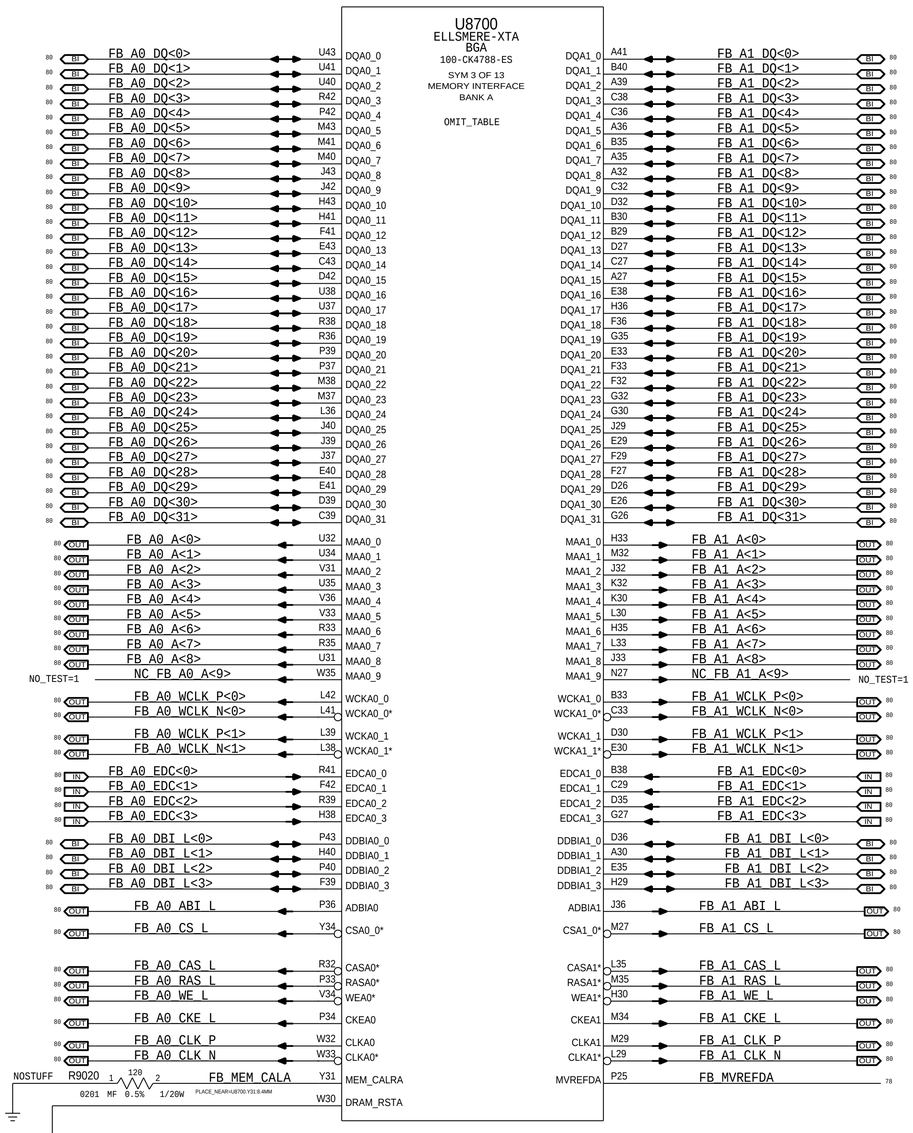

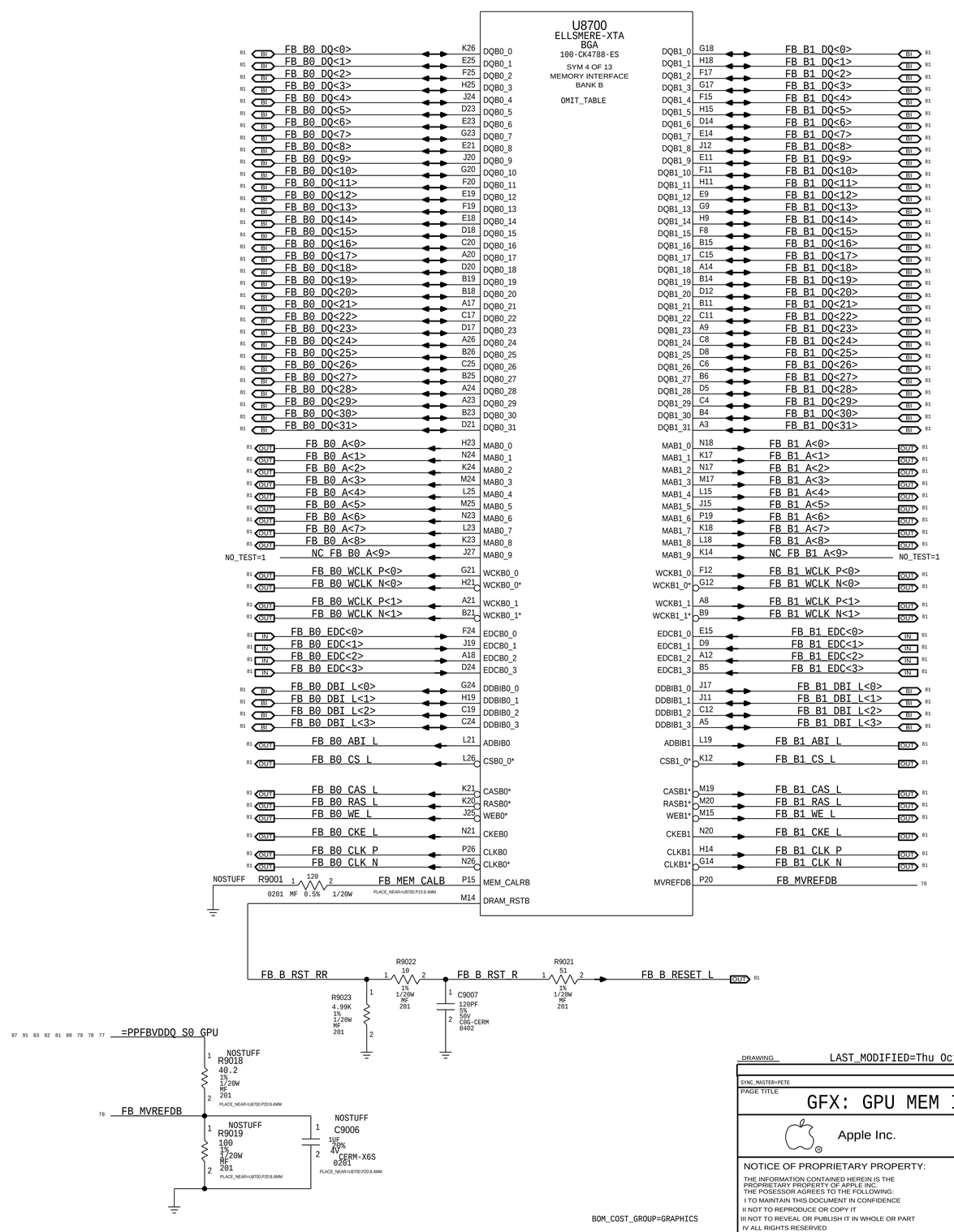

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DRAWING LAST_MODIFIED=Thu Oct 6 17:15:08 2016

SYNC_MASTER=PETE SYNC_DATE=02/04/2016

PAGE TITLE

GFX: GPU MEM INTF A/B

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REVISION 5.19.0

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PAGE 90 OF 121

SHEET 78 OF 102

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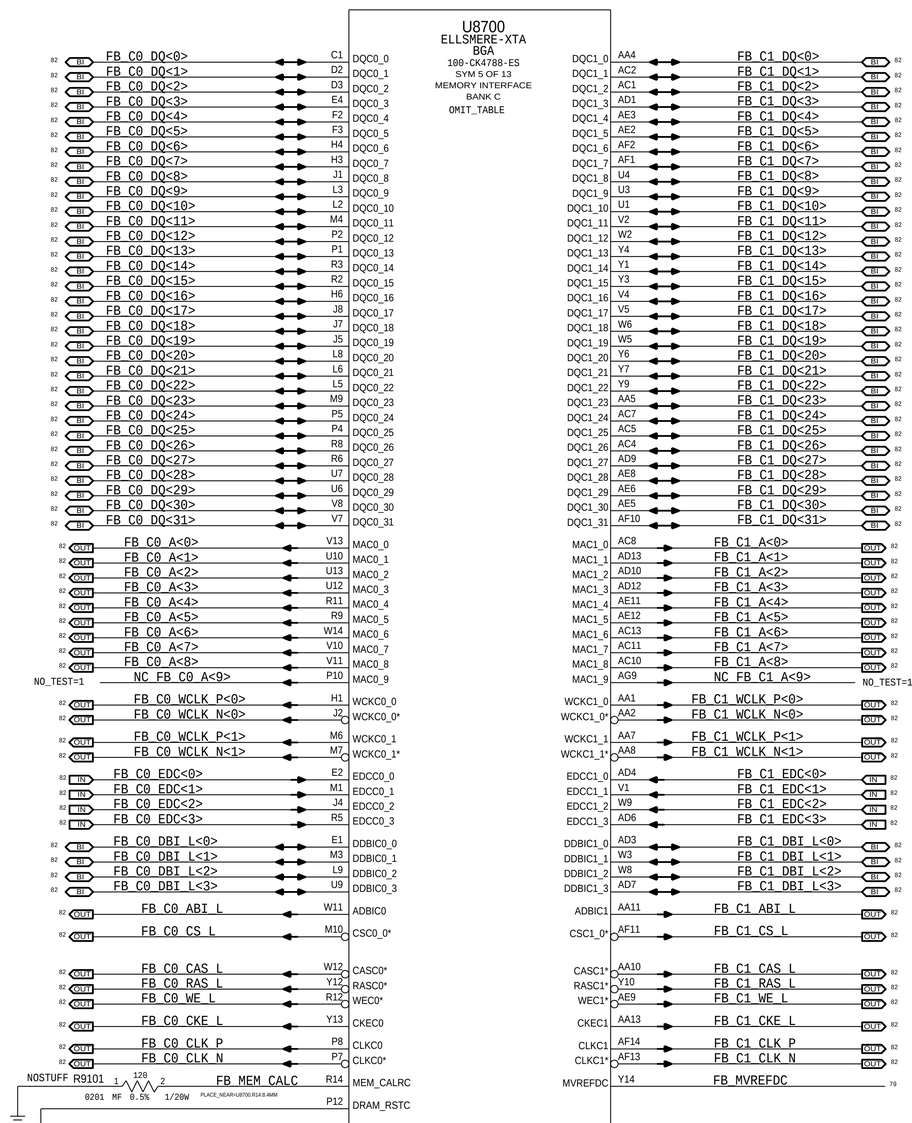

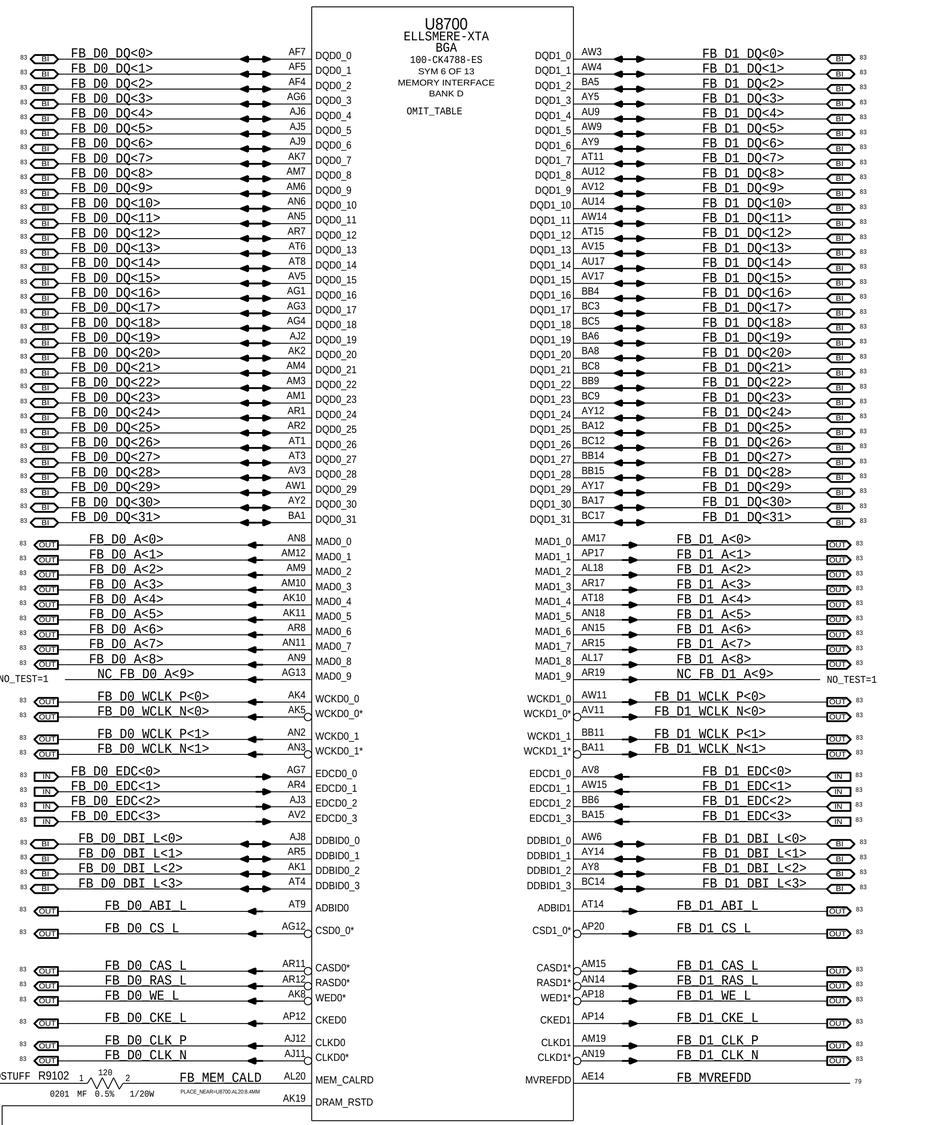

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SYNC_MASTER=PETE SYNC_DATE=02/04/2016

PAGE TITLE

GFX: GPU MEM INTF C/D

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REVISION 5.19.0

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PAGE 91 OF 121

SHEET 79 OF 102

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Page Notes

Power aliases required by this page:

- =PPFBVDDQ_S0_GPU

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

CAPS FOR DESENSE

NORMAL

U9200
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(1 OF 2)
OMIT_TABLE

MIRRORED

U9250
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(1 OF 2)
OMIT_TABLE

U9200
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(2 OF 2)
OMIT_TABLE

U9250
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(2 OF 2)
OMIT_TABLE

BOM_COST_GROUP=GRAPHICS

PAGE TITLE	GFX: GDDR5 FRAME BUFFER A
DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	92 OF 121
SHEET	80 OF 102

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Page Notes

Power aliases required by this page:

- =PPFBVDDQ_S0_GPU

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

NORMAL

U9300
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(1 OF 2)

OMIT_TABLE

MIRRORED

U9350
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(1 OF 2)

OMIT_TABLE

CAPS FOR DESENSE

U9300
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(2 OF 2)

OMIT_TABLE

U9350
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(2 OF 2)

OMIT_TABLE

BOM_COST_GROUP=GRAPHICS

PAGE TITLE

GFX: GDDR5 FRAME BUFFER B

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PAGE 93 OF 121

SHEET 81 OF 102

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Page Notes

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- =PPFBVDDQ_S0_GPU

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

NORMAL

U9400
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(1 OF 2)
OMIT_TABLE

MIRRORED

U9450
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(1 OF 2)
OMIT_TABLE

U9400
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(2 OF 2)
OMIT_TABLE

U9450
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(2 OF 2)
OMIT_TABLE

CAPS FOR DESENSE

BOM_COST_GROUP=GRAPHICS

PAGE TITLE	GFX: GDDR5 FRAME BUFFER C
Apple Inc.	
DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	94 OF 121
SHEET	82 OF 102

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Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

NORMAL

U9500
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(1 OF 2)

OMIT_TABLE

MIRRORED

U9550
32MX32-1.5GHZ-MFH
BGA
K4G10325FG-HC03
(1 OF 2)

OMIT_TABLE

CAPS FOR DESENSE

U9500
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(2 OF 2)

OMIT_TABLE

U9550
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BGA
K4G10325FG-HC03
(2 OF 2)

OMIT_TABLE

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PAGE TITLE	GFX: GDDR5 FRAME BUFFER D
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	REVISION 5.19.0
	BRANCH
	PAGE 95 OF 121
	SHEET 83 OF 102

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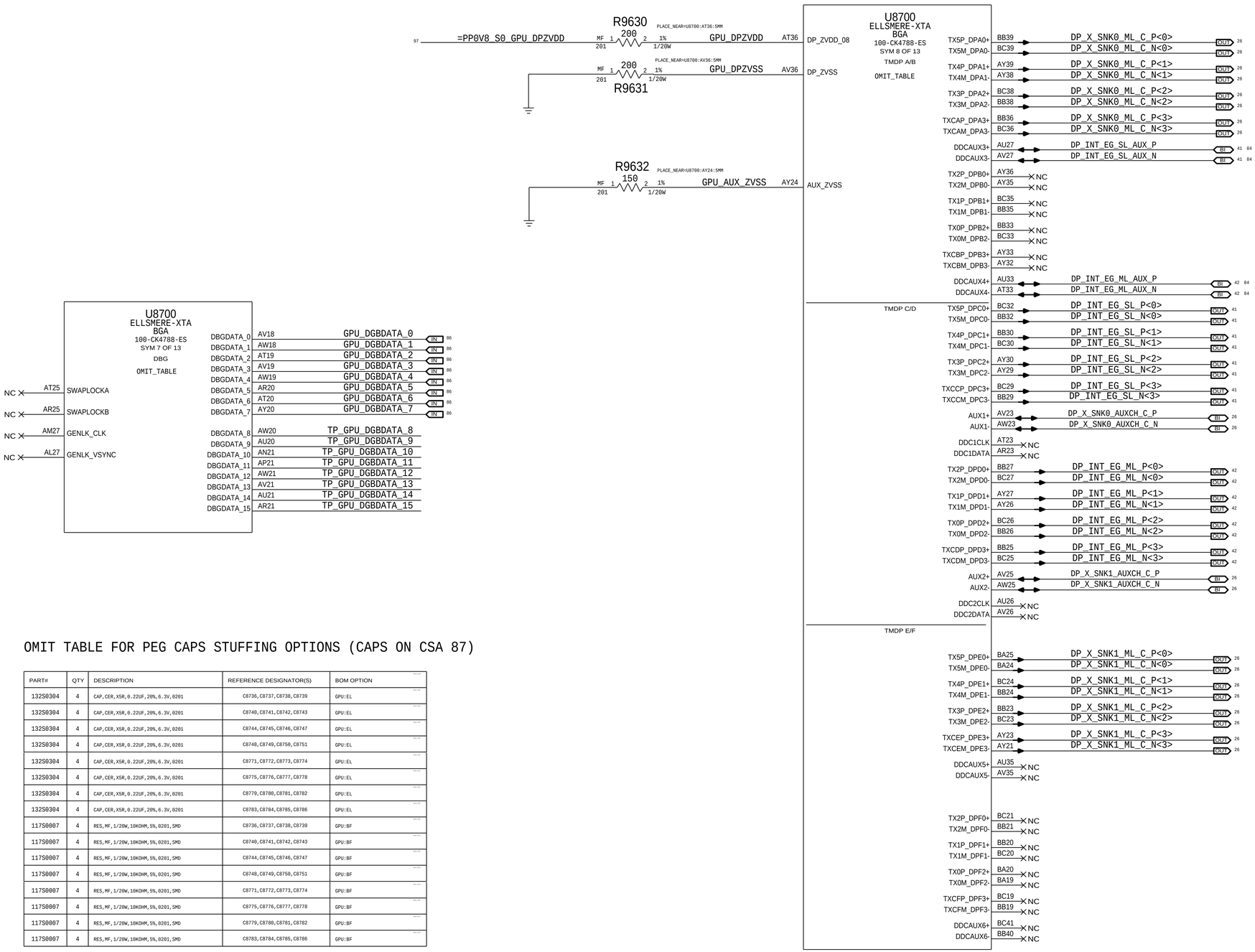

OMIT TABLE FOR PEG CAPS STUFFING OPTIONS (CAPS ON CSA 87)

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8736,C8737,C8738,C8739	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8740,C8741,C8742,C8743	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8744,C8745,C8746,C8747	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8748,C8749,C8750,C8751	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8771,C8772,C8773,C8774	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8775,C8776,C8777,C8778	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8779,C8780,C8781,C8782	GPU:EL
132S0304	4	CAP,CER,X5R,0.22UF,20%,6.3V,0201	C8783,C8784,C8785,C8786	GPU:EL
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8736,C8737,C8738,C8739	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8740,C8741,C8742,C8743	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8744,C8745,C8746,C8747	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8748,C8749,C8750,C8751	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8771,C8772,C8773,C8774	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8775,C8776,C8777,C8778	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8779,C8780,C8781,C8782	GPU:BF
117S0007	4	RES,MF,1/20W,10KOHM,5%,0201,SMD	C8783,C8784,C8785,C8786	GPU:BF

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GFX: DisplayPort INTF

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PAGE 96 OF 121

SHEET 84 OF 102

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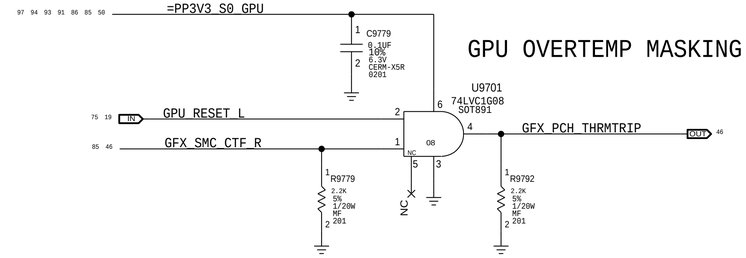
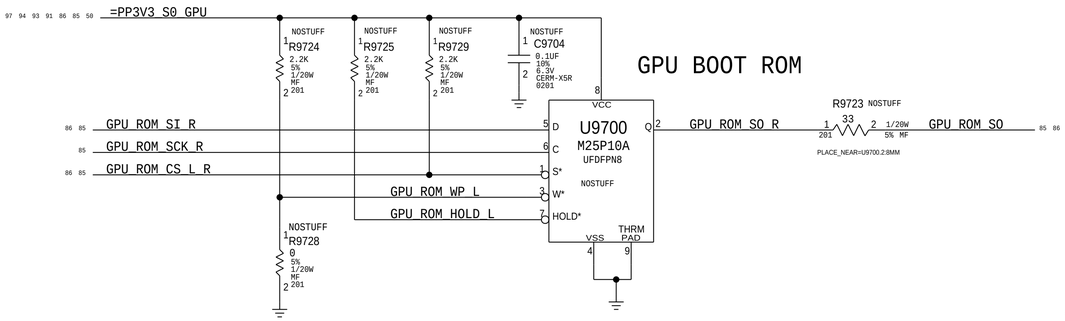
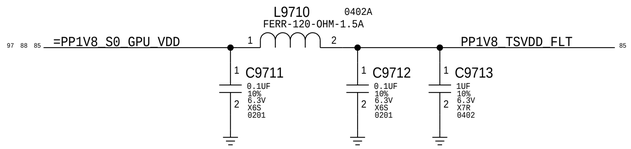
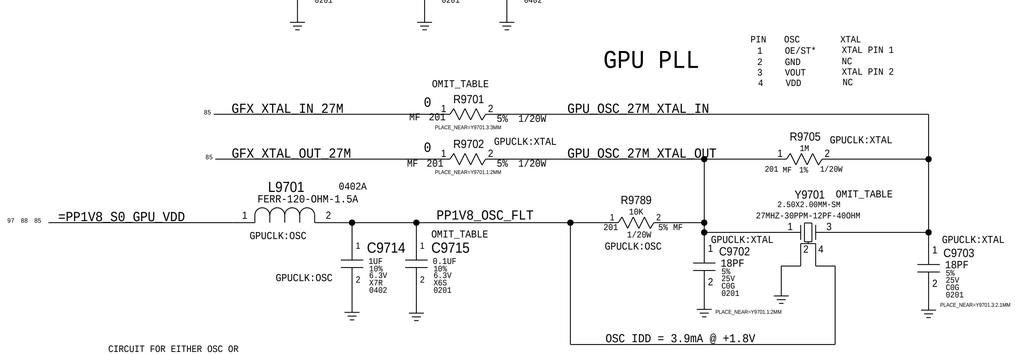


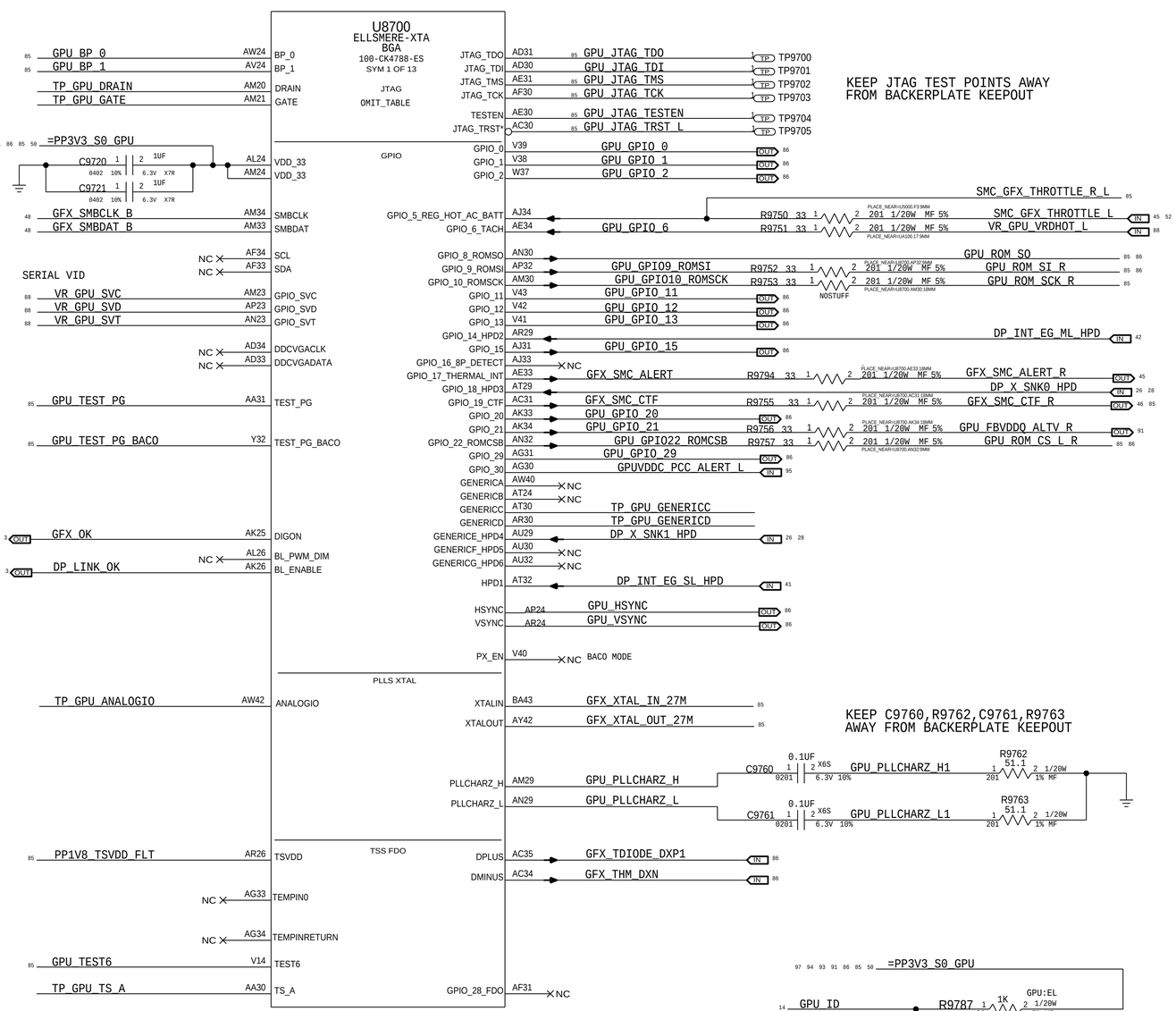


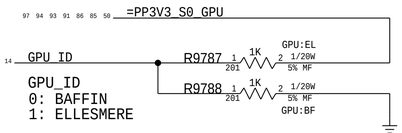


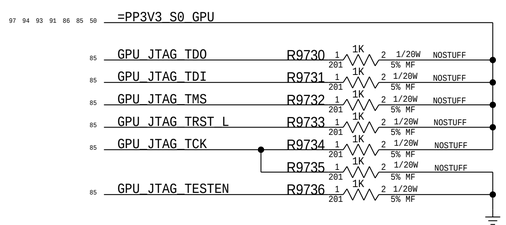


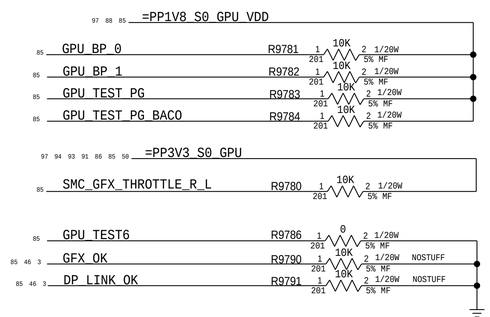
PIN	OSC	XTAL
1	OE/ST*	XTAL PIN 1
2	GND	NC
3	VOUT	XTAL PIN 2
4	VDD	NC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
197S00056	1	OSC,MEMS,100MHZ,+/-20%,1V8,2520	Y9701	GPUCLK:OSC
197S0499	1	XTAL,27MHZ,30PPM,12PF,2.5X2.0MM	Y9701	GPUCLK:XTAL
117S0080	1	RES,33OHM,5%,0201	R9701	GPUCLK:OSC
117S0201	1	RES,0 OHM,5%,0201	R9701	GPUCLK:XTAL
117S0201	1	RES,0 OHM,5%,0201	C9715	GPUCLK:XTAL
132S0395	1	CAP,0.1UF,10%,0201	C9715	GPUCLK:OSC









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PAGE TITLE

GFX: GPU GPIOS/PLLS/VBIOS ROM

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PAGE 97 OF 121

SHEET 85 OF 102

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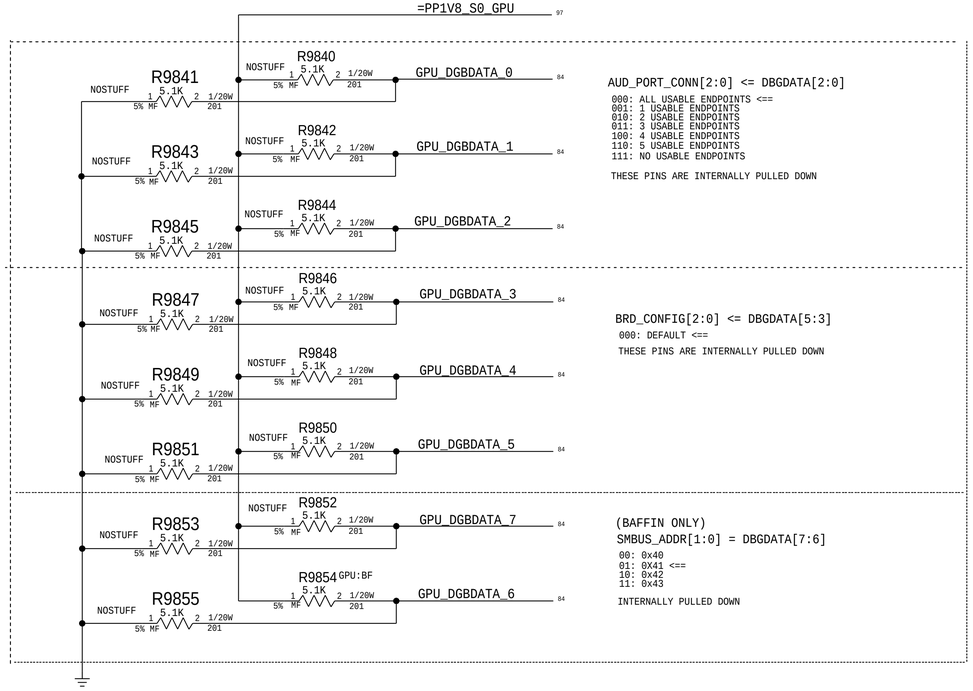
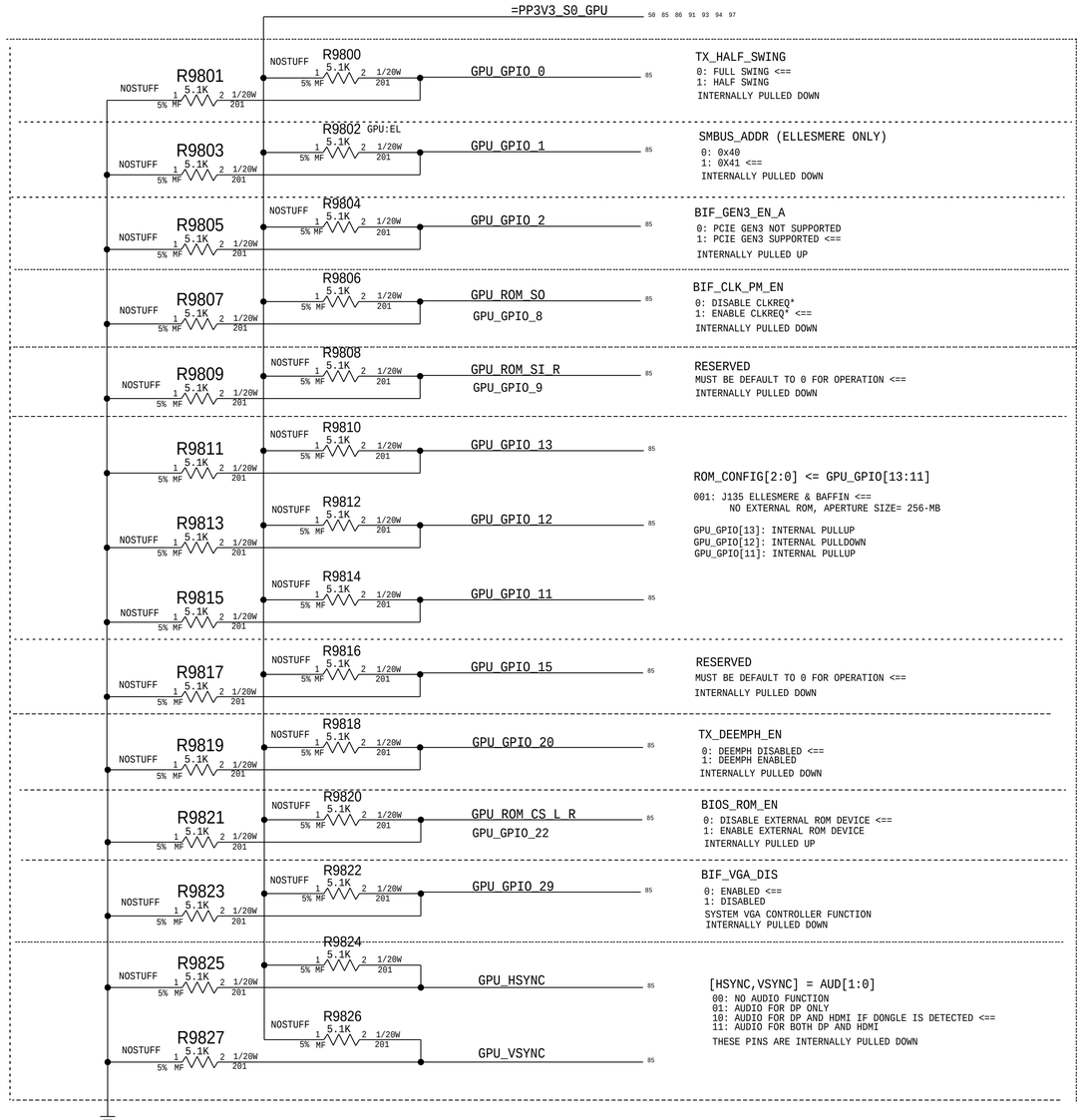
PIN-BASED CONFIG STRAPS

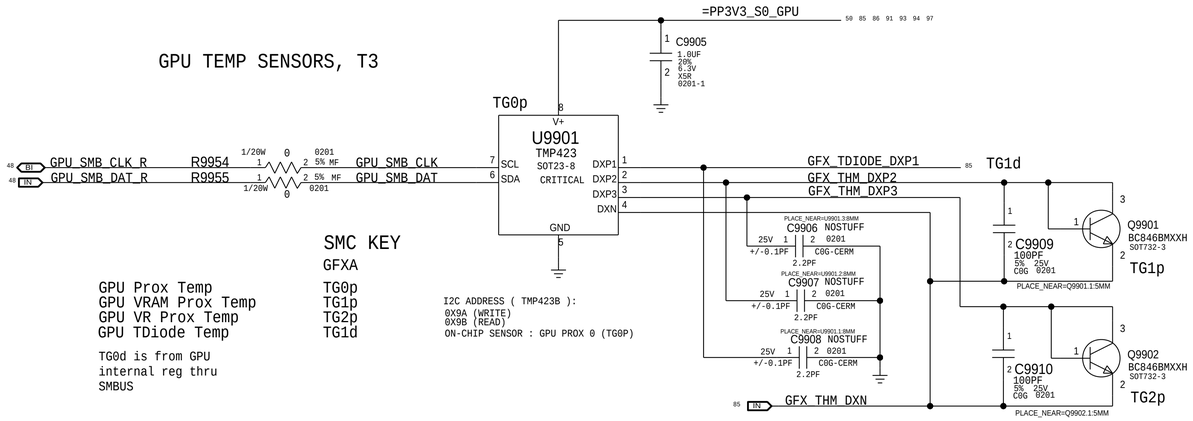
+1.8V PIN STRAPS



+3.3V PIN STRAPS



GPU TEMP SENSORS, T3



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SYNC_DATE=02/19/2016

PAGE TITLE

GFX: GPU STRAPS/TEMP SENSORS

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SIZE D
REVISION 5.19.0
BRANCH
PAGE 98 OF 121
SHEET 86 OF 102

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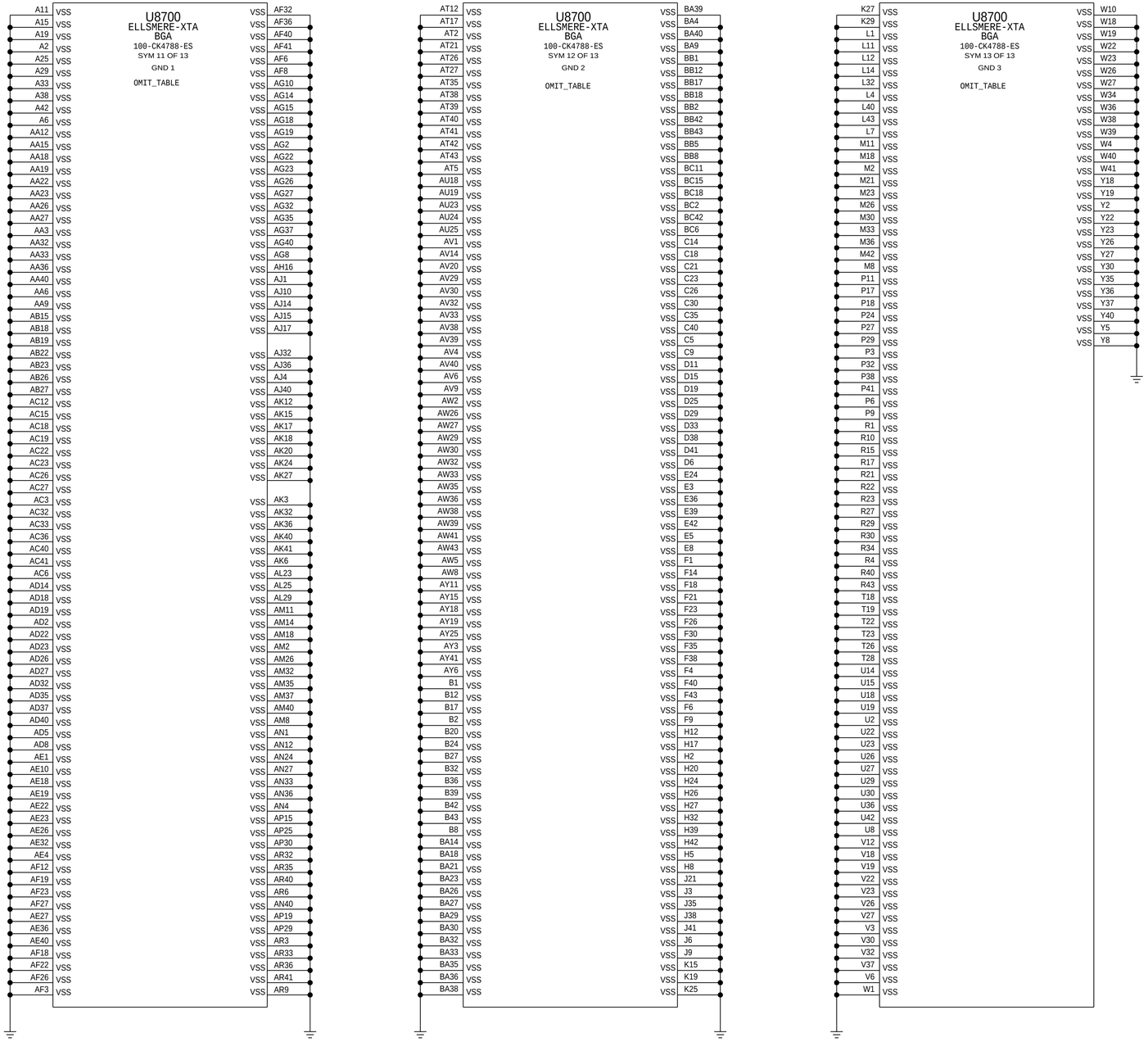

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PAGE TITLE
GFX: GPU GNDS

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SIZE D
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PAGE 99 OF 121
SHEET 87 OF 102

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O/P= PPGPUCORE_S0_REG

GPUCORE

VR_VSNS_GPU_VDDCI

XWAA105 SM NO_XNET_CONNECTION=1

VR_GPU_VDDCI_VSEN_L

RA197 0 402 MF-LF

XWAA110 SM NO_XNET_CONNECTION=1

CA145 3300PF 10% 25V X7R-CERM 0402

NO_XNET_CONNECTION=1

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CA130 1UF 10% 10V X6S-CERM 0402

CA131 0.1UF 10% 16V X7R-CERM 0402

=PP5V_S0_REG_GPUCORE

CA132 1UF 10% 10V CER-X6S 0402

UA101 INA331IDGK MSOP

VR_GPU_IMON

0.01V = 1A 3V @ 300A

RA117 0 5% 1/10W MF-LF 402

VR_GPU_IMON_R

RA108 232K 1% 1/10W MF-LF 402

VR_GPU_IMON_RG

RA109 100K 1% 1/10W MF-LF 402

NO_XNET_CONNECTION=1

OMIT_TABLE

RA102 10 1% 1/10W MF-LF 402

VR_GPU_RCSP_R

RA101 1.43K 1% 1/10W MF-LF 402

CA101 560PF 10% 50V X7R-CERM 0402

RA103 4.99 1% 1/10W MF-LF 402

VR_GPU_RCSM_R

RA196 348 1% 1/10W MF-LF 402

VR_GPU_VDDCI_RCSM_R

CA144 150PF 5% 25V C0G-CERM 0402

RA194 348 1% 1/10W MF-LF 402

RA195 4.12K 1% 1/10W MF-LF 402

VR_GPU_VDDCI_RCSP_R

RA105 75 5% 1/10W MF-LF 603

REG_V18A_GPUCORE

CA135 0.47UF 10% 6.3V X6S-CERM 0402

CA136 10UF 20% 6.3V CERM-X6S 0402

VR_GPU_VDDCI_VRTN_R

VR_GPU_VDDCI_VSEN_R

VR_GPU_RCSP_L2

VR_GPU_RCSM_L2

XWA120 SM OMIT

VR_VSNS_GPU_VDDC

MIN_LINE_WIDTH=0.1000 MIN_NECK_WIDTH=0.1000

VR_GPU_VSEN_L

RA190 0 5% 1/10W MF-LF

CA115 3300PF 10% X7R-CERM 0402 50V 2

VR_GPU_VSEN_R

VR_VSNS_GPU_VSS

XWA130 SM OMIT NO_XNET_CONNECTION=1

UA100 111-4168PBF QFN CRITICAL OMIT_TABLE

VCC V18A

RCSP_L2 RCSM_L2 VSEN_L2 VRTN_L2 RCSP RCSM VSEN VRTN ISEN1 IRTN1 ISEN2 IRTN2 ISEN3 IRTN3 ISEN4 IRTN4 ISEN5 IRTN5 ISEN6 IRTN6 ISEN1_L2 IRTN1_L2 TSEN1 VINSEN SM_DIO SM_CLK SV_DIO/VIDSEL0 SV_CLK/VIDSEL1 EN ADDR_PROT EN_L2/INMODE/PWROK RRES THRM_PAD 49

PWM1 PWM2 PWM3 PWM4 PWM5 PWM6 PWM1_L2

REG_PWM_GPUCORE_1 REG_PWM_GPUCORE_2 REG_PWM_GPUCORE_3 REG_PWM_GPUCORE_4 REG_PWM_GPUCORE_5 REG_PWM_GPUCORE_6 REG_PWM_GPU_VDDCI

REG_GPUCORE_ISNS_1_P REG_GPUCORE_ISNS_1_N REG_GPUCORE_ISNS_2_P REG_GPUCORE_ISNS_2_N REG_GPUCORE_ISNS_3_P REG_GPUCORE_ISNS_3_N REG_GPUCORE_ISNS_4_P REG_GPUCORE_ISNS_4_N REG_GPUCORE_ISNS_5_P REG_GPUCORE_ISNS_5_N REG_GPUCORE_ISNS_6_P REG_GPUCORE_ISNS_6_N

RA170 RA171 RA172 RA173 RA174 RA175 RA176 RA177 RA178 RA179 RA180 RA181 RA198 RA199

REG_GPUCORE_ISNS_1_P_R REG_GPUCORE_ISNS_1_N_R REG_GPUCORE_ISNS_2_P_R REG_GPUCORE_ISNS_2_N_R REG_GPUCORE_ISNS_3_P_R REG_GPUCORE_ISNS_3_N_R REG_GPUCORE_ISNS_4_P_R REG_GPUCORE_ISNS_4_N_R REG_GPUCORE_ISNS_5_P_R REG_GPUCORE_ISNS_5_N_R REG_GPUCORE_ISNS_6_P_R REG_GPUCORE_ISNS_6_N_R

GPU:EL GPU:BF

REG_GPUCORE_ISNS_1_N_R REG_GPUCORE_ISNS_2_P_R REG_GPUCORE_ISNS_2_N_R REG_GPUCORE_ISNS_3_P_R REG_GPUCORE_ISNS_3_N_R REG_GPUCORE_ISNS_4_P_R REG_GPUCORE_ISNS_4_N_R REG_GPUCORE_ISNS_5_P_R REG_GPUCORE_ISNS_5_N_R REG_GPUCORE_ISNS_6_P_R REG_GPUCORE_ISNS_6_N_R

REG_GPUCORE_ISNS_1_P_R REG_GPUCORE_ISNS_2_P_R REG_GPUCORE_ISNS_3_P_R REG_GPUCORE_ISNS_4_P_R REG_GPUCORE_ISNS_5_P_R REG_GPUCORE_ISNS_6_P_R REG_GPUCORE_ISNS_1_N_R REG_GPUCORE_ISNS_2_N_R REG_GPUCORE_ISNS_3_N_R REG_GPUCORE_ISNS_4_N_R REG_GPUCORE_ISNS_5_N_R REG_GPUCORE_ISNS_6_N_R

VR_GPU_RCSP VR_GPU_RCSM VR_GPU_VSEN_R VR_GPU_RGND VR_GPU_TSEN1 SMB_VR_GPU_SDA_R SMB_VR_GPU_SCL_R VR_GPU_SVD_R VR_GPU_SVC_R VR_GPU_ADDR_PROT VR_GPU_RRES

VR_GPU_TSEN1_R

RA191 0 402 MF-LF

RTA107 47KOHM-1% 0603

RA107 13K 1% 1/16W MF-LF 402

CA126 0.1UF 10% 16V X7R-CERM 0402

OMIT SM XWA107

VR_GPU_TSEN_GND

RA140 13K 1% 1/10W MF-LF 402

VR_GPU_VINSEN

RA121 1K 1% 1/16W MF-LF 402

CA121 0.01UF 20% 16V X7R-CERM 0402

ACCESS POINT FOR I2C DEBUG HOOKUP PLACE GND TP NEARBY

TP_VR_GPU_SDA TP_VR_GPU_SCL

RA186 RA187 RA188 RA189 RA182 RA183 NOSTUFF

=SMB_VR_GPU_SDA =SMB_VR_GPU_SCL

VGD/TSEN2/VAUXSEN VR_GPU_VAUXSEN

CA124 0.01UF 10% 16V X7R-CERM 0402

RA124 1K 1% 1/16W MF-LF 402

RA142 13K 1% 1/16W MF-LF 402 =PP5V_S0_VRD

=PP1V8_S0_GPU_VDD

CA125 0.1UF 10% 16V X7R-CERM 0402

SV_ADDR/VDDIO SM_ALERT* SV_ALERT*/SVT VRHOT_ICRIT* VRDY1 VRDY2

NC

VR_GPU_SVT_R

RA184 0 402 MF-LF

VR_GPU_SVT

CA140 0.1UF 10% 6.3V X6S 0201

RA1101 4.99K 1% 1/16W MF-LF 402

RA143 4.99K 1% 1/16W MF-LF 402

RA144 4.99K 1% 1/16W MF-LF 402

=PP3V3_S0_REG_GPUCORE

RA145 4.99K 1% 1/16W MF-LF 402

NOSTUFF

RA145 SHOULD BE STUFFED IF RA185 IS NOSTUFF

RA185 0 402 MF-LF

VR_GPU_PHSFLT_L

VR_GPU_VRDHOT_L

PM_PGOOD_REG_GPUCORE_S0

PM_PGOOD_REG_GPU_VDDCI_S0

=PP12V_S0_GPUCORE_FLT

CA141 0.1UF 10% 6.3V X6S 0201 NOSTUFF

CA142 0.1UF 10% 6.3V X6S 0201

VR_GPU_SVD

VR_GPU_SVC

RA150 10K 5% 1/20W MF 201 NOSTUFF

RA151 10K 5% 1/20W MF 201

RA152 10K 5% 1/20W MF 201 NOSTUFF

RA153 10K 5% 1/20W MF 201

SVC	SVD	GPUCORE
0	0	1.1 V
0	1	1.0 V
1	0	0.9 V
1	1	0.8 V

DEFAULT->

PM_EN_REG_GPUCORE_S0

RA122 100K 1% 1/20W MF 201

CA122 0.001UF 20% 50V CERM 402

=PP3V3_S0_REG_GPUCORE

CA143 0.1UF 10% 6.3V X6S 0201

=PP1V8_S0_GPU_VDD

RA125 10K 1% 1/10W MF-LF 402

VR_GPU_SYS_PWROK

PM_PCH_PWROK

UA102 74LVC1G00GF SOT891

VR_GPU_RESET_L

VR_GPU_SYS_PWROK_L

QA102 2N7002TXG SOT-523-3

NC

RA123 845 1% 1/16W MF-LF 402

CA123 0.01UF 10% 16V CERM 402

GPU CORE INPUT Filtering

CRITICAL

LA100

0.105UH-20%-31A-0.00032OHM

=PP12V_S0_REG_GPUCORE PP12V_S0_GPUCORE_FLT

SSL0705M-SM

152-0110

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S01143	1	IR3566B/111-4204PBF, 6+1,EL	UA100	GPU:EL
353S01144	1	IR3566B/111-4205PBF, 4+1,BF	UA100	GPU:BF
132S0385	1	CAP, 560PF, X7R, 10%, 50V	CA101	GPU:EL
131S0463	1	CAP, 330PF	CA101	GPU:BF
114S0232	1	RES, 1.43K	RA101	GPU:EL
114S0248	1	RES, 2.1K	RA101	GPU:BF
114S0022	1	RES, 4.99 OHMS	RA103	GPU:EL
114S0138	1	RES, 154 OHMS	RA103	GPU:BF
114S0446	1	RES, 232K OHMS	RA108	GPU:EL
114S0411	1	RES, 100K OHMS	RA108	GPU:BF
114S0023	1	RES, 10 OHMS	RA102	GPU:EL
114S0138	1	RES, 154 OHMS	RA102	GPU:BF

BOM_COST_GROUP=GRAPHICS

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:11 2016

PAGE TITLE

GRAPHICS: GPU CORE VR

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DRAWING NUMBER 051-01506 SIZE D

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PAGE 101 OF 121

SHEET 88 OF 102

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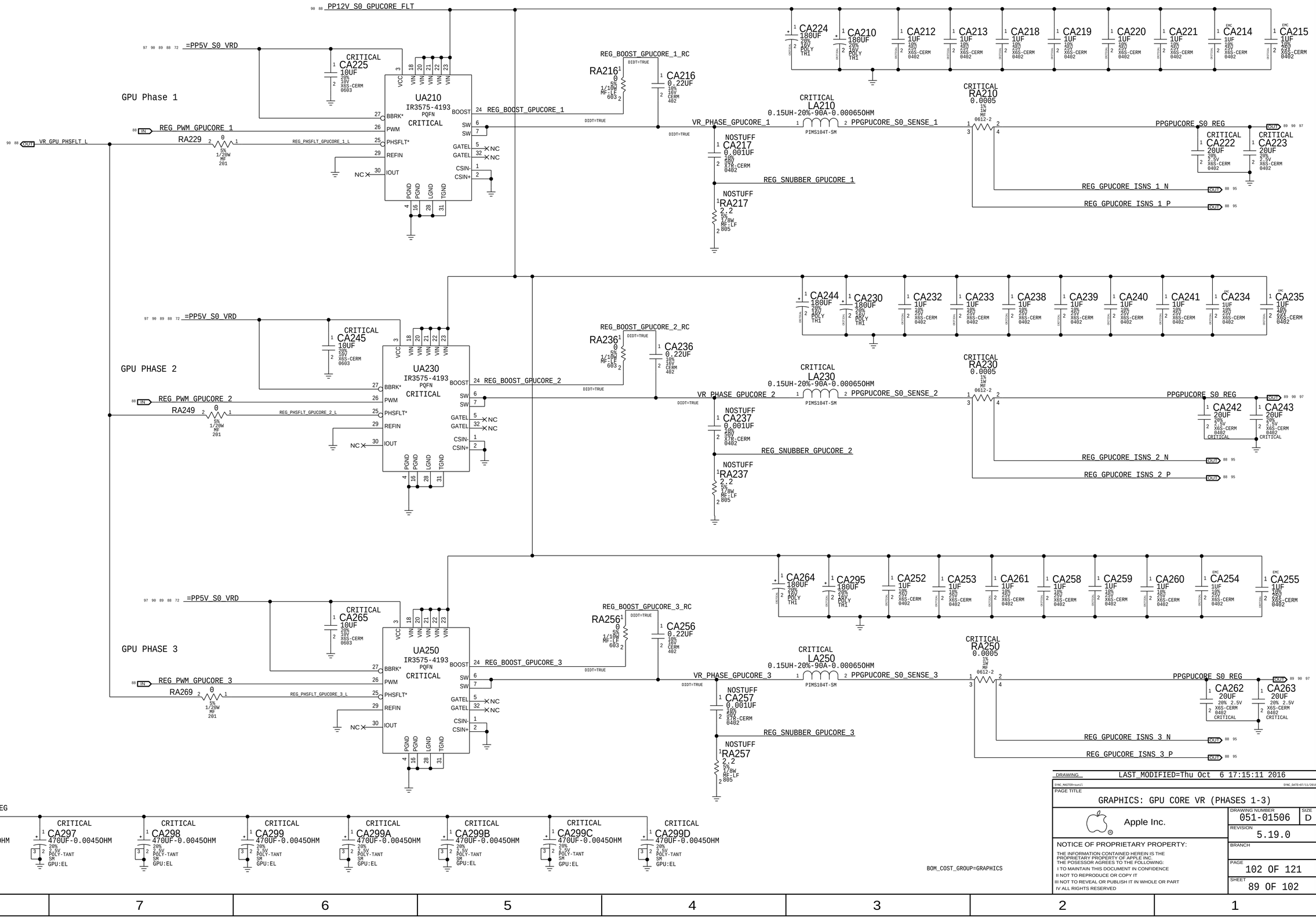
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GPU Phase 1
=PP5V_S0_VRD
CRITICAL CA225 10UF 20% 10V X6S-CERM 0603
UA210 IR3575-4193 PQFN CRITICAL
REG_BOOST_GPUCORE_1
REG_BOOST_GPUCORE_1_RC
RA216 0 5% 1/10W MF-LF 603
CA216 0.22UF 10% 16V CERM 402
REG_PWM_GPUCORE_1
RA229 0 5% 1/20W MF 201
VR_GPU_PHSFLT_L
REG_PHSFLT_GPUCORE_1_L
VR_PHASE_GPUCORE_1
CRITICAL LA210 0.15UH-20%-90A-0.000650HM PIMS104T-SM
PPGPUCORE_S0_SENSE_1
CRITICAL RA210 0.0005 1% 1W MF 0612-2
PPGPUCORE_S0_REG
CRITICAL CA222 20UF 20% 2.5V X6S-CERM 0402
CRITICAL CA223 20UF 20% 2.5V X6S-CERM 0402
NOSTUFF CA217 0.001UF 10% 50V X7R-CERM 0402
REG_SNUBBER_GPUCORE_1
NOSTUFF RA217 2.2 5% 1/8W MF-LF 805
REG_GPUCORE_ISNS_1_N
REG_GPUCORE_ISNS_1_P
CA224 180UF 20% 16V POLY TH1
CA210 180UF 20% 16V POLY TH1
CA212 1UF 10% 25V X6S-CERM 0402
CA213 1UF 10% 25V X6S-CERM 0402
CA218 1UF 10% 25V X6S-CERM 0402
CA219 1UF 10% 25V X6S-CERM 0402
CA220 1UF 10% 25V X6S-CERM 0402
CA221 1UF 10% 25V X6S-CERM 0402
CA214 1UF 10% 25V X6S-CERM 0402
CA215 1UF 10% 25V X6S-CERM 0402
GPU PHASE 2
=PP5V_S0_VRD
CRITICAL CA245 10UF 20% 10V X6S-CERM 0603
UA230 IR3575-4193 PQFN CRITICAL
REG_BOOST_GPUCORE_2
REG_BOOST_GPUCORE_2_RC
RA236 0 5% 1/10W MF-LF 603
CA236 0.22UF 10% 16V CERM 402
REG_PWM_GPUCORE_2
RA249 0 5% 1/20W MF 201
REG_PHSFLT_GPUCORE_2_L
VR_PHASE_GPUCORE_2
CRITICAL LA230 0.15UH-20%-90A-0.000650HM PIMS104T-SM
PPGPUCORE_S0_SENSE_2
CRITICAL RA230 0.0005 1% 1W MF 0612-2
PPGPUCORE_S0_REG
CA242 20UF 20% 2.5V X6S-CERM 0402 CRITICAL
CA243 20UF 20% 2.5V X6S-CERM 0402 CRITICAL
NOSTUFF CA237 0.001UF 10% 50V X7R-CERM 0402
REG_SNUBBER_GPUCORE_2
NOSTUFF RA237 2.2 5% 1/8W MF-LF 805
REG_GPUCORE_ISNS_2_N
REG_GPUCORE_ISNS_2_P
CA244 180UF 20% 16V POLY TH1
CA230 180UF 20% 16V POLY TH1
CA232 1UF 10% 25V X6S-CERM 0402
CA233 1UF 10% 25V X6S-CERM 0402
CA238 1UF 10% 25V X6S-CERM 0402
CA239 1UF 10% 25V X6S-CERM 0402
CA240 1UF 10% 25V X6S-CERM 0402
CA241 1UF 10% 25V X6S-CERM 0402
CA234 1UF 10% 25V X6S-CERM 0402
CA235 1UF 10% 25V X6S-CERM 0402
GPU PHASE 3
=PP5V_S0_VRD
CRITICAL CA265 10UF 20% 10V X6S-CERM 0603
UA250 IR3575-4193 PQFN CRITICAL
REG_BOOST_GPUCORE_3
REG_BOOST_GPUCORE_3_RC
RA256 0 5% 1/10W MF-LF 603
CA256 0.22UF 10% 16V CERM 402
REG_PWM_GPUCORE_3
RA269 0 5% 1/20W MF 201
REG_PHSFLT_GPUCORE_3_L
VR_PHASE_GPUCORE_3
CRITICAL LA250 0.15UH-20%-90A-0.000650HM PIMS104T-SM
PPGPUCORE_S0_SENSE_3
CRITICAL RA250 0.0005 1% 1W MF 0612-2
PPGPUCORE_S0_REG
CA262 20UF 20% 2.5V X6S-CERM 0402 CRITICAL
CA263 20UF 20% 2.5V X6S-CERM 0402 CRITICAL
NOSTUFF CA257 0.001UF 10% 50V X7R-CERM 0402
REG_SNUBBER_GPUCORE_3
NOSTUFF RA257 2.2 5% 1/8W MF-LF 805
REG_GPUCORE_ISNS_3_N
REG_GPUCORE_ISNS_3_P
CA264 180UF 20% 16V POLY TH1
CA295 180UF 20% 16V POLY TH1
CA252 1UF 10% 25V X6S-CERM 0402
CA253 1UF 10% 25V X6S-CERM 0402
CA261 1UF 10% 25V X6S-CERM 0402
CA258 1UF 10% 25V X6S-CERM 0402
CA259 1UF 10% 25V X6S-CERM 0402
CA260 1UF 10% 25V X6S-CERM 0402
CA254 1UF 10% 25V X6S-CERM 0402
CA255 1UF 10% 25V X6S-CERM 0402
PPGPUCORE_S0_REG
CA296 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA297 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA298 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA299 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA299A 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA299B 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA299C 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
CA299D 470UF-0.00450HM 20% 2.5V POLY-TANT SM GPU:EL
BOM_COST_GROUP=GRAPHICS
DRAWING
LAST_MODIFIED=Thu Oct 6 17:15:11 2016
PAGE TITLE
GRAPHICS: GPU CORE VR (PHASES 1-3)
Apple Inc.
DRAWING NUMBER 051-01506
SIZE D
REVISION 5.19.0
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PAGE 102 OF 121
SHEET 89 OF 102
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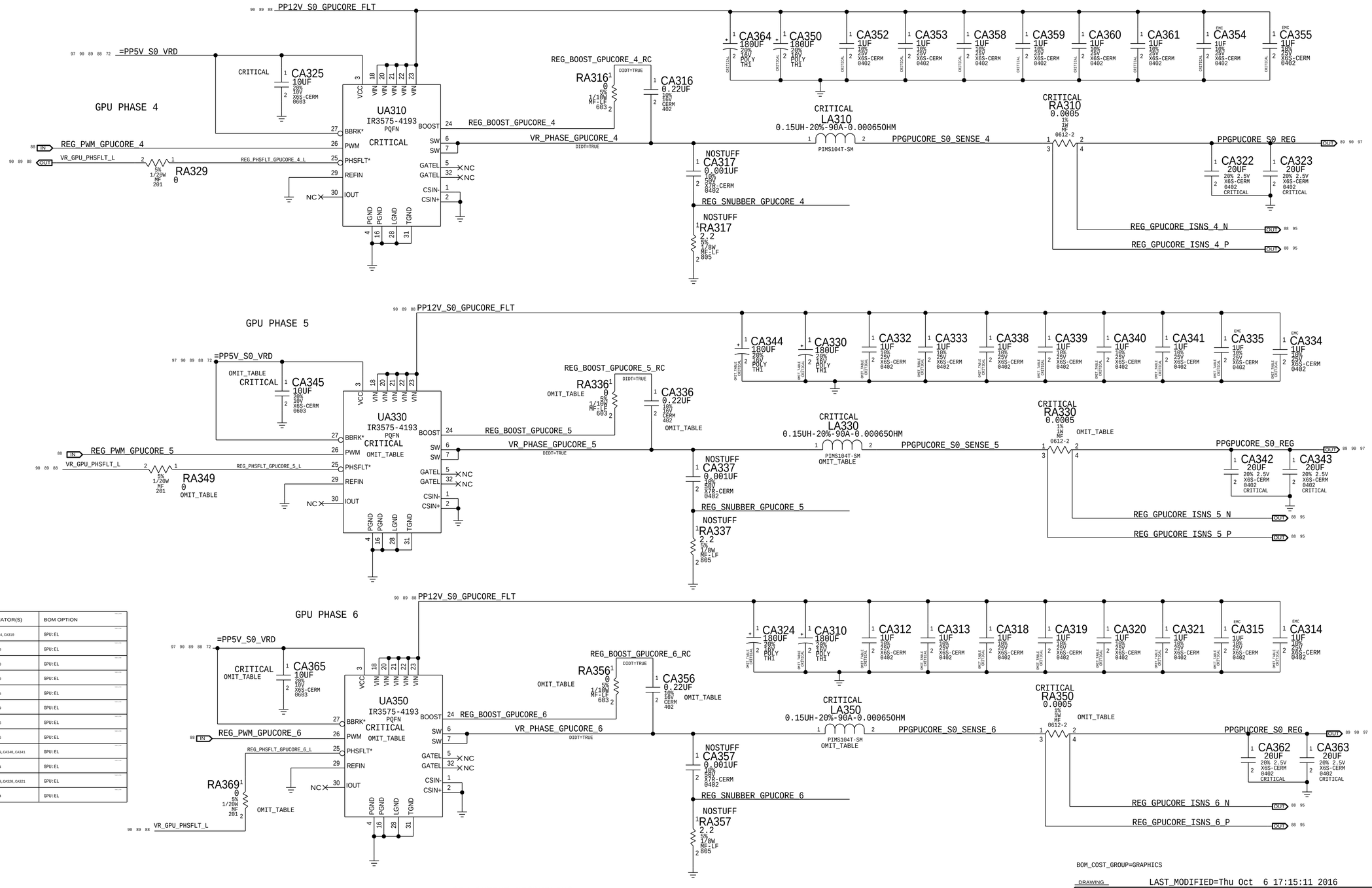

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
124-0403	4	CAP,180UF,16V, POLY, TH1	CA344,CA330,CA324,CA310	GPU:EL
107S00039	2	RES,0.0005,1%,1W,0612-2	RA330,RA350	GPU:EL
152S00302	2	IND,150NH,0.00064	LA330,LA350	GPU:EL
353S01013	2	IC,IR3575M	UA330,UA350	GPU:EL
138S0768	2	CAP,10UF,10V,X6S,0603	CA345,CA365	GPU:EL
117S0002	2	RES,0,5%,201	RA349,RA369	GPU:EL
113S0022	2	RES,0,5%,603	RA336,RA356	GPU:EL
132S0301	2	CAP,0.22UF,16V,402	CA336,CA356	GPU:EL
138S0775	6	CAP,1UF,25V,X6S,402	CA332,CA333,CA338,CA339,CA340,CA341	GPU:EL
138S0775	2	CAP,1UF,25V,X6S,402	CA335,CA334	GPU:EL
138S0775	6	CAP,1UF,25V,X6S,402	CA312,CA313,CA318,CA319,CA320,CA321	GPU:EL
138S0775	2	CAP,1UF,25V,X6S,402	CA315,CA314	GPU:EL

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DRAWING	LAST_MODIFIED=Thu Oct 6 17:15:11 2016
SYNC_MASTER=SUN1L	SYNC_DATE=07/15/2016
PAGE TITLE	GRAPHICS: GPU CORE VR (PHASES 4-6)
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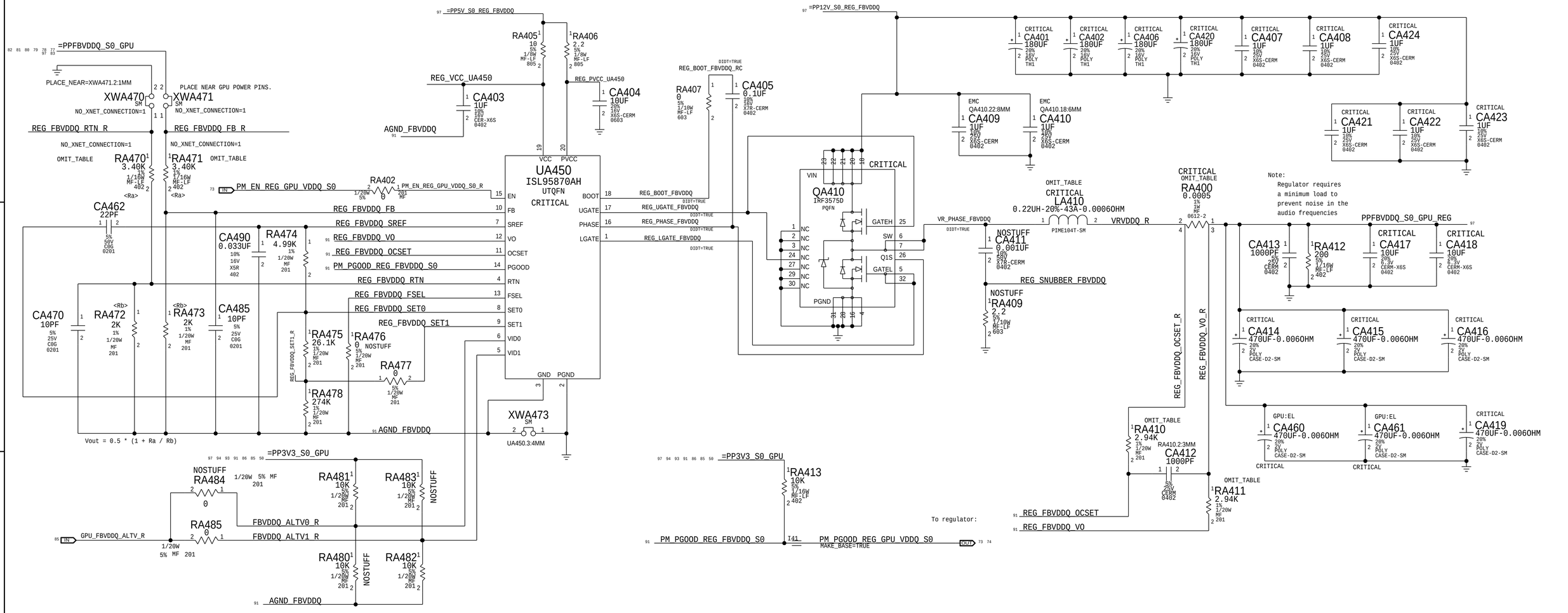
	SHEET 90 OF 102

GPU VDDQ SUPPLY



VID 1	VID 0	GPU VDDQ
1	1	1.35 V
DEFAULT -> 0	1	1.5 V

GPU VDDQ	RA470 = RA471
1.5 V	3.4K
1.55 V	3.57K
1.6 V	3.74K

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S00176	1	IND,0.22UH,43A,0.64mOHM	LA410	GPU:EL
152S1799	1	IND,0.36UH,40A,0.8mOHM	LA410	GPU:BF
107S00039	1	RES,0.0005OHM,1%,0612-4T	RA400	GPU:EL
107S0255	1	RES,0.001OHM,1%,0612-4T	RA400	GPU:BF
114S0269	2	RES,3.4K,1%,0402	RA470, RA471	VR_VRAM_HYN_1V50
114S0271	2	RES,3.57K,1%,0402	RA470, RA471	VR_VRAM_HYN_1V55
114S0269	2	RES,3.4K,1%,0402	RA470, RA471	VR_VRAM_SAM_1V50
114S0271	2	RES,3.57K,1%,0402	RA470, RA471	VR_VRAM_SAM_1V55
114S0271	2	RES,3.57K,1%,0402	RA470, RA471	VR_VRAM_ELP_1V55
114S0273	2	RES,3.74K,1%,0402	RA470, RA471	VR_VRAM_SAM_1V60
114S0273	2	RES,3.74K,1%,0402	RA470, RA471	VR_VRAM_ELP_1V60
118S0269	2	RES,2.94K	RA410, RA411	GPU:EL
118S0269	2	RES,2.94K	RA410, RA411	GPU:BF
114S0269	2	RES,3.4K	RA470, RA471	VR_VRAM_ELP_1V50

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PAGE TITLE
GRAPHICS: GPU VDDQ VR

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REVISION 5.19.0
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PAGE 104 OF 121
SHEET 91 OF 102

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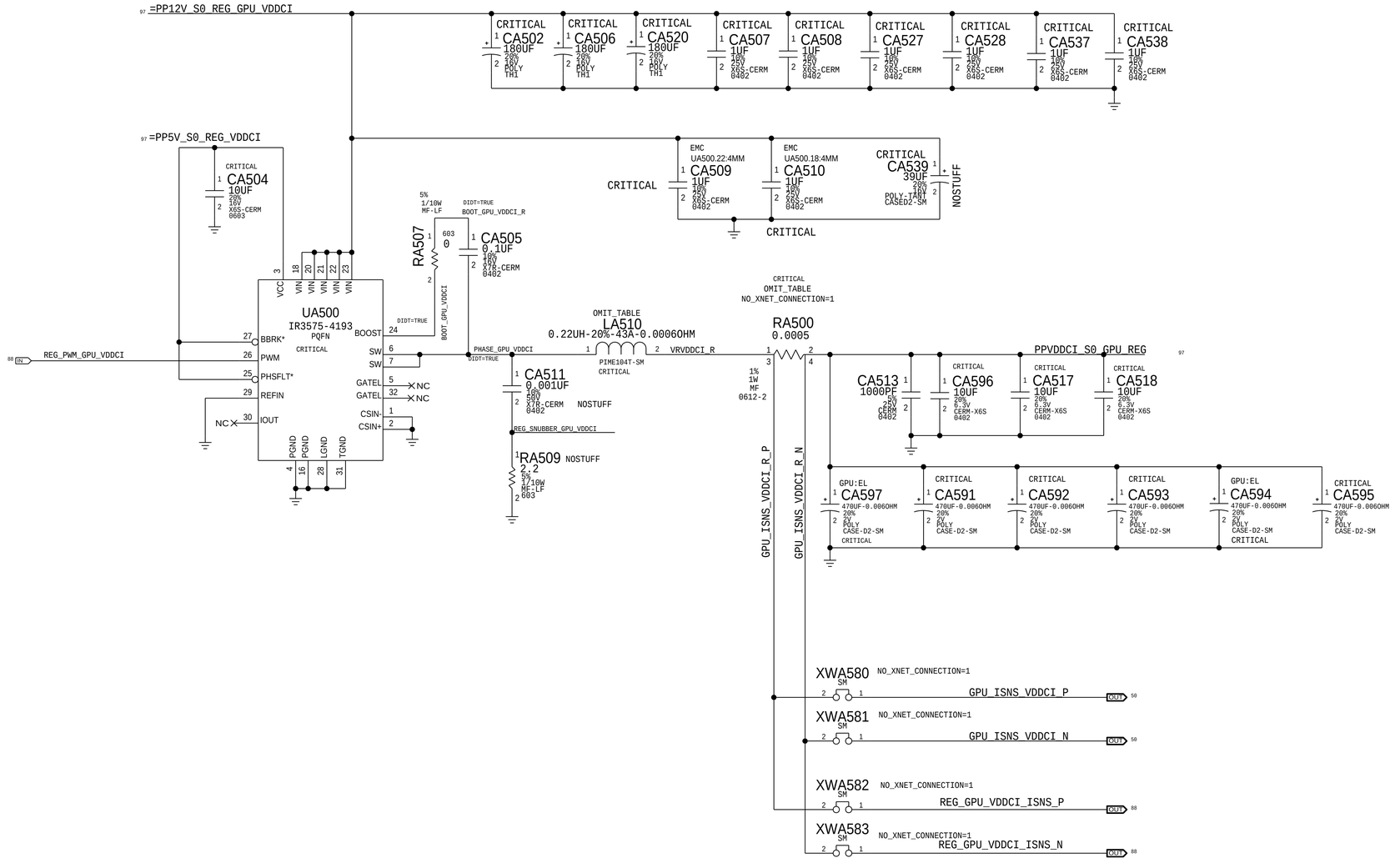
GPU VDDCI SUPPLY



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S00176	1	IND,0.22UH,43A,0.64mOHM	LA510	GPU:EL
152S1799	1	IND,0.36UH,40A,0.8mOHM	LA510	GPU:BF
107S00039	1	RES,0.0005OHM,1%,0612-4T	RA500	GPU:EL
107S0255	1	RES,0.001OHM,1%,0612-4T	RA500	GPU:BF

BOM_COST_GROUP=GRAPHICS

PAGE TITLE
GRAPHICS: GPU VDDCI VR

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SIZE D
REVISION 5.19.0
BRANCH
PAGE 105 OF 121
SHEET 92 OF 102

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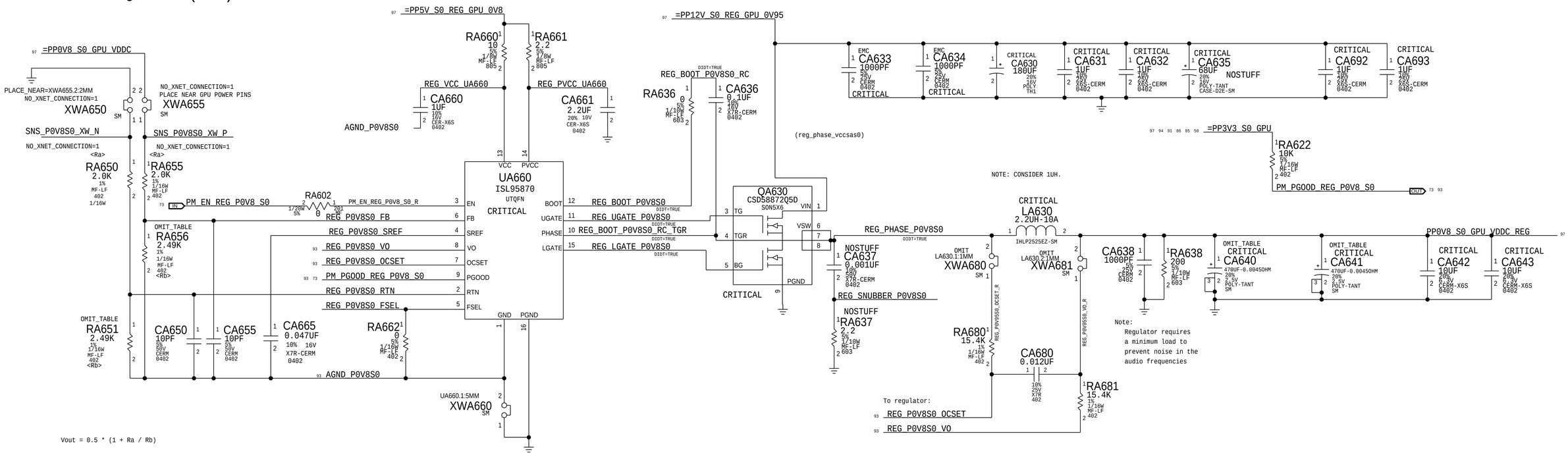
GPU VDDC (0.8V) S0 REGULATOR

GPU VDDC = 0.8V @ 4.6A (EDP)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0358	2	CAP,470uF,0.006OHM,2V,D2	CA640,CA641	VR_BULKCAP:CURRENT
128S0381	2	CAP,470uF,0.0045OHM,2.5V,SM	CA640,CA641	VR_BULKCAP:FUTURE
114S0256	2	RES, 2.49K	RA651, RA656	GPU:EL
114S0262	2	RES, 2.87K	RA651, RA656	GPU:BF

BOM_COST_GROUP=GRAPHICS

DRAWING LAST_MODIFIED=Thu Oct 6 17:15:12 2016

PAGE TITLE: GRAPHICS: GPU 0V8 VR

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BRANCH:

PAGE: 106 OF 121

SHEET: 93 OF 102

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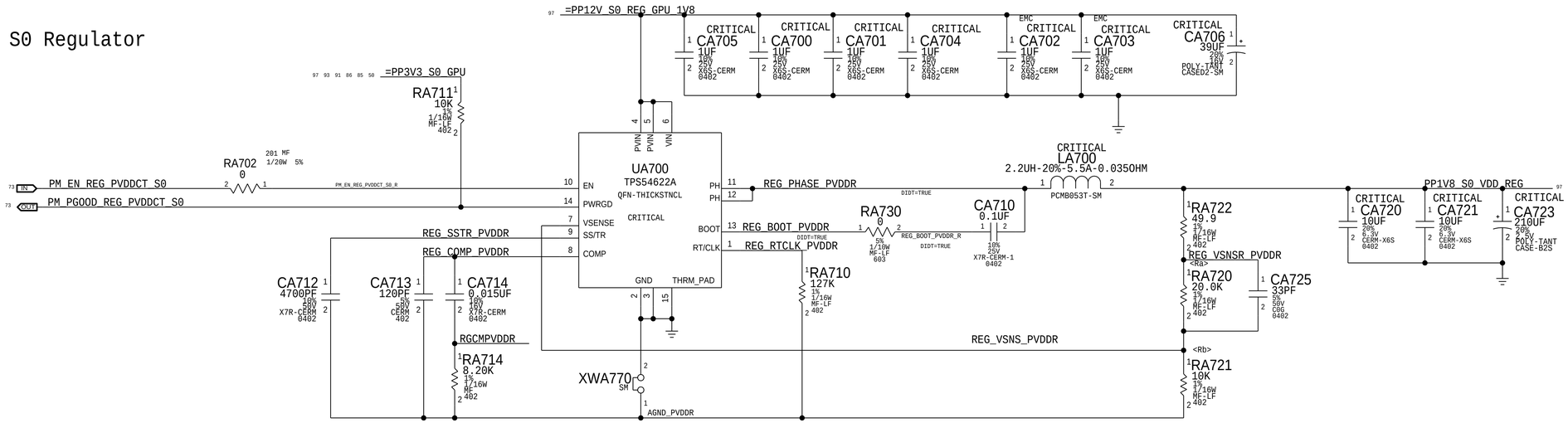
1.8V S0 Regulator



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DRAWING LAST_MODIFIED=Thu Oct 6 17:15:12 2016

SYNC_MASTER=SUNCL SYNC_DATE=07/14/2016

PAGE TITLE

GRAPHICS: GPU 1V8 VR

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PAGE 107 OF 121

SHEET 94 OF 102

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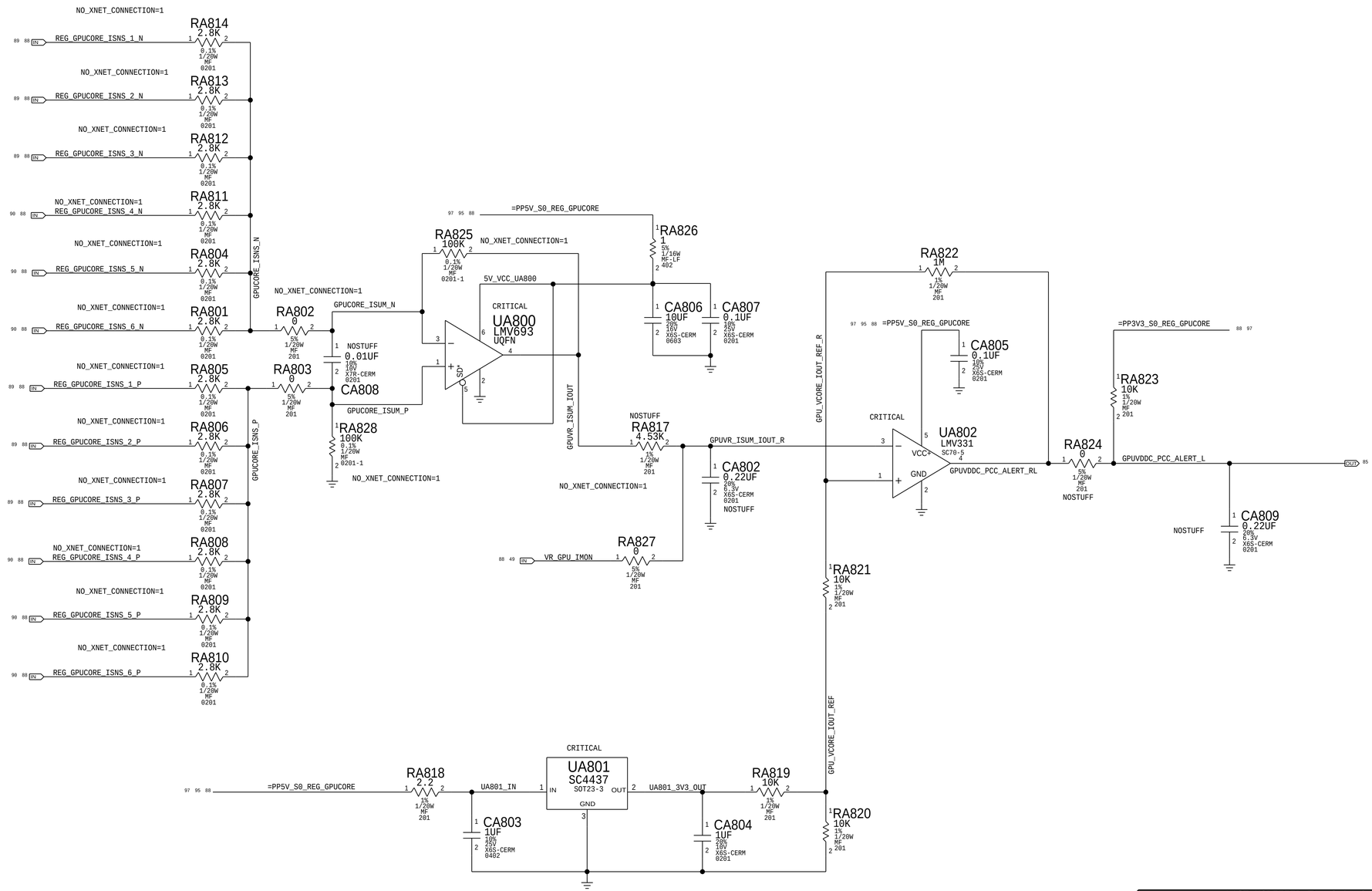

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DRAWING NUMBER	051-01506
SIZE	D
REVISION	5.19.0
BRANCH	
PAGE	108 OF 121
SHEET	95 OF 102

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LAST_MODIFIED=Thu Oct 6 17:00:43 2016

SYNC_MASTER=sunil

SYNC_DATE=01/13/2016

PAGE TITLE

GRAPHICS: GPU VR MISC

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PAGE 109 OF 121

SHEET 96 OF 102

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G3 Rails

Always on: Keeps the PCH RTC alive

PP3V3_G3 (MAKE_BASE=TRUE)
- PP3V3_G3_RTC — 18
- =PP3V3_G3_PCH_RTC — 12 15

G3H Rails

On with AC/DC plugged in

PP12V_ACDC (MAKE_BASE=TRUE)
- PP12V_G3H_ACDC — 61
- =PP12V_G3H_SNS_R — 49

PP12V_G3H (MAKE_BASE=TRUE)
- PP12V_G3H_SNS — 49
- =PP12V_G3H_P3V3 — 61
- =PP12V_G3H_FET_P12V_S0 — 72
- =PP12V_G3H_FET_P12V_S5 — 72

PP3V3_G3H (MAKE_BASE=TRUE)
- PP3V3_G3H_REG — 61
- =PP3V3_G3H_BT — 33
- =PP3V3_G3H_RTC_D — 18
- =PP3V3_G3H_SMC — 45 46 53
- =PP3V3_G3H_SMC_USBMUX — 43
- =PP3V3_G3H_T112 — 47
- =PP3V3_G3H_LED — 3
- =PP3V3_G3H_UPC_XA — 29
- =PP3V3_G3H_UPC_XB — 30
- =PP3V3_G3H_SMB — 48
- =PP3V3_G3H_PWRCTL — 73

Ground/Common

GND (MAKE_BASE=TRUE)

S5 Rails

On when in S5

PP12V_S5 (MAKE_BASE=TRUE)
- PP12V_S5_FET — 72
- =PP12V_S5_FET_P3V3_S5_P5V_S4 — 67
- =PP12V_S5_REG_VDDQ_S3 — 65
- =PP12V_S5_PWRCTL — 73
- =PPHV_SW_TBT_X — 28
- =PP12V_S5_SNS — 49
- =PP12V_S5_REG_P1V0_S5 — 68

PP5V_S5 (MAKE_BASE=TRUE)
- PP5V_S5_LDO — 67
- =PP5V_S5_REG_P1V0 — 68
- =PP5V_S5_PCC — 52

PP3V3_S5_SNS (MAKE_BASE=TRUE)
- PP3V3_S5_REG — 67
- =PP3V3_S5_REG_SNS_R — 50

PP3V3_S5 (MAKE_BASE=TRUE)
- PP3V3_S5_REG_SNS — 50
- =PP3V3_S5_FET_P3V3_S0 — 72
- =PP3V3_S5_FET_P3V3_S4 — 72
- =PP3V3_S5_SMC — 34 46
- =PP3V3_S5_PCH_VCC — 12 13 14 15 19
- =PP3V3_S5_PWRCTL — 34 72 73 74
- =PP3V3_S5_ROM — 17 47
- =PP3V3_S5_VRD — 67 68
- =PP3V3_S5_SENSE — 49 52
- =PP3V3_S5_RSTBUF — 19
- =PP3V3_S5_LDO_P1V8_S5 — 68
- =PP3V3_S5_SSD — 34
- =PP3V3_S5_REG_P2V5_S3 — 65

PP1V8_S5 (MAKE_BASE=TRUE)
- PP1V8_S5_LDO — 68
- =PP1V8_S5_PCH_VCCHDA — 15 18
- =PP1V8_S5_PWRCTL — 73

PP1V0_S5 (MAKE_BASE=TRUE)
- PP1V0_S5_REG — 68
- =PP1V0_S5_PCH_VCC — 11 15
- =PP1V0_S5_XDP — 17
- =PP1V0_S5_FET_PP1V0_S3 — 72

S4 Rails

Enabled when system has AC and is in run or sleep

PP5V_S4_SNS (MAKE_BASE=TRUE)
- PP5V_S4_REG — 67
- =PP5V_S4_REG_SNS_R — 50

PP5V_S4 (MAKE_BASE=TRUE)
- PP5V_S4_REG_SNS — 50
- =PP5V_S4_REG_VDDQ_S3 — 65
- =PP5V_S4_FET_P5V_S0 — 72
- =PP5V_S4_CPUCOREVR — 62
- =PP5V_S4_USB — 43 44
- =PP5V_S4_USBC — 28
- =PP5V_S4_PWRCTL — 72

PP3V3_S4 (MAKE_BASE=TRUE)
- PP3V3_S4_FET — 72
- =PP3V3_S4_FET_ENET — 37
- =PP3V3_S4_PWRCTL — 65 72
- =PP3V3_S4_LED — 3
- =PP3V3_S4_MEMRESET — 6
- =PP3V3_S4_PCH_VCC — 12 13 14
- =PP3V3_S4_SDCARD — 38
- =PP3V3_S4_SMC — 46
- =PP3V3_S4_TBT — 19
- =PP3V3_UPC_XA_AUX — 29
- =PP3V3_UPC_XB_AUX — 30
- =PP3V3_S4_AP — 33 46

97 34 PP3V3_S4_FET_SSD (MAKE_BASE=TRUE)
- PP3V3_S4_FET_SSD — 34 97
- =PPVSSD_S4_REG_SNS_R — 50

PP3V3_S4_SSD (MAKE_BASE=TRUE)
- PPVSSD_S4_REG_SNS — 50
- =PP3V3_S4_SSD_CONN — 34

S3 Rails

Enabled when system is in run or sleep

PP3V3_ENET (MAKE_BASE=TRUE)
- PP3V3_ENET_FET — 37
- =PP3V3_ENET_PHY — 36 37

PPVDDQ_S3 (MAKE_BASE=TRUE)
- PPVDDQ_S3_REG — 65
- =PPVDDQ_S3_SNS_DDR_R — 50

THIS IS 1.2V RAIL
- =PP1V2_S3_SNS_R — 50

PPVDDQ_S3_DDR (MAKE_BASE=TRUE)
- PPVDDQ_S3_SNS_DDR — 50
- =PPDDR_S3_MEMVREF — 20

THIS IS 1.2V RAIL
- =PPVDDQ_S3_MEM_A — 21 22
- =PPVDDQ_S3_MEM_B — 23 24
- =PPVDDQ_S3_MEMRESET — 12
- =PPVDDQ_S3_LDO_DDRVTT — 65

PP1V2_S3_CPU (MAKE_BASE=TRUE)
- PP1V2_S3_SNS — 50
- =PPVDDQ_S3_CPU — 8 10

PP1V0_S3_CPU_VCCST (MAKE_BASE=TRUE)
- PP1V0_S3_FET — 72
- =PP1V0_S3_VCCST — 12 62
- =PP1V0_S3_SMC — 46
- =PP1V0_S3_XDP — 17
- =PP1V0_S3_CPU — 6 8 10

PP2V5_S3 (MAKE_BASE=TRUE)
- PP2V5_S3_REG — 65
- =PPDDRVPP_S3_MEM_A — 21 22
- =PPDDRVPP_S3_MEM_B — 23 24

S0 Rails

Enabled when system is in run

PP12V_S0 (MAKE_BASE=TRUE)
- PP12V_S0_FET — 72
- =PP12V_S0_AUDIO_SPKRAMP — 56 57
- =PP12V_S0_FAN — 53
- =PP12V_S0_SNS_GPUCORE_R — 49
- =PP12V_S0_LCD — 41
- =PP12V_S0_CPUCORE — 62
- =PP12V_S0_SNS_FBVDDQ_R — 49
- =PP12V_S0_HDD — 35
- =PP12V_S0_BKLT — 70

PP12V_S0_GPU_AUX (MAKE_BASE=TRUE)
- PP12V_S0_SNS_GPU_AUX — 49
- =PP12V_S0_REG_GPU_0V95 — 93
- =PP12V_S0_REG_GPU_1V8 — 94

PP12V_S0_GPUCORE (MAKE_BASE=TRUE)
- PP12V_S0_SNS_GPUCORE — 49
- =PP12V_S0_REG_GPUCORE — 88
- =PP12V_S0_REG_GPU_VDDCI — 92

97 62 PP12V_S0_CPUCORE_FLT (MAKE_BASE=TRUE)
- PP12V_S0_CPUCORE_FLT — 62 97
- =PP12V_S0_SNS_CPUCORE_R — 49

PP12V_S0_CPUCORE (MAKE_BASE=TRUE)
- PP12V_S0_SNS_CPUCORE — 49
- =PP12V_S0_REG_CPUCORE — 62 63 64 66 69

PPHDD12_S0 (MAKE_BASE=TRUE)
- PP12V_S0_HDD_FET — 35
- =PP12V_S0_HDD_SNS_R — 50

PP12V_S0_HDD (MAKE_BASE=TRUE)
- PP12V_S0_HDD_SNS — 50
- =PP12V_S0_HDD_PWR — 35

PP12V_S0_FBVDDQ (MAKE_BASE=TRUE)
- PP12V_S0_SNS_FBVDDQ — 49
- =PP12V_S0_REG_FBVDDQ — 91

PP5V_S0 (MAKE_BASE=TRUE)
- PP5V_S0_FET — 72
- =PP5V_S0_BKLT — 70
- =PP5V_S0_ISENSE — 49
- =PP5V_S0_REG_CPUVCC_S0 — 63 64 66 69
- =PP5V_S0_REG_FBVDDQ — 91
- =PP5V_S0_HDD — 35
- =PP5V_S0_VRD — 72 88 89 90
- =PP5V_S0_REG_VDDCI — 92
- =PP5V_S0_REG_GPU_0V8 — 93
- =PP5V_S0_CAMERA — 39
- =PP5V_S0_REG_GPUCORE — 88 95

PPHDD_S0 (MAKE_BASE=TRUE)
- PP5V_S0_HDD_FET — 35
- =PPHDD_S0_SNS_R — 50

PP5V_S0_HDD (MAKE_BASE=TRUE)
- PPHDD_S0_SNS — 50
- =PP5V_S0_HDD_PWR — 35

PP3V3_S0 (MAKE_BASE=TRUE)
- PP3V3_S0_FET — 72
- =PP3V3_S0_AUDIO — 55 58
- =PP3V3_S0_BKLT — 70
- =PP3V3_S0_DP — 41
- =PP3V3_S0_ENET — 36
- =PP3V3_S0_FAN — 53
- =PP3V3_S0_GPU — 50 85 86 91 93 94
- =PP3V3_S0_INTDPMUX — 41 42
- =PP3V3_S0_LED — 3
- =PP3V3_S0_LED_SATA — 35
- =PP3V3_S0_MEM_A_SPD — 21 22
- =PP3V3_S0_MEM_B_SPD — 23 24
- =PP3V3_S0_PCH_VCC — 11 12 14 15 18
- =PP3V3_S0_PWRCTL — 72
- =PP3V3_S0_RSTBUF — 19
- =PP3V3_S0_SDCARD — 38
- =PP3V3_S0_SENSE — 35 49 50 51
- =PP3V3_S0_SMBUS — 48
- =PP3V3_S0_SMBUS_SMC_0 — 48
- =PP3V3_S0_SMBUS_SMC_1 — 48
- =PP3V3_S0_SMBUS_SMC_3 — 48
- =PP3V3_S0_SMC — 46
- =PP3V3_S0_VRD — 61 62 66
- =PP3V3_S0_CAMERA — 39 40
- =PP3V3_S0_ALS — 39
- =PP3V3_S0_SMBUS_SMC_2 — 48
- =PP3V3_S0_REG_GPUCORE — 88 95
- =PP3V3_TBT_X_S0_R — 28

PPDDRVTT_S0 (MAKE_BASE=TRUE)
- PPDDRVTT_S0_LDO — 65
- =PPDDRVTT_S0_MEM_A — 21 22
- =PPDDRVTT_S0_MEM_B — 23 24

GPU Rails (S0)

Enabled when system is in run

PPGPUCORE_S0 (MAKE_BASE=TRUE)
- PPGPUCORE_S0_REG — 89 90
- =PPGPUCORE_S0 — 76
- =PPGPUCORE_S0_SNS — 49

FBVDDQ IS 1.5V

PPFBVDDQ_S0_GPU (MAKE_BASE=TRUE)
- PPFBVDDQ_S0_GPU_REG — 91
- =PPFBVDDQ_S0_GPU — 77 78 79 80 81 82 83 91

PPVDDCI_S0_GPU (MAKE_BASE=TRUE)
- PPVDDCI_S0_GPU_REG — 92
- =PPVDDCI_S0_GPU — 77

PP0V8_S0_GPU_VDDC (MAKE_BASE=TRUE)
- PP0V8_S0_GPU_VDDC_REG — 93
- =PP0V8_S0_GPU_VDDC — 93
- =PP0V8_S0_GPU_PCIEVDDC — 75
- =PP0V8_S0_GPU_DPZVDD — 84

PP1V8_S0_MISC (MAKE_BASE=TRUE)
- PP1V8_S0_VDD_REG — 94
- =PP1V8_S0_AUD_DIG_TOP — 39 54 56
- =PP1V8_S0_AUD_DIG_BOT — 54 55 57 58
- =PP1V8_S0_CAMERA — 39 40
- =PP1V8_S0_DP — 42
- =PP1V8_S0_GPU_VDD — 85 88
- =PP1V8_S0_GPU_PEX_IO — 75
- =PP1V8_S0_GPU — 86

CPU/PCH Rails (S0)

Enabled when system is in run

PPVCC_S0_CPU (MAKE_BASE=TRUE)
- PPVCC_S0_CPU_REG — 63
- =PPVCC_S0_CPU — 8 10
- =PPVCC_S0_SNS_CPU — 49

PPVCCGT_S0_CPU (MAKE_BASE=TRUE)
- PPVCCGT_S0_CPU_REG — 64
- =PPVCCGT_S0_CPU — 8 10
- =PPVCCGT_S0_SNS_CPU — 49

PPVCCSA_S0_CPU (MAKE_BASE=TRUE)
- PPVCCSA_S0_CPU_REG — 69
- =PPVCCSA_S0_CPU — 8 10

PPVCCIO_S0 (MAKE_BASE=TRUE)
- PPVCCIO_S0_CPU_REG — 66
- =PPVCCIO_S0_SNS_CPU_R — 49

PPVCCIO_S0_CPU (MAKE_BASE=TRUE)
- PPVCCIO_S0_SNS_CPU — 49
- =PPVCCIO_S0_CPU — 5 8 10 17

SYNC_MASTER=sunil SYNC_DATE=11/06/2015

PAGE TITLE: Power Connectors/Aliases

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REVISION: 5.19.0
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PAGE: 110 OF 121
SHEET: 97 OF 102

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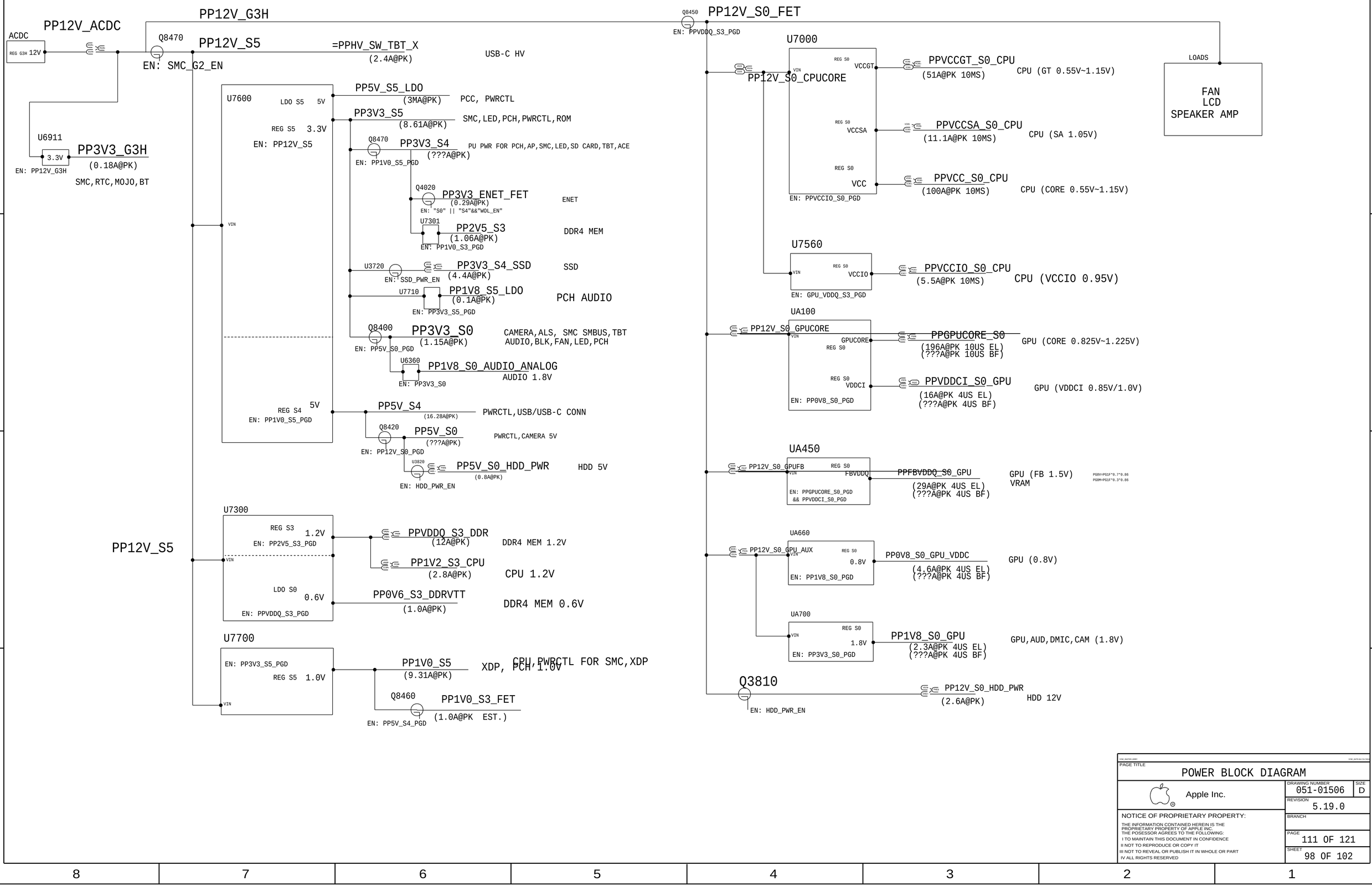

PAGE TITLE

POWER BLOCK DIAGRAM

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PAGE 111 OF 121

SHEET 98 OF 102

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PEG aliases

75 PEG_D2R_N<0..15> ⟹ =PEG_D2R_N<15..0> 5
MAKE_BASE=TRUE

75 PEG_D2R_P<0..15> ⟹ =PEG_D2R_P<15..0> 5
MAKE_BASE=TRUE

75 PEG_R2D_C_N<0..15> ⟹ =PEG_R2D_C_N<15..0> 5
MAKE_BASE=TRUE

75 PEG_R2D_C_P<0..15> ⟹ =PEG_R2D_C_P<15..0> 5
MAKE_BASE=TRUE

SYNC_MASTER=ANDREW SYNC_DATE=07/14/2016

PAGE TITLE

Signal Aliases

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PAGE 112 OF 121

SHEET 99 OF 102

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CPU Reserved

17 6 CPU_CFG<15..12> —— TP_CPU_CFG<15..12>
MAKE_BASE=TRUE

12 TP_CLINK_DATA —— CLINK_DATA
MAKE_BASE=TRUE NO_TEST=1

12 TP_CLINK_CLK —— CLINK_CLK
MAKE_BASE=TRUE NO_TEST=1

12 TP_CLINK_RESET_L —— CLINK_RESAT_L
MAKE_BASE=TRUE NO_TEST=1

12 TP_PCH_SLP_LAN_L —— PCH_SLP_LAN_L
MAKE_BASE=TRUE NO_TEST=1

11 TP_PCH_2 —— PCH_2
MAKE_BASE=TRUE NO_TEST=1

12 TP_PM_SLP_A_L —— PM_SLP_A_L
MAKE_BASE=TRUE

PCH Clocks

TP_ITPXDP_CLK100MP —— ITPXDP_CLK100M_P 11 17
MAKE_BASE=TRUE

TP_ITPXDP_CLK100MN —— ITPXDP_CLK100M_N 11 17
MAKE_BASE=TRUE

SYNC_MASTER=ANDREW SYNC_DATE=04/14/2010

PAGE TITLE

Unused Signal Aliases

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PAGE 114 OF 121

SHEET 100 OF 102

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J135 ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
377S0147	377S0126		ALL	USB Diode Array	=> USB-A
197S0481	197S0480		ALL	25MHX PCH XTAL	=> ETHERNET
197S0479	197S0478		ALL	12MHZ CAM/NXP XTAL	=> CAMERA,SMC
372S0186	372S0185		ALL	Alternate Temp Diode	=> SENSOR(TEMP)
107S00030	107S00029		ALL	Alt 5mOhm sense	=> SENSORS(V/I)
341S00016	341S3912		U3990	IC,ENETROM,ADESTO,V1,15,J78	=> ETHERNET
138S0681	138S0638		ALL	10uF Caps	=> AUDIO
138S0860	138S0775		ALL	Single-source 1uF 402	=> CPU&CHIPSET,USBC, PLATFORM POWER,GRAPHICS
378S0391	378S0390		ALL	DEGUG LEDS	=> CPU&CHIPSET(LED)
311S00008	311S0271		ALL	AND GATE	=> GRAPHICS(GPU)
155S0830	155S0316		ALL	FERR BD,600 OHM,25,0.5A	=> ETHERNET
376S00016	376S0644		ALL	MOSFET,P-CH,20V,1A	=> CPU&CHIPSET(RTC)
376S00001	376S0659		ALL	MOSFET,P-CH,20V	=> CPU&CHIPSET(RTC)
376S00015	376S0610		ALL	XSTR,FET,DL NCH,60V	=> CPU&CHIPSET(LED), ETHERNET
311S00004	311S0370		ALL	Config. Logic Gate	=> CPU&CHIPSET(ROM)
132S0447	132S0448		ALL	0.1uF 0201	=> GRAPHICS (GPU)
138S0855	138S0750		ALL	4.7uF 0402	=> GRAPHICS (GPU)
311S00072	311S0657		ALL	Dual 2 INP AND GATE	
376S00122	376S0843		Q8450	NFET,30V.49A,PQFN8	=> PLATFORM POWER(PASS FET)
376S00124	376S00200		Q8470	NFET,30V.40A,8P DFN	=> PLATFORM POWER(PASS FET)
311S00118	311S0489		Q8601	DUAL INPUT AND GATE	=> POWER SEQUENCE
128S0444	128S0323		C4602,C4702	330UF, 6.3V D2E	=> USB-A
371S0718	371S0531		ALL	DIODE,BAT54,SHOTTKY,30V	=> POWER SEQUENCE
377S0124	377S0057		ALL	VARISTOR,6VB,2UA,100PF,	=> AC/DC POWER
138S0854	138S0762		ALL	CAP,1UF,20%,4V,X6S,0201	=> GRAPHICS (GPU)
138S00015	138S0777		ALL	CAP,10UF,20%,16V,X6S,0603	=> CPU&CHIPSET (CPU POWER) PLATFORM POWER (G3H_3V3.GPU)
138S0759	138S0762		ALL	CAP,10UF,20%,16V,X6S,0603	=> GRAPHICS (GPU)
138S0676	138S0691		C4100	CAP,22UF,20%,6.3V,X5R,0603	=> CAMERA

132S0325	132S0291		CA115,CA145	CAP,3300PF,10% 50V,X7R,0402	=> PLATFORM POWER (GPU)
131S00144	131S0825		C7818,C7831	CAP,CER,NP0/C0G,560PF,+/-2%	=> CPU&CHIPSET (CPU POWER)
152S1499	152S0864		L4102	IND,47NH,2%,1.3OHM,0402	=> SD CARD
107S00015	107S00011		R5400	RES,MTL FLM,0.00075OHM,1%	=> SENSORS(V/I)
107S0255	107S0240		ALL	RES,MTL FILM,.001OHM,1%	=> SENSORS(V/I)
107S0251	107S0249		R5520	RES,SENSE,0.002OHM,1W	=> SENSORS(V/I)
155S00115	155S00114		L3700	FERR BD,10 OHMZ,25%,8A	=> SSD
155S0659	155S0382		L5100	FLTR,EMI,FERR BD,30 OHMS	=> SMC
155S00007	155S0667		L3100/3200	FLTR,COMMON MODE,90 OHM	=> USB-C
155S0935	155S0667		L3100/3200	FLTR,COMMON MODE,90 OHM	=> USB-C
197S0588	197S0591		Y1101	XTAL,24.000MHZ,20PPM,9.5PF	=> CPU&CHIPSET (PCH)
197S0590	197S0591		Y1101	XTAL,24.000MHZ,20PPM,9.5PF	=> CPU&CHIPSET (PCH)
353S00711	353S2073		ALL	IC,INA210,CURRENT SENSE	=> SENSORS(V/I)
					=> SENSORS(V/I)
353S00712	353S2216		ALL	IC,INA211,CURRENT SENSE	=> SENSORS(V/I)
138S0684	138S0660		C3832	CAP,CER,10UF,20%,10V,X5R	=> HDD
353S00854	353S4342		U5100	IC,REF3330,V-REF,3.0V,5MA	=> SMC
353S4376	353S3384		U3800,U5700	IC,LMV331,COMP,OPEN COL,2.7V	=> HDD & PSU PCC
376S0977	376S00078		Q0600,Q5127	XSTR,FET,N-CH,DUAL,20V,	=> CPU & SMC
376S1129	376S00078		Q0600,Q5127	XSTR,FET,N-CH,DUAL,20V,	=> CPU & SMC
376S0972	376S00075		ALL	MOSFET,N-CH,30V,100MA	=> BT.SMC,USB,CAMERA,AUDIO
376S0855	376S00074		ALL	MOSFET,DUAL N-CH30V,100MA	=> HDD & SMC
132S00091	132S0429		C8453,C8492	CAP,CER,0.022UF,10%,50V,	=> PLATFORM POWER
371S00096	371S0595		DZ5550	DIODE,ZENER,3.0V,3.22V	=> SENSORS(V/I)
376S00167	376S0910		ALL	MOSFET NCH,30V,11.8MOHM	=> HDD & PLATFORM POWER
371S00095	371S0567		D6020	DIODE,SW,100V,.25A,2P	=> FAN
311S00116	311S0324		U2070,U2075	74LVC2G34GF/S500	=> CPU&CHIPSET (PCH)
353S00992	353S2541		ALL	IC,OPA348,OP-AMP	=> SENSORS(V/I)
353S00993	353S2541		ALL	IC,OPA348,OP-AMP	=> SENSORS(V/I)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
311S0649	311S0541		ALL	2-INP AND GATE,SOT665	=> SMC
138S0748	138S0751		ALL	CAP,1UF,10%,6.3V,X7R,0402	=> GRAPHICS(GPU)
132S00057	132S0668		ALL	CAP,2200PF,10%,25V,X7R,0201	=> CPU&CHIPSET (CPU POWER)
132S00086	132S00045		C7006	CAP,X6S,0.022UF,6.3V,0201	=> CPU&CHIPSET (CPU POWER)
128S0368	128S0365		ALL	CAP,EL,AL,POLY,150UF,,BLK	=> PLATFORM POWER (3V3_S5/5V_S4)
138S00102	138S0773		CA130	CAP,1UF,10%,10V,X6S,0402	=> GRAPHICS (POWER)
138S0772	138S00013		C7561,CA661	CAP,2.2UF,20%,10V,X6S,0402	=> PLATFORM POWER
103S00185	103S0205		ALL	RES,1/16W,10KOHM,0.5%,0402	=> PLATFORM POWER (3V3_S5)
103S00186	103S0205		ALL	RES,1/16W,10KOHM,0.5%,0402	=> USB-C
103S00187	103S0205		ALL	RES,1/16W,10KOHM,0.5%,0402	=> USB-C
152S00403	152S00322		ALL	IND,0.68UH,20%,6.1A,	=> USB-C
138S0714	138S0713		ALL	CAP,10UF,6.3V,X5R,0402	=> USB-C
377S0178	377S00031		ALL	TVS,5.5V,7.3VBR,0.2PF,0201	=> USB-C
128S0311	128S0329		ALL	CAP,POLY,220UF,6.3V,35MOHM,82	=> USB-C
138S00084	138S00060		ALL	CAP,47UF,20%,6.3V,X5R,0603	=> USB-C
152S00358	152S00208		ALL	IND,1.0UH,20%,2.0A,135 MOHM	=> USB-C
126-0312	126-0306		ALL	CAP,EL,100UF,35V,0.54A,THD	=> BLC
106S00017	106S4171		ALL	RES,MF,30.1OHM,0.125W,1,805	=> BLC
138S0758	138S0742		ALL	CAP,X5R,22UF,,MURATA	=> BLC
371S00043	371S00019		D6921	DIODE,SCHOTTKY,30V,100MA	=> PLATFORM POWER
155S00155	155S0441		L6000	FLTR,EMI,FER BD,220OHM,1.4A	=> FAN
371S0684	371S0495		D1900	DIODE,DUAL SCHOTTYKY,30V	=> CPU&CHIPSET (PCH)
155S0660	155S0513		ALL	FERR BD,22 OHM,25%,1A	=> AUDIO

138S0641	138S0700		ALL	CAP,CER,X5R,2.2UF,10%,10V	=> AUDIO
138S0739	138S0706		ALL	CAP,CER,1UF,20%,10V,X5R	=> AUDIO
107S0184	107S0185		RT7000/1	THERMISTOR,NTC,200KOHM,1%,	=> CPU&CHIPSET (CPU POWER)
353S2970	353S2781		UA801	IC,SC4437,VOLT REF,3.3V	=> GRAPHICS(GPU)
107S00112	107S00006		RTA107	THERMISTR,NTC,47K OHM,1%	=> GRAPHICS(GPU)
311S00115	311S0302		UA102	IC,74LVC1G00,SNGL 2-NAND	=> GRAPHICS(GPU)
107S00041	107S00039		ALL	RES,MF,.0005 OHM,1%,150PPM	=> POWER & SENSORS
152S00388	152S00182		L6961	IND,PWR,2.2UH,20%,4.5A,43MOH	=> POWER
353S00831	353S00519		Q71X,72X,78X	IC,SIC621,DR MOS,IMVP-8,60A,	=> CPU&CHIPSET (CPU POWER)
107S0240	107S0255		R7800	RES,SENSE,0.001OHM,1W,4-TERM	=> CPU&CHIPSET (CPU POWER)
116S00007	116S00006		R2981	RES,MTL FILM,0 OHM,TIGHT TOL	=> USB-C
197S00046	197S00036		Y3000	XTAL,25MHZ,25PPM,20PF,50OHM	=> USB-C
197S00047	197S00036		Y3000	XTAL,25MHZ,25PPM,20PF,50OHM	=> USB-C
197S00048	197S00036		Y3000	XTAL,25MHZ,25PPM,20PF,50OHM	=> USB-C
353S00769	353S4398		U3510	IC,TPS22966,LOADSW,6A,5.5V	=> WIRELESS
155S0588	155S0367		ALL	FLTR,EMI BEAD,120OHM,3A	=> USB-A
138S0703	138S0648		C6302, C6305	CAP,CER,X5R,4.7UF,20%,6.3V	=> AUDIO
152S0098	155S0137		ALL	IND,CHP EMIFIL,1000OHM,0.2A	=> AUDIO & CAMERA
152S00543	152S00484		L8900	IND,MLD,0.24UH,20%,18A	=> GRAPHICS(GPU)
377S0184	377S00011		ALL	DIODE,TVS,BIDIR,5.3V,0.2PF	=> USB-A
138S1103	138S0719		C6309, C6310	CAP,CER,4.7UF,20%,10V,X5R	=> AUDIO
155S00076	155S0546		C6309, C6310	FER.BD,600OHM,25,300MA,	=> ETHERNET & CAMERA

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PAGE TITLE

ALTERNATE PARTS

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DRAWING NUMBER 051-01506 SIZE D

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PAGE 115 OF 121

SHEET 101 OF 102

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X646 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT,BGA_VRAM	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	0.070 MM	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

_____	Top	Signal	0.5 oz (Cu plated)
_____		Prepreg	0.071 MM
_____	2	Plane	1 oz
_____		Core	0.101 MM
_____	3	Signal	0.5 oz
_____		Prepreg	0.115 MM
_____	4	Plane	1 oz
_____		Core	0.076 MM
_____	5	Signal	0.5 oz
_____		Prepreg	0.380 MM
_____	6	Plane	1 oz
_____		Core	0.076 MM
_____	7	Plane	1 oz
_____		Prepreg	0.380 MM
_____	8	Signal	0.5 oz
_____		Core	0.076 MM
_____	9	Plane	1 oz
_____		Prepreg	0.115 MM
_____	10	Signal	0.5 oz
_____		Core	0.101 MM
_____	11	Plane	1 oz
_____		Prepreg	0.071 MM
_____	Btm	Signal	0.5 oz (Cu plated)

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

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PAGE TITLE: X646 RULE DEFINITIONS

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BRANCH:

PAGE: 121 OF 121

SHEET: 102 OF 102

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